

**PROPOSED (B+G+2+R) COMMERCIAL OFFICE BUILDING
ON PLOT NO. 2514674 C2_O/2 AT MIRDIF, DUBAI.
FOR MR. ALIAHMED MOHAMMAD ALHABBAI**



TENDER DOCUMENT

VOLUME IV

MEP SPECIFICATIONS

PART 2 – PLUMBING SPECIFICATIONS

JULY 2025

TABLE OF CONTENTS
DIVISIONS 01 - MECHANICAL SERVICES

S. NO.	SECTION	DESCRIPTION
1	01 00 00	General Requirements (Mechanical)
2	01 88 45	Noise & Vibration Control Requirement

TABLE OF CONTENTS
DIVISIONS 21 & 22 - MECHANICAL SERVICES

S. NO.	SECTION	DESCRIPTION
DIVISION – 21 : FIRE SUPPRESSION		
1	21 10 00	Water Based Fire Suppression Systems
2	21 13 13	Wet Pipe Sprinkler System
3	21 13 39	Foam Water Sprinkler System
4	21 22 10	Clean Agent Fire Suppression Systems – FM 200 System
5	21 30 00	Fire Fighting Pumps
DIVISION – 22 : PLUMBING		
6	22 05 16	Expansion Fittings And Loops
7	22 05 17	Sleeves for Plumbing
8	22 05 19	Meters And Gages for Plumbing
9	22 05 29	Hangers And Supports for Plumbing
10	22 05 53	Identification for Plumbing
11	22 05 78	Plumbing Specialties
12	22 05 80	Cleaning of Piping Systems
13	22 09 13	Valves for Plumbing
14	22 10 00	Plumbing Pipe work
15	22 11 23	Domestic Water Pumps
16	22 12 00	Water Storage Tanks
17	22 13 19	Drainage Specialties
18	22 14 29	Submersible Sump Pumps
19	22 32 00	Domestic Water Filtration and Softening
20	22 33 30	Electric Water Heater
21	22 40 00	Plumbing Fixtures
25		Vendors List

MECHANICAL SECTION: 01 00 00
GENERAL REQUIREMENTS (MECHANICAL)

PART -1 GENERAL

1.1 RELATED SECTIONS

- 1.1.1 The General and Supplementary Conditions are a part of the requirements for the Work under this Division of the Specifications.
- 1.1.2 This section applies to all sections in Divisions 21, 22, 23 & 25.

1.2 SUMMARY

- 1.2.1 The Specifications require the furnishing and installing of all items, including every article, device or accessory (whether or not specifically called for by item) reasonably necessary to facilitate each system's functioning as indicated by the design and the equipment specified. Elements of the Work include, but are not limited to, materials, labour, supervision, supplies, equipment, transportation, storage, utilities and all required permits, licenses, fees and royalties.

1.3 QUALITY ASSURANCE

- 1.3.1 The mechanical and public health installation shall be of the highest quality in accordance with the standards as specified and shall conform to:
 - 1.3.1.1 Local Code/Regulations
 - 1.3.1.2 Rafal specifications and standard details drawings
 - 1.3.1.3 Local Authorities Regulations related to Mechanical Services Installations.
 - 1.3.1.4 Local Civil Defence Regulations.
 - 1.3.1.5 Health and Safety at Work Act.
 - 1.3.1.6 Relevant British, European, American and International Standards and/or Codes of Practice for materials and equipment and method of installation.
All materials shall bear 3rd party assurance stamp such as UL, FM, LPC, British standards Kitemark, or other marks stated within the specs.
 - 1.3.1.7 Chartered Institution of Building Services Engineers guides and memos.
- 1.3.2 Should any change in Drawings or Specifications be required to comply with governmental regulations, the Contractor shall notify the Construction Administrator prior to execution of the Work. The Work shall be carried out according to the requirements of such code in accordance with the instruction of the Engineer and at no additional cost to the Employer.
- 1.3.3 The provisions of Standards, Codes, Laws, Ordinances, etc., shall be considered minimum requirements. In case of conflict between their published requirements, the Construction Administrator shall determine which is to be followed and his decision shall be binding. Specific requirements of this Specification and/or the Drawings, which exceed the published requirements, shall take precedence over them.

- 1.3.4 Execute the Work in strict accordance with the best practices of the trades in a thorough, substantial, workmanlike manner by competent workmen. Provide a competent, experienced, full-time Superintendent who is authorized to make decisions on behalf of the Contractor.

1.4 DEFINITIONS

- 1.4.1 Use the following definitions for discernment within the Drawings and Specifications.
- 1.4.1.1 "PROVIDE" means to supply, purchase, transport, place, erect, connect, test and turn over to Employer, complete and ready for regular operation, the particular work referred to.
- 1.4.1.2 "INSTALL" means to move from property line, set in place, join, unite, fasten, link, attach, set up or otherwise connect together before testing and turning over to Employer of equipment supplied under another division. Installation to be complete and ready for regular operation, the particular work referred to.
- 1.4.1.3 "FURNISH" means to supply all materials, labour, equipment, testing apparatus, controls, tests, accessories and all other items customarily required for the proper and complete application for the particular work referred to.
- 1.4.1.4 "WIRING" means the inclusion of all wire ways, fittings, conductors, connectors and outlet boxes, connections, joints and all other items necessary and/or required in connection with such work.
- 1.4.1.5 "CONDUIT" means the inclusion of all fittings, hangers, supports, sleeves, etc.
- 1.4.1.6 "AS DIRECTED" means as directed by the Construction Administrator, or his representative.
- 1.4.1.7 "CONCEALED" means embedded in masonry or other construction, installed behind wall furring or within double partitions, or installed within hung ceilings.

1.5 ABBREVIATIONS

- | | | |
|--------|--------|---|
| 1.5.6 | CD | Civil Defence |
| 1.5.8 | BS | British Standards |
| 1.5.9 | ASHRAE | American Society of Heating, Refrigeration and Air Conditioning Engineers |
| 1.5.10 | CIBSE | Chartered Institution of Building Services Engineers |
| 1.5.11 | BMS | Building Management System |
| 1.5.12 | HVCA | Heating and Ventilation Contractors Association |
| 1.5.13 | NFPA | National Fire Protection Association |
| 1.5.14 | ANSI | American National Standards Institute |
| 1.5.15 | AMCA | Air Moving and Conditioning Association |

1.5.16	ASTM	American Society for Testing and Materials
1.5.17	SMACNA	Sheet Metal and Air Conditioning Contractors National Association
1.5.18	ARI	American Refrigeration Institute
1.5.19	ASME	American Society of Mechanical Engineers
1.5.20	AWS	American Welding Society
1.5.21	AWWA	American Water Works Association
1.5.22	MSS	Manufacturers Standardization Society of Valve and Fitting Industry
1.5.23	IPC	International Plumbing Code
1.5.24	ASPE	American Society for Plumbing Engineers
1.5.25	UL	Under Writers Laboratory
1.5.26	EN	Euro Norms
1.5.27	IEE	Institution of Electrical Engineers
1.5.28	IEC	International Electro Technical Commission
1.5.29	ISO	International Organization of Standardization
1.5.30	C	Degree Celsius
1.5.31	F	Degree Fahrenheit
1.5.32	DB	Dry Bulb
1.5.33	WB	Wet Bulb
1.5.34	Kg	Kilogram
1.5.35	Kpa	Kilopascals
1.5.36	Pa	Pascals
1.5.37	Mm	Millimeter
1.5.38	NC	Noise Criteria
1.5.39	RPM	Revolution per minute
1.5.40	SS	Stainless Steel
1.5.41	W.G.	Water gauge

PART 2 - PRODUCTS

1.0 GENERAL.

- 1.1 The Instructions to Tenderers, Form of Tender, Main Conditions of contract, Note on Pricing, the Preliminary Section and any Supplementary Conditions of Contract apply to and are part of this Division and all subsections.
- 1.2 Sections of this Services Specification which relate to other Divisions are not intended to supersede the specifications of these divisions, but to guide the Contractor in tendering and performance. In no event shall a quality less than that indicated here be acceptable.
- 1.3 The precedence of various Documents and Drawings shall be as determined in the General Contract Documents, subject to the limitation that any item shown or required in any of the documents or drawings shall be considered by the Contractor to be included in all Documents.
- 1.4 The scope of works includes design and build by EPC contract including detailed design of the schematic drawings provided by the consultant, supply of material & equipment, delivery to site, installation, testing, commissioning, demonstrating and staff training of the various complete systems outlined in this specification and on the drawings and documentation including liaison with the Main Contractor, Authorities, Master Plan Developer and their representatives and Consultants as necessary for the duration of the works and the defects liability period.
- 1.5 The MEP contractor will be responsible for reaching an agreement on their Program with the Main Contractor and coordinating every aspect of their design, submissions, approvals as well as installation, testing, commissioning and handover works with them.
- 1.6 Wherever the term authority is indicated, it shall mean statutory authorities, the master plan developer or their representatives and services provider.

2.0 INTENT.

- 2.1 Carry out Detailed design drawings as per the schematic design drawings part of EPC contract, and accept responsibility for proper performance of all mechanical and plumbing systems in accordance with relevant standards and codes and design parameters listed hereinafter, AND in accordance with Rafal specifications and standard details drawings.
- 2.2 Provide all items, articles, materials, samples, operations, sundries, labor, supervision, guarantees, allowances for overhead and profit, etc., to achieve a fully functionable and acceptable system.
- 2.3 Consider the specifications as an integral part of the work together with the drawings. Consider any item or subject omitted from one, but mentioned or reasonably implied on the other as properly and sufficiently indicated and provide the same under the work of this division.
- 2.4 The Contractor is responsible for developing his own take off of materials and is to make this available as requested to the Consultant.

3.0 EXAMINATION OF SITE AND DRAWINGS.

- 3.1 Visit the site of the proposed works and obtain all information as to existing conditions and limitations and all proposed works on adjacent sites and in adjacent areas which might affect the works on this site, whether by Private Individuals or by Government Authorities or others.
- 3.2 Examine the documents including the Specifications and Drawings of all other Divisions before bidding and again before commencing any portion of the works.
- 3.3 Neither the Owner nor the Consultant will be responsible for any claim for extra work or expense resulting from the failure of the Contractor to be fully aware of Site Conditions, Drawing implications, etc.

4.0 PERMITS, FEES AND INSPECTIONS.

- 4.1 Arrange for inspection of all work by the Authorities having jurisdiction over the works. This is to occur on an on-going basis throughout construction to avoid delays at the termination of the project. On completion of the work present to the Consultant, for the Owner, final unconditional approval certificates of the Inspecting Authorities and the master plan developer or their representatives.
- 4.2 Pay fees / charges as levied by statutory authorities for inspections, approvals, services connections, rerouting of existing services, etc. at no additional cost to contract.
- 4.3 Before commencing any work on site, submit the required number of copies to the relevant Authorities (after coordination with present building Contractor) for checking and approval. Follow the progress of such drawings to permit the timely approval of them by the Authorities. Comply with any changes requested by the Authorities, but notify the Consultant immediately of any such change for proper processing of these requirements. The Contractor, working on a regular basis with the Authorities is expected to know which changes are likely at the time he bids the job, and no extra cost will be allowed for such changes.

5.0 CONTRACT DRAWINGS

- 5.1 The drawings for services works are performance drawings, diagrammatic, and intended to convey the scope of work and indicate general arrangement and approximate locations of apparatus, fixtures, pipe and duct runs, etc. The drawings do not intend to indicate architectural or structural details, nor do they show any fabrication or installation details.
- 5.2 Do not scale drawings. Obtain accurate dimensions to structure and architectural items from drawings of those trades. Confirm by site measurement. Verify location and elevation of all services (Water, Electrical, Telephone, Sanitary, Storm Drainage, Gas, etc.) before proceeding with the work.
- 5.3 Make at no extra cost, any changes or additions to materials, and/ or equipment necessary to accommodate structural conditions (pipes or ducts around beams, columns, etc.)
- 5.4 Alter, at no additional cost, the location of materials and/ or equipment as directed, provided that the changes are made before installation and do not necessitate additional material.

- 5.5 Install all ceiling mounted components (Diffusers, grilles, detectors, light fixtures, emergency lights, fire detectors, loudspeakers, camera points, etc.) in accordance with the reflected ceiling drawings which are to be prepared by the Contractor and coordinated with all trades. These must be submitted for approval and be approved before any work commences on site.
- 5.6 Leave space clear and install all work to accommodate future materials and/or equipment as indicated and/or supplied by another division of work of the contract. Install all pipe runs, conduit runs, cable trays, etc., to maintain maximum headroom and clearances, and to conserve space in shafts and ceiling spaces and under floors, and to provide adequate space for service and maintenance.
- 5.7 Confirm on the site the exact location of outlets and fixtures. Confirm also location of outlets and fixtures provided by any other division of work under the contract.

6.0 CONSTRUCTION DRAWINGS.

- 6.1 Prepare fully coordinated drawings in conjunction with all trades concerned, showing sleeves and openings for all passages through structure and all insert sizes and locations.
- 6.2 Prepare composite construction drawings in color fully dimensioned, of piping and equipment in tunnels, shafts, mechanical equipment rooms and areas, and all other critical locations to avoid a conflict of trades. Composite drawing shall include all MEP services in coordination with the structure and false ceiling heights given in the Architectural drawings.

Base equipment drawings upon shop drawings and include but do not necessarily limit to, all details pertaining to access, cleanouts, tappings, sleeves, electrical connections, drains, location and elevation of pipes, ducts, conduits, etc., obtained from consultation with, and agreement of, all trades involved.

The location of final fix items such as supply / return air diffusers and sprinkler heads shown in the contract drawings are indicative only. Contractor shall co-ordinate with the reflected ceiling plans and Interior design drawings and agree the actual location in relation to other items such as light fittings, smoke detectors, speakers etc. Final location should be based on the approved reflected ceiling plans prepared by Contractor and obtain approval from the Architect / Engineer.

- 6.3 Prepare drawings of equipment bases, pump pits, anchors, inertia slabs, floor and roof curbs, wall openings, trenches, pertaining to mechanical work.
- 6.4 Prepare all drawings to scale as agreed with the Consultant. Forward these drawings, approved by all trades concerned to the Consultant for his records. Provide transparencies and provide printed copies in a number as specified elsewhere in the Contract but not less than four sets.
- 6.5 Bind one complete set of construction drawings showing "as built" conditions in each operating and maintenance instruction manual. The extent of these drawings will be indicated to the Contractor in advance by the Consultant.

7.0 SHOP DRAWINGS & MATERIAL SUBMITTAL

- 7.1 Submit shop drawings and samples for materials and equipment as listed in this and in each subsequent section.

Provide transparencies wherever possible and provide printed copies in a number as specified elsewhere in the contract documents. The Consultant may retain the sample of each item at his discretion until the completion of the contract.

- 7.2 Present a schedule of shop drawings and material submittal within two weeks after award of the contract, indicating the anticipated date when the drawings will be submitted for review. Assume full responsibility for timely submission of all drawings. Allow a minimum of three weeks for the Consultant's review. At time of submission indicate order deadlines and cost implications, etc., otherwise the Consultant will advise the Owner that cost and completion date will be unaffected.
- 7.3 Drawings production and presentation is a Contractual matter and any delay on the Contractor's behalf in making these submissions will be considered a Contractual delay and may be subject to Contractual penalties and other remedies as determined by the Consultant.
- 7.4 The Consultant will only consider shop drawings bearing the stamp of the Contractor and all Subcontractors involved. Check for all pertinent information such as physical dimensions, make, performance, electrical characteristics and indicate the intended use and location before submitting these drawings. Use reference symbols or enumeration to correspond to the design drawings.
- 7.5 Assume responsibility for accuracy of equipment dimensions related to space available, accessibility for maintenance and service, compliance with inspection authorities codes. Ensure that shop drawings indicate the shipping and working weights of all equipment.
- 7.6 The submission of samples will be subject to the same procedure as that of shop drawings. One set of such samples shall be required to be brought to site and kept there after approval till substantial completion.
- 7.7 The Consultant will mark the drawings "resubmit specified item", "rejected", "no exception taken", or "make correction noted". In the last case all revisions will be clearly marked on the returned print and corrected prints may be issued for manufacture and construction.

Make the revisions shown on the "make corrections as noted" prints onto the tracings as soon as practicable and forward copies to the Consultant for his records.
- 7.8 The Consultant is not responsible for any delays caused by the inadequacy of the Contractor's drawings or his failure to obtain initial or subsequent approval. Any time taken by the Contractor to obtain approval after the originally scheduled date will be considered as a delay to the contract caused by the Contractor.
- 7.9 When drawings are marked "resubmit specified item", the Contractor is to resubmit the item as originally specified or as may be determined to be equivalent by the Consultant. The Consultant is the sole arbiter of whether any item is satisfactory or equivalent.
- 7.10 When drawings are marked "rejected" a complete resubmission of the particular drawing is necessary, subject to the same conditions as outlined above.
- 7.11 The Consultant's review shall not relieve the Contractor from responsibility for deviations from the Contract documents, unless he has, in writing, called the Consultant's attention to such deviations at the time of submission of drawings. The Consultant's review shall be construed to apply to, and only to, general arrangements and shall not relieve the Contractor from the entire

responsibility. Any approval by the Consultant shall be on the understanding that any item submitted shall be ordered with options and modifications to fully meet the specification. Any fabrication, erection, setting out or other work done in advance of receipt of stamped drawings shall be done entirely at the Contractor's risk and cost.

- 7.12 Furnish prints of the reviewed details to all other parties who may require them for proper coordination of their work, and furnish all information necessary for the work as a whole.
- 7.13 Obtain Manufacturers' installation directions to aid in the proper execution of the work. Submit two copies of such directions to the Consultant prior to installation, for use in inspecting the work.
- 7.14 Bind one complete set of checked shop drawings into each operating and maintenance manual.

8.0 RECORD DRAWINGS

- 8.1 As the job progresses, mark on one set of prints to accurately indicate the status of installed work. Have the white prints available for inspection at the site at all times, and present them for scrutiny at all job meetings. Transfer all information onto the set of transparencies. Obtain the Contractor's approval before forwarding these to the Consultant. Submit two sets of white prints to the Consultant for checking and approval.

Upon approval, submit one set of transparencies and five sets of white prints to the consultant for onward transmission to the client.

- 8.2 Show on the record transparencies the installed inverts of all services entering and leaving the building and the property. Dimension underground services at key points of every run in relation to structure and building. Record all elevations for underground services in relation to floor level of the building and give reference datums to Municipal benchmarks.
- 8.3 Indicate exact location of all services left for future work. Show and dimension all work embedded in the structure.

9.0 TEMPORARY SERVICE.

- 9.1 Do not use any of the permanent service facilities during construction, unless specific written approval is obtained from the Consultant and the Owner or where specifically allowed elsewhere in the Contract Documents.
- 9.2 The use of permanent facilities for temporary services will not affect, in any way, the commencement day of the period in which the correction after completion is the Contractor's responsibility. Such period will commence only when the overall project is completed and certificates of acceptance are issued.
- 9.3 In the event that mains power is not available or is not of sufficient capacity during construction, supply, install and operate diesel generator(s) of the correct capacity (which must be approved by the Consultant.)

Any generator must have full safety features and must be maintained regularly to ensure site power at all times.

- 9.4 The Contractor is to ensure that permanent services are operating within the show flat to allow its use for the Client sales teams at the earliest opportunity, this must be identified and agreed with the main contractor.

10.0 COOPERATION

- 10.1 Confer with all trades installing equipment which may affect the work of this division, and arrange equipment in proper relation with that equipment installed under all Divisions of the Contract.
- 10.2 Furnish all items to be built in by others, in time, complete with all pertinent information, commensurate with the progress of the work.
- 10.3 Store materials neatly and out of the way and clean up all refuse caused by the work daily.

11.0 PROTECTION AND STORAGE

- 11.1 Protect the buildings and structures from damage due to carrying out of this work.
- 11.2 Protect all mechanical, plumbing & electrical works from damage. Keep all equipment dry and clean at all times.
- 11.3 Cover all openings in equipment and materials. Cover all temporary openings in ducts and pipes with polyethylene sheets or caps until final connection is made.

The quality of such cover must be determined with due regard to how long it may be until final connection.

- 11.4 Be responsible for and make good any damages caused directly or indirectly to any walls, floors, ceilings, woodwork, brickwork, finishes, services, roads, gardens, etc.

12.0 EXISTING WORK AND EQUIPMENT

Before this Contractor undertakes work in any area he must prepare a list of deficiencies in that area which affect his works, or which could possibly be construed as being caused by himself if not noted. In the event that such deficiency list is not prepared, then he shall be deemed responsible for such deficiencies. Any list shall be brought to the attention of the Consultant forthwith.

13.0 SUPERVISION.

- 13.1 The Subcontractor will maintain at site, as necessary for the performance of the Contract, qualified personnel and supporting staff, with proven experience in erecting, testing, and adjusting projects of comparable nature and complexity.

The Contractor team is to be headed by a suitably qualified Engineer who will act as the MEP Project Manager, he is to be assisted by Senior Engineers for Electrical, Mechanical, Fire Alarm and Commissioning and covering each discipline as required and resident on the site full time supported by a suitable structure of supervision based on a standard hierarchy format.

- 13.2 Before commencing work the Contractor will submit details of the proposed Engineers and Supervisors, including copies of their Certificates. If in the Consultant's opinion the proposed

Engineers or Supervisors are not adequately qualified or are otherwise unacceptable, the onus is on the Contractor to submit alternates until such approval is given.

- 13.3 Where the Contractor's or Subcontractor's staff is, or becomes during the Contract deficient in the choice of adequate or acceptable personnel, the Contractor is to remedy the situation by appropriate measures.
- 13.4 Approval of the Contractor's or Subcontractor's Engineers or Staff shall in no way prevent the withdrawal of that approval at any time during the Contract should the Consultant so desire. In the event of such disapproval then the Contractor will be required to rectify the position as stated above within 14 days.
- 13.5 In the event of any negligent or severely detrimental behavior the Consultant has the right to order the removal from site of any Engineer, Supervisor, or worker on a "forthwith" basis.

The Contractor will remedy the situation on a similar forthwith basis and in any event will not be allowed to proceed with any area of work which as a result is left without proper supervision or technical backup until such time as the Consultant is again satisfied by the measures taken.

14.0 SCHEDULING.

- 14.1 In the event that a completion date has been set before tendering it will be the obligation of the Contractor to provide a schedule adequate for all trades to meet the completion date.
- 14.2 If no completion date is specified it will be the obligation of the Contractor to provide to the Consultant a proposed schedule of work including mechanical trades, within 21 days of award of Contract. This schedule will be either approved by, or negotiated with, the Consultant.

15.0 WORK TO CONTINUE.

- 15.1 Unless specifically instructed otherwise by the Consultant the works are to continue to completion in the most expeditious manner.

16.0 STANDARDS AND AUTHORITIES.

- 16.1 All equipment and installations shall be in accordance with local Codes and Regulations. In general, all equipment and installation shall also be in accordance with :

- Uniform plumbing code of USA.
- National Plumbing Code
- Institute of Plumbing (UK)
- ASTM
- ISO

- AWWA
- ASME
- UL Laboratories

16.2 Obtain approval of shop drawings from Authorities (Including the master plan developer or their representatives) having jurisdiction including but not limited to Drainage Department, Municipality, Ministry of Water and Electricity, Fire Brigade, Department of Health, and local telephone authority. All costs towards such approvals to be borne by the contractor.

17.0 WORKMANSHIP.

17.1 Install equipment, ductwork, conduit and piping in a workmanlike manner to present a neat appearance and to function properly to the satisfaction of the Consultant. Install ducts and pipes parallel and perpendicular to the building planes. Install all piping and ductwork concealed in chases, behind furring, or above ceiling, except in unfinished areas. Install all exposed systems neatly and group to present a neat appearance.

17.2 Install all gauges, thermometers, etc., to permit easy observance.

17.3 Install all equipment and apparatus which requires maintenance, adjustment, or eventual replacement with due allowance for this.

17.4 Install control valves to guarantee proper sensing. Shield elements from direct radiation and avoid placing them behind obstructions.

17.5 Install all panels and boards, etc., to permit easy operation.

17.6 Include in the work all requirements of Manufacturers as shown on their drawings.

17.7 Replace all work unsatisfactory to the Consultant without extra cost and to the standard required by the Consultant. This applies to any item which is found to be defective in service during the maintenance period, or extended maintenance period as appropriate.

17.8 Where height dimensions are given, the installation shall be reasonably expected to be within 1/8" (3mm) of the design dimensions. Where room centre or proportion dimensions are not shown these shall be reasonably expected to be located on site to within 1/2" (13mm) of actual position.

17.9 Items displaying a horizontal dimension or edge must be plumbed with a level, and must be to the Consultant's satisfaction. Any apparent discrepancy between "level" electrical items and adjacent items MUST be reported immediately to the Consultant.

18.0 MATERIALS AND EQUIPMENT.

18.1 All materials used on the project shall be brand new and of high quality. Obtain approval of all manufacturers from the Consultant and the Owner.

18.2 "New" is defined as newly manufactured, "state of the arts," tested and proven item of equipment. Items which have been held in stock for any extended period of time by either the Manufacturer or the Supplier will be rejected.

- 18.3 All materials used on this project shall bear the third party quality assurance stamp like British Standard kitemark, UL, FM, NFPA and/or other quality assurance authority as stated herein elsewhere.

19.0 CLEANING.

- 19.1 Each day as the work proceeds and on completion, clean up, vacuum, wipe down, remove labels and markings from equipment and remove from the premises all rubbish, surplus material, equipment, machinery, tools, scaffolds, and other items used in the performance of the work. Clean out dirt and debris and leave the buildings clear and free from rubbish, waste, dust, dirt, debris and sand. Equipment is to be spotless and in 'Brand New' condition at handover complete with new or thoroughly cleaned filters and in a condition acceptable to the Consultant.
- 19.2 Where electrical items form part of the visible finish in the rooms, protect from over-painting, etc. and give all items a final cleaning before handing over of the project.

20.0 ACCESSIBILITY.

- 20.1 Each item of equipment shall be located so as to be accessible for maintenance or repair without removing adjacent structures, equipment, piping, ducts, or other materials. For the large Air Handling units the Contractor shall ensure that these can be built up on site from components which can be taken into the area. Any openings, framing or special lifting equipment which may be required shall be included by the General Contractor.
- 20.2 Any item of equipment needing maintenance shall be located so as to be accessible for maintenance or repair without removing adjacent structures, equipment, piping, ducts, or other materials.
- 20.3 Cleanouts shall be located to permit rodding of all drain lines. These shall be located wherever possible external to occupied areas, and to minimize spillage problems during rodding.

21.0 CUTTING AND PATCHING.

- 21.1 Inform all other Divisions in time concerning required openings. Obtain the approval of the Consultant before doing any cutting.
- 21.2 Obtain the approval of the Consultant before doing any cutting. Supporting members of any floor, wall or the building structure shall only be cut and in such a manner as approved by the Consultant. All reinstatement work must be done to the same standard as the original work.

22.0 INSERTS, SLEEVES, ESCUTCHEONS AND CURBS.

- 22.1 Under this section supply necessary sleeves and other inserts to other trades.
- 22.2 Use only factory made, threaded or toggle type inserts as required for supports and anchors, properly sized for the load to be carried. Place inserts only in portions of the main structure and not in any finishing material.

- 22.3 Use factory made expansion shields where inserts cannot be placed, but only where approved by the Consultant and for light weights.
- 22.4 Do not use powder activated tools except with the written permission of the Consultant.
- 22.5 Supply and locate all inserts, holes, anchor bolts, and sleeves in good time when walls, floors, and roof are erected.
- 22.6 Pass insulation unbroken where pipe or duct is insulated. Size sleeves to provide adequate clearance all around.
- 22.7 Use the following materials for pipe sleeves:
- a) Through all interior walls above grade use uPVC Class D pipes, machine cut, flush with finished structure. For non metallic pipe crossing in fire wall use steel pipe (medium).
 - b) Through all exterior walls above grade use uPVC Class D pipes, machine cut, flush with finished structure inside and to suit flashing on outside.
 - c) Through all exterior walls below grade and all other waterproof walls use uPVC Class D sleeves, machine cut. Check flashing specification for further details.
 - d) Through all waterproof floors, janitor's closets, mechanical rooms, kitchens, roofs, use uPVC Class 150 sleeves, machine cut. Extend sleeves 100 mm above finished floor upwards and cut flush with underside of floor. Refer to flashing detail through waterproof floors.
 - e) Provide 100 mm high, 100 mm wide watertight concrete curbs with 20 mm chamfered edges around all pipes passing through waterproof floors except where furred in. Read Concrete specification before proceeding with curbs. All gaps to be filled up with waterproof type sealant.
 - f) Where uPVC pipes pass through fire rated walls/partitions, provide fire rated sleeves.
- 22.8 Pack all sleeves between the insulated pipe and the sleeve or where un-insulated between the pipe and the sleeve with loose fibreglass insulation. Seal the annular space as follows:
- a) For all horizontal sleeves in exposed areas, use a seal equal or better fire rated than the wall to be sealed.
 - b) For all vertical sleeves through roofs, janitor's closets, equipment rooms, use permanently resilient silicone based sealing compound, non- inflammable and waterproof.
- Ensure that the seal is compatible with floor and ceiling finishes. Check the room finishing schedules for further details and clarify if necessary with the Consultant.
- c) Cover exposed floor and wall pipe sleeves in finished areas with satin finish chrome or nickel plated solid brass or with satin finished stainless steel escutcheons with non-ferrous set screws. Split cast plates of the screw locking type may not be used. Do not use stamped steel friction type split plates

Through roofs, provide curbs and sleeves as shown on drawings and to suit flashing requirements.

22.9 Use the following sleeving for ducts:

For rectangular duct openings through walls and floors a removable hardwood box-out shall be provided of the required size. soft wood or plywood will not be acceptable. Hardwood shall be further protected with anti termite paint / solvent.

Through fire walls, build fire dampers into wall, or make detailed fixing in accordance with DW 144 / SMACNA Standards.

Through floors where ducts are not furred in or enclosed in a duct shaft, provide 100 mm high and 100 mm wide watertight concrete curbs, with 25 mm chamfered edges all around. Extend sleeves where used flush to top of curb. Read Concrete Specification before proceeding. Through floors where duct is enclosed in a duct shaft or furred in, provide the watertight curbs at the extreme top and bottom only.

22.10 After ducts are installed, pack the opening and seal the packing as follows:

- a) Use fiberglass insulation for packing except through curbed concrete floors where asbestos braid must be used.
- b) Seal the asbestos braid packing in openings through floors with permanently resilient silicone base non inflammable waterproof compound. Press duct supports firmly down into caulking before bolting it down to curb.
- c) Through all vertical walls seal the fiberglass packing using permanently resilient silicone based sealant.

22.11 Brace duct sleeves and box-outs to retain their position and shape during the pouring of concrete and other work.

22.12 Provide bracing for each duct at every passage through structure to prevent sagging.

22.13 Cover exposed pipe work, gullies, duct sleeves and openings in exposed areas. Use 100 mm galvanized steel escutcheons in the form of a duct collar or screw fixed pipework cap or similar approved protection. Over curbs extend the collar 30 mm down the side of the curb, similar to counter flashing. Fix collar in place with Cadmium plated screws.

23.0 ACCESS PANELS AND DOORS.

23.1 Install all concealed mechanical equipment requiring adjustment or maintenance in locations easily accessible through access panels or doors. Install systems and components to result in a minimum number of access panels. Indicate access panels on as-built drawings.

23.2 Provide the respective Division of work with panels, doors or frames, complete with all pertinent information for installation.

Ensure that access doors are installed in a manner to match the building grids where applicable

- 23.3 Prepare detail drawings showing location and type of all access doors in coordination with other trades before proceeding with installation and hand these to the Contractor to obtain approval.
- 23.4 Size all access doors to provide adequate access and commensurate with the type of structure and Architectural finish. Should it be necessary for persons to enter, provide a minimum opening of 600 x 450 mm.
- 23.5 Ensure proper fire rating of access doors in fire separations
- 23.6 Lay-in type ceiling tiles, if properly marked may serve as access panels.
- 23.7 Provide panels in glazed tile walls of 2.6mm thick 304 alloy stainless steel, no. 4 finish, with recessed frame secured with stainless steel, countersunk, flush-headed screws.
- 23.8 Provide panels in plaster surfaces with dish shaped door and welded metal lath, ready to take plaster. Provide a plastic grommet for door key access.
- 23.9 Provide other access doors of welded 2.6 mm thick steel, flush type with concealed hinges, lock and anchor straps, complete with factory prime coat. Submit details to Consultant for approval.

24.0 FLASHING.

- 24.1 Flash all mechanical and electrical parts passing through or built into a roof, an outside wall, or a waterproof floor.
- 24.2 Provide uPVC sleeves with puddle flanges for services crossing water proofed walls in basement.
- 24.3 Provide stainless steel puddle flanged connections where pipework penetrates walls of the water storage tank.
- 24.4 All flashing shall suit roof angle and shall extend a minimum of 400 mm on all sides. Leave the flashing as directed by the Contractor for him to build into the roofing system to render a completely watertight connection.
- 24.5 Provide counter flashing on all stacks, ducts, and pipes passing through roofs to fit over flashing or curb.
- 24.6 Flash pipes through waterproof floors as per approved detail.
- 24.7 Provide pipes and sleeves passing through outside walls with lead or copper flashings and as directed by the Consultant. All visual aspects of such sleeves to be approved by the Consultant.
- 24.8 Pay special attention to the waterproofing conditions of basements and walls and floors that may exist. Cooperate at all times with the waterproofing trade and do not cut or destroy any waterproofing seal without the consent of the waterproofing trade.

Provide piping sleeves passing through waterproof walls with asphalt roofing felt wrapped around so as to leave a 25 x 50 mm recess on both sides of the wall. These recesses and the space between pipe and sleeve will be caulked by the General Contract, Moisture protection, or Waterproofing, Division.

25.0 GENERAL WELDING REQUIREMENTS.

- 25.1 All welding shall be generally in accordance with ANSI B31. This will be modified where appropriate for other materials and may be relaxed or varied by order of the Consultant, provided that the Contractor has made a reasonably comprehensive request for an alternate.
- 25.2 Tack welds shall be performed by fully qualified welders and all tack welds shall be of a length equal to twice the pipe thickness and shall fully penetrate the pipe walls.
- 25.3 Where welding is carried out in the proximity of inflammable materials special precautions shall be taken to prevent risk of fire or other damage to the building fabric.
- 25.4 Where oxyacetylene cutting equipment or any welding plant is being used by an operative for any of the works defined in the contract documents, then fire extinguishers shall be supplied and carried as part of the equipment. The operators of cutting and welding equipment shall be trained in the use of the fire extinguishers which they carry and all extinguishers shall be fully charged and ready for use. In all cases, extinguishers shall be positioned immediately adjacent to the position where cutting and welding is being carried out and shall be readily accessible for use in the event of an emergency.
- 25.5 All accommodation, benches, tools, welding plant, acetylene, oxygen or electricity, filler rods and electrodes, which are necessary for installations where welding is required shall be provided as part of this contract.
- 25.6 Welding shall not be done at a temperature of 5 degrees Centigrade or below unless the parent metal is preheated by torches or other approved means until it is warm to the hand (about 27 degrees C) for about 150mm either side of the joint. No welding shall be done below minus 18 degrees C. After preheating, the heated portion including the welded joints shall be covered with muffs or suitable insulation materials to allow cooling free from draughts. Any open ends of the pipe or assembly shall be effectively sealed to prevent heat flow by convection.
- 25.7 All welded pipe assemblies shall be constructed so that individual welded joints do not affect each other. The distance between the centres of adjacent welds shall be not less than twice the bore diameter of the pipe.
- 25.8 No welded joints shall be left partially completed. Any joints tacked in position must be promptly finished within the working day. The Consultant will reject all work not done in accordance with this instruction.
- 25.9 Where work is rejected, pipes must be machine cut at least 150 mm either side of rejected welds and proper weld preparation must be used on the shortened sector. Where shorter fill-in sections are required because of such rejection and reworking, then new full sized lengths must be supplied by the Contractor.
- 25.10 Where pipes with longitudinal seams are specified, pipe seams shall be arranged such that adjacent seams are opposed 45 deg. from each side of top dead centre and branches shall be made only with weldable fittings
- 25.11 All filler metals which are coated shall be protected from excessive moisture changes. Filler materials or flexes which show any sign of deterioration shall not be used. If requested by the Consultant samples of filler rods to be used shall be submitted to him for approval before any work is done on site.

These may be submitted, at the Contractor's cost, to an independent testing laboratory for verification.

26.0 TESTS FOR WELDER QUALIFICATION.

- 26.1 The purpose of the welder's qualification tests is to determine the ability of the welders to make sound and acceptable welds. Before any site welding on the contract is allowed, each proposed welder shall carry out the tests required in the presence of the Consultant, or the Employer's Insurance Inspector.
- 26.2 Any weld test specimens which have been suitably marked and approved shall be kept on site by a responsible person, so that they can be produced at any time, at the request of the Consultant.
- 26.3 All accommodation, benches, tools, welding plant, acetylene, oxygen, electricity, test pieces, filler rods, electrodes, facilities for cutting and grinding, polishing, bending and examining, which are necessary for welders qualification tests shall be provided by the Contractor. In the absence of any items for inspecting the welds, the Consultant may submit the finished samples to an independent laboratory for testing at the Contractor's expense.
- 26.4 Under no circumstances shall a welder be employed on the contract, either on or off the site for welding operations other than those for which that welder is qualified.
- 26.5 Copies and records of all test reports shall be promptly given to and kept by the Consultant.
- 26.6 Even welders holding a certificate for welding are to be tested at this job site in full accordance with the specifications.
- 26.7 Tests for Qualification on Steel Pipes
 - 26.7.1 The Contractor shall be responsible for the quality of welding and brazing and shall provide the testing of the welders employed. Certificates of such proficiency test, together with stamped samples provided for Engineer's review before welding work is commenced. The welder's proficiency standard to be equal to Lloyd's Standard Class 2, certified by an authorized testing laboratory paid for the contractor.
 - 26.7.2 Qualification tests will be made for butt and branch welds in pipes. One test piece per welder will be required for the qualification of welding of pipes exceeding 100mm nominal diameter.
 - 26.7.3 In addition, test pieces will be required for each position of a pipe. These positions will be horizontal and rotated while welding, horizontal and fixed while welding, vertical and fixed while welding for each range of nominal diameters specified.

27.0 TESTING OF WELDED PIPEWORK INSTALLATION.

- 27.1 Testing of welded pipe work installation shall be by either destructive or non-destructive test method.
- 27.2 Each butt-welded test piece to be subjected to two root bend tests and two nick break tests of the welding.

Each branch welded test piece will be subjected to two root bend tests and two fillet break tests of the welding.

Each test piece weld to be radio graphically examined.

- 27.3 The Contractor shall assign an identification symbol to identify each welder and this will be used to stamp the work done by that welder.
- 27.4 The Contractor will compile a set of safety instructions concerning welding procedures for the welders and after review by the Engineer, these will be issued to the welding operatives and foremen.
- 27.5 The Contractor shall allow for two percent of all the welds to be radiographically examined.

Determination of piping sections to be tested will be designated by the Engineer.

The timing of the radiographic examination of a weld or welds designated to be so examined will be confirmed in writing by the Engineer during the installation and commissioning.

Radiographic examination of welds will be carried out by a specialist employed by the Contractor subject to review by the Engineer. The specialist so employed will be responsible for the proper indexing, interpretation and reporting of the results of the radiographic tests.

- 27.6 Any weld, whether radio graphically tested or not which are considered by the Engineer to be faulty, either in whole or in part, or which have failed under hydraulic or other tests so designated by the Engineer, will be removed and replaced by the Contractor.
- 27.7 The faulty part of a weld may be removed by flame gouging, chipping or grinding. Otherwise, the Engineer may instruct that the portion of the works containing the fault weld be cut out and replaced by the welding in of a new section of pipe or member.
- 27.8 Should any of the welder's work (as defined by the Engineer) be found to be faulty during radiographic tests, then that welder shall be removed from site and all other welds undertaken by that welder shall be checked by radiographic means and replaced as necessary at the expense of the contractor.
- 27.9 The extra radiography involved shall not be part of the 2% inspection previously stipulated.

28.0 PAINTING, IDENTIFICATION AND VALVE TAGS

- 28.1 All painting of plant, equipment, storage vessels, and the like, and all surfaces to be painted in the areas where such items are installed, including generator area, condenser areas, etc., shall be carried out by an approved specialist painting contractor.
- 28.2 Painting shall be applied to all exposed pipe work, ductwork and associated components, valves, fittings, etc., equipment, supports of any kind, insulation, plain mild steel, copper, or cast iron surfaces. Where movement is required between adjacent surfaces, request clarification from the Consultant.

28.3 The following items do not require site painting:

- (a) Insulation having any of the following finishes:
 - (1) Aluminium foil.
 - (2) Hot Dip Galvanized angles / channels
 - (3) Polyisobutylene sheeting / bitumen coating.
 - (4) Vinyl-glass/ resin-aluminium foil laminate.
- (b) Equipment or plant or supports or frames delivered to site with any of the following finishes:
 - (1) Painted finish, other than primed only, provided that the finish is not damaged in any way.
 - (2) Stainless steel or plastic coated steel.
 - (3) Stoved enamel.

28.4 All surfaces to be painted shall be prepared by thoroughly cleaning and removing all rust, grease, oil, dirt and surface corrosion, using wire brush, emery paper and/ or degreasing medium as required. The paint shall be applied in accordance with the maker's instructions and the type of paint to be used shall be in accordance with the following:

Ferrous surface, one coat of Calcium Plum bate metal primer plus wash primers as necessary, followed by 2 undercoats and one finishing coat.

Non-Ferrous surface, one coat of zinc chromate primer plus wash primers as necessary, followed by 2 undercoats, and one finishing coat.

All paint shall be manufactured by approved manufacturer and type of paint shall be selected by the manufacturer based on the location i.e. Indoor / Outdoor.

28.5 For factory applied finishes, repainting or refinishing of any surfaces damaged during shipping, erection or construction shall be done using only factory supplied materials.

28.6 After insulation or final painting is completed, identify each piped and ducted service. Locate identification bands, stickers and arrows :

- (a) Behind each access door.
- (b) At each change of direction on all joining pipes and ducts.
- (c) At not more than 10 meters apart in straight runs of exposed pipes and ducts, but on both sides of sleeves.
- (d) At not more than 10 meters apart in straight runs of pipes and ducts behind removable enclosures such as lay-in ceiling but on both sides of sleeves.

- (e) Above each floor or platform for pipes in shaft and vertical exposed pipes, preferably 1.5 meters above floor or platform level.
- 28.7 Identification Bands, stickers and arrows shall be made of self adhesive plastic long lasting type. The stickers when used outdoor shall be weatherproof type suitable for the temperature in KSA.
- 28.8 Identification of ductwork shall be carried out as per DW 144 standards coloured triangle with the type of service. Additionally the AHU / FAN reference no shall be identified in the ducts.
- 28.9 All insulated or un-insulated pipe work shall be identified as detailed above by means of PVC tape identification bands in accordance with British Code to BS 1710.

In addition the name of the service and flow direction arrow shall be provided.

- 28.10 The access doors and fire dampers in duct shall be identified by a suitable sticker.
- 28.11 All identifying band colours and finishing colour of services shall be agreed with the Consultant prior to application.
- 28.12 All equipment shall have a nameplate secured to the item giving the following information:

Equipment reference number (as indicated on the record drawings.)

Make / Model No.

System.

Room / Area served.

Duty / output information.

Power fed from :

The name plate shall be 100 x 100 mm approximately, of white plastic 3 mm thick with the above information engraved in black lettering and the plate shall be secured by screws, bolts, clips, etc., as appropriate to the item concerned.

This plate is in addition to any name plate supplied by the manufacturer of the item giving detailed specification information for the equipment.

- 28.13 All main valves in plant room, roof, risers shall be numbered system wise, each with a unique number. The number shall be engraved in a tag 30mm dia and suspended with a stainless steel chain from the valve handle.

The tag shall be of aluminium material for outdoors and plastic indoor.

The piping schematic diagrams shall have the valve numbers with valve chart and the same shall be enclosed as a part of operating instruction and shall be framed with glass and displayed in the relevant plant rooms.

29.0 FLUSHING AND CLEANING.

- 29.1 Flush all fluid-carrying systems after completion with a stream of water or appropriate fluid at the highest obtainable pressure and velocity. Discharge the flushing fluid through all strainers

and out through line sized valves with hose ends. Clean all strainers. Repeat flushing operation to the approval of the Consultant until the water is running clear, colorless and odorless and the conductivity is no more than 30 ms/cm above the mains water.

- 29.2 If the water pressure normally available on site is, in the Consultant's opinion, inadequate to perform proper flushing, then the Contractor shall provide at his own expense, pumps, tanks, etc., to give pressure and volume flows to the Consultant's satisfaction.
- 29.3 For layer systems, the flushing and cleaning shall be carried out in sections of every three floors.
- 29.4 During flushing and cleaning all cooling coils, heat exchangers shall be bypassed in order to avoid debris / dirt going to coils.

30.0 INSPECTION, TESTING AND ADJUSTING.

- 30.1 All the works provided as part of this Contract shall be inspected and commissioned in accordance with all relevant Specifications, and Codes of Practice and to the entire satisfaction of the Consultant.
- 30.2 The Contractor shall employ the services of a specialist company approved by the Engineer regularly engaged in providing a testing balancing and commissioning service and who has been in continuous business for not less than 7 years employing fully trained staff having not less than 3 years dedicated experience.

A senior experienced Commissioning Engineer with minimum dedicated experience of 7 years shall be responsible for supervising and directing the activities for the testing and commissioning team.

- 30.3 Carryout all tests specified. Carry out all tests required by Authorities having jurisdiction. Test equipment to the requirements of, and where necessary, in the presence of the Manufacturer.
- 30.4 Provide all equipment, labour, instruments, loading devices, incidentals, and pay for all fuel, power and sundries required to carry out the tests.
- 30.5 All installations shall be inspected and tested in sections as the work proceeds and on completion as composite systems and it shall be noted that the Consultant or any of the other relevant Authorities may require to inspect or test any equipment during manufacture at the Manufacturer's works. All necessary arrangements shall be made as part of this Contract.

This will generally not apply to specified items unless specifically noted in these documents, but may be necessary for alternate equipment, should this be considered at all.

- 30.6 All tests shall be arranged in cooperation with the Consultant and his Engineer and all other concerned parties and shall be subject to at least five (5) days notice in writing of the time, location and nature of the test to be performed. No test shall be considered valid unless the Consultant is present.
- 30.7 All necessary skilled and unskilled labour shall be provided for attendance during the tests (including pre- and post- test activities,) and the test media shall be provided and subsequently disposed of except where specifically stated otherwise.

- 30.8 The testing and adjusting is the contractual responsibility of the Contractor but actual performance of the tests is expected to be the sole responsibility of the Subcontractor.
- 30.9 Have all testing and balancing performed by only persons who are thoroughly versed in this type of testing and balancing and with proven ability. Submit names, complete with experience records, and references for approval of the Consultant.
- 30.10 Any defects occurring at any time during the test duration shall be made good and a complete re-test shall be carried out, all at no cost to the Contract.
- 30.11 Where failure occurs during a test, inspection or commissioning procedure which results in damage to the building fabric and/ or any services not provided as part of this Contract, or requires subsequent builder's work to be carried out, carry out all such repair work to the entire satisfaction of the Consultant at no cost to the Owner or the Consultant.
- 30.12 All test points shall be provided which are necessary to carry out the specified tests and commissioning procedures including facilities for measuring or monitoring temperature, pressure, pressure drop, volume flow, in-duct sound power or sound pressure, humidity, or other relevant conditions in both air side and water side systems. Such points shall be fitted with removable plugs, flanges, or other appropriate and approved devices.
- 30.13 Prepare test report forms for each test to be performed and submit these to the Consultant at least two weeks prior to the commencement of any tests.
- 30.14 Test only after the system installation has been completed and the system has been put into continuous operation. Perform the testing, adjusting, and balancing when outside conditions are commensurate with the design conditions for the given system. Add dummy loads to the system if outside and inside conditions are less severe than the specified points.
- 30.15 Prepare a complete list of instruments for each test containing for each instrument:
- Name of instrument Manufacturer.
 - Scale and full scale accuracy.
 - Date of last calibration test.
 - Name of last calibrating company.

All instruments and consumables, such as recording paper, necessary for conducting the tests shall be provided, including but not limited to the following:

- Tube type velometer.
- Sloping tube manometer.
- Pitot tubes of various lengths.
- Mercury in glass thermometers.
- Weekly recording thermometers & digital thermometers.
- Weekly recording relative humidity meter.
- Anemometer for diffusers, with collector.
- Ammeter, voltmeter, wattmeter.
- Power factor meter.
- Insulation tester.
- Earth loop impedance tester.
- Tachometer.

30.16 Duplicate signed test certificates shall be provided after each test which will be countersigned by the Consultant who witnesses the test. The test certificate shall give the following particulars:-

- Apparatus or section under test.
- Maker's number (if any).
- Nature, duration and conditions of test.
- Result of test.

No test shall be valid until the test certificate is provided.

30.17 Duplicate copies of test certificates carried out at Manufacture's works shall be forwarded to the Consultant for approval prior to dispatch of the article to site.

30.18 No section of the works shall be insulated or in any other way concealed prior to testing and inspection and subsequent concealment where applicable shall only take place following written authority from the Consultant.

30.19 All necessary facilities, measuring and recording instruments including test pumps and gauges for inspection, testing and commissioning requirements shall be provided and shall be checked or calibrated as necessary before use.

The Consultant reserves the right to call for a demonstration of the accuracy of any instruments provided.

30.20 All representatives present during inspection, testing and commissioning shall be fully conversant with the system concerned and the method of system and instrument operation.

Manufacturer's or Specialist Subcontractors' representatives shall attend where specifically indicated elsewhere in the specification or where necessary to ensure full service and cooperation is available to the Consultant to enable the works to be tested and commissioned in accordance with the requirements of the specification.

30.21 All necessary precautions shall be taken to safeguard against damage during inspection, testing or commissioning. Any damage so caused shall be made good at no cost to the contract.

30.22 All tests shall last for the minimum time period stated or longer if necessary to ensure all sections have been fully examined as required by the test.

One set of test instruments shall be made available during the warranty period to check the system whenever required.

30.23 Third Party testing and Commissioning for MEP systems is waived off except for HVAC Systems. HVAC System third party testing and Commissioning shall be conducted as specified. All the pumps shall be factory assembled , tested and witnessed by client/engineer.

31.0 TEMPORARY USAGE.

31.1 The Consultant reserves the right to use any piece of electrical and mechanical equipment, device, or material installed under this section for such reasonable lengths of time as may be required to make a complete and thorough test of same before the final completion and acceptance of the work.

Such tests shall not be construed as acceptance of any part of the work, and it is agreed and understood that no claim for damage will be made for any injury or breakage to any part or parts of the above due to these tests, where caused by weakness, inaccuracy of parts, or by defective materials or workmanship.

32.0 INSPECTION OF THE WORK.

32.1 The representatives of the Owner and the Consultant will make periodic visits to the site during construction to ascertain that the work is being executed in reasonable conformity with all plans and specifications, but will not execute quality control at all times. Each Subcontractor must maintain the quality control as intended in the contract documents.

32.2 Correct all deficiencies immediately as noted during field inspections.

32.3 Request in writing that a final inspection of all services.

Do not issue this written request until:

All deficiencies noted during job inspections have been corrected.

All systems have been balanced and tested and are ready for operation.

All balancing reports have been submitted and reviewed.

All instruction manuals have been submitted and reviewed.

The cleaning up is finished in all respects.

All spare parts and replacement parts specified have been provided and receipt of the same acknowledged.

33.0 OPERATING INSTRUCTIONS TO OWNER.

33.1 Instruct the Owner's staff how to make minor adjustments, carry out necessary maintenance and how to operate each system.

For new equipment, provide the Consultant with three copies of complete operating and maintenance instructions for equipment at the time of delivery of the equipment.

Equipment will be rejected unless accompanied by instructions. Such documents must be received at least one month prior to the completion date of the relative section of the works.

Instructions shall be bound in a suitable loose leaf booklet or binder, and shall include:

- a) Prints of the following drawings:
 - 1) List of all equipment installed.
 - 2) General layout.
 - 3) Wiring diagram of control panels.

- 4) Non-dimensional layout, showing location of all electrical devices (Limit Switches etc.).
- b) Operation instructions, include start-up and shutdown procedure.
- c) Maintenance instructions, including preventative maintenance instructions for components of the equipment.
- d) Lubricating instruction and recommended cycle of lubrication.
- e) Complete parts list, showing manufacturer's name and catalogue number.
- f) Complete list of recommended spare parts for 2-year operation with price list for each item of equipment.

The contractor is to prepare a draft for discussion with the Consultant prior to finalizing the documents.

Portable tools and spare parts shall be correctly labeled and handed over to the Consultant.

The Contractor is to provide operating and maintenance instruction and training presentations and demonstration for the maintenance staff, these should be based on the content of the Operation and Maintenance manuals and in sufficient detail to talk the Contactor through the practicalities of the work on the site in situ.

- 33.2 The section dealing with complete systems shall be subdivided into each service with a ready means of reference and detailed index. The function and manner of operation of each system shall be clearly described together with illustrations and line diagrams in schematic form showing the location and function of control valves, items of equipment and spaces or areas which are served by these items. The colour coding and identification systems employed shall be explained.

34.0 COMMISSIONING OF SERVICES.

The Commissioning of HVAC and other mechanical systems shall be carried out by one of the specialized commissioning agencies approved by the Engineer. The agency shall be appointed by the MEP Contractor at an early stage of the project. The commissioning agent shall be an approved company by client.

- 34.1 All piping systems shall be flushed, chemically cleaned and then filled with treated water, or appropriate fluid, vented as necessary, and brought to operating conditions and the flows then regulated to the design values.
- 34.2 All water circuits shall be balanced by means of the regulating valves provided and flow rates shall be determined on a temperature and pressure drop basis.
Flow through pumps shall be measured by relating the pressure drop across the pump to manufacturers test curves. A copy of the test curve indicating the final operating point shall be forwarded to the Consultant.

- 34.3 All refrigerant systems shall be vacuumed, purged and charged according to A.S.H.R.A.E. standard methods. All safety controls and interlocks will be tested as part of the commissioning procedures.
- 34.4 Flow through coils shall be adjusted to give the design temperature difference. In general water systems shall be balanced in accordance with the guidelines specified in NEBB Manual. Upon completion of balancing and testing operations for heating and cooling systems, temperature measurements shall be taken in all rooms and the readings tabulated in schedule form together with hourly ambient temperature readings taken over the measuring period. Two copies of the schedule shall be forwarded to the Consultant.
- 34.5 All ventilating systems shall be commissioned in accordance with the procedures recommended in the AABC / CIBSE Manual. Duplicate schedules of commissioning results shall be forwarded to the Consultant and shall detail the recorded air volume and percentage deviation from design air volume, for each air input and extract terminal. Wet and dry bulb temperature measurements shall be taken in all rooms served by air supply systems and the results indicated on the schedule of commissioning results.
- These shall be recorded together with external ambient wet and dry bulb temperatures measured at hourly intervals over the measuring period.
- 34.6 Sound pressure levels (dBA) shall be measured in all rooms containing supply or extract terminals, all plant rooms, all rooms immediately adjacent to plant rooms and all rooms located above or below plant rooms. The Consultant may at his discretion request a spectrum band analysis of sound pressure in any locations.
- 34.7 All automatic controls shall be commissioned by the controls supplier.
- 34.8 Fault conditions shall be simulated and all alarms and safety devices shall function correctly. Such proving tests shall be carried out in the presence of the Consultant and certificates shall be provided specifically detailing all check procedures which have been carried out.
- 34.9 Following proving tests all installations shall be left in working order ready for handover.
- 34.10 All electrical systems shall be tested and commissioned in accordance with IEE regulations and local authority codes.

35.0 SYSTEM ACCEPTANCE.

- 35.1 The ultimate condition for system acceptance is that the Owner and Consultant have inspected the system and found it to be acceptable, and indicated this in writing.

Issuance of the final payment certificate does not necessarily indicate system acceptance, neither does release of final payment holdback in whole or in part. The Consultant's acceptance may be contingent on any or all of the following if applicable:

- 35.1.1 Submit original copies of letters from manufacturers of all systems indicating that their technical representatives have inspected and tested the respective systems and are satisfied with the methods of installation, connections and operation.
- 35.1.2 Submit "as built" drawings and operation and maintenance manuals.

35.1.3 Train owner's maintenance staff.

35.1.4 Satisfactory completion of 12 months guarantee period.

36.0 CORRECTION AFTER COMPLETION.

36.1 Remedy all work in accordance with the General Conditions of Contract for a period of one year from the date of acceptance of the completed work, or if so noted, for longer periods for parts of the work.

36.2 Attend immediately to any and all the defects occurring during the period defined above and repair in a manner to prevent recurrence. This contractor is responsible for all work required by other trades necessary to repair the works of this section, or necessary to repair damage caused by the failure of any part of this section.

36.3 Instruct all Suppliers and Manufacturers that guarantees on equipment will commence when the completed work is accepted and not from the date the equipment is put into operation. In the event that this condition is omitted by the supplier, or if subsequent cost to the Owner is involved, this Contractor shall be liable to the full extent of the guarantee as indicated in the specification.

37.0 GUARANTEES

37.1. The Contractor will guarantee all material and workmanship for at least one year after TOC (taking over certificate).

37.2. All warranties from equipment suppliers will be vested in the Owner, regardless of whether the Contractor who supplied the equipment is still associated with the project or not.

37.3. Guarantees will be full guarantees and will include all overhead, profit, incidental charges and sundries.

37.4. Where damage is caused to any other item by any failure of the item guaranteed, then the guarantee shall also include the costs incurred in rectifying that damage.

38.0 MAINTENANCE.

38.1. Maintenance is defined as the Contractual Liability to maintain the equipment in working condition, PLUS the regular checks and servicing of equipment during the maintenance period to keep the equipment in best working order.

38.2. Regular maintenance shall be as necessary, but in any event not less frequently than monthly.

38.3. In the event that the Owner has his own staff, the Contractor is still to check monthly and advise on any problems and is still to assume responsibility.

PART 3 - EXECUTION

3.1 GENERAL

3.1.1 Refer to individual sections for requirements for execution of the work.

3.1.2 The flame spread rating for all materials shall be in accordance with the specifications mentioned in each relevant section and in accordance with developer Requirements.

END OF SECTION

MECHANICAL SECTION: 01 88 45
NOISE AND VIBRATION CONTROL REQUIREMENT

1. **PART 1 - GENERAL**

1.01 SECTION INCLUDES

- A. Requirements for noise and vibration control of Mechanical, Electrical, and Public Health (MEP) services.
- B. To prevent transfer of noise and vibration to occupied spaces and to prevent sound insulation between spaces being degraded by services penetrations.
- C. This Specification shall be read in conjunction with all Equipment Data Sheets and Schedules and all relevant Contract Drawings and Specifications.
- D. Where there are discrepancies between this Specification and other Specifications, with regard to noise and vibration control, this Specification shall take preference, unless there are specific technical or practical reasons that must take precedence over this Specification, or unless otherwise instructed by the Employer or Engineer.

1.02 REFERENCES

- A. References shall include, but not necessarily be limited to, the following:

ASID – American Society of Interior Designers (United States).

AHRI – Air-conditioning, Heating and Refrigeration Institute (United States).

ANSI – American National Standards Institute (United States).

ASHRAE – American Society of Heating, Refrigerating and Air-Conditioning Engineers (United States).

ASTM – American Society for Testing and Materials (United States).

BCO – British Council for Offices (United Kingdom).

BREEAM – Building Research Establishment Environmental Assessment Method (United Kingdom).

BSI – British Standards Institute.

CENELEC – European Committee for Electro technical Standardization.

CIBSE – Chartered Institute of Building Services Engineers (England).

DIN – Deutsches Institut für Normung. V. (Germany).

EC – European Commission .EN–European Norms.

EPA – Environmental Protection Agencies (United States & United Kingdom).

HVCA – Heating & Ventilating Contractors' Association.

IEC – International Electrical Commission.

ISO – International Organization for Standardization

LEED – Leadership in Energy and Environment Design (United States).

NFPA – National Fire Protection Association (United States).
SMACNA – Sheet Metal and Air Conditioning Contractors' National Association.
WHO – World Health Organization.

- B. Related Specifications include Sections contained in the following Divisions, as prepared specifically for this Project:

Division 1 –General Requirements.
Division 10 – Specialties.
Division 11 – Equipment.

Division 13 – Special Construction.
Division 14 – Conveying Equipment.
Division 22 – Plumbing.
Division 23 – Heating, Ventilating and Air Conditioning.
Division 26 – Electrical.
Division 27 – Communications.

- C. Related Acoustic Specifications include the following:

Section 01 88 45-1 – Acoustic Performance Requirements.
Section 23 05 48-1 – Vibration and Seismic Controls for HVAC.
Section 23 05 93-1 – Testing, Adjusting and Balancing for HVAC.
Section 23 33 19 – Duct Silencers.
Section 23 33 53 – Duct Acoustic Lining.

Section 26 05 48-1 – Noise and Vibration Control of Electrical Systems.

1.03 CONTRACT OR REQUIREMENTS

- A. Tender Proposals:

The Contractor shall submit at the time of Tender the names and qualifications of the noise and vibration control manufacturers. If a manufacturer is not one of the pre-approved manufacturers, the submittal shall be accompanied by a complete catalogue of that manufacturer's products and samples of each proposed vibration and/or noise control hardware.

The Contractor shall supply at the time of tender the design sound power levels (L_w in dB re 10⁻¹²W) for the entire MEP device, or the design sound pressure levels (L_p in dB re 2x10⁻⁵Pa) at specified distances from each side of the device. This information shall be submitted for each air moving, noise-generating and vibration-generating device including all those described in this Specification and all others as described in the Contract Documents and related Specifications.

If the actual sound power or sound pressure generated by any device exceeds the levels for the equipment specified in the Contract Documents and related Schedules and Specifications, the Contractor shall include in its price the cost for modifications to compensate for the extra noise or vibration at no additional expense to others. Any such modification shall be subject to the review and approval by the Engineer.

- B. Tender Submissions:

The Contractor shall submit detailed information on noise and vibration levels

associated with each equipment item to the Engineer prior to ordering plant and the noise and vibration control hardware. This information shall demonstrate that these elected plant and attenuating hardware can achieve the specified noise and vibration criteria under the required design duties.

Prior to Contract Award, the Tendered shall provide, as a minimum, the following information with the Tender:

A statement of compliance with all related Specifications. Manufacturer's details of the products and systems being offered.

Test certificates (in English or accompanied by a full English translation) from an independent acoustic test laboratory to show compliance with the specified acoustic performances for the products offered. Failing this, the Tendered shall allow in the Tender for the expense of such testing as necessary to show compliance with this Specification and all other related Specifications.

After Contract Award, the Supplier shall provide, as a minimum, the following information:

Shop drawings showing the methods for installation of all equipment and associated noise and vibration control measures.

Details for the complete noise and vibration control systems including frames, seals, isolators, panels, etc., and all materials and methods for acoustically sealing and fixing all elements and components.

C. Laboratory Testing:

Certified and verifiable tests shall be provided in accordance with the standards stated in this Specification and all other related Specifications.

Laboratory measurements shall be obtained from an independent acoustic test laboratory, as approved by the Engineer, under test conditions on representative samples (including joints, seals, bases, isolators, attenuators, etc.) of the products and systems being offered. Test laboratories shall have certifications for their facilities, including instrumentation, traceable to any of the following (or acceptable equivalent):

NAMAS (National Measurement Accreditation Service)

UKAS (United Kingdom Accreditation Service)

NVLAP (National Voluntary Laboratory Program)

NIST (National Institute of Science and Technology)

DINENISO/IEC17025 General requirements for the competence of testing and calibration laboratories.

Measurements obtained in accordance with equivalent standards acceptable to the Engineer shall also be permitted, subject to their applicability.

Where existing data is available from a cognized independent test laboratory it shall be considered acceptable providing the tests have been carried out for the exact same product and system offered.

If test data is not available, within 8 weeks of the award of contract, or other period to be advised by the Engineer, tests shall be undertaken in accordance with the specified standards to demonstrate that the proposed products and systems, together with the associated elements shown in the relevant Contract Drawings and described in all related Specifications, comply with the specified performance requirements.

D. Mock-up Testing:

Where required and stated in the Contract Documents, Mock-up tests shall be carried out for Air Handling Units (AHUs), Fans, Fan Coil Units (FCUs), Constant Volume Units (CVs), Variable Air Volume Units (VAVs), Variable Refrigerant Volume Units (VRVs) and any other MEP equipment as advised by the Engineer, to confirm compliance with the acoustic performance requirements, as described below and in accordance with all other related Specifications.

The Mock-up room shall be constructed on site, at the manufacturer's or Subcontractor's factory, or within an acoustic test laboratory. The relative merits of each option shall be evaluated to ensure the following specified parameters are achieved.

Are preventative unit of each proposed size shall be tested.

The unit(s) selected, shall be from the upper volume range for each size.

The unit(s) shall be complete with filters, coils, low velocity duct work, plenum box and diffusers, etc., to simulate a typical room/space layout.

All details of the proposed units and duties shall be submitted to the Engineer for comment and acceptance.

The Mock-up room shall have the same nominal floor area and height as the area to be served by that unit. The room / space shall be provided with a false ceiling of the same type as proposed for the project, along with typical weakness stop simulate return air paths. Return air light fittings, diffusers, grilles and floor finishes shall be fitted, as proposed for the Project, with the unit in its installed position within the building. Shop Drawings shall be submitted to show the room/space in plan and section along with the unit for comment and acceptance by the Engineer.

Where applicable, the Subcontractor shall ensure that noise from the discharge/outlet side of the unit does not affect acoustic readings in the Mock-up room.

Prior to recording any noise levels, the background noise in the Mock-up room shall preferably be at least 10dB below the Mock-up room noise criterion, with a minimum of 6dB below.

The unit shall be set at its design speed and run on a 3hour on/1hour off basis for a minimum of 200hours prior to any acoustic testing, or other operational timing

sequence to be advised by the Engineer.

The units shall be operated at the agreed air volumes, which shall be verified by separate measurement and accepted by the Engineer.

For VAV units, noise levels generated when the damper is closed to half flow at the same total inlet pressure shall also be measured.

A minimum of 4 Nos. measurements shall be taken in the Mock-up room, which shall be at least 1m from a wall, or other measurement positions at least 1.2m above the floor. No individual measurement shall be exceeded by more than 2dB and the logarithmic average of all the measurements shall not exceed the Mock-up room noise criterion.

Measurements shall be carried out by a competent person using a sound level meter and octave band filters to complying with BSEN60651 Type 1 which shall be calibrated before and after use.

The Subcontractor shall provide are port listing all test details, and measurements, to show compliance with the Specifications.

A full method statement shall be prepared along with Shop Drawings, to show the proposals for the Mock-up room and shall be submitted to the Engineer for acceptance at least 20 working days prior to the day of the tests, or other period to be advised by the Engineer.

E. Site Testing:

All noise and vibration site tests and measurements shall be carried out in accordance with Section 230593-1 (Testing, Adjusting and Balancing for HVAC).

All other acoustic tests and measurements shall be carried out in accordance with Section 018845-1 (Acoustic Performance Requirements).F.

Contractors and Manufacturers Responsibilities for Tests:

Noise data for items of plant not covered in this Specification or other related Specifications shall also be provided by the Manufacturer or Supplier, to allow the Contractor to demonstrate the following:

Confirm that the individual items of equipment offered shall achieve the specified noise and vibration criteria in each space.

Identify which individual items of equipment offered require additional noise and/or vibration control measures to ensure that the specified noise and vibration criteria shall be achieved in each space.

All acoustic testing referred to in this Specification shall be allowed for in the Contractor's Tender. A Method Statement for the test shall be provided in writing, not less than 14 working days before the test, or other period to be advised by the Engineer, for final acceptance by the Engineer. The Engineer shall be advised of the date of the test, in writing, not less than 20 working days before the test, or other period to be advised by the Engineer. The Contractor shall be responsible for testing

and securing approval for operation from the responsible legal authority having jurisdiction per any item in this Specification and all other related Specifications.

The Contractor shall responsible for reviewing this Specification and considering any special requirements from officials responsible for enforcing applicable codes and standards. Where necessary, the Contractor shall make corrections, changes and additions to the proposed methods and procedures for testing at no increase in the Contract Price.

The Contractors hall be responsible for verifying the completeness of the installations and the overall suitability of the equipment to meet the intent of this Specification and all other related Specifications. Any additional equipment needed to meet the intent of this Specification, even if not specifically mentioned here in or in the Contract Documents, shall be supplied by the Contractor without claim for additional payment. The Contractor shall not construe performance or waiving of inspection, testing or surveillance by the Employer or Engineer to relieve the Contractor from total responsibility to perform in strict accordance with the Contract Documents.

During any Laboratory, Mock-up or Site Tests, if any pure tones are audible, or of there is doubt as to whether a pure tone exists, discrete frequency components tests shall be undertaken. These shall be assessed as being significant and present in a one-third octave band if the

measured sound power or pressure level in that one-third octave band exceeds the arithmetic average of the measured sound power or pressure level in the two adjacent one third octave bands by at least the values stated below:

Discrete Frequency Sound Level Difference Sound Pressure Levels in $\text{dBLeq}(\text{re } 2 \times 10^{-5} \text{ Pa})$ Third Octave Band Centre Frequency(Hz)					
<100	125	160	200	250	315-10k
6dB	5dB	4dB	3dB	3dB	2dB

If measurable ton also uncharacteristic exist in the tested sound levels (as determined by the discrete frequency test specified above), the Contractor shall make every effort to eliminate them. Failure to achieve this shall require attenuation of the pure tones by the amounts for each octave band given below:

Pure Tone Attenuation Requirements Sound Pressure Levels in $\text{dBLeq}(\text{re } 2 \times 10^{-5} \text{ Pa})$ Octave Band Centre Frequency(Hz)					
63	125	250	500	1k	$\geq 2\text{k}$
3dB	5dB	9dB	12dB	14dB	15dB

1.04 NOISE CONTROL REQUIREMENTS

A. Acoustic Flexible Connections:

Where flexible connections are provided between a fan and its associated attenuator/silencer (and at other positions shown on the Contract Drawings), they shall be complete with the noise containment properties of the adjoining duct work system inclusive of any specified lagging.

Flexible connections shall typically be structured from sound barrier mat having a minimum superficial density of at least 5kg/m³.

Flexible connections shall be at least 50 mm long, straight but not rigid, with no offset in order to prevent turbulence.

The acoustic flexible connections shall be measured in attest laboratory, accepted by the Engineer, and designed to achieve the minimum performances tabulated below:

Sound Reduction Index/Sound Transmission Loss Octave Band Centre Frequency(Hz)					
125	250	500	1k	2k	4k
7	8	10	13	13	13

The above performances shall be verified by measurements in a test laboratory in accordance with the following, or acceptable equivalent, standards, which shall be approved by the Engineer:

ASTME90: Standard Test Method for Laboratory Measurement of Airborne Sound Transmission Loss of Building Partitions and Elements.

ISO10140-1:2010 Acoustics-Laboratory measurement of sound insulation of building elements –Part1: Application rules for specific products

B. Acoustic Louvres:

Where shown on the Contract Drawings, or specified elsewhere in the Contract Documents,

or required as an additional means to control noise break out to the environment, acoustic louvres shall be purpose -built units constructed by a specialist manufacturer with a minimum of 10 years of experience in acoustic louver manufacture.

The acoustic louvres shall be constructed as single-banked or double-banked systems, as required to meet the acoustic performance requirements, and designed to prevent the ingress of rain and sand, under normally encountered meteorological conditions in the Kingdom of Saudi Arabia.

Sound transmission loss (STL) or sound reduction index (SRI) test data shall be provided for acoustic louvers identical in every significant detail to those proposed, for acceptance by the Engineer, before an order is placed / This refers to air flow velocity, pressure drop and, as far as is reasonable, entry and exit conditions, to demonstrate that the following performance is achieved:

Sound Reduction Index/Sound Transmission Loss				
250	500	1k	2k	4k
15	18	27	30	33

The above STL or SRI performances shall be verified by measurements in a test laboratory in accordance with ISO10140 -1,ASTME90 or acceptable equivalent standards which shall be approved by the Engineer. Any extrapolation or calculation used to produce the data shall be clearly stated. STL or SRI test data shall include either, the static insertion loss and regenerated noise for the given air flow conditions, or both static and dynamic insertion losses for the given air flow conditions.

The louvre frames shall be constructed from a suitable gauge of galvanized mild steel, or aluminum, supporting louvre blades of similar material. The acoustic material in the blades shall have a density of not less than 48 kg/m³, be inert, rot and vermin proof, non-hygroscopic and composed of incombustible mineral fiber, faced with mineral fiber tissue and retained on the lower blade by perforated, galvanized mild steel or aluminum.

When the louvres are actually manufactured, each unit shall be shop assembled as a whole unit unless the Contractor or Engineer instructs to the contrary.

If necessary, louvres shall be supplied with an integral bird screen of galvanized mild steel or aluminum mesh, fixed in its internal face. The mesh pitch shall be a maximum of 25 mm. The louvres shall be supplied complete with all necessary fittings, flanges etc., for fitting in to the louvered openings as required by the Contractor or Engineer.

All gaps between the outside of the louvre frames and walls or ducts shall be made good and sealed with a heavy grout and/or non-hardening, dense mastic.

C. Acoustic Panels:

Where shown on the Contract Drawings, or specified elsewhere in the Contract Documents, or required as an additional means to control noise break out, acoustic panels shall be fitted to walls and, if required, ceilings/soffits. These treatments apply to large plena, plant rooms and builders work shafts associated with, but not limited to, chiller plant rooms, generator rooms, boiler/dilution flue rooms, transformer rooms and exhaust fan rooms. These panels can also be used to build acoustic enclosures around noisy plant items such as fans, motors, pumps and other plant equipment.

The internal surfaces of the enclosure panels shall be designed to achieve the minimum performances tabulated below using a Type Mounting as described in ASTM E795:

Random Incidence Sound Absorption Coefficient(α)					
Octave Band Centre Frequency(Hz)					
125	250	500	1k	2k	4k
0.50	0.70	0.90	0.90	0.90	0.90

The above performances shall be verified by measurements in a test laboratory in accordance with the following, or acceptable equivalent, standards, which shall be approved by the Engineer:

ASTM C423: Determination of Sound Absorption Coefficients in a Reverberation Room.

ASTM E795: Standard Practices for Mounting Test Specimens During Sound Absorption Tests.

ISO 354: Acoustics-Measurement of sound absorption in a reverberation room.

The acoustic enclosure panel systems (but excluding the effects of services penetrations) shall be designed to achieve the following minimum performance when measured in a test laboratory, accepted by the Engineer, in accordance with ISO 10140-1, ASTM E90 or acceptable equivalent standard which shall be approved by the Engineer:

Sound Reduction Index/Sound Transmission Loss					
Octave Band Centre Frequency(Hz)					
125	250	500	1k	2k	4k
16	26	32	32	32	31

Where specified, the acoustic enclosure panels shall be constructed from galvanized mild steel sheet at least 1.6 mm thick (16 swg) or as otherwise specified.

The absorbent internal lining shall be faced with glass fiber cloth or other infill protection membrane and retained by perforated galvanized mild steel sheet having an open area preferably in excess of 23%, with the whole panel not less than 50 mm thick. Provision shall be made inside the panel to prevent settling of the acoustic medium.

Full allowance shall be taken of any loss of insulation due to doors, windows, ventilation openings through louvres/vents and panel joints.

Sections shall be designed to allow unskilled personnel to easily disassemble and reassemble the parts without compromising the acoustic performance.

The Supplier shall ensure that the assembled enclosure is designed and constructed to withstand site operating conditions such as wind or sand storm loads, roof-mounted plant, etc., and to be suitably weather proofed for use outdoors.

D. Air Cooled Condensers and Similar Types of Cooling Equipment:

Where shown on the Contract Drawings, or specified elsewhere in the Contract Documents, Air Cooled Condensers, Dry Air Coolers or similar types of cooling equipment shall be independently tested, in accordance with AHRI, ISO, Euro vent or other accepted equivalent standards, in order to provide octave band noise data as stated elsewhere in this Specification and/or other related Specifications. The tests shall be carried out for the full range of operational volumes at pressure drops equivalent to those expected in the installed systems.

To ensure minimal tonal noise is generated from the fan operations, electronic noise filters or other noise-attenuating measures shall be provided. The noise filters shall be factory-fitted items that consist of heavy copper coils inside a weather proof IP55 enclosure where the copper coil will cut voltage peaks out of the controller and smooth out tonal noises. Each noise filter shall have its dedicated capacitor supplied that will act as a compensator for the filters.

Based on the acoustic test data provided, noise and vibration attenuation requirements shall be incorporated by the Contractor as required to achieve the specified acoustics, noise and vibration.

E. Air Handling Units (AHUs), Fresh Air Handling Units (FAHUs) and Computer Room Air Handling Units (CRAHs):

Where shown on the Contract Drawings, or specified elsewhere in the Contract Documents, all AHUs, FAHUs, CRAHs or similar types of equipment shall be independently tested, in accordance with AHRI, ISO, Euro vent or other accepted equivalent standards, in order to confirm compliance with the octave band sound power levels (LwindBre10-12W) specified in the Schedules and other related Specifications. The tests shall be carried out for the full range of operational volumes at pressure drops equivalent to those expected in the installed systems. The in-duct sound power levels specified shall include the effect of regenerated noise from attenuators integral to the AHU.

Based on the acoustic test data provided, noise and vibration attenuation requirements shall be incorporated by the Contractor as required to achieve the specified acoustics, noise and vibration criteria.

The casing insulating material shall be a thermal, sound retarding and absorbing media which is non-combustible, non-hygroscopic, inorganic, mildew resistant vermin proof and shall comply with the requirements of BS476-6 and BS476-7, or other acceptable equivalent standard advised by the Engineer. The media shall be securely fixed to internal surfaces with suitable adhesives or mechanical fixings such that voids are eliminated cold bridging is avoided and to ensure that it is held rigid during shipment. The insulation shall be totally encapsulated to prevent migration of particles into the airstream.

The minimum thickness of insulation shall be 25 mm for rigid polyurethane foam or 50mm for units insulated with mineral fibre. The insulation thickness shall also be of adequate thickness to ensure that condensation does not form and to

achieve a minimum overall thermal resistance for external air handling units or for those located within naturally ventilated plant rooms of $1.0\text{m}^2\text{C/W}$ and $1.25\text{m}^2\text{C/W}$ for internal units. (BS EN 1886 Table 5 Class T3). Polyurethane insulating materials are to be CF and HCFC free and blown using agents with an ODP of zero.

The Sub contractor shall establish insulation thicknesses based on the worst combination of external and internal conditions.

The insulation shall be either naturally impermeable to water or provided with a continuous vapour barrier with due consideration given to insulation edges and cold bridges.

F. Air Terminal Devices (Miscellaneous):

Noise levels generated by air terminal devices, including Grilles, Registers and Diffusers (GRDs), shall be specified in the schedule of air terminal devices either as limiting sound power levels or as room noise level selection figures. Where no limiting sound power levels are specified, the air terminal devices shall be chosen to achieve a noise rating at least 10dB less than the specified room noise criterion for the total number of terminal devices serving the space.

The noise data described above shall be derived from actual tests performed using devices similar in every significant detail to those proposed and operating under the same conditions as those proposed, e.g. in terms of air flow velocity, pressure drop, angle of diffuser blades, volume control damper positioning and connecting duct work geometry.

All diffuser and grille plenums shall be constructed to suit the Specialist Suppliers' requirements. In general supply diffusers shall be fitted with purpose made plenum boxes to ensure stable air discharge flow patterns under all flow specified conditions. Spigot connection sizes and their associated duct work connections shall be sized by the Sub-Contractor such that the specified room noise criteria are not exceeded.

Where grilles and diffusers are used for return air in to ceiling voids purpose designed acoustic plenums shall be provided in positions as indicated on the Contract Drawings.

In the event of any installed terminal devices failing to meet the specified room noise criteria, the Contractor shall carry out remedial works or provide replacement units, to the satisfaction of the Engineer, at no extra cost.

G. Chillers and Cooling Towers:

Noise attenuation requirements for all chillers, cooling towers and associated equipment shall be incorporated by the Contractor, as required, to achieve the specified acoustics, noise and vibration criterion.

Floating floors shall be considered by the Contractor, as required, in accordance with Section 096248 (Floating Floors).

H. Constant Volume Units (CVs), Fan Coil Units (FCUs), Variable Air Volume Units (VAVs) and Variable Refrigerant Volume Units (VRVs):

Suppliers of CVs, FCUs, VAVs and VRVs shall provide suitable acoustic data in order to determine compliance with the specified acoustics, noise and vibration criteria.

Casing radiated levels shall be a combination of noise from the air in take opening and the noise from the unit casing. The designs shall ensure that the motors are adequately isolated from the unit casings to ensure that electrically induced vibration does not result in excessive noise radiation from the units.

All noise data shall account for the specified external static pressures along with all coil/filter losses, etc., as free flow operation is not relevant. The designs shall also ensure that the motors are adequately isolated from the unit casings to ensure that electrically induced vibration does not result in excessive noise radiation from the units.

Unless otherwise agreed, the Sub-Contractor shall arrange for independent acoustic laboratory and/or acoustic mock-up performance tests to be carried out on one or more typical units, as selected by the Engineer, under taken in accordance with all relevant Clauses of this Specification and other related Specifications.

In-duct sound power levels($L_{windBre10-12W}$)shall be measured at the air inlet and outlet of the unit (after attenuation if necessary). Casing-radiated sound power

Levels shall also be measured and reported separately. Noise data shall be provided for low, medium and high design fan speeds. The noise at a shall be obtained in accordance with the following, or other accepted equivalent standards, which separate casing - radiated / in let noise from discharge / outlet noise:

AHRI Standard 260: Sound rating of ducted air moving and conditioning equipment.

BS4856 - 4: Methods for testing and rating fan coil units, unit heaters and unit coolers – Determination of sound power levels off an coil units, unit heaters and unit coolers using reverberation rooms (2nd Method).

ISO3741: Acoustics –Determination of sound power levels of noise sources using sound pressure - Precision methods for reverberation rooms. If tests are carried out to ISO 3741 (2nd Method -ducted discharge), the duct end correction shall be calculated using ISO 5135: Acoustics –Determination of sound power levels of noise from air - terminal devices, air - terminal units, dampers and valves by measurement in a reverberation room.

In the event of an installed unit failing to meet the specified acoustic performance, the Contractor shall include for provision of necessary remedial works or replacement units to the satisfaction of the Engineer.

I. Duct and Pipe Acoustic Lagging:

Cladding/lagging and thermal insulation requirements for ducts and pipes shall

be incorporated as required, to avoid break-out or break-back of noise through duct work, and to achieve the specified room noise criteria.

Where acoustic cladding / lagging is necessary and is located on both sides of flexible duct work coupling, the coupling shall also be clad

The acoustic lagging shall be designed to achieve the following minimum performance when measured / rated in a test laboratory, accepted by the Engineer, in accordance with ISO 10140-1, ASTM E90 or acceptable equivalent standard which shall be approved by the Engineer:

Sound Reduction Index/Sound Transmission Loss Octave Band Centre Frequency(Hz)					
125	250	500	1k	2k	4k
16	18	22	30	35	45

External pipe and duct acoustic lagging shall take the form of either of the following:

Construction using a minimum of 50mm (density 32kg/m³) fiber glass covered by a sound barrier material of adequate surface weight to resist the level of break-through envisaged (using dry construction techniques only).

Construction of a separate enclosure where by a support is taken from the duct work and the surface weight is again adequate to resist the expected break-through, or equivalent, all to the satisfaction of the Engineer.

The acoustic lagging shall be fixed to the duct wall using 100% coverage of fire retardant adhesive. The lagging shall additionally be fixed using GI steel bands at not less than 600 mm centres. No pin type fasteners linking the external layer of the lagging to the duct wall shall be used. No fixings which pierce the duct wall shall be used.

The fire rating of the lagging shall comply with BS5588, BS8313 or accepted equivalent, and shall have a BS476 Class 1 surface spread of flame rating, unless otherwise advised by the Engineer.

J. Duct Acoustic Lining:

Where additional measures are necessary to control duct-borne noise, acoustic duct lining shall be provided to provide the following minimum performance when measured in a test laboratory accepted by the Engineer, in accordance with ISO354, ASTM C423 or acceptable equivalent standard which shall be approved by the Engineer, using a Type Amounting as described in ASTM E795:

Random Incidence Sound Absorption Coefficient(α) Octave Band Centre Frequency(Hz)					
125	250	500	1k	2k	4k
0.10	0.25	0.50	0.75	0.80	0.80

The acoustic duct lining shall be dry and free from off-cuts, dust, grease,

solvents, etc.

When fitting the acoustic duct lining insulation, all junctions shall be tightly butted together to prevent heat leakage.

If the acoustic duct lining changes from internal to external, there shall be a minimum over lap of 300mm.

The lining facing shall be factory applied except when a sheet metal facing is specified.

Adhesives shall be restricted to those having the approval to the appropriate authority in relation to fire hazard and shall be used in accordance with the manufacturer's recommendations.

The acoustic duct lining insulation shall be supported against the surface by means of stud welded pins and speed clips. At corners, sheet metal angle can be used to provide a butted edge (to overlap the insulation joints) or the acoustic duct lining insulation can be folded and compressed into the corners.

K. Duct Attenuators:

Duct attenuators shall be purpose-built units constructed by a specialist manufacturer with a minimum of 10 years of experience in sound attenuator manufacture.

Duct attenuators shall be provided as indicated on the drawings and to meet the performance specification given in the attenuator schedules and shall be rectangular or circular(as required) in cross-section and shall be straight in length.

Where applicable, to prevent duct work and ceiling void penetrations reducing the sound insulation between adjacent rooms, cross talk attenuators shall be provided where indicated on the drawings. If common duct work feeds adjacent rooms, the Contractor's duct layouts shall be arranged to minimize the number of cross talk attenuators needed.

Insertion loss data shall be provided from tests under taken in accordance with ISO 7235: Acoustics – Laboratory measurement procedures for ducted silencers and air-terminal units-Insertion loss, flow noise and total pressure loss, or other acceptable equivalent standard approved by the Engineer. Any extrapolation or calculation used to produce the data shall be clearly stated. Tests shall be carried out for attenuators identical in every significant detail to those proposed, for approval by the Engineer, before an order is placed. This refers to air flow velocity, pressure drop and, as far as is reasonable, entry and exit conditions, to demonstrate that the installed insertion losses required in the schedule are met. The insertion loss test data shall include either, the static insertion loss and regenerated noise for the given air flow conditions, or both static and dynamic insertion losses for the given air flow conditions.

Attenuator casings shall be constructed from galvanized steel sheet. Joints shall be longitudinal, lock-formed and mastic sealed during construction. End flanges shall be welded to the casings, and shall be supplied with slotted bolt holes or other fixing

details as necessary.

Casing thickness and flange construction shall be in accordance with HVCA Specification DW/144, but with a minimum thickness of 0.8mm.

Sound absorbent materials used within attenuators shall be to BS5422 (clause4), inert, incombustible, non-hygroscopic, rot and vermin proof and packed under not less than 5% compression to eliminate voids due to settling. The materials shall withstand an air passage velocity of 25m/s without surface erosion or other forms of material migration. The air ways shall be free from projections into the air stream. The material shall achieve. If special attenuators are required inducts serving hygienic areas, the filler material shall be totally encapsulated and sealed with melinex or Mylar film with a thickness of at least 38 microns. The encapsulated filler material shall be totally separated from the internal perforated baffles using a non-combustible, erosion-resistant, acoustic stand-off which must be factory installed. Vapour barrier materials shall be resistant to chemicals as required for the application. Vapour barriers for use in aggressive atmospheres shall not exceed 0.07 mm thickness and be installed in a non-taut state. The material shall achieve Class I to BS476: Part7, and shall not emit toxic or hazardous fumes if ignited.

Attenuator insertion loss data provided shall include for the use of the vapour barrier. The direction of air flow shall be clearly marked on the attenuator outer casing. The airways shall be free from projections into the airstream.

Attenuators for site connection shall be delivered with blocked ends, and be undamaged on completion of installation.

Rectangular attenuators splitter elements shall be round-nosed and stand vertically and be close fitting within the casing and, where required, be easily removable for cleaning.

Where required, access panels shall be fitted with effective air tight seals which shall retain their performance after repeated use.

Notwithstanding the above requirements, all duct attenuators shall be designed, manufactured and installed in accordance with Section 233319 (Duct Silencers).

L. Fans:

Noise attenuation requirements for fans shall be incorporated as required to achieve the specified acoustics, noise and vibration criteria.

The units shall be independently certified to the following standards, or accepted equivalent standards to be advised by the Engineer:

AMCA Standard 301: Methods for Calculating Fan Sound Ratings from Laboratory Test Data.

AMCA Publication 303 (R2012): Application of Sound Power Level Ratings for Fans.

ANSI/AMCA Standard 300: Reverberant Room Method for Sound Testing of

Fans.

To minimize tonal noise generated by the fan operations, electronic noise filters shall be provided. These shall be factory fitted items that consist of heavy copper coils inside a weather proof IP55 enclosure where the copper coil will cut voltage peaks out of the controller and smooth out tonal noises. Each noise filter shall have its dedicated capacitor supplied that will act as a compensator for the filters.

Where additional means are required to control noise break out to the environment, fans shall be attenuated with purpose-built acoustic enclosures, as described in Clause 1.4c.

M. Generator Exhaust Systems:

Refer to Section 263213 (Engine Generators).N.

Grilles, Registers and Diffusers:

Refer to Clause 1.4F.O.

Motors:

Noise attenuation requirements for motors shall be incorporated as required to achieve the specified acoustics, noise and vibration criteria.

The maximum A-weighted sound power levels for factory acceptance testing shall be specified in accordance with EN60034-9 (IEC60034-9) - Rotating electrical machines Noise limits, and shall not exceed the following“ no-load” overall sound pressure levels (indBre2x10-5Pa) at 1m from any side of the motor:

Motors ≤ 22 kW ≤ 75 dBA Motors > 22 kW and ≤ 112 kW ≤ 80 dBA Motors > 112 kW ≤ 85 dBA

Where additional means are required to control noise break out to the environment, motors shall be attenuated with purpose-built acoustic enclosures, as described in Clause 1.4C.

All addition noise control requirements for Electric Motors shall be incorporated as specified in Section 260548-1 (Noise and Vibration Control of Electrical Systems).

P. Pumps:

Noise attenuation requirements for pumps shall be incorporated as required to achieve the specified acoustics, noise and vibration criteria.

Where additional means are required to control noise breakout to the environment, pumps shall be attenuated with purpose-built acoustic enclosures, as described in Clause1.4C.

Q. Transformers and Uninterrupted Power Supply (UPS) Systems:

Transformers and UPS systems shall comply with NEMA (National Electrical Manufacturers Association) Standards Publication No.TR1-1993 (R2000) Transformers, Regulators and Reactors, or other accepted equivalent standards or publications as advised by the Engineer.

Maximum permissible sound pressure levels (SPL's) at 1m from any side (or top) of each transformer shall not exceed the values specified below:

Two-Winding Size (kVA)	Liquid-Immersed Distributed & Network Transformers	Dry-Type Self-Cooled or Sealed Transformers	Dry-Type Forced Ventilation Transformers
0–1000	≤58dBA	≤65dBA	≤68dBA
1501–2000	61dBA	66dBA	69dBA
2001–3000	72dBA	68dBA	71dBA
3001–4000	74dBA	70dBA	73dBA
4001–5000	75dBA	71dBA	74dBA
5001–6000	76dBA	72dBA	75dBA
6001 – 7500	77 dBA	73 dBA	76 dBA

Noise attenuation for Substations, Transformers and UPS Equipment shall be incorporated as specified in Section 260548-1 (Noise and Vibration Control of Electrical Systems).

R. Transportation Systems:

The Specialist Manufacturers for Lifts/Elevators and Escalators shall be responsible for informing the Engineer of items concerning their packages for which the Main Contractor and/or the Electrical Contractor have responsibility. This shall include but not be limited to builders work requirements, delivery times, temporary power and access requirements, power supply locations and electrical power for testing and commissioning.

Noise and vibration levels from escalators and lifts shall comply with all requirements specified in Section 260548-1 (Noise and Vibration Control of Electrical Systems).

1.05 VIBRATION CONTROL REQUIREMENTS

A. Air Cooled Condensers and Similar Types of Cooling Equipment:

Vibration isolation requirements shall be incorporated as specified in this Specification and Section 230548-1 (Vibration and Seismic Controls for HVAC).

Rubber mounts or 'tico' pads shall be installed between the feet of the condenser and steel works or finished roof/floor finish level as a minimum. Vibration or spring type isolators shall not be provided unless the Engineer supplies specific information separately. If vibration/spring isolators shall be provided, pipe work shall use flexible connections to isolate the unit from rigid fixed pipe work. If used, spring isolators shall be located below the support beams, and not the feet which only provide stability.

To ensure no significant tonal noise is generated from the fan operation,

electronic noise filters shall be provided. These are factory fitted items that shall consist of heavy copper coils inside a weather proof IP55 enclosure where the copper coil will cut voltage peaks out of the controller and smooth out tonal noises. Each noise filter shall have its dedicated capacitor supplied that will act as a compensator for the filters

- B. Air Handling Units (AHUs), Fresh Air Handling Units (FAHUs) and Computer Room Air Handling Units (CRAHs):

Vibration isolation requirements shall be incorporated as specified in this Specification and Section 230548-1 (Vibration and Seismic Controls for HVAC).

- C. Compressors:

All floor mounted compressors shall be bolted and grouted to Type 3 or Type 4 concrete inertia bases and supported on Type B or Type C isolators, as specified in Section 230548-1 (Vibration and Seismic Controls for HVAC). Drain lines and air lines shall be supported on Type A hangers, as specified in Section 230548-1 (Vibration and Seismic Controls for HVAC), for the first three supports. Thereafter, isolated pipe clamps shall be used to prevent vibration entering the structure.

- D. Chillers and Cooling Towers:

Vibration attenuation requirements for all chillers, cooling towers and associated equipment shall be incorporated by the Contractor, as required, to achieve the specified acoustics, noise and vibration criteria.

Notwithstanding the above requirements, vibration isolation shall be incorporated, as required, in accordance with Section 230548-1 (Vibration and Seismic Controls for HVAC).

- E. Constant Volume Units (CVs), Fan Coil Units (FCUs), Variable Air Volume Units (VAVs) and Variable Refrigerant Volume Units (VRVs):

The designs shall ensure that motors are adequately isolated from the unit casings to ensure that electrically induced vibration does not result in excessive noise radiation from the units.

- F. Equipment without Motors or Pumps:

Expansion tanks, de aerators, heat exchangers, water heaters, etc., without pumps or motors which are floor-mounted, shall be supported on Type D, E or F mounts, as specified in Section 230548 -1 (Vibration and Seismic Controls for HVAC). Where piping on isolators is connected to these units the connection shall be made with a neoprene flexible connector.

Expansion tanks, de aerators, heat exchangers, water heaters, etc., without pumps or motors which are suspended, shall be supported on Type A or C hangers, as specified in Section 230548 -1 (Vibration and Seismic Controls for HVAC). Where piping on isolators is connected to these units the connection shall be made with a neoprene

flexible connector.

G. Expansion Joints or Arched-Type Connectors:

The tie rod systems on expansion joints used for vibration isolation shall be designed to achieve the vibration isolation required across the joint. The tie rod fixings shall use rubber or neoprene bushed washers to prevent metal to metal contact throughout the normal range of movement of the joint.

H. Fans:

Fan plena, air mixing plena and package air handling plena shall be installed on Type 3, 4 or 5 inertia bases, as specified in Section 230548 -1 (Vibration and Seismic Controls for HVAC), furnished with continuous Type D, E or F mounts located at the entire perimeter of each base, as specified in Section 230548-1 (Vibration and Seismic Controls for HVAC). The plenum manufacturer shall supply and furnish all factory fabricated panels, joiners, access doors, structural supports, channels, fasteners, sealing accessories, assembly drawings and all other items necessary for erection.

Coils and filters in built-up air handling units shall be supported from the housekeeping pads on neoprene pads. Any remaining gaps beneath the coil / filter frames shall be sealed air tight with approved non-hardening sealant.

Each floor-mounted fan or packaged air handling unit shall be mounted on a Type 2 steel base or Type 3, 4 or 5 concrete inertia base with Type B, C or I un-housed steel spring isolators, as specified in Section 230548 - 1 (Vibration and Seismic Controls for HVAC).

Drain pipes for air handling units shall be supported only from the isolated air handling unit frame. The condensate shall drip in to a funnel that is supported from the floor or floor drain. A gap of at least 50 mm shall be maintained between the end of the air handling unit drain pipe and funnel or floor drain.

Suspended fans and air handling units from over head structures shall be hung on Type B or C steel spring hangers, as specified in Section 230548 -1 (Vibration and Seismic Controls for HVAC).

Unless otherwise specified in the Vibration Isolator Schedule, the static deflection of the isolators shall be 50 mm. If the fan does not contain a frame with mounting brackets of suitable rigidity and strength, or if there exists an extreme over hang condition, then a structural steel base shall be used. Fans shall be suspended from above only if expressly noted as such on the Contract Drawings or advised by the Engineer. Any required thrust restraint shall only be by pre-compressed isolators.

I. Flexible Couplings:

Flexible couplings shall be fitted on connections to dynamic plant and equipment including circulating pumps, water pressure boosting sets and refrigeration plant and elsewhere as indicated on the Contract Drawings and instructed by the Engineer.

Couplings shall be suitable for the system fluid, temperatures and pressures and shall be selected to minimize the transmission of vibration to the connecting pipe work.

Couplings shall be constructed from terylene reinforced neoprene, corrosion resistant metal or other accepted equivalent material. Couplings shall be complete with flanged ends to match the pipe work system specification.

Vibration isolated restraining tie bars shall be provided on flexible couplings to refrigeration plant and other dynamic plant connections where recommended by the plant supplier.

All flexible couplings shall be selected with a natural frequency remote from the exciting frequency of the plant to minimize the risk of fatigue failure.

J. Flexible Hoses:

Flexible hoses shall be fabricated from EDPM rubber covered with a stainless steel braid and shall be suitable for use with the working fluid on which they are installed and for the specified system working and test pressures.

Flexible hoses shall be provided with a self-sealing, quick-connect coupling as TEMA or accepted equivalent.

Flexible hoses shall be installed in accordance with the requirements and recommendations of the manufacturer, the HMSA and BSRIACOP11 / 2002

Flexible pipe hose connectors of corrugated metal, or rubber, neoprene or other flexible liner with braided metal or other similar internal or external reinforcing, and intended for use without tie rods, shall have the minimum live lengths recommended by ASHRAE when used for anti-vibration purposes. When, however, the hose

connectors are used to connect equipment with high pressures > 16 bar (g), multiple grooved couplings (MGC) shall be used. The guide lines tabulated below shall be considered for hose connections to different plant equipment.

Equipment	Isolator Type
Pump connections (less than 16 bar(g))	Flexible connectors (FC)
Pump connections (greater than 16 bar(g))	Multiple grooved couplings (MGC) to provide adequate
AHU connections, less than 80mm	Direct connection
AHU connections, 80 mm and greater	Flexible connector (FC) or multiple grooved couplings to provide adequate vibration isolation
FCU connections less than 25 mm	Direct connection
FCU connections 25 mm and above	Flexible hose (FH)
Connections to expansion tanks, deaerators, heat exchangers, etc.	Multiple grooved couplings (MGC) to provide adequate vibration isolation

Connections to chillers, cooling towers and similar plant	Multiple grooved couplings (MGC) to provide adequate vibration isolation
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K. Flexible Joints:

Wherever the natural flexibility of the pipe work cannot practicably be used to accommodate differential expansion, Victaulic flexible joints shall be designed into a system to cater for pipe movement, thermal expansion and contraction, providing the design of the installation complies with the manufacturer's recommendations and the movement limits of the joint are not exceeded.

L. Flexible Pipe Connectors:

Reinforced flexible pipe connectors shall be used where pipes are to be located adjacent to acoustically sensitive spaces needing to achieve $\leq$ NR35. The correct selection of connector shall provide pipe flexibility so that isolators can function properly, protect equipment from strain due to misalignment and expansion or contraction of piping, and to reduce noise and vibration transmission along the piping.

All flexible connectors require end restraint to counter act pressure thrust. Over extension caused by pressure thrust on isolated equipment will cause failure which shall be avoided.

Manufacturer's recommendations on restraint, pressure and temperature limits shall be followed. Flexible neoprene pipe connectors manufactured of multiple plies of nylon tire cord fabric and neoprene both molded and cured in hydraulic rubber presses. No steel wire or rings shall be used as pressure reinforcement.

Straight connectors shall have two spheres. Neoprene elbows shall be manufactured with a single sphere forming the corner of the joint itself. Connectors up to and including 50 mm diameter may have threaded ends. Connectors 63 mm and larger shall be manufactured with floating steel flanges needed to lock the connector's raised face neoprene flanges.

All connectors shall be rated a minimum of 1000 kN/m² at 105 °C. All straight through connections shall be made with either flanged or screwed twin spheres properly pre-extended as recommended by the manufacturer to prevent additional elongation under pressure.

Size 300 mm and larger operating at pressures above 650 kN/m² shall employ control cables with end fitting isolated from the anchoring plates by means of 12 mm thick bridge bearing neoprene washer bushings designed for a maximum of 0.7 kN/mm².

Flexible pipe connectors of corrugated metal, or rubber, neoprene or other flexible liner with braided metal or other similar internal or external reinforcing, and intended for use without tie rods, shall have the minimum live lengths tabulated below (as per ASHRAE guide lines) when used for anti-vibration purposes.

Nominal Pipe Bore Diameter	Live Length
0–65mm	300mm
80–100mm	450mm
125–250mm	600mm
300mm or greater	900mm

Note that the above live length applies to the end-to-end length for flanged flexible rubber hose. For other types of hose, the live length applies to the overall end-to-end length less the fitting length.

M. Generators:

Refer to Section 263213 (Engine Generators).N. Inertia Bases:

Refer to Section 230548-1(Vibration and Seismic Controls for HVAC).O. Isolation of Ducts, Fittings, Pipes and Valves:

Vibration isolation to all distributed ducted and piped services shall follow the requirements specified in Section 230548-1 (Vibration and Seismic Controls for HVAC), and the following additional requirements.

The table below recommends where Resilient Inserts (RI) within pipe clamps, neoprene hangers (NH), spring hangers (SH) and neoprene mounts (NM) shall be used for different building services.

Service	Pipe Diameter	Isolator	Static
Heating	Less than 50mm	NH,NM	8mm
	50–150mm	SH,SM	25mm
	More than 150mm	SH,SM	50mm
Chilled Water	Less than 50mm	NH,NM	8mm
	50–150mm	SH,SM	25mm
	More than 150mm	SH,SM	50mm
Domestic Water	Less than 50mm	NH,NM	8mm
	More than 150mm	SH,SM	25mm
Heat Recovery	Less than 50mm	NH,NM	8mm
	More than 150mm	SH,SM	25mm
Condenser Cooling	Less than 50mm	NH,NM	8mm
	More than 150mm	SH,SM	25mm
Rainwater	Less than 50mm	R	Not applicable
	More than 150mm	R	Not applicable
Sprinklers, Dry Risers/Vents, Hose Reels & Sanitary	All	R	Not applicable
Compressed Air	All Steel	R	Not applicable

The above requirements shall be read in conjunction with Section 230548-1 (Vibration and Seismic Controls for HVAC), all other related Specifications, and the following specific requirements.

Within plant rooms, all pipes connected to equipment installed on spring vibration isolators shall be suspended, supported and guided by means of appropriate spring vibration isolators. This includes domestic water, heating, compressed air, hydraulic fluid, steam and chilled water/glycol pipes, etc. The only exclusions are roof and floor drains and sprinkler piping which shall be supported by Type SM isolators or hung from the structure on Type SH isolators.

The first isolator both upstream and downstream of equipment on springs shall have a static deflection equal to 1.5 times that of the equipment isolators, up to a maximum of 50 mm. The static deflection of the remaining SM/SH pipe isolators in the plant room shall be 25mm.

Piping in the plant room that is connected only to machinery installed on neoprene isolators shall be either supported from the floor on Type NM mounts or suspended from the structure on Type NH hangers.

Where a pipe run connects multiple items of equipment in the plant room the pipe isolators for the entire run shall be chosen to suit the connected equipment of greatest static deflection.

Piping, conduit or mechanical equipment shall not be hung from or supported on other equipment, pipes, or duct work installed on vibration isolators. It shall be supported on or suspended from the building structure.

Where pipes are ganged on a trapeze the pipes shall rest on the trapeze and the trapeze shall be isolated from the structure with the appropriate isolators. Neoprene pipe riser guides shall be used where lateral restraint is required.

All pipes outside the plant room with a diameter greater than 50mm (including domestic water, sanitary sewer and HVAC pipes, excluding roof/floor drain pipes and vent pipes) shall be supported on Type SM isolators or suspended on Type SH isolators, for the first 15m.

Outside plant rooms all HVAC, compressed air and domestic hot and cold water pipes with a diameter less than or equal to 50mm shall be isolated on Type NM and NH isolators for the first 15m. All drain pipes and vent pipes shall be isolated from the structure with closed cell neoprene, felt or glass/mineral fiber sleeves between the pipe and pipe clamp or with neoprene pads between the clamp and the structure.

When it is compressed, the sleeve shall be not less than 3mm in thickness. This provision shall apply to all piping not otherwise required to be installed on vibration isolators. In-line domestic water pumps and the surrounding piping shall be installed on isolators as described above.

Where pipes or flues rise in a vertical chase and are supported from a structure with Type SH or NH isolators and require lateral bracing, Neoprene riser guides shall be

mounted around the pipe to limit lateral movement and to prevent direct contact with the supporting structure.

Riser guides for isolated pipes and flues shall be Type NM mounts based on the vertical edge of a steel angle. The guide shall allow the pipe to move axially, but shall limit lateral movement to $\pm 25\text{mm}$.

Drainage pipe work, including rain water pipes and soil pipes located within spaces of NR 35 or below shall be cast iron and shall be lagged and boxed-in. All pipe supports shall incorporate neoprene inserts with a minimum thickness of 1/10th of the pipe diameter.

Pipes, ducts, and their contents, or other services connected, shall be supported to avoid load on equipment items.

Flexible duct connectors at connections to fans or AHU's shall be a coated fabric with 200mm across a clear gap of 100mm.

The first five supports for duct work on either side of the vibration source shall also be provided with vibration isolators equal in static deflection to that of the fan or AHU.

Electrical connections to equipment mounted on vibration isolation bases shall be made through flexible conduit which changes direction by at least 90° in a minimum length of 25 conduit diameters.

Mineral insulated cables shall be looped through at least 360° at 75mm radius or double the permissible minimum radius, whichever is larger.

For conduit, where isolated containment is ganged on a trapeze it shall rest on the trapeze and the trapeze shall be isolated from the structure with the appropriate isolators. Neoprene pipe riser guides shall be used where lateral restraint is required. The first three containment support vibration isolators shall provide the same static deflection as the equipment vibration isolators.

P. Motors:

Vibration isolation for Electric Motors shall be incorporated as specified in Section 260548-1 (Noise and Vibration Control of Electrical Systems).

Q. Pumps:

Rotating impellers of fans, pumps, chillers, etc., shall be balanced to standards not inferior to the balance quality grades of BS6861 (ISO 1940/1) or to the corresponding residual eccentricities. The criteria shall apply to finished impeller and shaft assemblies at delivery. Where the balancing quality grades are not indicated they shall be to the following minimum standards:

Grade G6.3: Where the product (impeller mass in kg x operating speed in Hz) is less than 250.

Grade G2.5: Where the product (impeller mass in kg x operating speed in Hz) is greater than 250.

Grade G2.5: For impellers on equipment located in occupied areas, on fan convectors, packaged ventilation extract units and warm air heaters with

motor power inputs greater than 100W, or canned rot or pumps or similar.

Impellers operating below 10Hz shall be balanced as if they operated at 10 Hz. Impellers of multi-speed or variable speed machines shall be balanced to the quality grade determined by their highest operating speed.

Vibration caused by fluid dynamic effects of any kind shall not be greater than the equivalent permissible vibration due to an impeller rotating at the frequency of the vibration.

Base mounted pumps $\geq 1.1\text{kW}$ shall be bolted and grouted to prefabricated, reinforced concrete inertia bases, which shall weigh ≥ 1.5 times the combined weight of the fluid filled pump (s) and motor (s). Minimum thickness of the concrete base shall be 225mm for pumps $< 2.2\text{kW}$ and 300mm for pumps $> 2.2\text{kW}$. Any rigid pipe elbows at the pump suction and discharge connections shall be supported from the inertia base. The inertia base shall be supported on Type B or Type C isolation hangers.

Base mounted pumps $< 1.1\text{kW}$ and installed on floors above-grade shall be installed on Type CIB concrete inertia bases with at least 4 Nos. Type D or Type E isolation mounts.

Base mounted pumps $< 1.1\text{kW}$ and located on grade shall be mounted directly on a concrete housekeeping pad, and shall be supported on at least four Type D or Type E isolation mounts.

In-line pumps $\geq 1.1\text{kW}$ shall be supported by the piping which is isolated on Type B isolation hangers. If support is required below the pump, the pump shall be supported on Type B or Type C isolation hangers.

In-line pumps $< 1.1\text{kW}$ and adjacent pipe work shall be supported on Type A isolation hangers, unless other provisions of this Specification or other Specifications call for greater deflections.

If an associated circulation pump or chemical feed pump is to be mounted on the unit, and if the pump is $\leq 0.4\text{ kW}$, the pump shall be mounted to the unit through Type H restrained isolation hangers. Such pumps $> 0.4\text{kW}$ shall be supported from the building structure on Type A isolation hangers or Type E isolation mounts.

Circulation pumps $> 1.2\text{kW}$ shall be supported on Type C concrete inertia bases which are supported on Type B or Type C isolation hangers. All connections to circulation pumps shall be with flexible connections.

Provide motor slide rails, pump base elbow supports, height saving brackets, equipment bolting provisions and isolators as required by the application.

Concrete inertia bases shall be formed in structural steel perimeter bases and reinforced to prevent flexure, misalignment of drive and driven unit, and stress transfer to the equipment. Inertia bases shall be designed to the following thicknesses:

Motorize	Inertia Base Minimum Thickness
Less Than 15kW	150mm
15–39kW	200mm
40–69kW	250mm
70–99kW	300mm
100–149kW	350mm
150–199kW	400mm
Above 200kW	450mm

Concrete inertia bases for pumps shall be sized to support the suction elbow of end suction pumps and both the suction and discharge elbows of horizontal split-case pumps. The bases shall be T-shaped where necessary to conserve space.

All inertia bases shall be sized to extend a minimum of 100mm beyond the base of the equipment, and in the case of belt-driven equipment, 100mm beyond the end of the drive shaft.

R. Substations, Transformers and UPS Equipment:

Vibration isolation for Substations, Transformers and UPS Equipment shall be incorporated as specified in Section 260548-1 (Noise and Vibration Control of Electrical Systems).

S. Vertical Transportation Systems:

Vibration levels from escalators and lifts shall comply with all requirements specified in Section 260548-1 (Noise and Vibration Control of Electrical Systems).

1.06 SERVICES PENETRATION SA.

General:

Where ducts or pipes penetrate the building structure it is essential to prevent such penetrations being either vibration flanking paths or sources of noise leakage. The penetrations shall, therefore, be sleeved with a 25mm thickness of mineral wool having a density of at least 50kg/m³. Care shall be taken to seal any gaps by means of heavy grout and the hole shall be finished with a liberal application of dense non-setting mastic.

All openings around pipes and ducts in the structure surrounding the plant rooms and surrounding noise-critical spaces of NR40 and below, and any structure with a sound insulation requirement, shall be sealed air tight as described herein and on the drawings. This includes all slab penetrations and penetrations of noise sensitive walls indicated on the architectural drawings. It also includes penetrations of all dry wall ceilings and concrete slabs suspended on isolators, and entry/exit of all enclosed shafts. These a ling of penetrations shall be the responsibility of the Contractor.

Where each duct passes through a wall, floor or ceiling, there shall be a clear annular space of 20mm between the duct and structure. After all of the duct work is installed the Contractor shall check the clearance, pack the voids full depth with glass/mineral fibre, and caulk both ends with a non-ageing, non-hardening sealant backed by a polyethylene foam rod or permanently flexible fire stop material. Where there is not sufficient access space to pack around all sides of a duct (e.g., at the underside of a slab), first place a short stub duct in the wall, pack and caulk around it and then attach the inlet and outlet ducts to each end.

Where ducts enter or leave a shaft or pass through a wall or slab in insufficient numbers and density that individual pack-and-caulk details are not possible, a special isolation detail shall be developed.

Before the shaft is fully enclosed the penetration shall be sealed with a heavy membrane surrounding the ducts on each side of the wall or slab being penetrated. Each membrane may be 2 layers of plaster board, 25mm plaster or 14 gauge lead sheet. Other materials may be acceptable. The void between the two membranes shall be packed with mineral fibre or pumped full of resilient closed cell fire stop foam. All edges shall be caulked air tight. (This detail shall be assigned to the Contractor unless otherwise specified. In all cases the proposed details shall be approved by the Engineer.

B. HVAC, Domestic Water, Sewer and Vent Piping:

Where a pipe passes through a sound-sensitive wall, ceiling or floor slab, a steel sleeve shall be cast or grouted into the structure. The internal diameter of the sleeve shall be 40mm larger than the external diameter of the pipe passing through it.

After all of the piping is installed in a specific area, the Contractor shall check the clearance and correct it, if necessary, to within 12mm. Then the void shall be packed full depth with glass/mineral fiber and sealed at both ends, 20mm deep, with sealant backed by foam rod.

Where pipes pass through a masonry wall in insufficient numbers and density that individual pack-and-caulk details are not possible, a special isolation detail shall be developed.

Cast the pipe sleeves in a block of concrete with the sleeves located at least 50mm apart and then build the sleeved block into the wall. The block shall be at least as thick as the surrounding wall construction. Each sleeve shall have a diameter 40mm larger than the external diameter of the pipe passing through it. After the pipes are installed, the voids shall be packed and caulked as described above.

C. Compressed Air and Sprinkler Piping:

Fire protection and compressed air pipes may be sleeved and sealed as described above, or may be grouted and caulked into the structure as follows:

Before grout has set, rake a groove around the pipe on each side of the wall or slab. The groove shall be 12mm wide and 12mm deep. After grout has set, fill the groove full depth with sealant.

Penetration of sound isolating ceilings (concrete or multi-layer plaster board) by sprinkler pipes and heads shall be sleeved and sealed as described herein. There shall be no rigid connection between ceilings and pipes or heads.

1.07 ACOUSTIC COMMISSIONING

- A. If required by the Contract, noise and vibration commissioning shall be undertaken by the Main Contractor and MEP Sub-Contractor to confirm compliance with the specified acoustics, noise and vibration criteria.
- B. Method statements shall be prepared as necessary, for the following site measurements to be completed:
 - Section 230593 - 1 (Testing, Adjusting and Balancing for HVAC).
 - Section 230800 – Commissioning of HVAC.
- C. The type and number of noise and vibration measurements and tests to be completed shall be discussed and agreed with all interested parties.
- D. The Contractor shall be responsible for payment of tests and inspections (inclusive of retests and re-inspections) required because of failures and shall be responsible for all remedial works required to meet the specified acoustics, noise and vibration criteria.

END OF SECTION

MECHANICAL SECTION: 21 10 00
WATER-BASED FIRE SUPPRESSION SYSTEMS

PART 1 – GENERAL

1.1 WORK INCLUDED

1.1.1 Compliance with sections of section1, General Requirements and documents referred to therein.

1.1.2 Conform to General Requirements for Mechanical Services

1.2 DESCRIPTION OF WORK

1.2.1 The work Includes supply, installation, testing and commissioning of combined Fire Sprinkler and Wet Riser system as an integral part of the complete Fire Services Installation.

1.2.2 The work shall also include not only the major items of the system and equipment but also all the incidental sundry components necessary together with the resources for installing such components, for the complete execution of the systems and for trouble free operations, whether or not these sundry components are stated in detail in this contract document.

1.2.3 Particularly related sections:

- a) 220529: Hangers and supports
- b) 220553: Mechanical Identification
- c) 220578: Plumbing Specialties

All related electrical works shall be provided to comply with the requirements of Electrical Divisions.

1.3 STANDARDS AND CODES

- A) All applicable local authority regulations and codes
- B) International Building Code – 2009 edition.
- C) International Plumbing Code – 2009 edition
- D) International Fire Code – 2009 edition
- E) NFPA 10 – Standard for Portable Fire Extinguishers 2007 edition.
- F) NFPA 11 - Low, Medium and High Expansion Foam
- G) NFPA 13 – Installation of Sprinkler Systems 2007 edition.
- H) NFPA 14 – Installation of Standpipe and Hose Systems 2007 edition.
- I) NFPA 15 - Water Spray Systems for Fire Protection
- J) NFPA 16- Installation of Deluge Foam-Water Sprinkler Systems.
- K) NFPA 20 – Installation of Fire Pumps 207 edition.
- L) NFPA 22- Water Tanks for Private Fire Protection
- M) NFPA 24 – Installation of Private Fire Services Mains and Their Appurtenances 2007 edition
- N) NFPA 30 – Flammable and Combustible Liquids Code 2008 edition

- O) NFPA 45 – Standard on Fire Protection for Laboratories Using Chemicals 2004 edition.
- P) NFPA 70 – National Electric Code 2008 edition.
- Q) NFPA 72 –Automatic Fire Detectors 2007 edition
- R) NFPA 99 – Standard for Health Care Facilities 2005 edition.
- S) NFPA 101 – Life Safety Code 2006 edition.
- T) NFPA 291 – Recommended Practice for Fire Flow Testing and Marketing of Hydrants 2007 edition.
- U) Factory Mutual (FM) Requirements – for Fire Protection Equipment
- V) Underwriters Laboratories (UL) listings
- W) British Standards: BS EN 671-1:2001
- X) Fire protection system shall be approved by DCD

1.4 DESIGN CRITERIA

- 1.4.1 The combined fire suppression system shall be selected to meet the required duties stated in the drawings and schedules.
- 1.4.2 The entire Fire Protection System shall be hydraulically analyzed and installed by a specialist Civil Defence approved Subcontractor only.
- 1.4.3 The sprinkler system shall meet all requirements of National Fire Protection Association and shall also satisfy local Civil Defence Authorities regulations and requirements.
- 1.4.4 Sprinkler Occupancy Hazard Classifications:

Office Areas	: Light Hazard
General Storage Areas and Equipment Rooms	: Ordinary Hazard – Group 2
Medical Centre & Laboratory Areas	: Ordinary Hazard – Group 2
Main Kitchen Block	: Ordinary Hazard – Group 2

1.4.5 Minimum Density for Automatic-Sprinkler Piping Design:

- a) Light-Hazard Occupancy : 4.1 LPM/SQ.M over 139-sq. m area.
- b) Ordinary-Hazard-Group-1 : 6.5 LPM/SQ.M over 139-sq. m area.
- c) Ordinary-Hazard-Group-2 : 7.5 LPM/SQ.M over 139-sq. m area

1.4.6 Maximum Protection Area per Sprinkler:

Light Hazard : 20.9 sq. m.

Ordinary Hazard : 12.1 sq. m.

1.5 SUBMISSIONS

1.5.1 Submit the complete fire protection system data sheets. List piping material types, ASTM number, schedule or pressure class, joint type, manufacturer and model number where appropriate. List valves, specialties and equipment with manufacturer and model number.

1.5.2 Submit product data for each fire stop system. Submittals shall include product characteristics, performance and limitation criteria, test data, installation details and procedures for each method of installation applicable to this project.

1.5.3 Shop drawings shall indicate water supply location and size, piping layout and size, sprinkler locations and type, hanger locations and type, equipment locations and type, valve locations and type, occupancy classes, hydraulic calculations reference points, node references of remote area and discharge densities. Fire suppression data, classification, and fire and smoke partitions shall also be indicated.

1.5.4 Submit hydraulic calculations for water supply and sprinkler systems. Include summary sheet and detailed work sheets. Describe characteristics of water supply and location of effective point used in calculations. Include graph illustration of water supply, hose demand, sprinkler demand

1.5.5 Together with Shop Drawings and Product Data, Plans and Specifications shall be submitted in conformance with NFPA. Contractor shall be responsible for obtaining approvals from Civil Defence Authority.

1.5.6 Test Reports shall be submitted in accordance with the approved format giving all required test details.

1.5.7 Steel pipes: Statement from manufacturer on his letterhead that the pipe furnished meets the BS and/or ASTM specification contained in this section. Contractor shall submit the Mill certification papers, also known as material test reports, for the pipe furnished for this project, in English. Heat numbers on these papers to match the heat numbers stenciled on the pipe. Chemical analysis indicated on the mill certification papers to meet or exceed the requirements of the referenced BS and/or ASTM specification.

1.6 QUALITY ASSURANCE AND CONTROL

1.6.1 Contractor shall use all Steel pipes with heat numbers rolled, stamped, or stenciled to each length or each bundle, depending on the size of the pipe, and in accordance with the appropriate specifications.

1.6.2 Any installed material not meeting the specification requirements must be replaced with material that meets these specifications.

1.6.3 All equipment and components used for fire fighting installation shall be UL / FM approved.

1.7 PRODUCT DELIVERY, STORAGE AND HANDLING

- 1.7.1 Handle equipment and components carefully to prevent damage, breaking, denting and scoring. Do not install damaged pumps or components, replace with new.
- 1.7.2 Store equipment and components in clean dry place. Protect from weather, dirt, fumes, water, construction debris, and physical damage.

PART 2 - PRODUCTS

2.1 FIRE PIPE WORK MATERIAL

- 2.1.1 All above ground fire pipe work shall be of seamless type piping to ASTM A53 SCH 40, screwed and socketed. The fittings shall be of malleable iron or wrought iron galvanized and of steam quality or Grooved type fitting (UL/FM approved).
Pipe joints shall be threaded for pipe sizes 50 mm and smaller, and for large sizes grooved with mechanical coupling and gaskets or flanged.
- 2.1.2 All below ground (buried underground) fire pipe work shall be in HDPE material withstanding the working pressure of 1200 KPA (12 bars) and test pressure of 1800 KPA.
- 2.1.3 Material and works with specified Steel Pipework, Grooved Steel Pipes with Couplings, Mechanical Joints and Gaskets, Flange Adapters, Fire Protection Fittings shall conform to requirements as specified within Section, pipes and tubes.
- 2.1.4 All materials and products shall be installed in accordance with NFPA Standards, ANSI/AWWA and other applicable Standards and manufacturer's published recommendations. Grooved end product manufacturer to be ISO-9001 certified.
- 2.1.5 All pipework and fittings shall conform to the relevant clauses in this specification.
- 2.1.6 Use only new material, free of defects, rust and scale, and meeting the latest revision of BS and/or ASTM specifications as listed in this section of specifications.
- 2.1.7 Construct all piping for the highest pressures and temperatures in the respective system in accordance with ANSI B31, but not less than 8 bar unless specifically indicated otherwise.
- 2.1.8 Where weld fittings or mechanical grooved fittings are used, use only long radius elbows having a centre line radius of 1.5 pipe diameters.
- 2.1.9 No elbows will be allowed on fire service systems where all changes of direction shall be by means of standard bend springs or long radius bends.
- 2.1.10 Where pipes cross fire rated walls and slabs, beams and waterproof membranes, galvanized steel sleeves shall be used and the penetrations shall be fireproofed (using flexible materials) to the same rating as the wall. The fire proofing materials shall be applied strictly as per the manufacturer's recommendations.

2.2 SPRINKLER HEADS

- 2.2.1 Only approved Sprinkler Heads specified by model number identifying the type and style of head shall be acceptable for installation. The temperature classification rating shall meet NFPA requirements based on the area classification.

- 2.2.2 Sprinkler Head Connections: Direct sprinkler head connections, branch connections, drop nipples and sprigs shall be made with outlet connections with locating collar engaging into hole.
- 2.2.3 Glass bulb type, cast brass or bronze construction. Provide heads with nominal 13mm discharge orifice except where greater than normal density requires large orifice.
- 2.2.4 Select sprinkler head temperature rating to not exceed maximum ambient temperature rating allowed under normal conditions at installed location. Provide ordinary temperature classification rating of 790C / 68 where exposed to ambient temperature such as skylights, sealed display windows, unventilated attics and roof spaces. The sprinkler head for different areas shall be selected based on the area applicable and shall fully comply with the regulations.
- 2.2.5 Provide 24 Nos of spare sprinkler heads and 1 wrench for each type of sprinkler head and temperature range installed. The steel cabinet shall be provided to store the sprinkler heads.
- 2.2.6 Sprinkler head types shall be as indicated on drawings. Guards shall be selected where mechanical damage is likely. the following finishing are required as a minimum:
 - a) Pendant: chrome plated finish and escutcheon.
 - b) Upright: brass finish.
 - c) Horizontal Sidewall: chrome plated finish and escutcheon.
 - e) Quick Response Pendant: chrome plated finish and escutcheon.
 - f) Quick Response Sidewall: chrome plated finish and escutcheon.

In all areas with false ceiling, sprinkler heads shall be concealed type with flat cover plate at false ceiling level. Colour of cover plate shall be approved by the Consultant.

2.3 VALVES

2.3.1 Ball valves:

1. 50mm and smaller: body material to be brass (Dezincification resistant copper alloy (CW602N)), threaded or sweat ends, standard port, blowout proof stem, chrome plated ball, glass reinforced seats, UL approved at 2500 kPa (25 bar).

2.3.2 Gate valves:

50mm and smaller: Outside screw and yoke gate valves 2500 kPa (25 bar)., b: body material to be brass (Dezincification resistant copper alloy (CW602N)), screwed bonnet, rising stem, solid wedge, with normally open supervisory/tamper switch with double wire leads.

65mm and larger: Outside screw and yoke gate valves, 2500 kPa (25 bar).cast iron body / ductile iron mounted, bolted bonnet, rising stem, solid wedge, with normally open supervisory/tamper switch with double wire leads and rising stem.

2.3.3 Butterfly valves:

50mm and smaller body material to be brass (Dezincification resistant copper alloy (CW602N)), butterfly valve, 2500 kPa (25 bar), geared operator, visible position indicator, normally open supervisory/tamper switch with double wire leads, stainless steel disc and stem.

50mm and larger: Cast or ductile iron body butterfly valve, lug style, 2500 kPa (25 bar).geared operator, visible position indicator, normally open supervisory/tamper switch with double wire leads, EPDM resilient seat, EPDM seals, nickel plated ductile iron disc. Valve assembly to be bubble tight to 12bar with no downstream flange/pipe attached. Use cap screws for removal of downstream piping while using the valve for system shutoff.

2.3.4 OS &Y valve :

50mm and smaller body material to be brass (Dezincification resistant copper alloy (CW602N)), butterfly valve, 2500 kPa (25 bar), geared operator, visible position indicator, normally open supervisory/tamper switch with double wire leads.

Standard: UL 262, End Connections: Threaded.

50mm and larger: Cast or ductile iron body OS &Y valve, 2500 kPa (25 bar).geared operator, visible position indicator, normally open supervisory/tamper switch with double wire leads.

Standard: UL 262, End Connections: Flanged or grooved.

2.3.5 Supervisory/tamper switches:

For O S & Y valve or butterfly valve installations, UL/FM listed/approved, to monitor position of valve, tamper resistant cover screws, single or double SPDT switch contacts, corrosion resistant, for indoor or outdoor use, NEMA 4 & 6P enclosures.

2.3.6 Check valves:

50mm and smaller: body material to be brass (Dezincification resistant copper alloy (CW602N)), threaded end, Y-pattern, regrindable bronze seat, renewable bronze disc, 2500 kPa (25 bar), suitable for installation in a horizontal or vertical line with flow upward.

65mm and larger: Cast or ductile iron body, flanged or grooved ends, bronze trim, bolted cap, renewable bronze seat and disc, 2500 kPa (25 bar).suitable for installation in a horizontal or vertical line with flow upward.

Provide 12mm automatic drip drain on inlet of fire dept. connection check valve.

2.3.7 Pressure-Restricting Valves: The valve shall be rated for 2500 kPa (25 bar). minimum, brass construction. Include, female NPS inlet and outlet; adjustable setting feature; and straight or 90-degree angle pattern design.

2.3.8 Drain valves:

25 MM min. two or three piece bronze body ball valve; threaded ends, chrome plated bronze ball; glass filled teflon seat; teflon packing and threaded packing nut; blowout-proof stem; 2,760 kPa (27.6 bar), with hose thread outlet and cap.

2.3.10 Alarm check valves:

Cast or ductile iron body, flanged or grooved ends, 2500 kPa (25 bar) minimum working pressure, bronze grooved seat with o-ring seal, single hitch pin and latch design. Provide trim for bypass, drain, electric sprinkler alarm switch, pressure gages, precision retarding chamber, drip cup assembly piped to floor or hub drain, fill line attachment with strainer.

2.4 FLOW SWITCHES

A vane type water flow switch with metal enclosure with adjustable pneumatic retard and electrical characteristics to be compatible with alarm system. Flow switches shall be UL/FM approved.

2.5 PRESSURE SWITCHES

Pressure actuated switch with field adjustable settings, metal enclosure and electrical characteristics compatible with alarm system. Pressure switches shall be UL/FM approved.

2.6 LOCAL ALARM BELL

Weatherproof electric alarm bell with red painted metal housing, mounting base and gong; solenoid operator; weatherproofing o-ring seal and electrical characteristics compatible with alarm system. Alarm bells shall be UL/FM approved.

2.7 PRESSURE GAUGES

Cast aluminium, stainless steel or brass case of not less than 9 cm in diameter, double strength glass window, black lettering on a white background, phosphor bronze bourdon tube with bronze bushings, recalibration from the front of the dial, 99% accuracy over the middle half of the scale, 98.5% accuracy over the remainder of the scale. Include bronze 3-way globe valve with plugged outlet for Fire Inspector's test gauge. Pressure gauges shall be UL/FM approved.

2.8 FIRE PUMP TEST METER(Where applicable)

Fire Pump Test Meter, UL/FM approved, incorporating a calibrated venturi and attached GPM meter, to be installed on the discharge side of the fire pump, to accurately measure pump performance. Test Meter shall be supplied with grooved ends for installation with grooved end couplings.

2.9 WATER MOTOR ALARM

Factory assembled water motor shall consist of body, impeller, gasket and cover. Body shall be cast from highly corrosion resistant aluminum magnesium alloy.

2.10 LANDING VALVES

Hydrant outlets should be of a type acceptable to the public fire brigade and should comprise a valve 65mm bore constructed in gun metal, screwed or flanged for attachment to the riser and fitted with a 65 mm instantaneous female coupling and a blank cap secured by a suitable length of chain. The valve gate should lift clear of the waterway and the valve cover should be securely fitted to the valve body so that it does not unscrew when operated. The valve spindle should not be less than 22mm diameter and fitted with a gun metal hand wheel about 150mm in diameter which should be anti-clockwise. The whole fitting should be of sound construction and shall be suitable for a working pressure of 16 bar. Valve shall comply with BS 5041/ UL1468. Landing valve nozzle shall be stainless steel finish.

The hydrant outlet valves used shall be of the type incorporating a device to limit the outlet pressure to a maximum of 4.5 bar. The instantaneous coupling shall be of the single-lug twist release instantaneous pattern with vulcanized rubber blank cap. Valves shall be UL listed / KITEMARK

2.11 EXTERNAL FIRE HYDRANTS

Hydrants shall be located in strategic locations generally at 60-70 meters intervals. Hydrants shall be a standpipe/pillar type with 2 outlets/landing valves per standpipe and shall be of a type approved by the Local Civil Defence. The hydrant valves shall be provided with pressure reducing valve to limit the pressure to manageable levels.

Hydrants shall not be located closer than 12.2 metres to the structure being protected or be separated from the structure with materials providing a 2 hour fire rating, except with the specific permission of the Client.

The design of hydrant connections to fire mains shall take into account the reactive forces that will be experienced during hydrant operation. Provide all necessary anchors and thrust blocks.

Bollards shall be provided around the hydrants to protect them from any mechanical damage by vehicles.

2.12 FLEXIBLE FIRE HOSES AND NOZZLE

Hoses shall be 65mm diameter suitable for a maximum working pressure of 12 kg / cm². Hoses shall be all synthetic nylon wrap and weft totally encapsulated in PVC/N to form a unified lining and cover. Hoses shall be UL/FM / Kitemark approved or certified by an equal certification authority for use on fire service. Hose length shall be 30M and shall terminate in a nozzle at one end. Nozzle shall be light weight of aluminum alloy with three operating positions-OFF, STRAIGHT JET and WATER SUPPLY. Fire hose reel nozzle shall be stainless steel finish.

The other end of the flexible hose shall have a quick connect coupling-to suit the landing valve.

2.12 FIRE BREECHING CONNECTIONS

4 ways breeching connections for use as alternative water supply entries by the Local Fire Department shall be of 65 mm male inlet type in accordance with BS 5041 Part 3: 1975 (1987) and local civil defense requirement. The pressure rating shall comply with the system pressure.

Breeching connections shall be housed in glass fronted boxes to BS 5041: Part 5: 1974 (1987) marked "Wet Riser Inlet" in English and Arabic. Cabinet shall be of stainless steel / construction.

2.13 FIRE HOSE CABINET

2.13.1 Fire Hose Cabinet

Fire hose cabinets, valves, hose and accessories shall be approved by and shall be in conformance with the requirements of the Consultant.

2.13.2 Unless otherwise indicated on drawings, fire hose and landing valve cabinet shall be a vertical combined unit with two compartments, one for hose reel and the other for landing valve, hose, nozzle and fire extinguishers.

2.13.3 All cabinets shall be MS powder coated.

- 2.13.4 Cabinets shall be suitable either for recessed or surface installation in locations as indicated on drawings.
- 2.13.5 Painted on the door of the fire hose cabinet with minimum 7mm block letters shall be the following notice in English and Arabic.

**IMPORTANT
PULL HOSE FROM REEL
WATER WILL FLOW AUTOMATICALLY**

- 2.13.6 Hose reel cabinet shall contain an automatic red powder coated swing hose reel complete with 30m of 25mm rubber fire hose to BS-EN 671-1 with chrome plated brass shutoff nozzle clamped ends, automatic valve to open water flow after 2 ½ turns of the reel and 25 mm key operated shut-off valve. All metal parts shall be polished and chrome plated.
- 2.13.7 Where recessed enclosure with timber door is provided on the architectural drawings as a fire cabinet, only the fire equipments shall be provided inside with signs on the timber door in Arabic and English.

2.14 PORTABLE FIRE EXTINGUISHERS

- 2.14.1 Fire extinguishers shall be 5 kg carbon dioxide type and 6 Kg dry powder type to the approval of the local authorities.
- 2.14.2 Fire extinguishers shall be provided in each Fire Cabinet, and in each Mechanical, telephone and Electrical Rooms and where required by the local Civil Defense Authority.
- 2.14.4 Extinguishers for use in Mechanical, Electrical and telephone Rooms shall be furnished with wall bracket and shall be mounted on walls.
- 2.14.5 The pressure test on the extinguishers shall remain valid for 5 years.
- 2.14.6 Diesel generator room shall be equipped with 6Kg. Foam extinguisher mounted on wall brackets.
- 2.14.7 Mount all fire extinguishers in offices and public areas in Fire Extinguisher Cabinets similar to the cabinets described for those reels with die cut lettering label 'Fire Extinguisher'.

2.15 AUTOMATIC FIRE HOSE REELS

Fire hose reel shall be slim type, recess mounted, swinging type to comply or exceed with BS 5274/EN 671 and shall be housed in cabinets.

The main contractor to Architectural specification shall install installation of doors and identification marks.

All the internal parts used shall be made of material according to BS 1490 LM 16/24. The hose tubing should be 25mm internal diameter rubber reinforced and 30mtr long to comply with BS 3169 type 1 class B or EN 694 with the exception of car parking areas where hose reels, with extended coverage tubing to be used.

The hose reel nozzle to comply with EN 671-PN 16/DN 25/6.

The unit shall be complete with lock shield valve and 25mm pressure reducing valve to maintain pressure of 3 bar at outlet.

For testing and maintenance drain line with non return valve to be used. Brief operating instructions should be clearly displayed on each hose reel.

2.16 FITTINGS FOR SITE FIRE SERVICES

2.16.1 External Fire Hydrants

External Fire hydrant shall be of the pillar type with protecting shell and shall meet or exceed the specifications of the local civil defence authority and French Standard NFS 61213. The hydrant shall include following features:

Upper barrel shall be of best quality ductile iron. Lower barrel shall be of cast iron. All barrel castings shall be painted inside and outside with 300 micron thick non toxic fusion bonded epoxy. Aluminum shell shall be painted with 60 micron thick polyester paint to colour as approved by the local Civil Defence Authority.

Hydrant rod shall be of steel, zinc electroplated in two part construction held together by a rod connector made of hot dip galvanized ductile iron.

A cast bronze nut shall directly actuate the hydrant rod. Grease groove shall ensure complete lubrication and long life 'O' rings shall seal the lubrication chamber and prevent water entering the operating mechanism.

Drain ring shall be of bronze.

Hydrant shall be with one front outlet size 100mm diameter and two side outlets each 65mm diameter. Outlets shall be provided with quick connect couplings, protective caps held in place by chains. Fire hydrants shall be with enclosed outlets in an aluminium casing as approved by the Civil Defence Authority and shall be suitable for a maximum working pressure of 16 Bars. Hydrants outlets shall be provided with quick connect couplings to the approval of the Local Civil Defence Authority.

2.16.2 Flexible fire hoses and Nozzle

Hoses shall be 65mm diameter suitable for a maximum working pressure of 12. Hoses shall be all synthetic nylon wrap and weft totally encapsulated in PVC/N to form a unified lining and cover. Hoses shall be UL/FM/ Kitemark approved or certified by an equal certification authority for use on fire service. Hose length shall be 30M and shall terminate in a nozzle at one end. Nozzle shall be light weight of aluminium alloy with three operating positions - OFF, STRAIGHT JET and WATER SPRAY.

The other end of the flexible hose shall have a instantaneous coupling to suit the fire hydrant.

Two flexible hoses along with nozzles shall be housed within a weather proof fire cabinet located near each fire hydrant.

Fire cabinet's construction shall be of 1.2mm thick steel. Each cabinet shall be fitted with internal fire hose support 300 mm from the top and reinforced space to facilitate mounting of 610mm tubular pedestal support finished in black powder coat.

2.17 Kitchen Hood Fire Extinguishment System

An automatic suppression system shall be provided for the kitchen hood areas as indicated on the drawings. In a fire situation the system shall be automatically actuated to discharge a spray of wet chemical to suppress the fire. A supervised alarm siren shall sound in the hazard area until manually silenced at the control panel. Warning signs outside hazard door shall be illuminated and the local air conditioning system shall be shut down.

The system shall also be interlocked with the fire alarm system to indicate an alarm both at the local fire alarm panel and at the fire station.

2.18 Foam Proportioning System

Wherever indicated on the drawings the contractor shall provide foam proportioning system with full design from specialist supplier, design and calculations shall be approved by Consultant prior to the selection of equipment. The system shall include the following:

Foam concentrate stored within butyl rubber fixed inside steel pressure vessel connected to balanced pressure foam proportioners. Upon operation, water at fire main pressure enters the tank between the vessel wall and the rubber bag. Foam concentrate is then expelled from the bag into the balanced pressure proportioner.

The system shall be complete with the following facilities:

Automatic operation pressure operated valve.

Sight glass within the bladder.

PART 3 - EXECUTION

3.1 PREPARATION

Piping and Fittings. Ream all pipe to full inside diameter after cutting and thoroughly clean before erection. All material shall have proper identification marks that can be verified prior installation.

3.2 INSTALLATION

- Install fire protection systems in accordance with NFPA rulings, listings and manufacturers recommendations. Locate where accessible for servicing and replacement.
- Install piping so that system can be drained. Where possible, slope to main drain valve. Where piping cannot be fully drained, install nipple and cap for drainage of less than 20 lit or valve/nipple/cap for drainage over 20 lit. Pipe drain valves to grade or to air gap sewer receptor.
- General installation works shall be in accordance with section Pipes and Tubes UL,FM and civil defense .
- Run all piping as direct as possible, avoiding unnecessary offsets. Conceal piping in finished rooms unless shown or specified otherwise.
- Where piping is embedded in masonry or concrete, provide protective sleeve covering of elastomeric pipe insulation. Maintain piping in clean condition internally during

construction. Provide clearance for access to valves. Provide anchors, expansion joints, swing joints and/or expansion loops so that piping may expand and contract without damage to itself, equipment or the building.

- Riser clamps: Support vertical piping with clamps secured to the piping and resting on the building structure or secured to the building structure below at each floor.
- Support for firefighting pipes shall be approved by UL, FM and civil defense
- Sprinkler heads shall locate sprinkler heads as indicated on fire protection plan and reflected ceiling plan maintaining minimum clearances from obstructions, ceilings and walls. Install sprinkler heads level in locations not subject to spray pattern interference.
- Switches: Locate flow and pressure switches where indicated and where required to obtain specified zoning to isolate floors and major areas of floors. Provide valved test connection for flow switch adjacent to flow switch. Pipe to floor drain. Test flow switch to verify proper operation.
- Gauges: Provide a valved pressure gauge in main fire protection riser, at the top of each piping riser, at inlet and outlet of pump and elsewhere as indicated.
- Valves: Properly align piping before installation of valves. Do not support weight of piping system on valve ends. Mount valves in locations where access for operation are available for servicing and replacement. Install all valves with the stem in the upright or horizontal position. Valves installed with the stems down will not be accepted. Provide a riser shutoff valve and a capped hose thread drain valve at the bottom of each riser. Provide capped hose thread drain valves to allow draining of each portion of piping.
- Specialty Valves: Install in vertical position fire protection riser. Install trim recommended by manufacturer including drain and test valves. Pipe drains to hub or floor drains. Test and adjust operation of valves, alarms and pressure maintenance devices.
- Fire Department Connections: Mount on wall where indicated. Support from structure independent of piping. Fill wall penetration with insulation and caulk exterior and interior face of wall opening weathertight. Provide an automatic ball drip installation in the inlet piping between the F.D.C. and the check valve, for drainage.

3.3 PAINTING

All fire fighting pipework (whether exposed or concealed) shall be given one coat of approved primer (similar to Jotun Jotamastic 80 and Hardtop XP as primer and paint) which can meet the LEED requirement for VOC content, and two coats of approved post office red enamel paint. Identification and marking shall be in accordance with Section 15075 Mechanical Identification and shall fully comply with NFPA regulations.

3.4 SIGNAGE

Careful consideration shall be given to the location of fire protection signage. A set of layouts shall be produced advising the type of fire protection signage and their location.

Signage shall be supplied as of type approved by the Local Civil Defence. Details of signage shall be submitted to Client and approved prior to the commencement of construction.

All signs shall be Arabic and English, with the Arabic (medium Naskh) being above

the English (Helvetica medium).

The floor area beneath each fire extinguisher shall be delineated to stop the storage of articles below them so that there is a clear access path should it be required to use the extinguisher.

Exact location of the fire extinguishers shall be agreed with the client during a visit on site.

3.5 PRE-COMMISSIONING CLEANING

Before installations are handed over or subjected to inspection and testing, the entire installation shall be thoroughly cleaned, both internally and externally.

All fire protection installation shall be flushed out with clean water. During the flushing out, provision/special care shall be made for water discharge and subcontractor should ensure that the cleaning operation avoids any damage to the plant and other services installed. The entire operation shall be carried out to the satisfaction of the Consultant. The contractor shall submit the method statement for Consultant's review prior to commencement of any works related with commissioning.

3.6 TESTING AND COMMISSIONING

Testing and Commissioning shall be as detailed in standards listed in this Section as well as per Section Testing, Adjusting and Balancing, Section Pipes and Tubes.

The testing and commissioning of diesel driven and electrical pumps shall fully comply with NFPA regulations and should also satisfy the local Civil Defence requirements.

The fire protection piping system shall be tested hydrostatically for not less than 2 hours at 1 ½ times the system pressure without leak.

Testing of pipe networks in sections or entire system shall be carried out as required by sequence of construction program. Do not conceal any pipework unless and until it has been successfully tested and certified. If required for the additional pressure load under test, provide temporary restraints at fittings or expansion joints. The entire test must be witnessed and certified by the Consultant. All pressure test reports are to be documented in accordance with NFPA Contractor's Material and Test Certificate forms.

Use clean water and remove air from the piping being tested where possible. Measure and record test pressure at the high point in the system.

END OF SECTION

MECHANICAL SECTION: 21 13 13
WET PIPE SPRINKLER SYSTEM

PART 1 GENERAL

1.1 RELATED DOCUMENTS

- A Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A Section Includes:

- 1 Pipes, fittings, and specialties.
- 2 Fire-protection valves.
- 3 Fire-department connections.
- 4 Sprinklers.
- 6 Alarm devices.
- 7 Manual control stations.
- 8 Control panels.
- 9 Pressure gages.

- B Related Sections:

- 1 Section 211000 "Water based fire suppression systems".

1.3 DEFINITIONS

- A High-Pressure Sprinkler Piping: Wet-pipe sprinkler system piping designed to operate at working pressure higher than standard 175 psig (1200 kPa), but not higher than 300 psig (2050 KPa).
- B Standard-Pressure Sprinkler Piping: Wet-pipe sprinkler system piping designed to operate at working pressure of 175 psig (1200 kPa) maximum.

1.4 SYSTEM DESCRIPTIONS

- A Wet-Pipe Sprinkler System: Automatic sprinklers are attached to piping containing water and that is connected to water supply through alarm valve. Water discharges immediately from sprinklers when they are opened. Sprinklers open when heat melts fusible link or destroys frangible device. Hose connections are included if indicated.

1.5 PERFORMANCE REQUIREMENTS

- A Standard-Pressure Piping System Component: Listed for 175-psig (1200-kPa) minimum working pressure.
- B High-Pressure Piping System Component: Listed for 300-psig (2070-kPa) working pressure.
- C Delegated Design: Design sprinkler system(s), including comprehensive engineering analysis by a qualified professional engineer, using performance requirements and design criteria indicated.

- D Sprinkler system design and equipment shall be approved by civil defense.
- 1 Margin of Safety for Available Water Flow and Pressure: 10 percent, including losses through water-service piping, valves, and backflow preventers.
 - 2 Sprinkler Occupancy Hazard Classifications shall be as per design drawings and NFPA-13, and civil defense fire code.
 - 3 Minimum Density for Automatic-Sprinkler Piping Design shall be as per design drawings and NFPA standards and civil defense fire code.
 - 4 Minimum Density for Deluge-Sprinkler Piping Design shall be as per design drawings and NFPA standards and civil defense fire code.
 - 5 Maximum Protection Area per Sprinkler: As per design drawings and UL listing.
 - 6 Total Combined Hose-Stream Demand Requirement: According to NFPA 13 unless otherwise indicated:
 - a. Light-Hazard Occupancies: 100 gpm (6.3 L/s) for 60 minutes.
 - b. Ordinary-Hazard Occupancies: 250 gpm (15.75 L/s) for 60 minutes.
 - c. Extra-Hazard Occupancies: 500 gpm (31.5 L/s) for 90 minutes.

- E Seismic Performance: Sprinkler piping shall withstand the effects of earthquake motions determined according to NFPA 13.

1.6 SUBMITTALS

- A Product Data: For each type of product provide manufacturers catalog sheet indicating size, capacity and accessories.
- B Shop Drawings: For wet-pipe sprinkler systems. Include plans, elevations, sections, details, and attachments to other work.
- 1 Wiring Diagrams: For power, signal, and control wiring.
- C Delegated-Design Submittal: For sprinkler systems indicated to comply with performance requirements and design criteria, including analysis data signed and sealed by the qualified professional engineer responsible for their preparation.
- D Coordination Drawings: Sprinkler systems, drawn to scale, to be coordinated with all MEP services.
- E Qualification Data: For qualified Installer.
- F Approved Sprinkler Piping Drawings: Working plans, prepared according to NFPA 13, that have been approved by Local Authorities, including hydraulic calculations.
- G Welding certificates.
- H Fire-hydrant flow test report.

I Field Test Reports and Certificates: Indicate and interpret test results for compliance with performance requirements and as described in NFPA 13. Include "Contractor's Material and Test Certificate for Aboveground Piping."

J Field quality-control reports.

K Operation and Maintenance Data: For sprinkler specialties to include in emergency, operation, and maintenance manuals.

1.7 QUALITY ASSURANCE

A Installer Qualifications:

1 Installer's responsibilities include designing, fabricating, and installing sprinkler systems and providing professional engineering services needed to assume engineering responsibility. Base calculations on results of fire-hydrant flow test.

a Engineering Responsibility: Preparation of working plans, calculations, and field test reports by a qualified professional engineer.

2 Company specializing in performing the work of this section with minimum 7 years experience approved by manufacturer.

B Welding Qualifications: Qualify procedures and operators according to ASME Boiler and Pressure Vessel Code.

C Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

D NFPA Standards: Sprinkler system equipment, specialties, accessories, installation, and testing shall comply with the following:

1 NFPA 13, "Installation of Sprinkler Systems."

2 NFPA 13R, "Installation of Sprinkler Systems in Residential Occupancies up to and Including Four Stories in Height."

3 NFPA 24, "Installation of Private Fire Service Mains and Their Appurtenances."

1.8 PROJECT CONDITIONS

A Interruption of Existing Sprinkler Service: Do not interrupt sprinkler service to facilities occupied by Owner or others unless permitted under the following conditions and then only after arranging to provide temporary sprinkler service according to requirements indicated:

1 Notify Construction Manager no fewer than two days in advance of proposed interruption of sprinkler service.

2 Do not proceed with interruption of sprinkler service without Construction Manager's written permission.

3 system design and material shall be approved by civil defense

1.9 COORDINATION

A Coordinate layout and installation of sprinklers with other construction that penetrates ceilings, including light fixtures, HVAC equipment, and partition assemblies.

1.10 EXTRA MATERIALS

A Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

- 1 Sprinkler Cabinets: Finished, wall-mounted, steel cabinet with hinged cover, and with space for minimum of six spare sprinklers plus sprinkler wrench. Include number of sprinklers required by NFPA 13 and sprinkler wrench. Include separate cabinet with sprinklers and wrench for each type of sprinkler used on Project.

PART 2 RODUCTS

2.1 PIPING MATERIALS

- 2.1.1 All above ground fire pipe work shall be of standard weight galvanized steel piping to ASTM A53 Grade B SCH 40, screwed and socketed. The fittings shall be of malleable iron to ASME B16.3 or wrought iron galvanized and of steam quality or Grooved type fitting (UL/FM approved).

Pipe joints shall be threaded for pipe sizes 50 mm and smaller, and for large sizes grooved with mechanical coupling and gaskets or flanged.

Mechanical Grooved Couplings: Malleable iron or DUCTILE IRON housing clamps to engage and lock, "C" shaped elastomeric sealing gasket, GI bolts, nuts, and washers.

Mechanical Formed Fittings: Carbon steel housing with integral pipe stop and O-ring pocked and O-ring, uniformly com-pressed into permanent mechanical engagement onto pipe.

- 2.1.2 All below ground (buried underground) fire pipe work shall be in HDPE material withstanding the working pressure of 1200 KPA (12 bars) and test pressure of 1800 KPA.

- 2.1.3 Material and works with specified Steel Pipe work, Grooved Steel Pipes with Couplings, Mechanical Joints and Gaskets, Flange Adapters, Fire Protection Fittings shall conform to requirements as specified within Section, pipes and tubes.

- 2.1.4 All materials and products shall be installed in accordance with NFPA Standards, ANSI/AWWA and other applicable Standards and manufacturer's published recommendations. Grooved end product manufacturer to be ISO-9001 certified.

- 2.1.5 All pipe work and fittings shall conform to the relevant clauses in this specification.

- 2.1.6 Use only new material, free of defects, rust and scale, and meeting the latest revision of BS and/or ASTM specifications as listed in this section of specifications.

- 2.1.7 Construct all piping for the highest pressures and temperatures in the respective system in accordance with ANSI B31, but not less than 8 bar unless specifically indicated otherwise.

2.1.8 Where weld fittings or mechanical grooved fittings are used, use only long radius elbows having a centre line radius of 1.5 pipe diameters.

2.1.9 No elbows will be allowed on fire service systems where all changes of direction shall be by means of standard bend springs or long radius bends having centre line radius of 1.5 times of pipe dia .

2.1.10 Where pipes cross fire rated walls and slabs, beams and waterproof membranes, galvanized steel sleeves shall be used and the penetrations shall be fireproofed (using flexible materials) to the same rating as the wall. The fire proofing materials shall be applied strictly as per the manufacturer's recommendations.

2.2 PIPING JOINING MATERIALS

A Pipe-Flange Gasket Materials: AWWA C110, rubber, flat face, 1/8 inch (3.2 mm) thick or ASME B16.21, nonmetallic and asbestos free.

1 Class 125, Cast-Iron Flanges and Class 150, Bronze Flat-Face Flanges: Full-face gaskets.

2 Class 250, Cast-Iron Flanges and Class 300, Steel Raised-Face Flanges: Ring-type gaskets.

B Metal, Pipe-Flange Bolts and Nuts: ASME B18.2.1, carbon steel unless otherwise indicated.

C Brazing Filler Metals: AWS A5.8/A5.8M, BCuP Series, copper-phosphorus alloys for general-duty brazing unless otherwise indicated.

D Welding Filler Metals: Comply with AWS D10.12M/D10.12 for welding materials appropriate for wall thickness and chemical analysis of steel pipe being welded.

2.3 COVER SYSTEM FOR SPRINKLER PIPING

A Description: System of support brackets and covers made to protect sprinkler piping.

B Brackets: Glass-reinforced nylon.

C Covers: Extruded PVC sections of length, shape, and size required for size and routing of CPVC piping.

2.4 LISTED FIRE-PROTECTION VALVES

Refer Section 211000

2.5 SPRINKLER SPECIALTY PIPE FITTINGS

A Branch Outlet Fittings:

1 Standard: UL 213.

2 Pressure Rating: 175 psig (1200 kPa) minimum.

3 Body Material: Ductile-iron housing with EPDM seals and bolts and nuts.

- 4 Type: Mechanical-T and -cross fittings.
- 5 Configurations: Snap-on and strapless, ductile-iron housing with branch outlets.
- 6 Size: Of dimension to fit onto sprinkler main and with outlet connections as required to match connected branch piping.
- 7 Branch Outlets: Grooved, plain-end pipe, or threaded.

B Flow Detection and Test Assemblies:

- 1 Standard: UL's "Fire Protection Equipment Directory" listing or "Approval Guide," published by FM Global, listing.
- 2 Pressure Rating: 175 psig (1200 kPa) minimum.
- 3 Body Material: Cast- or ductile-iron housing with orifice, sight glass, and integral test valve.
- 4 Size: Same as connected piping.
- 5 Inlet and Outlet: Threaded.

C Branch Line Testers:

- 1 Standard: UL 199.
- 2 Pressure Rating: 175 psig (1200 kPa).
- 3 Body Material: Brass.
- 4 Size: Same as connected piping.
- 5 Inlet: Threaded.
- 6 Drain Outlet: Threaded and capped.
- 7 Branch Outlet: Threaded, for sprinkler.

D Sprinkler Inspector's Test Fittings:

- 1 Standard: UL's "Fire Protection Equipment Directory" listing or "Approval Guide," published by FM Global, listing.
- 2 Pressure Rating: 175 psig (1200 kPa) minimum.
- 3 Body Material: Cast- or ductile-iron housing with sight glass.
- 4 Size: Same as connected piping.
- 5 Inlet and Outlet: Threaded.

2.6 SPRINKLERS

A General Requirements:

- 1 Standard: UL's "Fire Protection Equipment Directory" listing or "Approval Guide," published by FM Global, listing.
- 2 Pressure Rating for Automatic Sprinklers: 175 psig (1200 kPa) minimum.

B Suspended Ceiling Type: Concealed recessed type with face plate to match ceiling finish.

- 1 Finish: powder coated.
- 2 Face Plate Finish: To match ceiling or as instructed by Engineer.
- 3 Fusible Link: Fusible solder link type temperature rated for specific area hazard.

- C Exposed Area Type: Standard upright type with guard.
- 1 Finish: Brass.
 - 2 Fusible Link: Fusible solder link type temperature rated for specific area hazard.
- D Sidewall Type: Standard horizontal sidewall type with matching push on escutcheon plate and guard.
- 1 Finish: Brass.
 - 2 Escutcheon Plate Finish: Brass.
 - 3 Fusible Link: Fusible solder link type temperature rated for specific area hazard.
- E Dry Sprinklers: Standard pendant type with matching push on escutcheon plate.
- 1 Finish: Brass.
 - 2 Escutcheon Plate Finish: Brass.
 - 3 Fusible Link: Fusible solder link type temperature rated for specific area hazard.
- F Guards: Finish to match sprinkler finish.
- G Spray Nozzles: Brass with solid cone discharge, 30 degrees of arc with blow-off dust cap.

Note: All sprinklers heads for residential application shall be of the quick response type.

2.7 ALARM DEVICES

Refer section 211000.

2.8 MANUAL CONTROL STATIONS

- A Description: UL listed or FM approved, hydraulic operation, with union, NPS 1/2 (DN 15) pipe nipple, and bronze ball valve. Include metal enclosure labeled "MANUAL CONTROL STATION" with operating instructions and cover held closed by breakable strut to prevent accidental opening.

2.9 CONTROL PANELS

- A Description: Single-area, two-area, or single-area cross-zoned control panel as indicated, including NEMA ICS 6, Type 1 enclosure, detector, alarm, and solenoid-valve circuitry for operation of deluge valves. Panels contain power supply; battery charger; standby batteries; field-wiring terminal strip; electrically supervised solenoid valves and polarized fire-alarm bell; lamp test facility; single-pole, double-throw auxiliary alarm contacts; and rectifier.
- 1 Panels: UL listed and FM approved when used with thermal (SMOKE /HEAT) detectors and Class A / CLASS B detector circuit wiring. Electrical characteristics are 120-V ac, 60 Hz, with 24-V dc rechargeable batteries.
 - 2 Manual Control Stations: Electric operation, metal enclosure, labeled "MANUAL CONTROL STATION" with operating instructions and cover held closed by breakable strut to prevent accidental opening.

- 3 Manual Control Stations: Hydraulic operation, with union, NPS 1/2 (DN 15) pipe nipple, and bronze ball valve. Include metal enclosure labeled "MANUAL CONTROL STATION" with operating instructions and cover held closed by breakable strut to prevent accidental opening.

PART 3 EXECUTION

3.1 PREPARATION

- A Perform fire-hydrant flow test according to NFPA 13 and NFPA 291. Use results for system design calculations required in "Quality Assurance" Article.
- B Report test results promptly and in writing.

3.2 PIPING INSTALLATION

- A Locations and Arrangements: Drawing plans, schematics, and diagrams indicate general location and arrangement of piping. Install piping as indicated, as far as practical.
- 1 Deviations from approved working plans for piping require written approval from authorities having jurisdiction. File written approval with Architect before deviating from approved working plans.
- B Piping Standard: Comply with requirements for installation of sprinkler piping in NFPA 13.
- C Install seismic restraints on piping. Comply with requirements for seismic-restraint device materials and installation in NFPA 13.
- D Use listed fittings to make changes in direction, branch takeoffs from mains, and reductions in pipe sizes.
- E Install unions adjacent to each valve in pipes NPS 2 (DN 50) and smaller.
- F Install flanges, flange adapters, or couplings for grooved-end piping on valves, apparatus, and equipment having NPS 2-1/2 (DN 65) and larger end connections.
- H Install sprinkler piping with drains for complete system drainage.
- I Install sprinkler control valves, test assemblies, and drain risers adjacent to standpipes when sprinkler piping is connected to standpipes.
- K Install alarm devices in piping systems.
- L Install hangers and supports for sprinkler system piping according to NFPA 13. Comply with requirements for hanger materials in NFPA 13.
- M Install pressure gages on riser or feed main, at each sprinkler test connection, and at top of each standpipe. Include pressure gages with connection not less than NPS 1/4 (DN 8) and with soft metal seated globe valve, arranged for draining pipe between gage and valve. Install gages to permit removal, and install where they will not be subject to freezing.
- N Pressurize and check pre action sprinkler system piping and air-pressure maintenance devices.
- O Fill sprinkler system piping with water.

- P Install electric heating cables and pipe insulation on sprinkler piping in areas subject to freezing. Comply with requirements for heating cables in Division 21 "Heat Tracing for Fire-Suppression Piping" and for piping insulation in Division 21 Section "Fire-Suppression Systems Insulation."
- Q Install sleeves for piping penetrations of walls, ceilings, and floors. Comply with requirements for sleeves specified in Division 21 Section "Sleeves and Sleeve Seals for Fire-Suppression Piping."
- R Install sleeve seals for piping penetrations of concrete walls and slabs. Comply with requirements for sleeve seals specified in Division 21 Section "Sleeves and Sleeve Seals for Fire-Suppression Piping."
- S Install escutcheons for piping penetrations of walls, ceilings, and floors. Comply with requirements for escutcheons specified in Division 21 Section "Escutcheons for Fire-Suppression Piping."

3.3 JOINT CONSTRUCTION

- A Install couplings, flanges, flanged fittings, unions, nipples, and transition and special fittings that have finish and pressure ratings same as or higher than system's pressure rating for aboveground applications unless otherwise indicated.
- B Install unions adjacent to each valve in pipes NPS 2 (DN 50) and smaller.
- C Install flanges, flange adapters, or couplings for grooved-end piping on valves, apparatus, and equipment having NPS 2-1/2 (DN 65) and larger end connections.
- D Ream ends of pipes and tubes and remove burrs. Bevel plain ends of steel pipe.
- E Remove scale, slag, dirt, and debris from inside and outside of pipes, tubes, and fittings before assembly.
- F Flanged Joints: Select appropriate gasket material in size, type, and thickness suitable for water service. Join flanges with gasket and bolts according to ASME B31.9.
- G Threaded Joints: Thread pipe with tapered pipe threads according to ASME B1.20.1. Cut threads full and clean using sharp dies. Ream threaded pipe ends to remove burrs and restore full ID. Join pipe fittings and valves as follows:
 - 1 Apply appropriate tape or thread compound to external pipe threads.
 - 2 Damaged Threads: Do not use pipe or pipe fittings with threads that are corroded or damaged.
- H Twist-Locked Joints: Insert plain end of steel pipe into plain-end-pipe fitting. Rotate retainer lugs one-quarter turn or tighten retainer pin.
- I Steel-Piping, Pressure-Sealed Joints: Join light wall steel pipe and steel pressure-seal fittings with tools recommended by fitting manufacturer.
- J Welded Joints: Construct joints according to AWS D10.12M/D10.12, using qualified processes and welding operators according to "Quality Assurance" Article.

- 1 Shop weld pipe joints where welded piping is indicated. Do not use welded joints for galvanized-steel pipe.
- K Steel-Piping, Cut-Grooved Joints: Cut square-edge groove in end of pipe according to AWWA C606. Assemble coupling with housing, gasket, lubricant, and bolts. Join steel pipe and grooved-end fittings according to AWWA C606 for steel-pipe joints.
 - L Steel-Piping, Roll-Grooved Joints: Roll rounded-edge groove in end of pipe according to UL STANDARD. Assemble coupling with housing, gasket, lubricant, and bolts. Join steel pipe and grooved-end fittings according to UL STANDARD. for steel-pipe grooved joints.
 - M Steel-Piping, Pressure-Sealed Joints: Join Schedule 5 steel pipe and steel pressure-seal fittings with tools recommended by fitting manufacturer.
 - Q Extruded-Tee Connections: Form tee in copper tube according to ASTM F 2104. Use tool designed for copper tube; drill pilot hole, form collar for outlet, dimple tube to form seating stop, and braze branch tube into collar.
 - R Dissimilar-Material Piping Joints: Make joints using adapters compatible with materials of both piping systems.
- ### 3.4 INSTALLATION OF COVER SYSTEM FOR SPRINKLER PIPING
- A Install cover system, brackets, and cover components for sprinkler piping according to manufacturer's "Installation Manual" and with NFPA 13 or NFPA 13R for supports.
- ### 3.5 VALVE AND SPECIALTIES INSTALLATION
- A Install listed fire-protection valves, trim and drain valves, specialty valves and trim, controls, and specialties according to NFPA 13 and authorities having jurisdiction.
 - B Install listed fire-protection shutoff valves supervised open, located to control sources of water supply except from fire-department connections. Install permanent identification signs indicating portion of system controlled by each valve.
 - C Install check valve in each water-supply connection. Install backflow preventers instead of check valves in potable-water-supply sources.
 - D Specialty Valves:
 - 1 General Requirements: Install in vertical position for proper direction of flow, in main supply to system.
 - 2 Alarm Valves: Include bypass check valve and retarding chamber drain-line connection.
 - 3 Deluge Valves: Install in vertical position, in proper direction of flow, and in main supply to deluge system. Install trim sets for drain, priming level, alarm connections, ball drip valves, pressure gages, priming chamber attachment, and fill-line attachment.
- ### 3.6 SPRINKLER INSTALLATION
- A Install sprinklers in suspended ceilings in center of acoustical ceiling panels.

- B Install dry-type sprinklers with water supply from heated space. Do not install pendent or sidewall, wet-type sprinklers in areas subject to freezing.
- C Install sprinklers into flexible, sprinkler hose fittings and install hose into bracket on ceiling grid.

3.7 FIELD QUALITY CONTROL

- A Perform tests and inspections.
- B Tests and Inspections:
 - 1 Leak Test: After installation, charge systems and test for leaks. Repair leaks and retest until no leaks exist.
 - 2 Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.
 - 3 Flush, test, and inspect sprinkler systems according to NFPA 13, "Systems Acceptance" Chapter.
 - 4 Energize circuits to electrical equipment and devices.
 - 5 Start and run excess-pressure pumps.
 - 6 Coordinate with fire-alarm tests. Operate as required.
 - 7 Coordinate with fire-pump tests. Operate as required.
 - 8 Verify that equipment hose threads are same as local fire-department equipment.
- C Sprinkler piping system will be considered defective if it does not pass tests and inspections.
- D Prepare test and inspection reports.

3.8 CLEANING

- A Clean dirt and debris from sprinklers.
- B Remove and replace sprinklers with paint other than factory finish.

3.9 DEMONSTRATION

- A Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain specialty valves and pressure-maintenance pumps.

END OF SECTION

MECHANICAL SECTION: 21 13 39
FOAM WATER SPRINKLER SYSTEM

PART 1 – GENERAL

1.1 RELATED DOCUMENTS

- A Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A Section Includes:

- 1 Concentrate piping and piping specialties.
- 2 Bladder tanks and proportioning devices.
- 3 Foam concentrate.
- 4 Discharge devices.
- 5 Monitoring and alarm devices.

1.3 DEFINITIONS

- A AFFF: Aqueous film-forming foam.
- B AR-AFFF: Alcohol-resistant aqueous film-forming foam.

1.4 SYSTEM DESCRIPTION

- A Description: Engineered, fixed, deluge, automatically actuated, low-expansion, AFFF fire-extinguishing system for flammable-liquid fires. System includes diaphragm proportioning tanks and devices as described in NFPA 16.

1.5 PERFORMANCE REQUIREMENTS

- A Standard Working Pressure of Piping-System Component: Listed for at least 175 psig (1200 kPa).

- B Unless authorities having jurisdiction have stricter requirements, minimum design parameters are as follows:

- 1 Solution: 3 percent foam-water solution.
- 2 Sprinkler Spacing: Maximum of 100 sq. ft. (9.5 sq. m) per sprinkler, and maximum 12-foot (3.7-m) spacing.
- 3 Design Density: Minimum 0.16 gpm/sq. ft. (0.108 L/s per sq. m).
- 4 Foam Supply: Minimum 10 minute discharge time.
- 5 Water Supply: Minimum 60 minutes.
- 6 Remote Area: Open-sprinkler systems shall discharge over the entire system area.
- 7 Sprinkler Temperature Rating: Maximum 250 to 300 deg F (121 to 149 deg C) at a roof or ceiling, and 135 to 170 deg F (57 to 77 deg C) for intermediate sprinklers.

- C Seismic Performance: Fire-suppression piping shall withstand the effects of earthquake motions determined according to NFPA 13.

1.6 SUBMITTALS

- A Product Data: For each type of product indicated. Include the following:

- 1 Piping, valves, fittings, and hangers.

- 2 Seismic restraints for all equipment.
 - 3 Bladder tanks and proportioning devices.
 - 4 Foam concentrate.
 - 5 Discharge devices. Include flow characteristics.
 - 6 Monitoring and alarm devices. Include electrical data and supervision method.
 - 7 Foam-concentrate pumps. Include power supply and controller.
 - 8 Foam-concentrate storage tanks.
 - 9 Strainers.
 - 10 Test connections.
- B Shop Drawings: For each hazard area, drawn to scale, and signed and sealed by a qualified professional engineer. Include plans, elevations, sections, details, and attachments to other work.
- 1 Detail equipment assemblies and indicate dimensions, weights, loads, required clearances, method of field assembly, components, and location and size of each field connection.
 - 2 Wiring Diagrams: For power, signal, and control wiring.
 - 3 Design Calculations: For amount of foam concentrate required for each hazard area.
 - 4 Plans: Show the following:
 - a Foam-solution proportioning tanks and devices, piping, discharge devices, monitoring and alarm devices, and accessories.
 - b Method of attaching hangers to building structure.
 - c Fire-alarm panel.
 - d Equipment and furnishings.
- C Permit-Approved Drawings: Working plans prepared according to NFPA 16 and approved by authorities having jurisdiction. Include hydraulic calculations complying with NFPA 13.
- D Welding certificates.
- E Field quality-control reports.
- F Operation and Maintenance Data: For foam fire extinguishing to include in emergency, operation, and maintenance manuals. In addition to items specified in Division 01 Section "Operation and Maintenance Data," include the following:
- 1 Valves and specialties.
 - 2 Bladder tanks and proportioning devices.
 - 3 Foam concentrate.
 - 4 Discharge devices. Include flow characteristics.
 - 5 Monitoring and alarm devices. Include supervision method.
 - 7 Foam-concentrate storage tanks.
 - 8 Strainers.
 - 9 Test connections.

1.7 QUALITY ASSURANCE

- A Welding Qualifications: Qualify procedures and operators according to ASME Boiler and Pressure Vessel Code: Section IX.
- B Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

- C FM Global Compliance: Provide components that are FM Approved and that are listed in FM's "Approval Guide."
- D UL Compliance: Provide equipment listed in UL's "Fire Protection Equipment Directory."

1.8 EXTRA MATERIALS

- A Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.
 - 1 Discharge Devices: Not less than 20 percent of amount of each type installed.
 - 2 Foam Concentrate: Not less than 200 percent of amount installed.

PART 2 PRODUCTS

2.1 CONCENTRATE PIPING MATERIALS

- A Comply with requirements specified in Division 21 Section 211000 for pipes, fittings, joining materials, and valves.

2.2 VALVES

- A General Valve Requirements:
 - 1 UL listed or FM Approved for use in fire-protection systems.
 - 2 Compatible with type of foam concentrate used.
- B Comply with requirements specified in Division-21 section "wet pipe sprinkler systems".

2.3 SPECIALTIES

- A Specialties shall comply with NFPA 16, be compatible with the foam concentrate, and be designed to be drained and cleaned.
- B Foam-Concentrate Storage Tanks: Buna-N, bladder-type proportioning tank complying with UL 162 and ASME Boiler and Pressure Vessel Code: Section VIII; designed for use with foam-concentrate pumps and for specific type of foam concentrate used. Include bladder, internal piping, fill and drain, glass sight gage, piping, and valves. Contain concentrate in the bladder.
 - 1 Orientation: Horizontal design with saddle support.
- C Foam-Concentrate Storage Tanks: Atmospheric type, complying with UL 162 and ASME Boiler and Pressure Vessel Code: Section VIII; designed for use with foam-concentrate pumps and for specific type of foam concentrate used. Include pressure-vacuum vent, fill and drain, glass sight gage, piping, and valves.
- E Proportioning Controllers: Venturi type complying with UL 162 and of capacity to match design at minimum and maximum flow.
- F Concentrate Control Valves: Water-operated ball or deluge valve designed to open with flow through the proportioning controller.

- G Concentrate Strainers: Bronze body and stainless-steel mesh strainer with minimum 0.125-inch (3.2-mm) perforations to remove solids that would block system components.
- H Pressure Gages: Comply with UL 393; with 3-1/2-inch (90-mm) minimum-diameter dial, 0- to 300-psig (0- to 2070-kPa) dial range, and caption "WATER" or "CONCENTRATE" on dial face.
- 2.4 FOAM CONCENTRATE
 - A Description: AFFF liquid concentrate, complying with NFPA 11 and UL 162, for making foam-water fire-extinguishing foam solution.
- 2.5 DISCHARGE DEVICES
 - A Discharge devices shall be UL listed or FM Approved.
 - B Sprinklers: Open, air-aspirating type; UL listed or FM Approved and listed for use with type of foam concentrate used.
 - C Spray Nozzles: For foam water; include foam generator and distributing deflector to distribute foam or water.
- 2.6 MONITORING DEVICES
 - A Valve Supervisory Switches: Single pole, double throw, with normally closed contacts complying with UL 753. Switch shall signal fire-alarm panel or releasing panel when valve is in other than fully open position.
 - B Pressure Switches: Single pole, double throw, UL listed or FM Approved and complying with UL 753. Switch shall signal an alarm condition at the fire-alarm panel or releasing panel when switch is in other than fully open position.
 - C Flow Switches: Single pole, double throw, UL listed or FM Approved and complying with UL 753. Switch shall signal an alarm condition at the fire-alarm panel or releasing panel when switch is in other than fully open position.
- 2.7 ALARM DEVICES
 - A Description: UL listed or FM Approved, low voltage, and surface mounting. Alarm and monitoring devices are specified in Division 28 Section "Digital, Addressable Fire-Alarm System" or Division 28 Section "Zoned (DC Loop) Fire-Alarm System."

PART 3 EXECUTION

- 3.1 CONCENTRATE STORAGE TANK INSTALLATION
 - A Install concentrate storage tanks on concrete bases.
 - B Install concentrate storage tanks anchored to substrate according to tank manufacturer's written instructions.
 - C Install tanks level and plumb, according to layout drawings, original design, and referenced standards. Maintain manufacturer's recommended clearances. Arrange units so controls and devices needing service are accessible.

- D Install seismic restraints for tanks.

3.2 PIPING INSTALLATION

- A Install piping and other components level and plumb.
- B Comply with requirements for basic pipe installation and joint construction in Division 21 Section "Common Work Results for Fire Suppression."
- C Install pipe and fittings, valves, and discharge devices according to requirements listed in NFPA 16, "Installation of Foam-Water Sprinkler and Foam-Water Spray Systems."
 - 1 Support piping using supports and methods according to NFPA 13.
 - 2 Install seismic restraints for concentrate storage tanks and piping systems.
 - 3 Install monitoring and alarm devices according to NFPA 16 and NFPA 72.
- D Install flanges, flange adapters, or couplings for grooved-end piping on piping, valves, apparatus, and equipment having NPS 2-1/2 (DN 65) and larger end connections.
- E Ream ends of pipes and tubes and remove burrs.
- F Remove scale, slag, dirt, and debris from inside and outside of pipes, tubes, and fittings before assembly.
- G Flanged Joints: Select appropriate gasket material in size, type, and thickness suitable for foam concentrate. Join flanges with gasket and bolts according to ASME B31.9.
- H Threaded Joints: Thread pipe with tapered pipe threads according to ASME B1.20.1. Cut threads full and clean using sharp dies. Ream threaded pipe ends to remove burrs and restore full ID. Join pipe fittings and valves as follows:
 - 1 Apply appropriate tape or thread compound to external pipe threads.
 - 2 Damaged Threads: Do not use pipe or pipe fittings with threads that are corroded or damaged.
- J Steel-Piping, Roll-Grooved Joints: Roll rounded-edge groove in end of pipe according to UL/FM. Assemble coupling with housing, gasket, lubricant, and bolts. Join steel pipe and grooved-end fittings according to UL/FM for steel-pipe grooved joints.
- L Dissimilar-Material Piping Joints: Make joints using adapters compatible with materials of both piping systems and with foam concentrate.

3.3 CONNECTIONS

- A Comply with requirements for piping specified in Division 21 Section "Wet-Pipe Sprinkler Systems" or Division 21 Section "Dry-Pipe Sprinkler Systems." Drawings indicate general arrangement of piping, fittings, and specialties.
- B Provide concentrate control and drain valves with piping to permit maintenance of the foam concentrate with continuous sprinkler-system service.
- C Install proportioning controller in fire-suppression piping to provide coverage to area indicated on Drawings.

- D Where installing piping adjacent to equipment, allow space for service and maintenance of equipment.
- E Connect electrical devices to building's fire-alarm system. Comply with requirements for wiring and connections in Division 28 Section "Digital, Addressable Fire-Alarm System" or Division 28 Section "Zoned (DC Loop) Fire-Alarm System."
- 3.4 LABELING
 - A Install labeling on piping, equipment, and panels according to Division 22 Section "Identification for Plumbing Piping and Equipment."
- 3.5 CHARGING SYSTEM
 - A Fill proportioning tanks with foam concentrate after field quality-control testing is complete and satisfactory results have been achieved.
- 3.6 FIELD QUALITY CONTROL
 - A Inspection: Engage the services of a qualified professional engineer to inspect installed fire-extinguishing systems, prepare installation report, and certify that installation complies with the Contract Documents, calculations, and requirements of authorities having jurisdiction.
 - B Comply with operating instructions and procedures in NFPA 16, "Acceptance Tests" Chapter. Include the following tests and inspections to demonstrate compliance with requirements:
 - 1 Check mechanical items.
 - 2 Inspect equipment and check mountings for adequate anchoring to substrate.
 - 3 Check electrical systems.
 - 4 Flush supply piping.
 - 5 Perform hydrostatic pressure test.
 - 6 Perform acceptance test.
 - 7 Perform proportioning system test.
 - 8 Perform discharge test.
 - 9 Flush system piping.
 - 10 Correct malfunctioning equipment; retest to demonstrate compliance. Replace equipment that cannot be satisfactorily corrected or does not perform as specified and indicated; retest to demonstrate compliance. Repeat procedure until satisfactory results are obtained.
 - C Perform tests and inspections.
 - 1 Manufacturer's Field Service: Engage a factory-authorized service representative to inspect components, assemblies, and equipment installations including connections, and to assist in testing.
 - D Tests and Inspections:
 - 1 After installing foam fire-extinguishing piping system and after electrical circuitry has been energized, test for compliance with requirements.
 - 2 Leak Test: After installation, charge system and test for leaks. Repair leaks and retest until no leaks exist.

- 3 Operational Test: After electrical circuitry has been energized, start systems to confirm proper unit operation.
 - 4 Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.
- E Foam fire-extinguishing piping system will be considered defective if it does not pass tests and inspections.
- F Prepare test and inspection reports.

END OF SECTION

MECHANICAL SECTION: 21 22 10
CLEAN AGENT FIRE SUPPRESSION SYSTEMS
(FM-200 SYSTEM)

GENERAL CONDITIONS :

1) SCOPE:

This specification outlines the requirements for a "Total Flooding" Clean Agent Fire Suppression System with automatic detection and control. The work described in this specification includes all engineering, labor, materials, equipment and services necessary, and required, to complete and test the suppression system.

2) APPLICABLE STANDARDS AND PUBLICATIONS:

The design, equipment, installation, testing and maintenance of the Clean Agent Suppression System shall be in accordance with the applicable requirements set forth in the latest edition of the following codes and standards:

- 1) NFPA No. 2001 - Clean Agent Fire Extinguishing Systems
- 2) NFPA No. 70 - National Electrical Code
- 3) NFPA No. 72 - Standard For Fire Alarm System
- 4) Factory Mutual Approval Guide
- 5) U.L Listings
- 6) Requirements of the Authority Having Jurisdiction (AHJ)

The standards listed, as well as all other applicable codes and standards, shall be used as "minimum" design standards. Also to be considered are good engineering practices.

I. REQUIREMENTS:

The FM 200 Fire Suppression System installation shall be made in accordance with the drawings, specifications and applicable standards.

II. SCOPE OF WORK

The Specialist contract shall include the following materials / works in their scope but not limited to:

1. Design & Engineering of clean Agent fire suppression system;
2. Prepare workshop drawing and technical submittals;
3. Perform Flow calculation;
4. Clean Agent gas storage cylinders filled with gas and all hardware accessories;
5. Cylinder Manifold and discharge piping;
6. Discharge Nozzles;
7. Gas low pressure switch & Gas discharge pressure switch;

8. Fire Detection and Alarm system c/w Optical smoke detectors, manual release, Auto / Manual selector switch, Abort unit, First stage Alarm bell, second stage Alarm sounder, strobe horn, Gas discharge warning sign and associated cabling etc;
9. Gas release control panel
10. Interlock wiring and conduit for shutdown of HVAC, dampers
and/or electric power supplies, relays or shunt trip breakers.
11. Providing Drop curtains for any permanent openings;
12. Integrity Test and Functional test;
13. Training to the client representatives;
14. Provide As Built drawings and Operation & Maintenance Manuals.

The work listed below shall be provided by others, or under other sections of this specification:

- 1) 120 VAC or 208/220 VAC power supply to the Fire suppression control panel.
- 2) Connection to local/remote fire alarm systems, listed central alarm station.
- 3) Sealing of openings to maintain the protected room enclosure as Air tight.

III. QUALITY ASSURANCE:

IV. MANUFACTURER:

- 1) The manufacturer of the Suppression System hardware and detection components shall have a minimum of 10 years experience in the design and manufacture of similar types of suppression systems and who can refer to similar installations providing satisfactory service.
- 2) The name of the manufacturer, part numbers and serial numbers shall appear on all major components.
- 3) All devices, components and equipment shall be new, standard products of the manufacturer's latest design and suitable to perform the functions intended.
- 4) All equipment shall be U.L listed and/or FM approved.

V. INSTALLER:

- 1) The installing contractor shall be trained by the supplier to design, install, test and maintain Suppression Systems.
- 2) The installing contractor shall be an experienced firm regularly engaged in the installation of automatic Clean Agent, or similar, fire suppression systems in strict accordance with all applicable standards.

- 3) The installing contractor must have a minimum of eight (8) years experience in the design, installation and testing of Clean Agent, or similar, fire suppression systems. A list of systems of a similar nature and scope shall be provided on request.
- 4) The installing contractor shall maintain, or have access to, a Clean Agent recharging station. The installing contractor shall provide proof of his ability to recharge the largest Clean Agent system within 24 hours after a discharge. Include the amount of bulk agent storage available.
- 5) The installing contractor shall be an authorized stocking distributor of the Clean Agent system equipment so that immediate replacement parts are available from inventory.
- 6) The installing contractor shall show proof of emergency service available on a twenty-four hour, seven-days-a-week basis.

VI. SUBMITTALS:

- 1) The installing contractor shall submit the following information for approval to the local Fire Prevention Agency, owners Insurance Underwriter, Architect/Engineer, and all other Authorities Having Jurisdiction before starting installation:
 - a) Field installation layout drawings scale shall be 1:100m detailing the location of all agent storage tanks, pipe runs, including pipe sizes and lengths, control panel(s), detectors, manual pull stations, abort stations, audible and visual alarms, etc.
 - b) Auxiliary details and information such as maintenance panels, door holders, special sealing requirements and equipment shutdowns.
 - c) Separate layouts, or drawings, shall be provided for each level, (i.e.; room, underfloor, and above ceiling) and for mechanical and electrical work.
 - d) A separate layout or drawing shall show isometric details of agent storage containers, mounting details and proposed pipe runs and sizes.

Electrical layout drawings shall show the location of all devices and include point-to-point conduit runs and a description of the method(s) used for detector mounting.

Internal control panel wiring diagram shall include power supply requirements and field wiring termination points.

Graphic Annunciator wiring schematics and dimensioned display panel illustration shall be provided. (Optional device)

Complete flow calculations shall be provided from a U.L. listed computer program, for all engineered Clean Agent systems. Calculation sheet(s) must include the manufacturers name and U.L. listing number for verification. The individual sections of pipe and each fitting to be used, as shown on the isometrics, must be identified and included in the calculation. Total agent discharge time must be shown and detailed by zone.

Provide calculations for the battery stand-by power supply taking into consideration the power requirements of all alarms, initiating devices and auxiliary components under full load conditions.

A complete sequence of operation detailing all alarm devices, shutdown functions, remote signaling, damper operation, time delay and agent discharge for each zone or system.

VII. CLEAN AGENT SYSTEM REQUIREMENTS :

1. SYSTEM DESCRIPTION AND OPERATION:

A) The system shall be a Total Flood Clean Agent Suppression System supplied & installed by an Authorised by the manufacturer.

B) The system shall provide minimum design concentration of 7.0%, by volume, in all areas and/or protected spaces, at the minimum anticipated temperature within the protected area. System design shall not exceed the NOAEL value of 9.0%, adjusted for maximum space temperature anticipated, unless provisions for room evacuation, before agent release, are provided.

C) The system shall be complete in all ways. It shall include all mechanical and electrical installation, all detection and control equipment, agent storage containers, FM 200 agent, discharge nozzles, pipe and fittings, manual release and abort stations, audible and visual alarm devices, auxiliary devices and controls, shutdowns, alarm interface, caution/ advisory signs, functional checkout and testing, training and all other operations necessary for a functional, U.L. Listed and/or F.M. approved FM 200 Clean Agent Suppression System.

D) Provide two (2) inspections during the first year of service. Inspections shall be made at 6 month intervals commencing when the system is first placed into normal service.

E) The general contractor shall be responsible for sealing and securing the protected spaces against agent loss and/or leakage during the 10 minute "hold" period.

F) The system(s) shall be actuated by cross zoned photoelectric detectors installed at a maximum spacing of 250 sq. ft. (23.2 sq. m) per detector, in both the room, underfloor and above ceiling protected spaces. If the air flow is one air change per minute, photoelectric detectors shall be installed at a spacing not to exceed 125 sq. ft. (11.6 sq. m) per detector. (Ref. NFPA No. 72 - current edition)

G) Detectors shall be wired in Sequential Detection method of operation, standard Cross-Zoned detection, or Single Detector Release, using either a Class "A" or Class "B" wiring arrangement. No other detection / wiring arrangements will be acceptable.

H) Automatic operation of each protected area shall be as follows:

1) Actuation of one (1) detector, within the system, shall:

a) Illuminate the "ALARM" lamp on the control panel face.

b) Energize an alarm bell and/or an optional visual indicator.

c) Transfer sets of 5 Amp rated auxiliary contacts which can perform auxiliary system functions such as:

- 1) Operate door holder/closures on access doors.
- 2) Transmit a signal to a fire alarm system.
- 3) Shutdown HVAC equipment.
- d) Light an individual lamp on an optional graphic annunciator.
- 2) Actuation of a 2nd detector, within the system, shall:
 - a) Illuminate the "PRE-DISCHARGE" lamp on the control panel face.
 - b) Energize an predischARGE horn or horn/strobe device.
 - c) Shut down the HVAC system and/or close dampers.
 - d) Start time-delay sequence (not to exceed 60 seconds). System abort sequence is enabled at this time.
 - f) Light an individual lamp on an optional graphic annunciator.
- 3) After completion of the time-delay sequence, the HFC-227ea Clean Agent system shall discharge and the following shall occur:
 - a) Illuminate a "SYSTEM FIRED" lamp on the control panel face.
 - b) Shutdown of all power to high-voltage equipment
 - c) Energize a visual indicator(s) outside the hazard in which the discharge occurred.
 - d) Energize a "System Fired" audible device. (Optional)
- 4) The system shall be capable of being actuated by manual discharge devices located at each hazard exit. Operation of a manual device shall duplicate the sequence description above except that the time delay and abort functions SHALL be bypassed. The manual discharge station shall be of the electrical actuation type and shall be supervised at the main control panel.

VIII. MATERIALS AND EQUIPMENT:

1. GENERAL REQUIREMENTS:

The Clean Agent System materials and equipment shall be standard products of the supplier's latest design and suitable to perform the functions intended. When one or more pieces of equipment must perform the same function(s), they shall be duplicates produced by one manufacturer.

- 1) All devices and equipment shall be **U/L Listed and or FM approved.**

2. CLEAN AGENT SYSTEM AND DISTRIBUTION:

Each system shall have its own supply of clean agent.

The system design can be modular, central storage, or a combination of both design criteria.

Systems shall be designed in accordance with the manufacturer's guidelines.

3) Each supply container shall be located within the hazard area, or as near as possible, to reduce the amount of pipe and fittings required to install the system.

4) The clean agent shall be stored in Agent Storage Containers. Containers shall be super-pressurized, with dry Nitrogen, to an operating pressure of 360 psi @ 70° F. (2482 kpa at 21° C). Containers shall be of high-strength alloy steel construction and conform to NFPA 2001.

5) Clean agent containers with an Impulse valve can be actuated by the following methods and **Solenoid valve is not allowed**:

Method 1: Electric actuation – Single container system with Impulse valve operator (IVO) & Impulse release Module (IRM)

Method 2: Electric actuation – Multi container system with Impulse valve operator (IVO) & Impulse release Module (IRM)

Method 3: Electric & Pneumatic actuation – Two container system with Impulse valve operator (IVO) & Impulse valve pneumatic operator (IVPO)

Method 4: Electric & Pneumatic actuation – Multi container system with Impulse valve operator (IVO) & Impulse valve pneumatic operator (IVPO)

6) Each container shall have a pressure gauge and low pressure switch to provide visual and electrical supervision of the container pressure. The low pressure switch shall be wired to the control panel to provide an audible and visual "Supervisory" alarm in the event the container pressure drops below 288 psi (1986 kpa). The pressure gauge shall be color coded to provide an easy, visual indication of container pressure.

7) Each container shall have a pressure relief provision that automatically operates when the internal temperature exceeds 150°F. (66°C).

8) Engineered discharge nozzles shall be provided, within the manufacturers guidelines, to distribute the clean agent throughout the protected spaces. The nozzles shall be designed to provide proper agent quantity and distribution.

a) Nozzles shall be available in pipe sizes 3/8" thru 2.0" (BPS 10mm thru 50mm). Each size shall be available in 180° and 360° distribution patterns.

b) Ceiling plates, can be used with the nozzles to conceal pipe entry holes through ceiling tiles.

9) Distribution piping, and fittings, shall be installed in accordance with the manufacturer's requirements, NFPA 2001 and approved piping standards and guidelines. All distribution piping shall be installed by qualified individuals using good,

accepted practices and quality procedures. All piping shall be adequately supported and anchored at all directional changes and nozzle locations.

a) All piping shall be reamed, blown clear and swabbed with suitable solvents to remove burrs, mill varnish and cutting oils before assembly.

b) All pipe threads shall be sealed with Teflon tape pipe sealant applied to the male thread ONLY.

IX. ELECTRICAL REQUIREMENTS :

1. CONTROL PANEL :

The control panel shall be a SHP (Single Hazard Panel) Control Panel.

The SHP Control System, and its components, shall be UL listed and FM approved for releasing service and be suitable for clean Agent release service.

The SHP Control System shall perform all functions necessary to operate the system detection, actuation and auxiliary functions, as outlined.

The SHP Control System shall be capable of providing 7AH or 18AH battery standby power supplies.

The SHP Control System shall be microprocessor based with hardware and software integration designed to guarantee reliability.

The SHP Control System shall support Cross Zoned, Sequential, Single Detector Release and Manual Release detection/actuation methods.

The SHP Control System shall provide the following capabilities and functions:

- a) Three (3) Class B (Style Y) indicating appliance circuits rated for 2.0 amps @ 24 VDC.
- b) Four (4) Class B (Style A) initiating circuits
- c) Optional Class A (Style D) module for initiating circuits
- d) Eight (8) Status LEDs plus alpha-numeric display for troubleshooting: AC normal; alarm; pre-discharge; release; supervisory; trouble; alarms silenced and system abort.
- e) Programmable pre-discharge and discharge timers
- f) Resettable and continuous auxiliary output power
- g) Five (5) optional Abort types
- h) Intelligent Transistor protection to prevent noise spikes and microprocessor failure from inadvertently activating release outputs
- i) A dedicated Disarm switch for release outputs
- j) Dedicated alarm and trouble contacts programmable for alarm, trouble, pre-discharge, discharge, abort, supervisory or waterflow functions, depending on panel configuration.

k) Two (2) Form "C" relays, rated at 5 amps, are provided on the SHP™ panel board. Installation of the optional SRM4 Relay Module (P/N 10-2176) will provide four (4) additional 5 amp relays.

l) Multiple input power source - 120 VAC or 208/220 VAC @ 50/60 Hz

m) 2.6 amp @ 24 VDC power supply to operate high current draw horns and strobes. Capable of allowing expansion to 8 amp external power supply for a total of 10.6 amps of available power.

n) Available in either Red or Gray finish

2. OPTICAL SMOKE DETECTOR:

The detectors shall be spaced and installed in accordance with the manufacturer's specifications and the guidelines of NFPA 72 – current edition.

The optical smoke detector shall continuously measures the smoke level in the air, and gives a proportional output. It shall comply with the requirements of EN54 part 7.

The detector shall be an optical light scattering type sensitive to visible smoke and be stable under all environmental conditions. The internal test and calibration circuits shall use the same optical elements as the smoke sensing circuits, to ensure reliability.

A light emitting diode (LED) on the base must illuminate when a detector is in alarm

3. MANUAL RELEASE (ELECTRIC):

The electric manual release switch shall be a dual action device which provides a means of manually discharging the Suppression System when used in conjunction with the Fike SHP Control System.

The Manual Release switch or Manual Pull station shall be a dual action device(s) requiring two /one distinct operations to initiate a system actuation.

Manual actuation shall bypass the time delay and abort functions, shall cause the system to discharge, and shall cause all release and shutdown devices to operate in the same manner as if the system had operated automatically.

A Manual Release switch shall be located at each exit from the protected hazard and shall have an advisory sign, provided at each location.

4. ABORT STATION :

The optional Abort Station shall be the "Dead Man" type and shall be located next to each manual switch.

The Abort Station shall be supervised and shall indicate a trouble condition at the SHP Control Panel, if depressed, and no alarm condition exists.

"Locking" or "Keyed" abort stations shall not be permitted.

The Abort Station shall be located adjacent to each manual station.

5. AUDIBLE AND VISUAL ALARMS:

Alarm audible and visual signal devices shall operate from the SHP Control Panel.

The Alarm Bell, Alarm Horn and Horn/Strobe devices shall operate on 24 volts Dc power to allow supervision of the circuit wires.

The alarm bell and horn shall deliver a minimum sound level of 88 decibels at a distance of 3m.

Horn and strobe light are to operate simultaneous from one power supply with flash rate of 1-3 flashes per second minimum over the listed input voltage range.

A Strobe device shall be placed outside, and above, each exit door from the protected space. Provide an advisory sign at each light location.

6. CAUTION AND ADVISORY SIGNS:

Provide signs, as required, to comply with NFPA 2001 and the recommendations of the equipment supplier:

Entrance sign: (1) required at each entrance to a protected space.

Manual Discharge sign: (1) required at each manual discharge station.

Flashing Light sign: (1) required at each flashing light over each exit from a protected space.

7. SYSTEM AND CONTROL WIRING:

All system wiring shall be furnished and installed by the contractor.

All wiring shall be installed in electrical metallic tubing (EMT), or conduit, and must be installed and kept separate from all other building wiring.

All system components shall be securely supported independent of the wiring. Runs of conduit and wiring shall be straight, neatly arranged, properly supported, installed parallel and perpendicular to walls and partitions.

The sizes of the conductors shall be those specified by the manufacturer. Color coded wire shall be used. All wires shall be tagged at all junction points and shall be free from shorts, earth connections (unless so noted on the system drawings), and crosses between conductors. Final termination's between the SHP control panel and the system field wiring shall be made under the direct supervision of a factory trained representative.

All wiring shall be installed by qualified individuals, in a neat and workmanlike manner, to conform to the National Electrical Code, Article 725, and Article 760, except as otherwise permitted for limited energy circuits, as described in NFPA 72 - current edition. Wiring installation shall meet all local, state, province and/or country codes.

The complete system electrical installation, and all auxiliary components, shall be connected to earth ground in accordance with the National Electrical Code.

8. SYSTEM INSPECTION AND CHECKOUT:

After the system installation has been completed, the entire system shall be checked out, inspected and functionally tested by qualified, trained personnel, in accordance with the manufacturer's recommended procedures and NFPA standards.

All containers and distribution piping shall be checked for proper mounting and installation.

All electrical wiring shall be tested for proper connection, continuity and resistance to earth.

The complete system shall be functionally tested, in the presence of the owner or his representative, and all functions, including system and equipment interlocks, must be operational at least five (5) days prior to the final acceptance tests.

- a) Each detector shall be tested in accordance with the manufacturer's recommended procedures, and test values recorded.
- b) All system and equipment interlocks, such as door release devices, audible and visual devices, equipment shutdowns, local and remote alarms, etc. shall function as required and designed.
- c) Each SHP control panel circuit shall be tested for trouble by inducing a trouble condition into the system.

9. TRAINING REQUIREMENTS:

Prior to final acceptance, the installing contractor shall provide operational training to each shift of the owners personnel. Each training session shall include system SHP Control Panel operation, manual and abort functions, trouble procedures, supervisory procedures, auxiliary functions and emergency procedures.

10. OPERATION AND MAINTENANCE:

Prior to final acceptance, the installing contractor shall provide complete operation and maintenance instruction manuals, four (4) copies for each system, to the owner. All aspects of system operation and maintenance shall be detailed, including piping isometrics, wiring diagrams of all circuits, a written description of the system design, sequence of operation and drawing(s) illustrating control logic and equipment used in the system. Checklists and procedures for emergency situations, troubleshooting techniques, maintenance operations and procedures shall be included in the manual.

11. AS-BUILT DRAWINGS:

Upon completion of each system, the installing contractor shall provide four (4) copies of system "As-Built" drawings to the owner. The drawings shall show actual installation details including all equipment locations (i.e.: control panel(s), agent container(s), detectors, alarms, manuals and aborts, etc.) as well as piping and conduit routing details. Show all room or facilities modifications, including door and/or damper installations completed. One (1) copy of reproducible engineering drawings shall be provided reflecting all actual installation details.

12. ACCEPTANCE TESTS:

- 1) At the time "As-Built" drawings and maintenance/operations manuals are submitted, the installing contractor shall submit a "Test Plan" describing procedures to be used to test the control system(s). The Test Plan shall include a step-by-step description of all tests to be performed and shall indicate the type and location of test apparatus to be employed. The tests shall demonstrate that the operational and installation requirements of this specification have been met. All tests shall be conducted in the presence of the owner and shall not be conducted until the Test Plan has been approved.
- 2) The tests shall demonstrate that the entire control system functions as designed and intended. All circuits shall be tested: automatic actuation, solenoid and manual actuation, HVAC and power shutdowns, audible and visual alarm devices and manual override of abort functions. Supervision of all panel circuits, including AC power and battery power supplies, shall be tested and qualified.
- 3) A room pressurization test shall be conducted, in each protected space, to determine the presence of openings which would affect the agent system concentration levels. The test(s) shall be conducted using the Retro-Tec Corp. Door Fan system, or equivalent, with integrated computer program. All testing shall be in accordance with NFPA 2001.
- 4) If room pressurization testing indicates that openings exist which would result in leakage and/or loss of the extinguishing agent, the installing contractor shall be responsible for coordinating the proper sealing of the protected space(s) by the general contractor or his sub-contractor or agent. The general contractor shall be responsible for adequately sealing all protected space(s) against agent loss or leakage. The installing contractor shall inspect all work to ascertain that the protected space(s) have been adequately and properly sealed. THE SUPPRESSION SYSTEM INSTALLING CONTRACTOR SHALL BE RESPONSIBLE FOR THE SUCCESS OF THE ROOM PRESSURIZATION TESTS. If the first room pressurization test is not successful, in accordance with these specifications, the installing contractor shall direct the general contractor to determine, and correct, the cause of the test failure. The installing contractor shall conduct additional room pressurization tests, at no additional cost to the owner, until a successful test is obtained. Copies of successful test results shall be submitted to the owner for record.
- 5) Upon acceptance by the owner, the completed system(s) shall be placed into service.

13. SYSTEM INSPECTIONS:

The installing contractor shall provide two (2) inspections of each system, installed under this contract, during the one-year warranty period. The first inspection shall be at the six month interval, and the second inspection at the 12 month interval, after system acceptance. Inspections shall be conducted in accordance with the manufacturer's guidelines and the recommendations of NFPA 2001.

Integrity Test must be done by qualified certified Engineer. The installer must have UL Approved filling station in U.A.E for refilling and maintenance purpose

Documents certifying satisfactory system(s) operation shall be submitted to the owner upon completion of each inspection.

14. WARRANTY:

All system components furnished, and installed under this contract, shall be guaranteed against defects in design, materials and workmanship for the full warranty period which is standard with the manufacturer, but in no case less than one (1) year from TOC (taking over certificate).

END OF SECTION

MECHANICAL SECTION: 21 30 00

FIRE FIGHTING PUMPS

PART 1 – GENERAL

1.1 WORK INCLUDED

1.1.1 Compliance with sections of Division 1, General Requirements and documents referred to therein.

1.1.2 Conform to General Requirements for Mechanical Services

1.2 DESCRIPTION OF WORK

1.2.1 Furnish and install pumps as indicated on drawings

1.2.2 This section specifies the requirements for the fire pumpset.

1.2.3 Provide the electrical works related to this section, in compliance with the requirements of related sections.

1.3 QUALITY ASSURANCE

1.3.1 Manufacturer's Qualifications: Firms regularly engaged in manufacture of general – use, centrifugal pumps with characteristics, sizes and capacities required, whose products have been in satisfactory use in similar service for not less than 5 years.

1.3.2 Certification, Pump Performance: Provide pumps whose performances, under specified operating conditions, are certified by manufacturer.

1.3.3 Fire pump and accessories shall be UL listed and FM and DCD approved.

1.3.4 fire pump electrical starting current shall be as per DEWA regulation and requirement.

1.3.4 **All Firefighting pumps to be imported from USA.**

1.4 SUBMITTALS

1.4.1 Product Data: Submit manufacturer's pump specifications, installation and start- up instructions, and current accurate pump characteristic performance curves with selection points clearly indicated.

1.4.2 Shop Drawings: Submit manufacturer's assembly-type shop drawings indicating dimensions, weight loadings, required clearances and methods of assembly of components.

1.4.3 Wiring Diagrams: Submit manufacturer's electrical requirements for power supply wiring to pumps. Submit manufacturer's ladder-type wiring diagrams for interlock and control wiring. Clearly differentiate between portions of wiring that are factory – installed and portions to be field- installed.

1.4.4 Full hydraulic calculations based on shop drawings to verify actual flow and head operating conditions. Calculations shall be approved by the Consultant prior to final selection of equipment.

- 1.4.5 Maintenance Data: Submit maintenance data and parts lists for each type of pump, control and accessory, including “trouble – shooting” maintenance guide. Include this data, product data, shop drawings and wiring diagrams in maintenance manual.

1.5 PRODUCT DELIVERY, STORAGE AND HANDLING

- 1.5.1 Handle pumps and components carefully to prevent damage, breaking, denting and scoring. Do not install damaged pumps or components, replace with new items.
- 1.5.2 Store pumps and components in clean dry place. Protect from weather, dirt, fumes, water, construction debris, and physical damage.

PART 2 - PRODUCTS

2.1 PACKAGED FIRE PUMPSET

2.1.1 Electric Pump

- a) Type - The pump shall be UL/FM approved, especially designed and constructed for quiet operation in compliance with NFPA-20.
- b) Pump shall be of horizontal SPLIT CASE type design.
- c) pump shall be single stage and casing shall be Axially split case, cast iron body
- d) Impellor shall be cast bronze statically and dynamically balanced and keyed to shaft
- e) Shaft shall be stainless steel
- f) Motor – shall be drip proof, 2900 RPM and shall be especially selected for quiet operation. The horse power of the motor shall be of such a size as to insure non overloading of the motor through the capacity range of the pump.
- g) Installation – A concrete base 200 mm higher than the surrounding floor shall be provided. Construct base and install as shown by the detail on the drawings. Provide a minimum of four 12 mm anchor bolts from concrete to pump base.
- h) Controls – The pump shall be controlled by a combined manual and automatic across the line type fire pump control panel. The control panel shall have an externally operated circuit breaker, a pilot light to indicate when power is on, switches or push button for manual or automatic operation, a pressure switch control for automatic operation and a remote alarm signal when pump is operating. The entire control panel shall be factory assembled in a steel cabinet. The panel shall have provision for connection to the BMS. Controllers to have auto-program for start/stop for future periodic maintenance. Controllers shall be UL listed and FM approved.

2.1.2 Jockey Pump

- a) Type - The jockey pump shall be vertical multi-stage type, especially designed and constructed for quiet operation.

- b) Motor - Shall be drip proof, 2900 RPM and shall be especially selected for quiet operation. The horse-power of the motor shall be of such a size as to insure non-overloading of the motor throughout the capacity range of the pump.
- c) Controls - The jockey pump shall have a relief valve set for the proper pressure and a pressure switch for automatic operation. There shall be a magnetic across-the-line starter with "Hand-Off-Automatic" selector switch in cover provided to start and stop the pump. Provision for connection to BMS shall be provided.

2.1.3 Diesel Driven Pump

- a) Fire pump shall constitute a diesel engine and fire pump of horizontal SPLIT CASE type. Pump specification shall be as mentioned above under item 2.1.1 except for pump speed which shall be accordingly to the choice of the diesel engine speed.
- b) Rating of diesel engine shall be chosen at the prevailing conditions at U.A.E, i.e. ambient temperature of 120 deg F. and a humidity ratio of more than 90%.
- c) Automatically Controlled Diesel Engine The diesel engine for driving the fire pump shall be UL listed, FM approved specifically designed for fire pump service. It shall develop sufficient horse power to drive the pump and shall have bare engine brake horse power not less than 20 percent greater than the maximum horsepower required to drive the pump at its rated RPM.
- d) The engine shall be an open type self contained power unit including the following accessories: -
 - 1) Stub shaft
 - 2) Fuel pumps and filters.
 - 3) Air Cleaner.
 - 4) Adjustable governor capable of regulating engine speed within a range of 10 percent between shut-off and maximum load conditions. The governor shall be set to maintain rated pump speed at maximum pump load.
 - 5) Lube oil pump.
 - 6) Lube oil filter.
 - 7) Over speed shut down device arranged to shut down engine at a speed approximately 20 percent above rated engine speed and with manual reset. The over speed shut down device shall be so supervised that the automatic engine controller will continue to show an over speed trouble signal until the device is manually reset to normal operating position.
 - 8) Electric Starter, generator or alternator and voltage regulator.
 - 9) Dual storage battery units. Each unit shall have capacity to maintain cranking speed through a 6-minute cycle (15 seconds cranking and 15 seconds rest, in 12 consecutive cycles). Provide certification that the battery unit complies with this requirement. Batteries shall be furnished in a dry charge condition, with electrolyte liquid in a separate container. Electrolyte shall be added at the time the engine is put in service, and the battery is given a conditioning charge.
 - 10) Dual battery chargers, automatically controlled.

Chargers shall be specifically designed for fire pump service.

The rectifier shall be a semiconductor type.

The charger for a lead acid battery shall be a type which automatically reduces the charging rate to less than 500 mill amperes when the battery reaches a full charge condition.

The control equipment incorporated in an "off-on" type of charger for a lead acid battery shall start the rectifier hourly. It shall shut off automatically when the battery has been fully charged.

The battery charger at its rated voltage shall be capable of so delivering energy into a full discharged battery in such a manner that it will not damage the battery. It shall restore to the battery 100 percent of the battery's ampere hour rating within 24 hours.

An ammeter with an accuracy of five percent of the normal charging rate shall be furnished to indicate the operation of the charger.

The charger shall be designed so that it will not be damaged or blow fuses during the cranking cycle of the engine when operated by an automatic or manual controller.

A manual charge switch with indicator light shall be provided or, in lieu thereof, the charger shall automatically charge at the maximum rate whenever required by the state of charge of the battery.

When not connected through a control panel it shall be arranged to indicate loss of current output on the load side of the d-c over-current protective device.

Storage batteries shall be substantially supported, secured against displacement, and located where they shall be readily accessible for servicing.

- 11) Engine cooling system
Cooling water shall be from the discharge of the pump, taken off prior to the pump discharge valve. Threaded rigid pipe shall be used for this connection. The pipe shall include a manual shut off valve, flushing type strainer, pressure regulating valve, automatic electric solenoid valve and a second manual shut off valve. A bypass line with manual shut off valve shall be installed around the manual shut off valve, strainer, pressure regulator and solenoid valve.
- 12) Engine jacket temperature regulating device.
- 13) Engine jacket water pump.
- 14) Instrument panel with water temperature gauge, oil pressure gauge, ammeter, tachometer of the totalizing type or an hour meter.
- 15) Exhaust mufflers and flexible connectors, stainless steel non-sparking.
- 16) The engine shall be arranged for automatic operation with all wiring terminating in a junction box suitable for field hook-up.
- 17) Exhaust piping from flexible connection at engine to muffler and from muffler to point of discharge to atmosphere shall be schedule 40 black steel

pipe and fittings, welded. A flap valve shall be provided at the external end of piping.

- 18) Pump and diesel engine shall be mounted on a common base plate, direct connected through a flexible coupling, and equipped with a coupling guard. Coupling shall be a spacer type to permit pump shaft and impellers to be removed without moving diesel engine or pump bottom casing.

2.1.4 Automatic Diesel Engine Control Panel

2.1.4.1 The automatic engine control panel shall be UL listed, FM approved specifically designed for fire pump service. The panel shall be enclosed in a floor mounted duct and moisture resistant housing. The controller shall have possibility to program the start/stop of equipment for periodic maintenance.

2.1.4.2 The controller shall be the combined manual and automatic type including a "Manual-Off-Automatic selector switch.

2.1.4.3 A pressure switch and single phase power failure relay with local visual indication and auxiliary contact for remote indication of 240-volt AC power failure shall be provided which will, upon drop in system pressure or loss of single phase AC power, cause the controller to open or close all circuits necessary to automatically start the engine by performing the following functions: -

- (a) Turn on fuel supply.
- (b) Open the cooling water solenoid valve.
- (c) Crank the engine in a series of crank-rest cycles automatically alternating between the dual batteries on each cycle.
- (d) Disconnect cranking motor upon engine start.
- (e) Once the engine is started, it shall remain in operation until shut down manually. A signal in the Synoptic Panel shall indicate that the Diesel Engine is operating.
- (f) If the engine should fail to start after several cranking attempts, the controller shall disconnect all starting circuits and energize an audible and visual alarm to indicate "over crank" in the synoptic panel as well as in the engine control panel.

2.1.4.4 Alarm and Signal Devices on Controller

A pilot lamp(s) shall be provided to indicate that the controller is in the "automatic" position. The lamp shall be accessible for replacement. Separate pilot lamps and a common bell shall be provided to indicate trouble caused by: -

Low oil pressure in the lubrication system. The controller shall provide means for testing the position of the pressure switch contacts without causing trouble alarms. Failure of engine to start automatically.

Shutdown from over speed.

Battery failure. Each battery shall be provided with separate lamps which shall be lighted or extinguished on battery failure on attempt to start.

High engine jacket water temperature.

- 2.1.4.5 Water Pressure Control : A pressure actuated switch having high and low calibrated adjustments and responsive to water pressure in the fire system shall be provided in the controller circuit. Suitable provision shall be made for relieving pressure to the pressure switch to test the operation of the controller and the pump.
- 2.1.4.6 Enclosure : The structure or panel shall be securely mounted in an enclosure(s) which will protect the equipment against mechanical injury and falling drops of water striking the enclosure from the downward vertical. All switches required to keep the controller in the "automatic" position shall be within the locked cabinets having to break glass panels.
- 2.1.4.7 A wiring diagram shall be provided and permanently attached to the inside of the enclosure showing exact wiring for the controller, including a legend of identifying numbers of individual components. All wiring terminals shall be plainly marked to correspond with the wiring diagram furnished.
- 2.1.4.8 Wiring Elements : Wiring elements of the controller shall be designed on a continuous-duty basis.
- 2.1.4.9 Field Wiring : All wiring leading from the panel to the engine and to the batteries shall have adequate carrying capacity. Such wiring shall be protected against mechanical injury. Controller manufacturer's specifications for distance and wire size shall be followed. Provision for connection to BMS shall be provided.
- 2.1.4.10 Marking : Each operating component of the controller shall be marked to plainly indicate an identifying number referenced to the wiring diagram. The markings shall be located so as to be visible after installation.

Instructions: Complete instructions covering the operation of the controller shall be provided and conspicuously mounted on the controller.

2.1.5 Fuel Storage Tank

The fuel storage tank shall be an integral part of the unit and shall be mounted within the support base of the unit. The tank shall have a capacity of 8 hours operation. Provide all piping, fittings and valves required.

The unit support base shall be designed to allow space for oil pan clearance.

2.1.6 Tests

- 2.1.6.1 Prior to shipment, the pump and engine shall be thoroughly shop-tested as a unit by the pump manufacturer. A characteristic curve showing the pump performance based upon the results of the shop test shall be submitted to the Consultant prior to shipment, and shall include the performance pump and engine at the set-governor's speed. Absence of these test reports, indicating that the complete unit has been tested in the factory by the driver, will be ample grounds for the equipment rejection.
- 2.1.6.2 Each control panel shall be fully assembled and factory tested by the control manufacturer prior to their shipment, and evidence of this test shall be furnished with the controller.

2.1.6.3 Field Acceptance Tests

- (a) The field acceptance test results shall be as good as the manufacturer's certified shop test characteristic curve for the pump being tested within the accuracy limits of the test equipment.
- (b) Overheating : The pump shall operate at peak load conditions without objectionable heating of the bearings or of the prime mover. The operating pump speed shall be the speed at which the pumping unit would be expected to operate during a fire.
- (c) The engine shall not show signs of overload or stress and its governor shall properly regulate the speed.
- (d) Controllers:

The automatic controller shall be put through not less than ten (10) automatic and ten (10) manual operations.

A running interval of at least five minutes at full speed shall be allowed before repeating the starting cycle.

In an automatic operation, the controller shall start the pump from all the provided starting features, such as pressure switches, valves, etc.

Length of Test. The pump shall be in operation not less than one hour (total time) during the foregoing tests.

2.1.7 Single Source Responsibility :

The pump, driver, and all controls and necessary attachments, specified herein, shall be purchased under a unit contract. The pump manufacturer shall assume unit responsibility and shall provide the services of a qualified Consultant to supervise the installation and alignment of equipment.

Field tests shall be conducted in the field to satisfy the Consultant and Local Authorities having jurisdiction of satisfactory operation of pump, driver and controller. The pump manufacturer's Engineer shall be at the job site, supervising the installation and testing. The control manufacturer shall have their representative at the job site to train operators in the use of the controls. All costs related to the services shall be born by the Contractor.

2.1.8 Spare Parts and Tools

Spare parts and tools necessary for two years operation, as recommended by the manufacturer, shall be provided as a part of the contract for the engine. A list of these spares shall be included in the offer.

PART 3 - EXECUTION

3.1 EXAMINATION

- 3.1.1 Examine areas, equipment foundations, and conditions with installer present for compliance with requirements for installation and other conditions affecting pump performance. Do not proceed with installation until unsatisfactory conditions have been corrected.

3.1.2 Examine the installed fire fighting system, verify actual locations of piping and connections prior to testing and commissioning.

3.2 CONCRETE EQUIPMENT BASES

3.2.1 Install concrete equipment bases of dimensions required for fire pumps, and controllers.

3.3 INSTALLATION

3.3.1 Comply with pump manufacturers written installation and alignment instructions.

3.3.2 Install pumps in locations indicated and arrange to provide access for periodic maintenance, including removal of motors impellers, couplings and accessories.

3.3.3 Set base-mounted pumps on concrete equipment bases as indicated on drawings. Disconnect couplings halves before setting. Do not reconnect couplings until alignment operations have been completed.

3.3.4 Support pump base plate on rectangular metal blocks and shims or on metal wedges having small taper, at points near foundation bolts to provide a gap of 19 to 38 mm between pump base and foundation for grouting.

3.3.5 Adjust metal supports or wedges until pump and driver shafts are level. Check couplings faces and pump suction and discharge flanges to verify that they are level.

3.3.6 Install OS & Y Gate valves on pump suction and discharge line

3.3.7 install Spring loaded check valve on pump's discharge line

3.3.8 Install valves of types and at locations indicated that are same size as the piping connecting fire pump, bypasses, test headers and other piping systems.

3.3.9 Install pressure gages on pump suction and discharge in such a manner to be easily readable and completely isolated from vibrations.

3.3.10 Support pumps and piping separately so that weight of piping system does not rest on pumps.

3.3.11 Install piping accessories, hangers and supports, anchors, valves meters and gages, and equipment support as indicated for complete installation.

3.3.12 Install flow meter and sensing elements where indicated, install flow-measuring system components and make connections according to manufacturers written installation instructions.

3.3.13 Electrical Wiring: Install electrical devices furnished by equipment manufacturers but not specified as factory mounted. Furnish copy of manufacturers wiring diagram for submittal to the Consultant.

3.3.14 Verify that electrical wiring is installed according to manufacturer's recommendations and installation requirements. Do not proceed with equipment start up until wiring installation is acceptable.

3.4 ALIGNMENT

Align pump and driver shafts after complete unit has been leveled on foundation and after grout has set and foundation bolts have been tightened.

Once the alignment is correct, tighten foundation bolts evenly but not too firmly. Fill base plate completely with non shrink, non metallic grout, with metal blocks and shims or wedges in place. After grout has hardened, fully tighten foundation bolts. Check alignment and take corrective measures if required.

3.5 CONNECTIONS

Connect suction and discharge side of the piping system to the pumps. Connect flow-measuring and other hydronic system components according to manufacturer's written installation instructions.

3.6 FIELD QUALITY CONTROL

Contractor to ensure services of pump manufacturer representative to supervise field assembly of components, installation of fire fighting pump units, including piping and electrical connections and field acceptance tests. Report test results in writing.

Check all connections for tightness.

Perform field-acceptance tests for the pump unit (pump, driver and controller) and system piping when pump unit installation is complete. Where possible, field correct malfunctioning equipment, then retest to demonstrate compliance. Replace equipment that cannot be satisfactorily corrected or that does not perform as specified and as indicated, then retest to demonstrate compliance. Verify that each pump unit performs as specified and as indicated.

3.7 TESTING AND COMMISSIONING

3.7.1 General.

After completion of the Work, Contractor shall run a test over a sufficient period of time to prove the proper capacity and performance of all apparatus, etc., and the system as a whole.

The Contractor shall carry out commissioning and testing of all equipment installed.

Startup Services: Contractors shall provide services of pump manufacturer representative to provide startup service and to demonstrate and train Employer's maintenance personnel as specified below:

Test and adjust controls and safeties. Replace damaged and malfunctioning controls and components.

Review data in the Operating and Maintenance Manual.

Final Checks Before Startup: perform the following preventive-maintenance operations and checks before startup wherever applicable:

Lubricate oil-lubricated bearings.

Remove grease-lubricated bearing covers and flush bearings with kerosene and thoroughly clean. Fill with new lubricant according to manufacturer's recommendations.

Disconnect coupling and check electric motor for proper rotation. Rotation shall match direction of rotation marked on pump casing.

Check that the pump is free to rotate by hand. Do not operate the pump if it is bound or if it drags even slightly until cause of trouble is determined and corrected.

3.7.5 Starting procedure for pumps (open circuit):

Prime pump by opening suction valve and closing drains, and prepare pump for operation. Open sealing liquid supply valve if pump is so fitted.

Start motor.

Open discharge valve slowly.

Observe leakage from stuffing boxes and adjust sealing liquid valve for proper flow to ensure lubrication of packing. Do not tighten gland immediately, but let packing run in before reducing leakage through stuffing boxes.

Check general mechanical operation of pump and motor.

3.7.6 Commissioning shall be deemed to include all operations required in order to correctly set the plant to work, adjust and calibrate to design conditions to the complete satisfaction of the Consultant. The above shall include but not limited to testing and balancing of all fire fighting systems, including associated plant and equipment, balancing and adjusting to achieve correct flow rates, including setting valves and other regulating devices.

3.7.7 Final setting positions for all regulating and controlling devices shall be recorded and such records shall form part of the manual of the operating and maintenance instructions.

3.7.8 All systems shall be field tested and run on load, with all automatic controls in operation and all circuits balanced to produce design conditions, in presence and to the satisfaction of the Consultant.

3.7.9 In case of unsuccessful test run, the Contractor shall take necessary step for the rectification and retested to the satisfaction of the Consultant.

3.7.10 The Contractor shall also provide at his own cost services of skilled and unskilled staff during the test run of all the systems as required by the Consultant.

3.8 TRAINING

3.8.1 Train Employer's maintenance personnel on procedures and schedules related to startup and shutdown, troubleshooting, servicing, and preventive maintenance.

3.8.2 Schedule training with at least 7 days' advance notice.

END OF SECTION

MECHANICAL SECTION: 22 05 16
EXPANSION FITTINGS AND LOOPS

PART 1 – GENERAL

1.1 RELATED DOCUMENTS

- A Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A Section Includes:
 - 1 Flexible-hose pack less expansion joints.
 - 2 Metal-bellows pack less expansion joints.
 - 3 Rubber pack less expansion joints.
 - 4 Grooved-joint expansion joints.
 - 5 Pipe loops and swing connections.
 - 6 Alignment guides and anchors.

1.3 PERFORMANCE REQUIREMENTS

- A Compatibility: Products shall be suitable for piping service fluids, materials, working pressures, and temperatures.
- B Capability: Products to absorb 200 percent of maximum axial movement between anchors.

1.4 SUBMITTALS

- A Product Data: For each type of product indicated.
- B Delegated-Design Submittal: For each anchor and alignment guide indicated to comply with performance requirements and design criteria, including analysis data signed and sealed by the qualified professional engineer responsible for their preparation.
 - 1 Design Calculations: Calculate requirements for thermal expansion of piping systems and for selecting and designing expansion joints, loops, and swing connections.
 - 2 Anchor Details: Detail fabrication of each anchor indicated. Show dimensions and methods of assembly and attachment to building structure.
 - 3 Alignment Guide Details: Detail field assembly and attachment to building structure.
 - 4 Schedule: Indicate type, manufacturer's number, size, material, pressure rating, end connections, and location for each expansion joint.

- C Welding certificates.

- D Product Certificates: For each type of expansion joint, from manufacturer.

- E Maintenance Data: For expansion joints to include in maintenance manuals.

1.5 QUALITY ASSURANCE

- A Welding Qualifications: Qualify procedures and personnel according to the following:
 - 1 AWS D1.1/D1.1M, "Structural Welding Code - Steel."
 - 2 ASME Boiler and Pressure Vessel Code: Section IX.

PART 2 PRODUCTS

2.1 PACKLESS EXPANSION JOINTS

A Flexible-hose Pack less Expansion Joints:

- 1 Description: Manufactured assembly with inlet and outlet elbow fittings and two flexible-metal-hose legs joined by long-radius, 180-degree return bend or center section of flexible hose.
- 2 Flexible Hose: Corrugated-metal inner hoses and braided outer sheaths.
- 3 Expansion Joints for Copper Tubing NPS 2 (DN 50) and Smaller: Copper-alloy fittings with solder-joint end connections.
 - a Bronze hoses and single-braid bronze sheaths with 450 psig at 70 deg F (3100 kPa at 21 deg C) and 340 psig at 450 deg F (2340 kPa at 232 deg C) ratings.
 - b Bronze hoses and double-braid bronze sheaths with 700 psig at 70 deg F (4830 kPa at 21 deg C) and 500 psig at 450 deg F (3450 kPa at 232 deg C) ratings.
- 4 Expansion Joints for Copper Tubing NPS 2-1/2 to NPS 4 (DN 65 to DN 100): Copper-alloy fittings with threaded end connections.
 - a Stainless-steel hoses and single-braid, stainless-steel sheaths with 300 psig at 70 deg F (2070 kPa at 21 deg C) and 225 psig at 450 deg F (1550 kPa at 232 deg C) ratings.
 - b Stainless-steel hoses and double-braid, stainless-steel sheaths with 420 psig at 70 deg F (2890 kPa at 21 deg C) and 315 psig at 450 deg F (2170 kPa at 232 deg C) ratings.
- 5 Expansion Joints for Steel Piping NPS 2 (DN 50) and Smaller: Stainless-steel fittings with threaded end connections.
 - a Stainless-steel hoses and single-braid, stainless-steel sheaths with 450 psig at 70 deg F (3100 kPa at 21 deg C) and 325 psig at 600 deg F (2250 kPa at 315 deg C) ratings.
 - b Stainless-steel hoses and double-braid, stainless-steel sheaths with 700 psig at 70 deg F (4830 kPa at 21 deg C) and 515 psig at 600 deg F (3550 kPa at 315 deg C) ratings.
- 6 Expansion Joints for Steel Piping NPS 2-1/2 to NPS 6 (DN 65 to DN 150): Stainless-steel fittings with flanged end connections.
 - a Stainless-steel hoses and single-braid, stainless-steel sheaths with 200 psig at 70 deg F (1380 kPa at 21 deg C) and 145 psig at 600 deg F (1000 kPa at 315 deg C) ratings.
 - b Stainless-steel hoses and double-braid, stainless-steel sheaths with 275 psig at 70 deg F (1900 kPa at 21 deg C) and 200 psig at 600 deg F (1380 kPa at 315 deg C) ratings.

- 7 Expansion Joints for Steel Piping NPS 8 to NPS 12 (DN 200 to DN 300): Stainless-steel fittings with flanged end connections.
 - a Stainless-steel hoses and single-braid, stainless-steel sheaths with 125 psig at 70 deg F (860 kPa at 21 deg C) and 90 psig at 600 deg F (625 kPa at 315 deg C) ratings.
 - b Stainless-steel hoses and double-braid, stainless-steel sheaths with 165 psig at 70 deg F (1130 kPa at 21 deg C) and 120 psig at 600 deg F (830 kPa at 315 deg C) ratings.

B Metal-Bellows Pack Less Expansion Joints:

- 1 Standards: ASTM F 1120 and EJMA's "Standards of the Expansion Joint Manufacturers Association, Inc."
- 2 Type: Circular, corrugated bellows with external tie rods.
- 3 Minimum Pressure Rating: 300ig (2070) unless otherwise indicated.
- 4 Configuration: double joint with base class unless otherwise indicated.
- 5 Expansion Joints for Copper Tubing: Single- or multi-ply phosphor-bronze bellows, copper pipe ends, and brass shrouds.
 - A End Connections for Copper Tubing NPS 2 (DN 50) and Smaller: Solder joint or threaded.
 - B End Connections for Copper Tubing NPS 2-1/2 (DN 65) and larger flanged.

C Rubber Pack Less Expansion Joints for PPR and PVC pipes:

- 1 Standards: ASTM F 1123 and FSA's "Technical Handbook: Non-Metallic Expansion Joints and Flexible Pipe Connectors."
- 2 Material: Fabric-reinforced rubber complying with FSA-NMEJ-703.
- 3 Arch Type: Single or multiple arches with external control rods.
- 4 Spherical Type: Single or multiple spheres with external control rods.
- 5 Minimum Pressure Rating: 300ig (2070) at 194g F (90deg C).
- 6 Material for Water: Buna-N.
- 7 End Connections: Full-faced, integral steel flanges with steel retaining rings.

2.2 GROOVED-JOINT EXPANSION JOINTS

- A Description: Factory-assembled expansion joint made of several grooved-end pipe nipples, couplings, and grooved joints.
- B Standard: AWWA C606, for grooved joints.
- C Nipples: Galvanized, ASTM A 53/A 53M, Schedule 40, Type E or S, steel pipe with grooved ends.

- D Couplings: Seven flexible type for steel-pipe dimensions. Include ferrous housing sections, Buna-N gasket suitable for diluted acid, alkaline fluids, and cold and hot water and bolts and nuts.

2.3 ALIGNMENT GUIDES AND ANCHORS

A Alignment Guides:

- 1 Description: Steel, factory-fabricated alignment guide, with bolted two-section outer cylinder and base for attaching to structure; with two-section guiding spider for bolting to pipe.

B Anchor Materials:

- 1 Steel Shapes and Plates: ASTM A 36/A 36M.
- 2 Bolts and Nuts: ASME B18.10 or ASTM A 183, steel hex head.
- 3 Washers: ASTM F 844, steel, plain, flat washers.
- 4 Mechanical Fasteners: Insert-wedge-type stud with expansion plug anchor for use in hardened Portland cement concrete, with tension and shear capacities appropriate for application.
 - a Stud: Threaded, zinc-coated carbon steel.
 - b Expansion Plug: Zinc-coated steel.
 - c Washer and Nut: Zinc-coated steel.
- 5 Chemical Fasteners: Insert-type-stud, bonding-system anchor for use with hardened Portland cement concrete, with tension and shear capacities appropriate for application.
 - a Bonding Material: ASTM C 881/C 881M, Type IV, Grade 3, two-component epoxy resin suitable for surface temperature of hardened concrete where fastener is to be installed.
 - b Stud: ASTM A 307, zinc-coated carbon steel with continuous thread on stud unless otherwise indicated.
 - c Washer and Nut: Zinc-coated steel.

PART 3 EXECUTION

3.1 EXPANSION-JOINT INSTALLATION

- A Install expansion joints of sizes matching sizes of piping in which they are installed.
- B Install metal-bellows expansion joints according to EJMA's "Standards of the Expansion Joint Manufacturers Association, Inc."
- C Install rubber packless expansion joints according to FSA-NMEJ-702.
- D Install grooved-joint expansion joints to grooved-end steel piping

3.2 PIPE LOOP AND SWING CONNECTION INSTALLATION

- A Install pipe loops cold-sprung in tension or compression as required to partly absorb tension or compression produced during anticipated change in temperature.

- B Connect risers and branch connections to mains with at least five pipe fittings including tee in main.
 - C Connect risers and branch connections to terminal units with at least four pipe fittings including tee in riser.
 - D Connect mains and branch connections to terminal units with at least four pipe fittings including tee in main.
- 3.3 ALIGNMENT-GUIDE AND ANCHOR INSTALLATION
- A Install alignment guides to guide expansion and to avoid end-loading and torsional stress.
 - B Install two guide(s) on each side of pipe expansion fittings and loops. Install guides nearest to expansion joint not more than four pipe diameters from expansion joint.
 - C Attach guides to pipe and secure guides to building structure.
 - D Install anchors at locations to prevent stresses from exceeding those permitted by ASME B31.9 and to prevent transfer of loading and stresses to connected equipment.
 - E Anchor Attachments:
 - 1 Anchor Attachment to Black-Steel Pipe: Attach by welding. Comply with ASME B31.9 and ASME Boiler and Pressure Vessel Code: Section IX, "Welding and Brazing Qualifications."
 - 2 Anchor Attachment to Galvanized-Steel Pipe: Attach with pipe hangers. Use MSS SP-69, Type 42, riser clamp welded to anchor.
 - 3 Anchor Attachment to Copper Tubing: Attach with pipe hangers. Use MSS SP-69, Type 24, U-bolts bolted to anchor.
 - F Fabricate and install steel anchors by welding steel shapes, plates, and bars. Comply with ASME B31.9 and AWS D1.1/D1.1M.
 - 1 Anchor Attachment to Steel Structural Members: Attach by welding.
 - 2 Anchor Attachment to Concrete Structural Members: Attach by fasteners. Follow fastener manufacturer's written instructions.
 - G Use grout to form flat bearing surfaces for guides and anchors attached to concrete.

END OF SECTION

MECHANICAL SECTION: 22 05 17

SLEEVES FOR PLUMBING

PART 1 – GENERAL

1.1 RELATED DOCUMENTS

- A Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A Section Includes:

- 1 Sleeves.
- 2 Stack-sleeve fittings.
- 3 Sleeve-seal systems.
- 4 Sleeve-seal fittings.
- 5 Grout.

1.3 SUBMITTALS

- A Product Data: For each type of product indicated.

PART 2 PRODUCTS

2.1 SLEEVES

- A Cast-Iron Wall Pipes: Cast or fabricated of cast or ductile iron and equivalent to ductile-iron pressure pipe, with plain ends and integral water stop unless otherwise indicated.
- B Galvanized-Steel Wall Pipes: ASTM A 53/A 53M, Schedule 40, with plain ends and welded steel collar; zinc coated.
- C Galvanized-Steel-Pipe Sleeves: ASTM A 53/A 53M, Type E, Grade B, Schedule 40, zinc coated, with plain ends.
- D PVC-Pipe Sleeves: ASTM D 1785, Schedule 40.
- E Galvanized-Steel-Sheet Sleeves: 0.0239-inch (0.6-mm) minimum thickness; round tube closed with welded longitudinal joint.
- F Molded-PE or -PP Sleeves: Removable, tapered-cup shaped, and smooth outer surface with nailing flange for attaching to wooden forms.
- G Molded-PVC Sleeves: With nailing flange for attaching to wooden forms.

2.2 STACK-SLEEVE FITTINGS

- A Description: Manufactured, cast-iron sleeve with integral clamping flange. Include clamping ring, bolts, and nuts for membrane flashing.
 - 1 Under deck Clamp: Clamping ring with setscrews.

2.3 SLEEVE-SEAL SYSTEMS

- A Description: Modular sealing-element unit, designed for field assembly, for filling annular space between piping and sleeve.
- 1 Sealing Elements: EPDM-rubber interlocking links shaped to fit surface of pipe. Include type and number required for pipe material and size of pipe.
 - 2 Pressure Plates: Carbon steel.
 - 3 Connecting Bolts and Nuts: Carbon steel, with corrosion-resistant coating, of length required to secure pressure plates to sealing elements.

2.4 SLEEVE-SEAL FITTINGS

- A Description: Manufactured plastic, sleeve-type, water stop assembly made for imbedding in concrete slab or wall. Unit has plastic or rubber water stop collar with center opening to match piping OD.

2.5 GROUT

- A Standard: ASTM C 1107/C 1107M, Grade B, post-hardening and volume-adjusting, dry, hydraulic-cement grout.
- B Characteristics: No shrink; recommended for interior and exterior applications.
- C Design Mix: 5000-psi (34.5-MPa), 28-day compressive strength.
- D Packaging: Premixed and factory packaged.

PART 3 EXECUTION

3.1 SLEEVE INSTALLATION

- A Install sleeves for piping passing through penetrations in floors, partitions, roofs, and walls.
- B For sleeves that will have sleeve-seal system installed, select sleeves of size large enough to provide 1-inch (25-mm) annular clear space between piping and concrete slabs and walls.
- 1 Sleeves are not required for core-drilled holes.
- C Install sleeves in concrete floors, concrete roof slabs, and concrete walls as new slabs and walls are constructed.
- 1 Permanent sleeves are not required for holes in slabs formed by molded-PE or -PP sleeves.
 - 2 Cut sleeves to length for mounting flush with both surfaces.
 - A Exception: Extend sleeves installed in floors of mechanical equipment areas or other wet areas 2 inches (50 mm) above finished floor level.
 - 3 Using grout, seal the space outside of sleeves in slabs and walls without sleeve-seal system.

- D Install sleeves for pipes passing through interior partitions.
- 1 Cut sleeves to length for mounting flush with both surfaces.
 - 2 Install sleeves that are large enough to provide 1/4-inch (6.4-mm) annular clear space between sleeve and pipe or pipe insulation.
 - 3 Seal annular space between sleeve and piping or piping insulation; use joint sealants appropriate for size, depth, and location of joint. Comply with requirements for sealants specified in Division 07 Section "Joint Sealants."
- E Fire-Barrier Penetrations: Maintain indicated fire rating of walls, partitions, ceilings, and floors at pipe penetrations. Seal pipe penetrations with fire stop materials. Comply with requirements for fire stopping specified in Division 07 Section "Penetration Fire stopping."

3.2 STACK-SLEEVE-FITTING INSTALLATION

- A Install stack-sleeve fittings in new slabs as slabs are constructed.
- 1 Install fittings that are large enough to provide 1/4-inch (6.4-mm) annular clear space between sleeve and pipe or pipe insulation.
 - 2 Secure flashing between clamping flanges for pipes penetrating floors with membrane waterproofing. Comply with requirements for flashing specified in Division 07 Section "Sheet Metal Flashing and Trim."
 - 3 Install section of cast-iron soil pipe to extend sleeve to 2 inches (50 mm) above finished floor level.
 - 4 Extend cast-iron sleeve fittings below floor slab as required to secure clamping ring if ring is specified.
 - 5 Using grout, seal the space around outside of stack-sleeve fittings.
- B Fire-Barrier Penetrations: Maintain indicated fire rating of floors at pipe penetrations. Seal pipe penetrations with fire stop materials. Comply with requirements for fire stopping specified in Division 07 Section "Penetration Fire stopping."

3.3 SLEEVE-SEAL-SYSTEM INSTALLATION

- A Install sleeve-seal systems in sleeves in exterior concrete walls and slabs-on-grade at service piping entries into building.
- B Select type, size, and number of sealing elements required for piping material and size and for sleeve ID or hole size. Position piping in center of sleeve. Center piping in penetration, assemble sleeve-seal system components, and install in annular space between piping and sleeve. Tighten bolts against pressure plates that cause sealing elements to expand and make a watertight seal.

3.4 SLEEVE-SEAL-FITTING INSTALLATION

- A Install sleeve-seal fittings in new walls and slabs as they are constructed.
- B Assemble fitting components of length to be flush with both surfaces of concrete slabs and walls. Position water stop flange to be centered in concrete slab or wall.
- C Secure nailing flanges to concrete forms.
- D Using grout, seal the space around outside of sleeve-seal fittings.

3.5 SLEEVE AND SLEEVE-SEAL SCHEDULE

A Use sleeves and sleeve seals for the following piping-penetration applications:

- 1 Exterior Concrete Walls above Grade:
 - a Piping Smaller Than NPS 6 (DN 150): Cast-iron wall sleeves.
 - b Piping NPS 6 (DN 150) and Larger: Cast-iron wall sleeves.
- 2 Exterior Concrete Walls below Grade:
 - a Piping Smaller Than NPS 6 (DN 150) : Cast-iron wall sleeves with sleeve-seal system.
 - 1) Select sleeve size to allow for 1-inch (25-mm) annular clear space between piping and sleeve for installing sleeve-seal system.
 - b Piping NPS 6 (DN 150) and Larger: Cast-iron wall sleeves with sleeve-seal system.
 - 1) Select sleeve size to allow for 1-inch (25-mm) annular clear space between piping and sleeve for installing sleeve-seal system.
- 3 Concrete Slabs-on-Grade:
 - a Piping Smaller Than NPS 6 (DN 150) : Cast-iron wall sleeves with sleeve-seal system.
 - 1) Select sleeve size to allow for 1-inch (25-mm) annular clear space between piping and sleeve for installing sleeve-seal system.
 - b Piping NPS 6 (DN 150) and Larger: Cast-iron wall sleeves with sleeve-seal system.
 - 1) Select sleeve size to allow for 1-inch (25-mm) annular clear space between piping and sleeve for installing sleeve-seal system.
- 4 Concrete Slabs above Grade:
 - a Piping Smaller than NPS 6 (DN 150) : Galvanized-steel-pipe sleeves.
 - b Piping NPS 6 (DN 150) and Larger: Galvanized-steel-pipe sleeves.
- 5 Interior Partitions:
 - a Piping Smaller than NPS 6 (DN 150): Galvanized-steel-pipe sleeves.
 - b Piping NPS 6 (DN 150) and Larger: Galvanized-steel-sheet sleeves.

END OF SECTION

MECHANICAL SECTION: 22 05 19
METERS AND GAUGES FOR PLUMBING

PART 1 – GENERAL

1.1 RELATED DOCUMENTS

- A Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A Section Includes:

- 1 Bimetallic-actuated thermometers.
- 2 Filled-system thermometers.
- 3 Liquid-in-glass thermometers.
- 4 Light-activated thermometers.
- 5 Thermo wells.
- 6 Dial-type pressure gages.
- 7 Gauge attachments.
- 8 Test plugs.
- 9 Test-plug kits.
- 10 Sight flow indicators.

- B Related Sections:

- 1 Division 21 Section 21 10 00
- 2 Division 22 Section 22 10 00
- 3 Division 23 Section 23 21 00

1.3 SUBMITTALS

- A Product Data: For each type of product indicated.
- B Product Certificates: For each type of meter and gage, from manufacturer.
- C Operation and Maintenance Data: For meters and gages to include in operation and maintenance manuals.

PART 2 PRODUCTS

2.1 STEM TYPE THERMOMETERS

- A Thermometer: ASTM E 1, red appearing mercury, lens front tube, cast aluminum case with enamel finish.
- 1 Size: 7 inch (175 mm) scale.
 - 2 Window: Clear glass.
 - 3 Accuracy: 2 percent, per ASTM E 77.
 - 4 Calibration: Degrees F.

- B Thermometer: ASTM E 1, adjustable angle, red appearing mercury, lens front tube, cast aluminum case with enamel finish, cast aluminum adjust-able joint with positive locking device.

- 1 Size: 7 inch (175 mm) scale.
- 2 Window: Clear glass.
- 3 Stem: 3/4 inch (20 mm) NPT brass.
- 4 Accuracy: 2 percent, per ASTM E 77.
- 5 Calibration: Degrees F.

2.2 DIAL THERMOMETERS

- A Thermometer: ASTM E 1, stainless steel case, bimetallic helix actuated with silicone fluid damping, white with black markings and black pointer hermetically sealed lens, stainless steel stem.

- 1 Size: 2 inch (50 mm) diameter dial.
- 2 Lens: Clear glass.
- 3 Accuracy: 1 percent.
- 4 Calibration: Degrees F.

- B Thermometer: ASTM E 1, stainless steel case, adjustable angle with front recalibration, bimetallic helix actuated with silicone fluid damping, white with black markings and black pointer hermetically sealed lens, stainless steel stem.

- 1 Size: 3 inch (75 mm) diameter dials.
- 2 Lens: Clear glass.
- 3 Accuracy: 1 percent.
- 4 Calibration: Degrees F.

- C Thermometer: ASTM E 1, stainless steel case, vapor or liquid actuated with brass or copper bulb, copper or bronze braided capillary, white with black markings and black pointer glass lens.

- 1 Size: 2-1/2 inch (60 mm) diameter dial.
- 2 Lens: Clear glass.
- 3 Length of Capillary: Minimum 5 feet (1500 mm).
- 4 Accuracy: 2 percent.
- 5 Calibration: Degrees F.

2.3 THERMOMETER SUPPORTS

- A Socket: Brass separable sockets for thermometer stems with or without extensions as required, and with cap and chain.

- B Flange: 3 inch (75 mm) outside diameter reversible flange, designed to fasten to sheet metal air ducts, with brass perforated stem.

2.4 PRESSURE GAGES

- A Gage: ASME B40.100, UL 393 drawn steel case, phosphor bronze bour-don tube, rotary brass movement, brass socket, with front recalibration ad-justment, black scale on white background.

- 1 Case: Steel with brass bourdon tube.
- 2 Size: 2 inch (50 mm) diameter.
- 3 Mid-Scale Accuracy: One percent.
- 4 Scale: Psi.

2.5 PRESSURE GAGE TAPPINGS

- A Gage Cock: Tee or lever handle, brass for maximum 150 psi (1034 kPa).
- B Needle Valve: Brass, 1/4 inch (6 mm) NPT for minimum 150 psi (1034 kPa).
- C Pulsation Damper: Pressure snubber, brass with 1/4 inch (6 mm) connections.
- D Syphon: Steel, Schedule 40, 1/4 inch (6 mm) angle or straight pattern.

2.6 TEST PLUGS

- A Test Plug: 1/4 inch (6 mm) or 1/2 inch (13 mm) brass fitting and cap for receiving 1/8 inch (3 mm) outside diameter pressure or temperature probe with neoprene core for temperatures up to 200 degrees F (93 degrees C).
- B Test Plug: 1/4 inch (6 mm) or 1/2 inch (13 mm) brass fitting and cap for receiving 1/8 inch (3 mm) outside diameter pressure or temperature probe with Nordel core for temperatures up to 350 degrees F (176 degrees C).
- C Test Plug: 1/4 inch (6 mm) or 1/2 inch (13 mm) brass fitting and cap for receiving 1/8 inch (3 mm) outside diameter pressure or temperature probe with Viton core for temperatures up to 400 degrees F (204 degrees C).
- D Test Kit: Carrying case, internally padded and fitted containing one 2-1/2 inch (60 mm) diameter pressure gages, one gage adapters with 1/8 inch (3 mm) probes, two 1 inch (25 mm) dial thermometers.

2.7 SIGHT FLOW INDICATORS

- A Description: Piping inline-installation device for visual verification of flow.
- B Construction: Bronze or stainless-steel body, with sight glass and paddle wheel indicator, and threaded or flanged ends.
- C Minimum Pressure Rating: 300 psig (2070 kPa).
- D Minimum Temperature Rating: 200 deg F (93 deg C).
- E End Connections for NPS 2 (DN 50) and Smaller: Threaded.
- F End Connections for NPS 2-1/2 (DN 65) and Larger: Flanged.

PART 3 EXECUTION

3.1 INSTALLATION

- A Install thermowells with socket extending one-third of pipe diameter and in vertical position in piping tees.

- B Install thermowells of sizes required to match thermometer connectors. Include bushings if required to match sizes.
- C Install thermowells with extension on insulated piping.
- D Fill thermowells with heat-transfer medium.
- E Install direct-mounted thermometers in thermowells and adjust vertical and tilted positions.
- F Install remote-mounted thermometer bulbs in thermowells and install cases on panels; connect cases with tubing and support tubing to prevent kinks. Use minimum tubing length.
- G Install direct-mounted pressure gages in piping tees with pressure gage located on pipe at the most readable position.
- H Install remote-mounted pressure gages on panel.
- I Install valve and snubber in piping for each pressure gage for fluids.
- J Install test plugs in piping tees.
- K Install thermometers in the following locations:
 - 1 Inlet and outlet of each water heater.
 - 2 Inlets and outlets of each domestic water heat exchanger.
 - 3 Inlet and outlet of each domestic hot-water storage tank.
 - 4 Inlet and outlet of each remote domestic water chiller.
- L Install pressure gages in the following locations:
 - 1 Building water service entrance into building.
 - 2 Inlet and outlet of each pressure-reducing valve.
 - 3 Suction and discharge of each domestic water pump.

3.2 CONNECTIONS

- A Install meters and gages adjacent to machines and equipment to allow service and maintenance of meters, gages, machines, and equipment.

3.3 ADJUSTING

- A Adjust faces of meters and gages to proper angle for best visibility.

3.4 THERMOMETER SCHEDULE

- A Thermometers at inlet and outlet of each domestic water heater shall be stem type.
- B Thermometers at inlets and outlets of each domestic water heat exchanger shall be stem type.
- C Thermometers at inlet and outlet of each domestic hot-water storage tank shall be stem type.
- D Thermometers at inlet and outlet of each remote domestic water chiller shall be stem type.
- E Thermometer stems shall be of length to match thermowell insertion length.

3.5 THERMOMETER SCALE-RANGE SCHEDULE

- A Scale Range for Domestic Cold-Water Piping: 30 to 240 deg F and 0 to plus 115 deg C.
- B Scale Range for Domestic Hot-Water Piping: 0 to 250 deg F and 0 to 150 deg C.
- C Scale Range for Domestic Cooled-Water Piping: 0 to 100 deg F and minus 20 to plus 50 deg

3.6 PRESSURE-GAGE SCHEDULE

- A Pressure gage location:

- 1 At discharge of each water service into building.
- 2 At inlet and outlet of each water pressure-reducing valve.
- 3 At suction and discharge of each domestic water pump.

3.7 PRESSURE-GAGE SCALE-RANGE SCHEDULE

- A Scale Range for Water Service Piping: 0 to 160 psi and 0 to 1100 kPa.
- B Scale Range for Domestic Water Piping: 0 to 300 psi and 0 to 2500 kPa.

END OF SECTION

MECHANICAL SECTION: 22 05 29
HANGERS AND SUPPORTS FOR PLUMBING

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Pipe and equipment hangers, supports, inserts and associated anchors.
- B. Sleeves and seals.
- C. Flashing and sealing equipment and pipe stacks.
- D. Escutcheons.
- E. Equipment bases and supports.

1.02 WORK FURNISHED BUT SPECIFIED UNDER OTHER SECTIONS

- A. Furnish hanger and support inserts and sleeves to Section 03301 for placement into form work.

1.03 Related Sections:

- 1 Division 21 Section 21 10 00 "Water based fire suppression systems".
- 2 Division 21 Section 21 13 13 "Wet pipe sprinkler system".
- 3 Division 22 Section 22 05 16 "Expansion Fittings and Loops for Plumbing Piping" for pipe guides and anchors.
- 4 Division 22 Section "Vibration and Seismic Controls for Plumbing Piping and Equipment" for vibration isolation devices.

1.04 REFERENCES

- A. ANSI B 31.- Power piping.
- B. MSS-SP-69 - Manufacturer's Standardization Society standard on support and hangers.
- C. ASPE Data Book, Volume 4, chapter 2.
- D. NFPA 13 & 14 - Standards for installation of automatic sprinkler and stand pipe systems.
- E. ASHRAE 2000 Handbook - HVAC systems and equipment or latest additions

1.05 QUALITY ASSURANCE

- A. Select and apply pipe hangers and supports complying with MSS-SP-69.
- B. Except as otherwise indicated; provide factory-fabricated hangers and supports.
- C. Specified field fabricated items shall be made from quality structural steel. All holes required shall be drilled and not gas cut. Give two (2) coats of primer and one coat of zinc rich paint prior to installation.
- D. Cutting, welding or any such operation, which damages the finish, is not permitted on galvanised or other factory finished surfaces.
- E. Hangers and supports for fire protection piping shall be FM or equal approved.
- F. All hangers, supports, anchors and guides shall be in compliance with the requirement of superseded version of BS 3974
- G. All Firefighting support shall be UL listed FM & DM approved.
- H. All Plumbing water pipes supports to be Hot dipped galvanized steel.

1.06 SUBMITTALS

- A. Submit shop drawings, product data and samples under provisions of Section 01340.
- B. Indicate hanger and support framing and attachment methods.
- C. Submit selection charts for expansion anchors, relating permissible working loads to types and shapes of concrete elements.

1.07 DELIVERY, STORAGE AND HANDLING

- A. Deliver products to site, store and protect under provisions of Section 01600

PART 2 PRODUCTS

2.1 PIPE HANGERS

- A. Ferrous piping upto 150 mm (6 inch) nom. dia.: Adjustable Hot dipped galvanised steel clevis complete with galvanised steel pipe covering shield for insulated piping.
- B. Firefighting pipes support shall be UL listed FM & DM approved. For pipe, size up to 50-dia Swivel ring support is acceptable and above 50-dia clevis, hanger support shall be provided.
- C. Copper piping (non-insulated): Adjustable Hot dipped galvanised steel band with rigid, preformed, 6 mm (1/4 inch) thick felt, pipe isolator.
- D. Copper piping (insulated): Adjustable Hot dipped galvanised steel band with steel pipe covering shield.
- E. For water supply pipe, support shall be with rubber lined split GI clamp.
- F. For drainage and water supply pipes use rubber lined split clamps
- G. For all main pipes with plant room shall be with spring type vibration isolators

2.03 FLOOR SUPPORTS

- A. Ferrous piping upto 150 mm (6 inch) nom. dia.: Adjustable pipe stanchion saddle support complete with U-bolt, stand pipe with adjuster and floor flange, use galvanised steel pipe covering shield for insulated piping.
- B. Copper piping (non-insulated): Adjustable pipe stanchion saddle support complete with U-Bolt, stand pipe with adjuster and floor flange and rigid preformed felt, 6 mm (1/4 inch) thick, pipe isolator.
- C. Copper piping (insulated): Adjustable pipe stanchion saddle support complete with U-bolt, stand pipe with adjuster and floor flange and galvanised steel pipe covering shield.
- D. Plastic piping: same as that of ferrous piping

2.04 VERTICAL PIPING SUPPORTS

- A. Ferrous piping upto 150 mm (6 inch) nom. dia.: Hot dipped Galvanised steel riser clamps or offset pipe clamp, use solid wood inserts (Ref. by: Elcen no. 216) for insulated piping. As manufactured by: Elcen no. 39 and or 44 or approved equal.
- B. Copper piping (non-insulated) upto 50 mm (2 inch) nom. dia.: Hot dipped Galvanised malleable iron wall clamps with rigid, preformed felt, pipe isolator.
- C. Copper piping (non-insulated) over 50-mm (2-inch) nom. dia.: Hot dipped Galvanised steel offset riser clamp with isolator.
- D. Copper piping (insulated): Hot dipped Galvanised steel offset riser clamps over insulation.
- E. Plastic piping: Hot dipped Galvanised steel riser clamps or offset pipe clamp.

2.05 DISTRIBUTION PIPING IN SHAFTS AND CHASES

- A. Copper piping: Semi circular copper straps with ears for attaching to walls.

2.06 PIPE ANCHORS

- A. Horizontal ferrous piping (non-insulated): Hot dipped Galvanized steel anchor chair and U-bolt attached to structural steel framework.
- B. Horizontal ferrous piping (insulated): Hot dipped Galvanized Steel plate (minimum thickness as that of the pipe and length equal to the diameter) welded to pipe, projecting beyond the insulation; attached to structural steel framing by bolting or welding. Seal the vapor barrier around the plate with mastic sealants.
- C. Vertical ferrous piping (non-insulated): Hot dipped Galvanized Steel hold down anchor clamp and base plate attached to RCC / structural steel supports. Back weld clamp to pipe after installation.
- D. Vertical ferrous piping (insulated): Same as the support of vertical insulated ferrous piping over 150 mm (6 inch) nom. dia.; excluding the vibration isolation pad. Bolt down / weld the bearing plate to the cross channel.
- E. Horizontal copper piping: Hot dipped Galvanized steel anchor chair and U- Bolt over 150 mm (6 inch) long wood inserts all round, attached to structural steel frame work.

As manufactured by: Elcen no. 60 and 216 or approved equal.
- F. Vertical copper piping: Hot dipped Galvanised steel offset riser clamps, over 6 inches long wood inserts all round.
- G. Horizontal plastic piping: Hot dipped Galvanised steel anchor chair and U-bolt attached to structural steel framework.
- H. Vertical plastic piping: Hot dipped Galvanised steel hold down anchor clamp and base plate attached to RCC/structural steel supports.

2.07 HANGERS ROD ATTACHMENTS

- A. Turnbuckles: Provide for adjusting the heights, where required.
- B. Ball hangers: Provide for allowing angular movement of the hanger rod, where required.
- C. Threaded rod couplers: Provide for joining two lengths of threaded hanger rods, where required. Install back nuts on both sides of the coupler.

2.08 BUILDING ATTACHMENTS

- A. Concrete inserts: Provide malleable iron or Hot dipped galvanised steel concrete inserts complete with adjustable insert nut, suitable for hanger rods of size 22 mm (7/8 inch) diameter and under. Provide galvanised steel continuous concrete insert at Contractor's option.
- B. Self drilling type concrete inserts: Provide Hot dipped galvanised steel self drilling expansion type concrete inserts for hanger rods upto 12 mm (1/2 inch) dia., at Contractor's option.
- C. Support rails (cast-in type): Electrozincd steel sections with anchor lugs and filler strip complete with hammer head nut or extension pieces.
- D. Support rails (surface mounted type): Electrozincd steel sections with anchor lugs and filler strip complete with hammer head nut or extension pieces. The rail shall be single or double as appropriate.

2.09 HANGER RODS

- A. Galvanised steel hanger rods: continuous threaded.

2.10 FLASHING

- A. Metal flashing: 0.5-mm (26 gauge) thick galvanised steel.
- B. Lead flashing: 24.5 kg/sq.M (5 lb/sq.ft.) sheet lead for waterproofing; 5 kg/sq.M (1 lb/sq.ft.) sheet lead for soundproofing.
- C. Flexible flashing: 1.25-mm (47 mil) thick sheet compatible with roof and wall waterproofing.
- D. Caps: Hot dipped Galvanised steel, 0.8 mm (22 gauge) minimum; 1.5 mm (16 gauge) at fire resistant elements.

2.11 EQUIPMENT CURBS

- A. Provide factory manufactured sound attenuator curbs to suit the building construction, for all roof exhaust fans and whenever specified.

2.12 SLEEVES

- A. Sleeves for pipes through fire rated and fire resistive floors and walls: Schedule 40 galvanised steel pipe sleeve; pack with 96 kg/cu.M (6 lbs/cu.ft.) density fiberglass and caulk both ends with approved fire rated sealant. (UL listed)

- B. Sleeves for pipes passing through non-fire rated beams, walls, footings and dry floors: Schedule 40 galvanised steel pipe sleeve; pack with 96 kg/cu.M (6 lbs/cu.ft.) density fiberglass and caulk both ends with approved type sealant (UL listed). For vertical piping, extend sleeve 50 mm (2 inch) above the finish floor level.
- C. Sleeves for ferrous pipes passing through external retaining walls: Cast iron or ductile iron with integral puddle; pack with 96 kg/cu.M (6 lbs/cu.ft.) density fiberglass; caulk internal side with approved sealant (UL listed); caulk external side with hemp yarn and lead.
- D. Sleeves for cast iron drain, waste and vent piping through wet floors: Provide PVC pipe sleeve or wooden box to form the opening; remove this sleeve or box after curing of the concrete. After completion of cast iron pipe installation, fill the opening with non-shrinkable grout; trowel both sides of slab, smooth and level.

2.13 ESCUTCHEONS

- A. Provide set screw type escutcheons, size sufficient to completely cover exposed penetrations. Prime coat finish where adjoining surface is to be field painted; Baked enamel white finish where adjoining surface is fiberboard. Chrome finish where adjoining surface is tiled with ceramic, marble, etc.

2.14 FINISH

- A. All hangers supports inserts, shields and saddles shall have galvanised finish.
- B. Hangers and supports for copper piping shall be copper / copper plated, when used in direct contact with the copper piping.

PART 3 EXECUTION

3.1 INSERTS

- A. Provide inserts for suspending hangers from reinforced concrete slabs and sides of reinforced concrete beams.
- B. Provide hooked rod to concrete reinforcement section for inserts carrying pipe over 150 mm (6 inch) nom. dia.
- C. Where concrete slabs form finished ceiling, provide inserts to be flush with slab soffit.
- D. Where expansion type anchors are used, select type and quality to suit the concrete into which it is to be installed.

3.02 PIPE HANGERS AND SUPPORTS (GENERAL)

- A. Provide hanger or support assembly to take up the filled weight of piping for half spans on either side plus 100 kgs, with minimum overall factor of safety of two.
- B. Install hangers to provide minimum 12-mm (1/2-inch) space between finished covering and adjacent work.
- C. Place a hanger within 300-mm (12 inch) of each horizontal elbow.
- D. Use hangers with 38 mm (1 1/2 inch) minimum vertical adjustment.

- E. Where several pipes can be installed in parallel and at same elevation, provide multiple or trapeze hangers.
- F. Support riser piping independently of connected horizontal piping.
- G. Anchor piping at either side of flexible connections (at the system side only on connections to equipment).
- H. No support, hanger or anchor shall be attached to the building structural elements whereby their installation, weight or expansion stress will result in damage to the structure.
- I. Piping buried in raft slabs or trenches shall be supported as detailed on the Contract drawings. The support shall be capable of holding the pipe in position against buoyant forces exerted during pouring of concrete.
- J. Provide fender washers and lock nuts to threaded hanger rods.
- K. Provide amply sized structural steel sections for multiple trapeze hangers, anchor braces, supporting frames etc. Submit calculations where asked for.
- L. Where special slab construction (like omni slab) is used, install support rails anchored to concrete portions for installing hangers.
- M. Do not weld supporting framework to building structural elements, reinforcement bars etc. under any circumstances.
- N. Provide riser clamps at mid span for vertical piping upto and including 25 mm (1 inch) nom. dia.
- O. Support fire protection piping independently of other piping.
- P. All Plumbing pipes supports to be Hot dipped galvanized steel.

3.03 HANGERS AND SUPPORTS FOR CAST IRON PIPING

- A. Support horizontal cast iron piping at either side of joints (at least one on every piece of pipe) at intervals not exceeding 2.50 Ms (8 feet).
- B. Provide supports for fittings grouped together in horizontal runs, as appropriate.
- C. Anchor the piping at horizontal changes in direction, at clean outs and at start of horizontal lines.
- D. Provide floor clamps at every floor and offset riser clamps below the branch take off joint in vertical risers.

3.04 HANGERS AND SUPPORTS FOR STEEL PIPING

- A. Support horizontal piping as scheduled below.

Pipe size	Maximum spacing	Hanger rod dia.
Upto 25 mm nom.dia.	2.0 Ms	8 mm (5/16 inch)
32 to 50 mm nom. dia.	3.0 Ms	10 mm (3/8 inch)
65 and 75 nom. dia.	3.6 Ms	12 mm (1/2 inch)
100 to 150 nom. dia.	4.0 Ms	16 mm (5/8 inch)
200 to 300 mm nom. dia.	4.5 Ms	20 mm (3/4 inch)
350 mm nom. dia. and over	5.0 Ms	22 mm (7/8 inch)

- B. Provide pipe riser clamps / supports at every floor, for vertical piping.

3.05 HANGERS AND SUPPORTS FOR COPPER PIPING

- A. Support horizontal piping as scheduled below.

Pipe size	Maximum spacing	Hanger rod dia.
Upto 25 mm nom. dia.	1.8 Ms	8 mm (5/16 inch)
32 to 50 mm nom. dia.	2.4 Ms	10 mm (3/8 inch)
65 and 75 mm nom. dia.	3.0 Ms	10 mm (3/8 inch)

- B. Provide riser clamps with shield 60 cm (2 feet) /above floor level and at branch take offs for vertical piping.

3.06 SUPPORTS AND HANGERS FOR PLASTIC PIPING

- A. Horizontal piping shall be supported as scheduled below.

Pipe size	Maximum spacing	Hanger rod dia.
Upto 25 mm nom.dia.	1.0 Ms	8 mm (5/16 inch)
32 to 50 mm nom. dia.	1.5 Ms	10 mm (3/8 inch)
65 and 75 nom. dia.	1.8 Ms	10 mm (3/8 inch)
100 to 150 nom. dia.	2.4 Ms	12 mm (1/2 inch)

- B. Risers of all sizes shall be supported at one quarter (1/4) points of floor to floor height. Support to support spacing of 2.0Ms shall not be exceeded.

- C. Manufacturer's recommendations (which will supersede this specification) shall be conformed with in major drainage systems.

3.07 SLEEVES, PUDDLE FLANGES ETC.

- A. Set sleeves in position in form work. Provide reinforcing around sleeves.
- B. Where sleeves in concrete structural elements are omitted due to error, core the required openings only after obtaining written permission of the Engineer. Provide sleeves with puddle flanges resting on the concrete surface, and non-shrink grout to fix the sleeve in position. Cutting of hole by chiselling is not permitted.
- C. Project all sleeves in wet areas by at least 50 mm (2 inch) above the finish floor level.
- D. Erect sleeves, puddle flanges etc. in the formwork, perpendicular to the wall face and rigidly to resist displacement during pouring of concrete.
- E. Ensure verticality of flange faces and alignment of bolt holes with connected equipment (valves etc.).
- F. Seal the openings around, so as to avoid leakage (and resultant honeycombing) and undesirable formation of concrete.
- G. Check alignment immediately after pouring of concrete. Make adjustment where required.
- H. Size sleeves large enough minimum 25 mm (1 inch) around the finished pipe to allow for movement of pipe due to expansion or contraction. Provide for continuous insulation wrapping.

3.08 FLASHING AND SEALING

- A. Pipes passing through external retaining walls: Clamp a layer of waterproofing membrane to the pipe and flash the wall waterproofing membrane over.
- B. Plumbing pipes passing through roof slab: Provide flashing sleeve as Josam 26430 / Smith 1761 or approved equal with integral flashing flange, clamp the roof waterproofing membrane in the flashing flange and fill around with approved type grout.
- C. Roof drains: Co-ordinate construction details, workout actual dimensions for elements like extension filter etc. for cutting the required size at factory.
- D. Area drains and planter drains: Co-ordinate construction details provide lead flashing, 900 mm (3 feet) diameter at the drain and direct the seepage into the drain.
- E. Water Storage tank drains: Provide flashing, 900 mm (3 feet) diameter at the drain, with non-toxic synthetic membrane and integrate the same with the tanking, where provided.
- F. Sanitary fixtures: Seal the joints of all sanitary fixtures with walls and floors with transparent, hard setting, age proof sealant.
- G. HVAC piping and ducting passing through weather or water proofed walls, floors, and roofs: Provide flexible flashing and metal counter flashing as appropriate.

- H. Sound control: Provide acoustical lead flashing around ducts and pipes, penetrating equipment rooms, installed in accordance with manufacturer's instructions for sound control.
- I. Roof curbs: Provide curbs for mechanical roof installations, 300-mm (12-inch) minimum high above roofing surface. Provide flexible sheet flashing and counter flash with galvanised sheet metal to make the assembly watertight.

3.09 EQUIPMENT BASES AND SUPPORTS

- A. See Section 15070 for details.

3.10 ADJUSTING AND CLEANING

- A. Hanger adjustment: Adjust hangers so as to distribute loads equally on attachments.
- B. Support adjustment: Provide grout under supports so as to bring piping and equipment to proper level and adjustments.
- C. Cleaning: Clean factory finished surfaces. Repair any marred or scratched surfaces with manufacturer's touch-up paint, to the approval of the Engineer

END OF SECTION

MECHANICAL SECTION: 22 05 53
IDENTIFICATION FOR PLUMBING

PART 1 – GENERAL

1.1 RELATED DOCUMENTS

- A Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A Section Includes:

- 1 Equipment labels.
- 2 Warning signs and labels.
- 3 Pipe labels.
- 4 Stencils.
- 5 Valve tags.
- 6 Warning tags.

1.3 SUBMITTALS

- A Product Data: For each type of product indicated.
- B Samples: For color, letter style, and graphic representation required for each identification material and device.
- C Equipment Label Schedule: Include a listing of all equipment to be labeled with the proposed content for each label.
- D Valve numbering scheme.
- E Valve Schedules: For each piping system to include in maintenance manuals.

1.4 COORDINATION

- A Coordinate installation of identifying devices with completion of covering and painting of surfaces where devices are to be applied.
- B Coordinate installation of identifying devices with locations of access panels and doors.
- C Install identifying devices before installing acoustical ceilings and similar concealment.

PART 2 PRODUCTS

2.1 EQUIPMENT LABELS

- A Metal Labels for Equipment:

- 1 Material and Thickness: Stainless steel, 0.025-inch (0.64-mm) minimum thickness, and having predrilled or stamped holes for attachment hardware.
- 2 Minimum Label Size: Length and width vary for required label content, but not less than 2-1/2 by 3/4 inch (64 by 19 mm).
- 3 Minimum Letter Size: 1/4 inch (6.4 mm) for name of units if viewing distance is less than 24 inches (600 mm), 1/2 inch (13 mm) for viewing distances up to 72 inches

- (1830 mm), and proportionately larger lettering for greater viewing distances. Include secondary lettering two-thirds to three-fourths the size of principal lettering.
- 4 Fasteners: Stainless-steel rivets or self-tapping screws.
- 5 Adhesive: Contact-type permanent adhesive, compatible with label and with substrate.

B Plastic Labels for Equipment:

- 1 Material and Thickness: Multilayer, multicolor, plastic labels for mechanical engraving, 1/8 inch (3.2 mm) thick, and having predrilled holes for attachment hardware.
- 2 Letter Color: White.
- 3 Background Color: Black.
- 4 Maximum Temperature: Able to withstand temperatures up to 160 deg F (71 deg C).
- 5 Minimum Label Size: Length and width vary for required label content, but not less than 2-1/2 by 3/4 inch (64 by 19 mm).
- 6 Minimum Letter Size: 1/4 inch (6.4 mm) for name of units if viewing distance is less than 24 inches (600 mm), 1/2 inch (13 mm) for viewing distances up to 72 inches (1830 mm), and proportionately larger lettering for greater viewing distances. Include secondary lettering two-thirds to three-fourths the size of principal lettering.
- 7 Fasteners: Stainless-steel rivets.
- 8 Adhesive: Contact-type permanent adhesive, compatible with label and with substrate.

C Label Content: Include equipment's Drawing designation or unique equipment number, drawing numbers where equipment is indicated (plans, details, and schedules), plus the Specification Section number and title where equipment is specified.

D Equipment Label Schedule: For each item of equipment to be labeled, on 8-1/2-by-11-inch (A4) bond paper. Tabulate equipment identification number and identify Drawing numbers where equipment is indicated (plans, details, and schedules), plus the Specification Section number and title where equipment is specified. Equipment schedule shall be included in operation and maintenance data.

2.2 WARNING SIGNS AND LABELS

A Material and Thickness: Multilayer, multicolor, plastic labels for mechanical engraving, 1/8 inch (3.2 mm) thick, and having predrilled holes for attachment hardware.

B Letter Color: White.

C Background Color: Black.

D Maximum Temperature: Able to withstand temperatures up to 160 deg F (71 deg C).

E Minimum Label Size: Length and width vary for required label content, but not less than 2-1/2 by 3/4 inch (64 by 19 mm).

F Minimum Letter Size: 1/4 inch (6.4 mm) for name of units if viewing distance is less than 24 inches (600 mm), 1/2 inch (13 mm) for viewing distances up to 72 inches (1830 mm), and proportionately larger lettering for greater viewing distances. Include secondary lettering two-thirds to three-fourths the size of principal lettering.

G Fasteners: Stainless-steel rivets.

- H Adhesive: Contact-type permanent adhesive, compatible with label and with substrate.
- I Label Content: Include caution and warning information, plus emergency notification instructions.

2.3 PIPE LABELS

- A General Requirements for Manufactured Pipe Labels: Preprinted, color-coded, with lettering indicating service, and showing flow direction.
- B Pretensioned Pipe Labels: Precoiled, semirigid plastic formed to partially cover circumference of pipe and to attach to pipe without fasteners or adhesive.
- C Self-Adhesive Pipe Labels: Printed plastic with contact-type, permanent-adhesive backing.
- D Pipe Label Contents: Include identification of piping service using same designations or abbreviations as used on Drawings, pipe size, and an arrow indicating flow direction.
 - 1 Flow-Direction Arrows: Integral with piping system service lettering to accommodate both directions, or as separate unit on each pipe label to indicate flow direction. Lettering Size: At least 1-1/2 inches (38 mm) high.

2.4 STENCILS

- A Stencils: Prepared with letter sizes according to ASME A13.1 for piping; and minimum letter height of 3/4 inch (19 mm) for access panel and door labels, equipment labels, and similar operational instructions.
 - 1 Stencil Material: Aluminum.
 - 2 Stencil Paint: Exterior, gloss, alkyd enamel black unless otherwise indicated. Paint may be in pressurized spray-can form.
 - 3 Identification Paint: Exterior, alkyd enamel in colors according to ASME A13.1 unless otherwise indicated.

2.5 VALVE TAGS

- A Valve Tags: Stamped or engraved with 1/4-inch (6.4-mm) letters for piping system abbreviation and 1/2-inch (13-mm) numbers.
 - 1 Tag Material: Aluminum, 0.032-inch (0.8-mm) or anodized aluminum, 0.032-inch (0.8-mm) minimum thickness, and having predrilled or stamped holes for attachment hardware.
 - 2 Fasteners: Brass wire-link or beaded chain; or S-hook.
- B Valve Schedules: For each piping system, on 8-1/2-by-11-inch (A4) bond paper. Tabulate valve number, piping system, system abbreviation (as shown on valve tag), location of valve (room or space), normal-operating position (open, closed, or modulating), and variations for identification. Mark valves for emergency shutoff and similar special uses.
 - 1 Valve-tag schedule shall be included in operation and maintenance data.

2.6 WARNING TAGS

- A Warning Tags: Preprinted or partially preprinted, accident-prevention tags, of plasticized card stock with matte finish suitable for writing.

- 1 Size: 3 by 5-1/4 inches (75 by 133 mm) minimum (100 by 178 mm).
- 2 Fasteners: Brass grommet and wire.
- 3 Nomenclature: Large-size primary caption such as "DANGER," "CAUTION," or "DO NOT OPERATE."
- 4 Color: Yellow background with black lettering.

PART 3 EXECUTION

3.1 PREPARATION

- A Clean piping and equipment surfaces of substances that could impair bond of identification devices, including dirt, oil, grease, release agents, and incompatible primers, paints, and encapsulants.

3.2 EQUIPMENT LABEL INSTALLATION

- A Install or permanently fasten labels on each major item of mechanical equipment.

- B Locate equipment labels where accessible and visible.

3.3 PIPE LABEL INSTALLATION

- A Piping Color-Coding: Painting of piping is specified in Division 09 Section.

- B Stenciled Pipe Label Option: Stenciled labels may be provided instead of manufactured pipe labels, at Installer's option. Install stenciled pipe labels complying with ASME A13.1, on each piping system.

- 1 Identification Paint: Use for contrasting background.
- 2 Stencil Paint: Use for pipe marking.

- C Locate pipe labels where piping is exposed or above accessible ceilings in finished spaces; machine rooms; accessible maintenance spaces such as shafts, tunnels, and plenums; and exterior exposed locations as follows:

- 1 Near each valve and control device.
- 2 Near each branch connection, excluding short takeoffs for fixtures and terminal units. Where flow pattern is not obvious, mark each pipe at branch.
- 3 Near penetrations through walls, floors, ceilings, and inaccessible enclosures.
- 4 At access doors, manholes, and similar access points that permit view of concealed piping.
- 5 Near major equipment items and other points of origination and termination.
- 6 Spaced at maximum intervals of 50 feet (15 m) along each run. Reduce intervals to 25 feet (7.6 m) in areas of congested piping and equipment.
- 7 On piping above removable acoustical ceilings. Omit intermediately spaced labels.

D Pipe Label Color Schedule:

- 1 Domestic Water Piping:
 - a Background Color: White.
 - b Letter Color: Black.
- 2 Sanitary Waste and Storm Drainage Piping:
 - a Background Color: Black.
 - b Letter Color: White.

3.4 VALVE-TAG INSTALLATION

A Install tags on valves and control devices in piping systems, except check valves; valves within factory-fabricated equipment units; shutoff valves; faucets; convenience and lawn-watering hose connections; and similar roughing-in connections of end-use fixtures and units. List tagged valves in a valve schedule.

B Valve-Tag Application Schedule: Tag valves according to size, shape, and color scheme and with captions similar to those indicated in the following subparagraphs:

- 1 Valve-Tag Size and Shape:
 - a Cold Water: 1-1/2 inches (38 mm) round.
 - b Hot Water: 1-1/2 inches (38 mm) round.
- 2 Valve-Tag Color:
 - a Cold Water: Green.
 - b Hot Water: Green.
- 3 Letter Color:
 - a Cold Water: White.
 - b Hot Water: White.

3.5 WARNING-TAG INSTALLATION

A Write required message on, and attach warning tags to, equipment and other items where required.

END OF SECTION

MECHANICAL SECTION: 22 05 78

PLUMBING SPECIALITIES

PART 1 – GENERAL

To be read in conjunction with Division 1 – General Requirements.

1.1 WORK INCLUDED

- 1.1.1 Suction strainers.
- 1.1.2 In line Y-type strainers.
- 1.1.3 Back flow preventers.
- 1.1.4 Vacuum Breakers.
- 1.1.5 Automatic relief valves.
- 1.1.6 Automatic pressure reducing valves.
- 1.1.7 Automatic air vents.
- 1.1.8 Hose bibs and hydrants.
- 1.1.9 Curb stops and surface boxes.
- 1.1.10 Floor drains.
- 1.1.11 Area drains.
- 1.1.12 Clean outs.
- 1.1.13 Roof drains.
- 1.1.14 Vent caps.
- 1.1.15 Rain water shoes.
- 1.1.16 Manhole covers and frames.
- 1.1.17 Fixture waste adaptors.
- 1.1.18 Trench and pit gratings.
- 1.1.19 Quick coupling valves.
- 1.1.20 Expansion compensators

1.2 RELATED WORK SPECIFIED IN OTHER SECTIONS

- 1.2.1 Brackets, Hangers, Anchors and Supports
- 1.2.2 Piping and Accessories
- 1.2.3 Valves

1.2.4 Plumbing Pipe work

1.3 CODES AND STANDARDS

- 1.3.1 ANSI A112.14.1 Backwater Valves.
- 1.3.2 ANSI A112.21.1 Floor Drains.
- 1.3.3 ANSI A112.21.2 Roof Drains.
- 1.3.4 ANSI A112.26.1/PDI WH-201 Water Hammer Arresters.
- 1.3.5 ASSE 1001 Pipe Applied Atmospheric Type Vacuum Breakers.
- 1.3.6 ASSE 1010 Water Hammer Arrestors
- 1.3.7 ASSE 1011 Hose Connection Vacuum Breakers.
- 1.3.8 ASSE 1012 Backflow Preventers with Intermediate Atmospheric Vent.
- 1.3.9 ASSE 1013 Reduced Pressure Principle Backflow Preventers.
- 1.3.10 ASSE 1018 Trap Seal Primer Valves.
- 1.3.11 ASSE 1019 Wall Hydrants, Frost Proof Automatic Draining, Anti-Backflow Type.

1.4 SUBMITTALS

- 1.4.1 Submit product data, samples and installation details.
- 1.4.2 Submit selection criteria and ordering schedule, listing types against use and location.

1.5 QUALITY ASSURANCE

- 1.5.1 Vacuum breakers, trap primers, hydrants etc. shall be tested and certified for conformity with the requirements of ASME standards.
- 1.5.2 Water hammer arrestors shall be tested and certified for conformity with ANSI standards.
- 1.5.3 Pressure relief valves and pressure and temperature relief valves shall be tested and certified for conformity with the requirements of ASME standards.
- 1.5.4 The construction of the specialties shall be of corrosion resistant materials like cast iron, stainless steel or dezincification resistant copper alloys (bronze, brass).
- 1.5.5 All castings used in the construction of the drainage specialties shall be of good quality, without blow holes or inclusions and shall be finished smooth by sand or shot blasting. All these castings shall be given appropriate coatings, internally and externally, at the factory.

- 1.5.6 Flashing clamps, where used shall be non-puncturing type and shall be fixed by corrosion resistant fasteners. Drainage flanges shall have seepage holes to conduct away the water collected.
- 1.5.7 As far as practicable items of similar nature shall be products of the same manufacturer.
- 1.5.8 Use heavy duty gratings / covers in areas subjected to vehicular traffic.

PART 2 - PRODUCTS

2.1 APPROVED MANUFACTURERS

- 2.1.1 Refer to approved manufacturers list in Appendix.

2.2 SUCTION STRAINERS

- 2.2.1 For Copper and UPVC Pipes: Fabricate from line size copper pipes (Type K Hard) and bronze flange. Minimum total area of perforations (6mm size) shall be four (4) times the sectional area of the pipe. Use straightened pipe fabric brazed to the pipe end as end cap for copper strainers.
- 2.2.2 All fasteners shall be of corrosion resistant material like stainless steel or high tensile brass.

2.3 IN LINE Y-TYPE STRAINERS

- 2.3.1 50 mm (2 inch) nom. dia. and smaller:
 - 2.3.1.1 Bronze body with threaded ends, screwed on retainer cap with threaded plug and twenty (20) mesh stainless steel screen. Provide blow off connection with valve on the retainer cap, where required.
- 2.3.2 65 mm (2 1/2 inch) and larger:
 - 2.3.2.1 Cast steel body with flanged ends, flanged cover with threaded bronze plug and stainless steel screen. Provide blow off connection with valve, where required.
 - 2.3.2.2 Screen shall have 1.5 mm perforations for sizes from 65 to 100 mm (2 ½ to 4 inch) nom. dia.
 - 2.3.2.3 Screen shall have 3.0 mm perforations for sizes over 100 mm (4 inch) nom. dia.

2.4 BACKFLOW PREVENTORS

- 2.4.1 50 mm (2 inch) nom. dia. and smaller:
 - 2.4.1.1 Bronze body, stainless steel trim with atmospheric vent, complete with strainer, gate valves, test cocks and air gap fitting at vent connection.
- 2.4.2 65 mm (2 1/2 inch) and larger:
 - 2.4.2.1 Bronze body, stainless steel trim with atmospheric vent, complete with flanged strainer and gate valves, test cocks and air gap fitting at vent connection.

2.5 VACUUM BREAKERS

- 2.5.1 Chrome plated brass, atmospheric type with flat pipet type, rubber discs and hose thread outlet.
- 2.5.2 Bronze body, pressure type with heat resistant disc opening to atmosphere at less than 12 mm (1/2 inch) mercury vacuum.

2.6 AUTOMATIC RELIEF VALVES

2.6.1 PRESSURE RELIEF VALVES

- 2.6.1.1 Bronze body, stainless steel spring and trim, with pipet type manual actuation lever. The unit shall be preset and sealed at the factory. The label shall contain the manufacturer's name, serial no., maximum working pressure and set pressure.

2.6.2 PRESSURE AND TEMPERATURE RELIEF VALVES:

- 2.6.2.1 Bronze body, stainless steel spring and trim, one ceramic coated temperature sensing element and pipet type manual actuation lever. The unit shall be rated for relieving the maximum heat input to the water heater and preset and sealed at the factory. The label shall contain the manufacturers name, Serial no., maximum working pressure and temperature, set pressure and temperature and relieving capacity in Btuh.

2.6.3 AUTOMATIC PRESSURE REDUCING VALVES

2.6.3.1 CONSTANT PRESSURE CONTROL VALVES FOR BOOSTER PUMP SETS

- 2.6.3.2 Pilot operated type of globe pattern cast iron construction with bronze trim, for performing the following functions.

- 2.6.3.3 Maintain a constant downstream pressure regardless of fluctuating demand.

- 2.6.3.3.1 Shut off the pump set in the event of very low upstream pressure created by rupture of piping or similar run off conditions.

- 2.6.3.3.2 The valve shall have flanged ends and the assembly shall be complete with size adjustment spools, dielectric flange unions etc.

2.6.4 AUTOMATIC AIR VENTS

- 2.6.4.1 Bronze body with non ferrous float and internal working parts. Construction shall be take apart type for easy maintenance. Outlet shall be tapped for connection to drain line.

2.7 HOSE BIBS AND HYDRANTS

- 2.7.1 Hose bibs: Chrome plated brass, globe type with integral vacuum breaker with 1/2 inch male threaded inlet and 3/4 inch hose thread outlet. The valve operation shall be by loose keys where required. The inlet shall be of male thread with wall flange.
- 2.7.2 Hydrants: Hose bibs as above in cast brass housing suitable for recessed installation. The door of the housing shall be lockable with the same key as the hose bib.

2.8 CURB STOPS AND SURFACE BOXES

- 2.8.1 All bronze ball valve with union or pack joint ends with extended spindle and base. The base shall have a telescopic extension with threaded top for receiving the lid and arch pattern or directly threaded on base. The lid shall have a brass pentagonal operating head

2.9 PRESSURE RATING

- 2.9.1 All specialty items mentioned above shall be rated for 25 BAR (364psig) working water pressure at 121 deg.C (250 deg.F)

2.10 QUICK COUPLING VALVE

- 2.10.1 Red brass construction with ¾"/1 inch female threaded inlet and valved, quick coupling outlet with self closing thermoplastic rubber cover. Furnish coupling adaptor (Key) and swivel elbow with ¾"/1 inch hose threaded outlet with each valve.

2.11 EXPANSION COMPENSATORS

- 2.11.1 Provide expansion compensators for the expansion and contraction of all pipes. Install expansion compensators to have sufficient flexibility to prevent end thrust and movements caused by thermal expansion or contraction causing detrimental distortion or damage of connected equipment. Provide offsets between mains and equipment of sufficient length to safely absorb the expansion of the main. Provide guides as necessary.
- 2.11.2 If expansion compensators in chilled water lines are installed at a time when the ambient air temperature is above 30°C, they shall be pre-compressed in accordance with manufacturer's installation instructions. Expansion compensator shall have sufficient number of corrugations to absorb the expansion between anchors in the pipe plus not less than 25% safety factor for the temperature range between 4°C to 50°C.
- 2.11.3 Provide one expansion compensator between every two rigid supports for all chilled water risers. Provide guides at distances of 4 pipe diameters, 14 pipe diameters and at 3 meter intervals after that, from the expansion compensator.
- 2.11.4 Provide expansion compensator for all pipes crossing building expansion joints.

2.12 FLEXIBLE PIPE CONNECTORS

- a) Provide flexible connectors at all air handling units, pumps, and chillers.
- b) Provide flexible hoses for all fan coil units
- c) Install units as per manufactures printed installation instructions.
- d) Support pipes and equipment such that connector carries no weight

2.13 FLOW SENSING DEVICES

- 2.13.1 Pitot tube flow sensors:
Install where indicated on the drawings and details for flow sensing in hydronic and/or

steam piping systems. Butterfly valves installed at the location of a flow sensing device are to have a memory stop.

2.13.2 Vortex shedding flow sensors:

Install where indicated on the drawings and details for flow sensing in hydronic and/or steam piping systems. Do not install close to elbows, valves, or other piping specialties which might affect the reading of the sensor; follow manufacturer's installation instructions. Butterfly valves installed at the location of a flow sensing device are to have a memory stop.

2.14 WATER METERS

Install water meters on cold water line to each building, on makeup water line to hot water and chilled water systems ahead of water pressure reducing valve and as shown on drawings.

Water meters shall be digital type and shall be linked to BMS for monitoring.

2.15 DRAIN POINTS

- 2.15.1 Adequate provisions shall be made for emptying sections of all services as required for maintenance purposes whether indicated on the drawing or not.
- 2.15.2 Particular attention shall be paid to the requirements of the Civil Defence and the NFPA regulations with respect to the provision of drain taps on fire service pipe work systems.
- 2.15.3 All cylinders, coils and pumps together with all low points of pipe work within plant rooms shall be provided with drain cocks.
- 2.15.4 All other low points shall be provided with drain taps.
- 2.15.5 Drain taps shall also be provided on the flow and return of all sub-circuits provided with isolating facilities and shall be located on the isolated side of the valves
- 2.15.6 Drain cocks shall be in accordance with BS 2879 : 1980
- 2.15.7 Schemes involving external distribution mains below ground level are to be provided with ample sized dirt pockets to which the drain cock shall be provided. Underground mains shall be provided with a 54 mm gland cock with hose union outlet at the lowest position for the purpose of flushing out.
- 2.15.8 The Contractor shall be responsible for the design and positioning of all drain points.

2.16 THERMOMETERS

2.16.1 Stem type:

Install in piping systems as indicated on the drawings and/or details using a separable socket in each location.

2.16.2 Thermometer Sockets

Install at each point where a thermometer or temperature control sensing element is located in a pipeline.

2.17 TEST WELLS

Install in piping systems as indicated on the drawings and/or details wherever provisions are needed for inserting a thermometer at a later date.

2.18 PRESSURE/TEMPERATURE TEST PLUGS

Install in piping systems as indicated on the drawings and/or details. Do not insulate over test plugs.

2.18 PRESSURE GAUGES

Install in locations where indicated on the drawings and/or details, including any gauge piping, with scale range appropriate to the system operating pressures.

2.19 PRESSURE SNUBBERS

Install in gauge piping for all gauges used on water services.

2.20 COIL SYPHONS

Install in gauge piping for all gauges beside vibrating equipment.

2.21 GAUGE VALVES

Install at each gauge location as close to the main as possible and at each location where a gauge tapping is indicated.

2.22 EXPANSION LOOPS

Install where indicated on the drawings or details, locating anchors and guides as detailed.

2.23 Leak Detection System

The water leak detection system wherever is indicated in the project document.

The water leak detection system shall be able to detect water and display location in meters at the leak detection panel. The system shall also communicate with the building management system for remote monitoring.

The leak detection cable shall be round and shall contain four conductors.

The cable materials shall be nonporous, plenum rated and resistant to breakage, cut-through and abrasion.

The cable shall be able to detect water immediately and shall be able to shed water quickly.

The cable shall provide a signal that will allow direct reading of the location where leak is detected.

If water leak is detected at two cable points simultaneously, the indicated reading shall be between the two points. The cable shall also provide a method of sensing continuity so that a cable fault can be detected.

The cable shall be fixed with adhesive clips.

The panel shall provide the following indications as a minimum:

- a) Digital LCD readout of water location
- b) Audible Alarm
- c) Silence/Update push button
- d) System normal LED indicator

- e) Water detected LED
- f) Cable fault LED indicator
- g) LED indication of meters on readout

2.24 WATER HAMMER ARRESTOR

2.24.1 Water hammer arrestors shall be installed in all toilet and kitchen fed from booster pump as indicated on the drawings or wherever is require to be installed for efficient working system.

2.24.2 Water hammer arrestors shall be with piston type bellows contained within casing having sufficient displacement volume to dissipate the calculated kinetic energy generated in the piping system. Both casing and bellows shall be constructed of stabilized 18-8 stainless steel.

Water hammer arrestors shall be selected for following conditions.

- b) Maximum working pressure: 8 Bar.
- c) Maximum temperature: 150 deg. C

PART 3 - EXECUTION

3.1 SUCTION STRAINERS

3.1.1 Install suction strainers on all supply connections from water storage tanks, with bend down elbows in low level connections.

3.1.1.1 Use bronze / fabricated copper suction strainers for copper piping.

3.1.2 IN-LINE Y STRAINERS

3.1.2.1 Use dielectric, brass mating flanges with cast steel strainers used in copper piping.

3.1.2.2 Install in line strainers before pumps, pressure reducing valves, meters and other sensitive equipment, where indicated on the drawings or otherwise.

3.1.3 BACKFLOW PREVENTERS

3.1.3.1 Install at makeup connections to HVAC system and elsewhere as indicated in drawings.

3.1.4 VACUUM BREAKERS

3.1.4.1 Install vacuum relief valves at inlet to all central water heaters (install in line check valves at inlet of wall mounted water heaters).

3.1.5 AUTOMATIC RELIEF VALVES

3.1.5.1 Provide automatic pressure and temperature relief valves on all central electric water heaters.

- 3.1.5.2 Provide pressure relief valves on unit electric water heaters.
- 3.1.5.3 Provide bypass type pressure relief valves between discharge and suction manifolds of pressure controlled booster sets.
- 3.1.5.4 Pipe the discharge of water heater relief valves to safe locations, to prevent injury to personnel.

3.1.6 PRESSURE REDUCING VALVES

- 3.1.6.1 Pressure reducing valves shall be selected for the flow rates rather than the line size (these need not necessarily be of the line size).
- 3.1.6.2 Selection, marked on the cavitations chart, shall be submitted for the approval of the Engineer.

3.1.7 AUTOMATIC AIR VENTS

- 3.1.7.1 Provide automatic air vents on top of all central electric water heaters, as detailed.
- 3.1.7.2 Provide automatic air vents at system high points, as indicated on drawings.

3.1.8 HOSE BIBS AND HYDRANTS

- 3.1.8.1 Provide hose bibs with operating handles or with loose operating keys as appropriate.

3.1.9 DRAINS – GENERALLY

- 3.1.10 Coordinate location of floor drains in tiled areas, so as to attain an acceptable pattern of cutting the tiles. Accommodate square tops in one tile and round tops centrally at tiling junctions.
- 3.1.11 Ensure that the floor drain / clean out tops are installed true to finished grade. Where box outs are provided for installation of floor drains use non-shrink type concrete mix for filling the same.
- 3.1.12 Provide fixture waste adapters of rubber or neoprene to connect tail pipes of sanitary fixtures / drains to the waste piping.
- 3.1.13 Repair damages to shop coats of drains, prior to installation.
- 3.1.14 Keep the drains plugged, by appropriate means, throughout the construction period to prevent closure by cement slurries, paint and similar construction wastes. Remove prior to commissioning.
- 3.1.15 Allow for localized slopes at parking area drains and similar items installed in level finished areas.
- 3.1.16 Protect the surfaces of exposed metallic surfaces from damage by covering with PVC tapes or similar material.

3.2 TESTING OF PIPEWORK AND PIPEWORK SPECIALITIES

- 3.2.1 Refer to relevant Section of TESTING ADJUSTING AND BALANCING

- 3.2.2 Unless specifically indicated otherwise, Contractor shall apply following general procedures.
- 3.2.3 All pipe work shall be tested hydraulically after assembly on site and before any insulation is applied or pipe work made inaccessible by building cladding, to a pressure equal to twice the working pressure. If the working pressure is greater than 600 kPa the test pressure may be one and a half times the working pressure. The minimum test pressure for water pipes shall be 700 kPa.
- 3.2.4 Hydraulic pressure tests shall be carried out by means of a purpose made test pump, which shall be locked after the desired pressure is obtained.
- 3.2.5 The test pressure shall be registered at the lowest part of the system being tested and maintained for a period of two hours during which time all parts must remain completely watertight and no pressure loss is indicated on the test gauge. In the case of Fire Mains wet risers and Hose reel supplies, the pressure test shall be maintained for 24 hours. Any leakage must at once be made good and the pressure test repeated until the desired test conditions are maintained. During this period all welds on steel pipe shall be well hammered. Any defects shall be made good and the relevant section shall be re-tested all at the Contractor's expense.
- 3.2.6 Chillers, Heat exchangers, Water Heaters, Boilers, Cylinders, Pumps and any other ancillary items of equipment which may be liable to damage by the test pressure, shall be isolated or suitably protected during the test, and all necessary precautions shall be taken. Any damaged equipment shall be replaced.
- 3.2.7. In order not to delay the building operations sections of the pipe work may be pressure tested independently as the work progresses. All necessary plugs, caps or blank flanges shall be provided for testing and these shall be removed after the pressure tests have been approved.
- 3.2.8 A test certificate shall be issued for each section of the pipe work satisfactorily.

3.21 ADJUSTMENT AND CLEANING

Valves and Specialities.

All valves and specialities shall be adjusted to operate smoothly and without binding or leaking. All vents shall be tested and proven to open freely for passage of air.

Locate strainers as necessary to allow easy cleaning.

END OF SECTION

MECHANICAL SECTION: 22 05 80

CLEANING OF PIPING SYSTEMS

PART 1 – GENERAL

1.1 WORK INCLUDED

- 1.1.1 Work of this Section includes all labour, materials, equipment and services necessary to complete the cleaning of piping systems as shown on the drawings and specified herein, including, but not limited to, the following:
- 1.1.2 Preliminary cleaning.
- 1.1.3 Cleaning and flushing.
- 1.1.4 Chemical cleanout.
- 1.1.5 Filling of systems.

1.2 RELATED WORKS SPECIFIED IN OTHER SECTIONS

- 1.2.1 Water Treatment

1.3 SUBMITTALS

- 1.3.1 Product Data:
- 1.3.2 Manufacturer's latest published data for materials, equipment and installation indicating chemical compatibility with exposed surfaces.

PART 2 - PRODUCTS

PART 3 - EXECUTION

3.1 GENERAL

- 3.1.1 All stages of flushing, cleaning and filling must be witnessed by the Main Contractor.
- 3.1.2 The Contractor shall not proceed with further stages of cleaning or filling until agreed with the Main Contractor.
- 3.1.3 The Contractor is to provide for freeze protection of filled pipes.
- 3.1.4 The Contractor is to check where effluent can be discharged and make all arrangements for its removal. Pipe cleaning chemicals cannot be discharged to the drainage system and must therefore be disposed off by a specialist company. A certificate of disposal will be required.
- 3.1.5 The Contractor is to allow for the treatment of all mains water to eliminate pseudomona bacteria before use for flushing and filling. Allow for taking regular samples and testing.

3.2 PRELIMINARY CLEANING

- 3.2.1 Clean new piping internally by flushing prior to the application of pressure tests, and before the chemical cleanout procedures specified herein. Provide temporary strainer baskets at the inlet to all pumps before the start of cleaning procedures.
- 3.2.2 Block off and isolate circulating pumps, heating coils, cooling coils, heat exchangers and other equipment during the preliminary flushing and draining process. Provide bypass to allow flushing of equipment separately.
- 3.2.3 Thoroughly flush piping clear of foreign matter with mains water under pressure, and then drain before proceeding with pressure testing. Provide tanks, pumps and required equipment to pressurize during flushing. Blow down accumulations of grit, dirt and sediment at each strainer and each low point in the piping systems. Flush in both directions to ensure removal of all trapped sediment. Drain ports shall be a minimum of 50mm.
- 3.2.4 Provide bypass flush valves and required piping to permit full circulation of water during the washout of the piping systems. Close shutoff and balancing valves on branch piping to the terminal equipment units during the washout operation to prevent water circulation through the automatic control valves. All equipment isolated from general flushing of main piping shall be individually flushed and cleaned.

3.3 CLEANING AND FLUSHING

- 3.3.1 Following the completion of and prior to the handing over of the domestic water systems, disinfect these systems, including water service connections and storage tanks, with chlorine before they are placed in service. The amount of chlorine supplied shall be such as to provide a dosage of not less than 50 parts per million. Following a contact period of not less than 39 hours, the chlorinated water shall be flushed from the system with clean water until the residual chlorine content is not greater than 0.2 parts per million. All valves, after they have been disinfected, shall be opened and closed several times during the eight-hour period. Disinfection shall be repeated until piping meets Main Contractor's and local bacteriological test criteria.
- 3.3.2 All other piping systems shall be flushed free of foreign matter with clean water and left with strainers clean upon completion of the work.
- 3.3.3 Bacteriological tests shall be conducted by an independent licensed testing laboratory at the Contractor's expense. Results of the test shall be furnished to the Main Contractor.

3.4 CHEMICAL CLEANOUT

- 3.4.1 After completion of pressure testing and just prior to filling and balancing, chemically clean internally each circulating water system. All water used for cleaning and flushing must be chemically treated to ensure no pseudomona bacteria is introduced into any system.
- 3.4.2 Provide temporary connections with valves to fill the piping and remaining equipment with water for the purpose of draining piping and equipment after completion of the chemical cleanout procedure. Provide temporary blind flanges and/or caps to isolate the piping and equipment noted herein.
- 3.4.3 Provide temporary piping connections, valves, strainers, bypasses, and blank connections where required to clean out systems. Line each strainer basket with a fine mesh nylon

screen, and replace the screens at the end of each day's circulation until each system is thoroughly cleaned.

- 3.4.4 Temporary pumps shall be capable of circulating full water volume once every 30 minutes, at required system operating pressures. (Permanent pumps shall not be used for this task.)
- 3.4.5 Utilize chemical detergents which will not adversely affect pipe coating, pipe liners, pipe materials, valve seats, plant equipment, etc. Isolate any portion of the system which may be adversely affected by chemical cleanout and ensure no adverse effect on plant and equipment.
- 3.4.6 All recirculating water systems, both open and closed, to be filled and flushed with a solution of a non-foaming chemical detergent, to remove all foreign matter. Circulate the solution for a minimum of 7 hours and drain as rapidly as possible to remove suspended matter. Flush the system with fresh water, drain a second time and refill. After final filling, the pH of the water must not exceed the pH of the fresh incoming water by more than 0.5 pH.
- 3.4.7 Introduce the chemical solution into the system gradually by injecting into the suction side of the circulating pump, or by means of a bypass chemical feeder located on the discharge side of the water system circulating pump.
- 3.4.8 While the water is being circulated, open each drain connection for a short flow. Repeat at hourly intervals. Replace any water drained during blow down with chemical solution as required until air is eliminated from the system. The chemical cleanout procedure to be continuous in this manner for 2 full 7-hour periods.
- 3.4.9 At the conclusion of the chemical cleanout period completely drain the entire system. Flush out with fresh mains water prior to final activation of the system. Remove temporary equipment and strainers, reconnect permanent pump and replace items previously removed.

3.5 FILLING OF WATER SYSTEM

- 3.5.1 After completion of chemical cleanout, fill each water system with fresh water, air vent, and add chemical treatment.
- 3.5.2 Allow for testing and commissioning of chillers prior to hand over of building. Include for re-testing of chillers during peak summer conditions, including attendance of manufacturer's representative, commissioning engineers and all relevant personnel.

3.6 CHLORINE DIOXIDE EQUIPMENT FOR POTABLE WATER SYSTEMS

- 3.6.1 Provide a chlorine dioxide dosing unit for the control of microbiological populations in the water supply.
- 3.6.2 The unit shall comprise of a floor standing polypropylene module complete with integral moulded bunds suitable for 200 litre drums of precursor.
- 3.6.3 All COSHH details are to be clearly displayed on each chemical housing in compliance with the Health and Safety at work act.
- 3.6.4 The unit shall come with a pre piped injection manifold incorporating but not limited to:

- 3.6.4.1 Isolating valves
- 3.6.4.2 Back flow prevention valve suitable for the fluid category risk
- 3.6.4.3 Bypass
- 3.6.4.4 Contacting head water meter to ensure dosing is proportional to the water usage
- 3.6.4.5 Reaction chamber
- 3.6.4.6 20 micron filter incorporating pressure drop sensors and automatic backwash facilities.
- 3.6.5 The unit is to be per assemble with pumps and electronics in separate enclosures.
- 3.6.6 Pulse monitors are to be installed in each chemical injection line to ensure the correct quantity of each chemical is being pumped to the reaction chamber. The monitors will also give an alarm and BMS output if a fault occurs
- 3.6.7 Chemical injection tubes to be double walled.
- 3.6.8 A programmable logic controller with LCD display of data shall be installed in a secure tamper proof enclosure.
- 3.6.9 Low level cut out switches are to be installed in the chemical drums which initiate alarms and BMS output when either chemical requires replacing.
- 3.6.10 Provide test kits for onsite monitoring of the chlorine dioxide and total oxidant level.
- 3.7 WATER TREATMENT CONTROL TESTING EQUIPMENT
 - 3.7.1 Provide test equipment described hereinbefore and as follows:
 - 3.7.1.1 Supply a one-year's quantity of testing chemicals to properly analyze the treated water for pH, molybdate chlorine nitrite, phosphate and sulfite. Furnish the necessary test kits for these tests complete with instruments, reagents, materials and supplies.
 - 3.7.1.2 Provide a Myron L "DS" meter, 5 range, 0-0.5, 0-5, 0-50, 0-500, 0-5000, MICROMHOS/CM auto-temp compensation 50°-160°F, 9-volt transistor batteries.
 - 3.7.1.3 Furnish a supply of log sheets on which to record the test results and bound copy of full test instructions.
 - 3.7.1.4 Provide and install a corrosion coupon test rack as directed with four (4) insert positions for steel and copper coupons furnished by the water treatment company, and to be inspected every 90 days. Test rack to comply with ASTM D-2688 Method B.

END OF SECTION

MECHANICAL SECTION: 22 09 13

VALVES FOR PLUMBING

PART 1 – GENERAL

1.1 WORK INCLUDED

- 1.1.1 Work of this Section includes all labour, materials, equipment and services necessary to complete the valves as shown on the drawings and specified herein.

1.2 RELATED WORK SPECIFIED IN OTHER SECTIONS

- 1.2.1 Piping and Accessories
- 1.2.2 Testing, Inspections and Commissioning

1.3 CODES AND STANDARDS

- 1.3.1 API 600 – general design specifications for gate, globe and check valves.
- 1.3.2 ANSI B16.34 – pressure temperature ratings for valves.
- 1.3.3 ANSI B16.10 – face to face dimensions of cast iron gate, globe and check valves.
- 1.3.4 ANSI B16.24 – end flange dimensions of bronze valves.
- 1.3.5 ANSI B16.1 – end flange dimensions of cast iron valves.

1.4 SUBMITTALS

- 1.4.1 Shop Drawings:
- 1.4.2 Valves, regulating valves.
- 1.4.3 Product Data:
- 1.4.4 Manufacturer's latest published data for materials, equipment and installation.

1.5 QUALITY CONTROL AND ASSURANCE

In accordance with the Conditions of Contract, and as specified herein.

1. All gate, globe and angle valves shall be designed for repacking under pressure when fully opened and shall be equipped with packing suitable for intended service. When valve is fully opened, back seat shall protect the packing and stem threads from fluid.
2. Bronze gate, globe, angle and check valves shall conform to BS 5154.
3. Pressure containing parts of valves shall be of material conforming to BS 5150 PN16 & PN25.
4. Gate valves shall conform to BS 5150.

5. Valve stems shall be copper silicon alloy conforming to BS 2874.
6. Wheel handles shall be non-heating style cast from malleable iron or commercial aluminium.
7. Check valves shall operate equally well in horizontal or vertical positions.

Balancing valves shall conform to the requirements of BS 7350.

8. All valves shall be full line size.
9. All valves on potable water supply shall be either bronze construction or iron bodies with fusion bonded epoxy coating on all parts in contact with water and shall carry the approval of WRAS certificate.

PART 2 - PRODUCTS

2.1 GENERAL

- 2.1.1 Valves and valve construction to be suitable for the pressure, temperature, and water quality of the service.

2.2 APPROVED MANUFACTURERS

Refer to approved manufacturers list in Appendix.

2.3 MANUAL VALVES

- 2.3.1 Construct manual air vents from short vertical sections of 2-inch (50 mm) diameter pipe to form air chamber. Provide 1/8 inch (3 mm) brass needle valve at top of chamber. Provide at system high points, and where indicated on the drawings.

2.4 AUTOMATIC AIR VENTS

- 2.4.1 Float type brass or semi-steel body, copper float, stainless steel valve, and valve seat. Suitable for system operating temperature and pressure.

Provide at system high points not readily accessible for servicing.

2.5 AIR SEPARATOR

- 2.5.1 Air separator shall be constructed of steel designed and fabricated per ASME Section VIII Division I with a maximum working pressure rating of 364 PSIG at 375 F units up to 2 1/2" connection sizes shall have threaded connections and units 3" and above connection sizes shall have flanged connections. Units shall have separate top connections for system expansion tank and air venting, and bottom connection for blow down cleaning. Air separator shall be furnished with a removal stainless steel strainer with a free area not less than five times the cross-sectional area of the connections pipe.

2.6 EXPANSION TANKS

- 2.6.1 Bladder type tanks shall be designed and constructed per ASME section VIII, Division I and rated for a maximum working pressure of 250 PSI at 240 F. Units shall be furnished with lifting rings, and have the system connection and charging valve at the tank top and a tank

drain connection at the bottom. Bladder type expansion tanks shall have an internal replaceable elastomer bladder with maximum acceptance volume and factory precharge pressure as shown on the plans. The bladder shall be suitable for a maximum system operating temperature of 240. Bladder type tanks shall be furnished with a steel base ring for vertical mounting. Bladder type CA tanks shall be furnished with galvanised cover for corrosion resistance. Bladder type tanks shall be furnished with a factory installed replaceable California code "Bulls Eye" sight glass. All bladder type tanks shall be furnished with seismic mounting clips. All materials of construction shall be compatible with water and ethylene and propylene glycol mixtures. Automatic Cold Water Fill Assembly: Pressure reducing valve, for installation of accessories.

2.7 GATE VALVES

- 2.7.1 2 inches (50mm) and under: brass body (Dezincification resistant copper alloy (CW602N)), rising or non-rising stem and union bonnet solid wedge or disc, threaded ends, manufactured to comply with MSSSP-80 standards.
- 2.7.2 2 1/2" inches (65mm) and over: Iron body, bronze mounted (IBBM), OS&Y, solid wedge, renewable seats, flanged ends, manufactured to comply with MSS-SP-70 standards.

2.8 GLOBE VALVES

- 2.8.1 2 inches (50mm) and under: brass body (Dezincification resistant copper alloy (CW602N)), rising or non-rising stem; union bonnet, renewable composition disc, screwed ends, manufactured to comply with MSS-SP-80 standards. 2 1/2 inches (65mm) and over: Iron body, bronze mounted (IBBM), rising or non rising stem, OS&Y, renewable seat flanged ends, manufactured to comply with MSS-SP-85 standards.

2.9 BALL VALVES

- 2.9.1 2 inches (50mm) and under: brass body (Dezincification resistant copper alloy (CW602N)), stainless steel ball, teflon seats and stuffing box ring, lever handle threaded ends, manufactured to comply with MSSSP- 110 standards. 2 1/2 inches (65mm) and over: Stainless steel, full bore, floating ball, teflon seat and stuffing box seals, lever or key operated handle flanged ends, manufactured to comply with MSS-SP-72 and 25 standards.

2.10 CHECK VALVES

- 2.10.1 2 inches (50mm) and under: brass body (Dezincification resistant copper alloy (CW602N)), resilient seat and disc, spring loaded, screwed ends. 2 1/2 inches (65mm) and over: Iron body, bronze trim, spring loaded, renewable composition disc and seat flanged ends, manufactured to ANSI B16.1 and MILV-18436 F standards.

2.11 DRAIN VALVES

- 2.11.1 Brass angle globe pattern, key operated, threaded inlet with nipple and cap for hose connection, manufactured to comply with MSS-SP-80 standards.

2.12 REGULATING (BALANCING) VALVES

- 2.12.1 Use for throttling for water services: 50 mm and smaller variable orifice double regulating brass valve with inside rising stem, screwed bonnet, hand wheel operated with micrometer

style indicator and pressure test points. 65 mm to 150 mm larger cast iron variable orifice double regulating valve with rising stem, outside screw and yoke and bolted bonnet. All valves to be back seating with adjustable packing nut and be capable for repacking under pressure. Moulded Teflon impregnated mineral wool packing. Provide valves for throttling service with renewable metallic seats of nickel alloy or stainless steel.

2.13 PRESSURE REDUCING VALVES

2.13.1 35 mm and Smaller:

2.13.2 Valve shall maintain a constant downstream pressure regardless of varying inlet pressures and/or changing flow rates.

2.13.3 Valve shall be capable of reducing a maximum inlet water pressure of 18 bars to a lower desired pressure.

2.13.4 The valve shall close drop-tight when the downstream pressure rises above the spring setting.

2.13.5 Valve shall be of the balanced single seat design and shall contain an integral strainer constructed of chrome-nickel stainless steel.

2.13.6 All necessary repairs shall be possible without removing the valve from the line.

2.13.7 The pressure reducing unit shall be replaceable without the requirement of a new pressure adjustment.

2.13.8 The diaphragm assembly shall be fully guided at both the top and the bottom.

2.13.9 All trim shall be chrome-nickel stainless steel throughout.

2.13.10 Valve shall be constructed with union ends at the inlet and the outlet.

2.13.11 It shall be possible to check the pressure on the downstream side of the valve from threaded tapping located on either side of the valve with an installed pressure gauge.

2.13.12 The body and the cover shall be constructed of heavy-duty brass.

2.13.13 50 mm and larger:

2.13.14 This valve shall maintain a constant downstream pressure regardless of fluctuations in demand, or varying inlet pressures.

2.13.15 Valve shall be a hydraulically operated pilot controlled valve.

2.13.16 No external packing glands shall be permitted, and there shall be no pistons operating the main valve or any pilot controls.

2.13.17 The pilot control shall be a direct-acting, adjustable, spring-loaded, normally open diaphragm valve, designed to permit flow when controlled pressure is less than the spring setting.

2.13.18 Valve shall be all bronze with stainless steel trim.

ENTIRE INSTALLATION SHALL BE ASSEMBLED AS SHOWN ON DRAWINGS.

2.14 NON RETURN VALVES

- 2.14.1 Use for water and low pressure general services: 50 mm and smaller with screwed bonnet. 65 mm and larger with bolted bonnet. Valves to have renewable bronze seat and disc.

2.15 BUTTERFLY VALVE

- 2.15.1 Use butterfly valves for shutoff purpose for all sizes from 6 inch and above in place of gate valves. Use gate valves only up to and including 4 inch. Valves shall be raised face class 250 flanges. Body to be ductile iron and manufactured in accordance with MSS-SP-67. Body to have at least 2-inch extended neck for insulating. Valves to have EPDM rubber encapsulated disc with polymer-coated body. Stem shall be 400 series stainless steel and shall not have exposed stem to disc fasteners. Valves shall be capable for use as isolation valves for Bi-directional dead-end service at full pressure. Sizes up to 6 inch shall be lever operated. Sizes 8 inch and above shall be gear operated.

- 2.15.2 All butterfly valves shall be of the lug body style with lugs drilled and tapped and have drip tight shutoff capabilities in either direction up to and including maximum system working pressure. Butterfly valves shall be capable of closing tight after long periods of inactivity. All valve bodies 600 mm and above to be dual flanged.

- 2.15.2.1 Lug body valves shall provide tight shutoff up to the full valve rating on dead end or isolation service without the use of downstream flanges.

- 2.15.2.2 All valves shall be furnished with self-lubricated bronze bearings. Shaft seals shall be provided to prevent leakage and to protect bearings from internal or external corrosion.

- 2.15.2.3 Seats shall be of the reinforced resilient type (or retained seat on high performance valves) and shall also act as a body liner to prevent flow from contacting the body casting. Resilient seats shall have flange sealing lips to provide a positive seal without use of flanges gaskets.

- 2.15.2.4 Seats shall be Nordel suitable for use with HVAC water to 120°C. Shafts shall be one piece and shall be of 416 stainless steel. Shafts shall be finish ground and polished to minimize bearing and shaft seal wear. Shafts of 200 mm and larger valves shall have a non-adjustable thrust collar.

- 2.15.2.5 Discs shall be semi-steel with welded nickel edge. The disc-to-shaft connections shall be type 316 stainless steel. Pins, shaft and disc of all valves shall be individually machined and completely interchangeable.

- 2.15.2.6 On manually operated valves, latch lock levers shall provide automatic, positive latching in the open, closed or eight intermediate positions. These levers shall allow locking in any position with a standard padlock. Infinite position levers shall allow manual throttling and locking in any position from open to close.

- 2.15.2.7 All manually actuated valves 200 mm and larger shall be operated using a cast iron housed hand wheel actuator with chain wheel. All units to have adjustable open and closed position stops with provision to prevent accidental adjustment changes. Operating shafts to be supported axially and radially at input end by permanently lubricated bronze thrust and sleeve bearings.

2.16 PRESSURE RATINGS

- 2.16.1 **All valves shall be rated for 364 P.S.I.G (25 BAR) working pressure (class 250) at 250-degree F (121-degree C). Contractor shall allow for selecting higher pressure rated valves to meet the valve construction features.**

2.17 RELIEF VALVES

- 2.17.1 ASME rated direct spring-loaded type, lever operated, non-adjustable factory set discharge pressure. Provide relief valves on pressure tanks, low-pressure side of reducing valves, heating converters, and compression tanks.

Select system relief valves capacity so that it is greater than make-up pressure reducing valve capacity. Select equipment relief valve capacity to exceed rating of connected equipment.

2.18 WAFER STYLE VALVES (100MM THROUGH TO 800MM)

- 2.18.1 Wafer style valves (100mm through 800mm): Class 150 or 300 Wafer valves shall consist of grey iron (ASTM A-126-61 T, Class 30), or ductile iron (ASTM, A536-65T, class 60-45-18) body and single or multiple, parallel-installed stainless steel cartridge assemblies; shall be rated at 1380 Kpa/120 C, (Class 150) or 3450 kPa 200 C, (Class 300); class 150 shall be mechanically compatible with ANSI B/16.1-1967 125 lb. cast iron flanges, or ANSI B16.5-1968 or MSS-SP-44 150 lb. steel flanges; class 300 shall be mechanically compatible with ANSI B16.1 1967 250 lb cast iron, ANSI B16.5-1968 or MSS-SP-44 300 lb. steel flanges; shall be supplied with dual pressure and temperature test valves.

- 2.18.2 All flow control valves shall be supplied by a single source responsibility.

- 2.18.3 Each automatic flow control valve shall be furnished with a valve kit consisting of 1/4-inch x 2-inch minimum size nipples, quick disconnect valves (to be located outside of insulation), and fittings suitable for use with the measuring instruments specified.

- 2.18.4 Provide a metal identification tag, with chain, for each installed valve. The tag to be metal stamped with zone identification, valve model number, rated flow in GPM and differential pressure range.

- 2.18.5 Flow measuring instrument (minimum two) to verify flow rates shall be furnished.

- 2.18.6 Correct flow shall be verified by establishing that the operating pressure differential across the valve tapings is within the range indicated on the identification tag.

- 2.18.7 The pressure measuring apparatus shall be portable and consist of a pressure gauge with a 4 1/2" diameter dial, instructions, hoses and connections; and push buttons 3-way valve which transmits either of two pressures to a pressure gauge and a carrying case.

2.20 WATER STRAINERS

- 2.20.1 Provide Y-Type pipe line strainers with removable stainless steel basket 1200 micron size as shown on drawings.

- 2.20.2 Strainers shall be of the same size as the pipe to which it is connected and end material will match with the material of connecting pipe.

- 2.20.3 Strainers shall be suitable for 364 psig working pressure with type 304 stainless steel screen.

2.21 WATER HAMMER ARRESTOR

2.21.1 Water hammer arrestors shall be installed in all toilet and kitchen fed from booster pump as indicated on the drawings or wherever is require to be installed for efficient working system.

2.21.2 Water hammer arrestors shall be with piston type bellows contained within casing having sufficient displacement volume to dissipate the calculated kinetic energy generated in the piping system. Both casing and bellows shall be constructed of stabilized 18-8 stainless steel.

Water hammer arrestors shall be selected for following conditions.

b) Maximum working pressure: 8 Bar.

c) Maximum temperature: 150 deg. C

2.22 PUDDLE FLANGES

2.22.1 Where pipework passes through the external walls of the buildings or trenches below ground level, the Contractor shall supply and cast or built puddle flanges into the structure.

2.22.2 Puddle flanges are to be manufactured from the same material as the pipework of which they form a part.

2.22.3 Each puddle flange shall comprise a length of pipe, flanged or screwed at end according to diameter with an undrilled slip on flange welded on the outside at a point where it will be located mid-way in the thickness of the wall. The puddle flange is to be painted externally with two coats of bituminous paint before being built into the structure.

2.23 FLOW SENSING DEVICES

2.23.1 Contractor shall provide all flow sensing devices as indicated on the drawings.

Proper schedule indicating the flow rate that is to be sensed, the type of flow sensing device, and the permanent pressure loss at the design flow rate shall be submitted for approval of the Consultant.

The flow sensors shall be one of following types:

2.24 PITOT TUBE FLOW SENSORS:

Multi-port averaging type flow sensor designed to

Sense the velocity of a fluid flowing in a pipe and produce a pressure output that is proportional to the fluid velocity VORTEX SHEDDING FLOW SENSORS: Wafer type, unit with an analog output.

2.25 WATER METERS

The contractor shall install the intelligent metering system arrangement as required by DEWA. The meters shall be supplied by DEWA and installed by contractor.

2.26 FLOAT VALVES

2.26.1 Float valves shall be installed as indicated in the drawings to provide consistent level control in reserve supply water storage tanks. The valve shall meet the requirements of the Water Byelaws for air gaps and shall be constructed throughout in approved materials and shall prevent back siphoning. Inlet flow shall be co-axial with the piston movement, to ensure high discharge capacity and smooth, quiet operation. The valve shall have 'O' ring piston seals, resilient seated disk, Dezincification resistant bronze fabricated approved solder and shall be hydraulically tested to ensure buoyancy and constructed in accordance with BS1968.

2.26.2 Bronze equilibrium float valves 80 and above shall be flanged end, flat faced and drilled to suit BS4504 PN16.

2.26.3 Brass equilibrium float valves up to 50 shall be screwed end BS2779 parallel and shall be provided complete with back nut.

2.26.4 Floats for valve sizes 80mm and above shall be of copper.

PART 3 - EXECUTION

3.1 INSTALLATION

3.1.1 Install valves so that they are accessible for repacking.

3.1.2 Install with stem vertical where possible, never with stem below horizontal position.

3.1.3 Install with operating clearance for handle and stem.

3.1.4 Install isolation valves on equipment so that valve and piping do not interfere with equipment removal or maintenance. Install unions or flanges on equipment side of valves unless valve is flanged type.

3.1.5 Provide valves of a design permitting packing while open and under pressure.

3.1.5.1 Provide an isolation valve in the common supply line and a double regulating valve or globe valve as shown on drawings in the return line from each water coil, water-to-water interchanger, and all heat transfer elements. For isolation service use ball type valve for pipe sizes 65 mm and smaller, and butterfly type on larger sizes.

3.1.5.2 Provide a gate valve in each supply riser near main and a lubricated tapered plug valve in each return riser near main.

3.1.5.3 Provide capped drain cocks with threaded ends for hose connections at all drain points.

- 3.1.5.4 Provide all valves 200 mm and larger having a rating of over 10 bar with a 25 mm bypass valve of same pressure rating as the bypassed valve.
- 3.1.5.5 Provide all required manual or automatic vent valves at all high points of piping systems to facilitate venting of air and to ensure quiet operation. Pipe vent valves to nearest gully.
- 3.1.5.6 Provide renewable bronze seat rings and bronze spindles for all cast iron body valves.
- 3.1.5.7 Use double regulating or lubricated tapered valves or plug cocks for all throttling service and where noted on the drawings.
- 3.1.5.8 Install valves of the lubricated tapered plug type for pump discharge application.
- 3.1.5.9 Provide lubricated tapered plug cocks with the manufacturer's proper lubricant for water service before shipment to the job site. Furnish 4 hand wrenches for each size valve, where gear operators are not required.
- 3.1.6 Provide chain operated sheaves and chains for all valves which are more than 2 m above the floor in Pump or Equipment Rooms or in Fan Rooms.
- 3.1.7 Furnish and install all other hand valves, non-return valves, cocks, and other equipment as required for the complete and proper valving of the entire installation.
- 3.1.8 Furnish and install tamper switches for all valves used in public health and fire protection systems.

END OF SECTION

MECHANICAL SECTION: 22 10 00

PLUMBING PIPEWORK

PART 1 – GENERAL

1.1 WORK INCLUDED

- 1.1.1 Pipes and pipe fittings.
- 1.1.2 Domestic cold and hot water piping.
- 1.1.3 Sanitary Soil, waste and vent piping.
- 1.1.4 Storm water piping.
- 1.1.5 Kitchen/Laundry waste piping.
- 1.1.6 AC condensate drain piping.

1.2 RELATED WORK SPECIFIED IN OTHER SECTIONS

- 1.2.1 Brackets, Hangers, Anchors and Supports
- 1.2.2 Piping and Accessories
- 1.2.3 Valves
- 1.2.4 Plumbing Specialties
- 1.2.5 Health care/ Lab Gas piping & Equipment

1.3 CODES AND STANDARDS

- 1.3.1 DIN 19535 - Pipes and fittings of high density PE for hot water resistant waste and soil installation within buildings.
- 1.3.2 DIN 19537 - Pipes and fittings of high density PE for drainage and sewerage.
- 1.3.3 DIN EN 1451 & DIN 19560 – Polypropylene pipes and fittings, acid resistant.
- 1.3.4 DIN 8074 - HDPE soil waste and vent pipe and fittings; dimensions.
- 1.3.5 DIN 8075 - High density polyethylene (HDPE) pipes, general quality requirements and testing.
- 1.3.6 ASTM D635 - Polypropylene pipe and fittings.
- 1.3.7 ASTM D3311 - Polypropylene heat fusion fittings.
- 1.3.8 AWWA C 601 - Standard methods for the examination of water and waste water.
- 1.3.9 ASTM A 120 - Pipe, steel, black and hot dipped zinc coated (galvanized), welded and seamless, dimensions.
- 1.3.10 ASTM A 106 - Wrought steel pipes for corrosion resistant service.

- 1.3.10 ANSI B 16.3 - Malleable iron threaded fittings class 150 and 300.
- 1.3.11 ANSI B 2.1 - Pipe threads.
- 1.3.12 ANSI B 16.1 - Cast iron flanges.
- 1.3.13 BS 4375 - PTFE tape for pipe thread sealing application.
- 1.3.15 DIN 16962 - Pipe Joint assemblies and fittings for polypropylene pressure pipes.
- 1.3.19 ASTM B 32 - Solder metal.
- 1.3.20 ASTM D 1785 - PVC pressure pipe, schedules 40, 80, and 120.
- 1.3.21 ASTM D 2466 - Socket type PVC fittings, schedule 40.
- 1.3.22 ASTM D 2467 - Socket type PVC fittings, schedule 80.
- 1.3.23 ASTM D 2564 - Solvent cement for PVC pipes and fittings.
- 1.3.24 ASTM D 2855 - Making solvent cemented joints with PVC pipes and fittings.
- 1.3.25 ANSI B 16.21- Non-metallic gaskets for pipe flanges.
- 1.3.26 ASME Sec. 9 - Welding and brazing qualifications.
- 1.3.27 AWS A 5.8 - Brazing filler metal.
- 1.3.28 AWS D 10.9 - Standard for building services piping.
- 1.3.29 DIN 16833 – Multilayer composite pipes.

1.4 QUALITY ASSURANCE

- 1.4.1 All valves and similar control gear shall have the manufacturer's name or logo, size, pressure rating and direction of flow cast-on, on the body.
- 1.4.2 Welding materials and procedures: Conform to AWS codes.
- 1.4.3 Employ welders certified in accordance with ASME Section 9.
- 1.4.4 Equivalent British (BS), German (DIN) or other international standards are acceptable, provided that the Contractor substantiates their equivalence and ensures their compatibility with other components of the system.
- 1.4.5 For each product specified, provide from same manufacturer throughout, where possible.
- 1.4.6 Test certificates from approved independent laboratories or testing agencies shall be furnished at no extra cost if required by the Engineer.
- 1.4.7 Maintain uniformity in respect of connection standards (threads or flanges) throughout. Where this is not practical due to justifiable reasons, ensure compatibility, by using appropriate adaptors, conversion spools, mating flanges and the like, with the written approval of the Engineer.

1.5 SUBMITTALS

- 1.5.1 Submit product data, samples and manufacturer's installation instructions or recommendations.
- 1.5.2 Include data on pipe materials, pipe fittings, valve and accessories and recommended coating materials.
- 1.5.3 Submit copies of valve ordering schedules for approval prior to ordering valves.

1.6 DELIVERY, STORAGE AND HANDLING

- 1.6.1 Deliver products to site, store and protect.
- 1.6.2 Store plastic pipes on elevated racks only.
- 1.6.3 Store plastic and rubber products out of direct sunlight.
- 1.6.4 Keep ends of pipes closed with factory manufactured plugs to prevent entry of foreign matter.
- 1.6.5 Apply two coats of approved primer to all untreated ferrous pipes prior to storage.
- 1.6.6 Repaint ductile iron and cast iron pipes and fittings with factory recommended paint, prior to installation; where factory coating has been damaged.
- 1.6.7 Deliver and store valves in shipping containers with labeling in place.
- 1.6.8 Store plug and ball valves in open position and gate / globe valves in closed position to prevent damage to valve seats. Retain flange protectors on the valves, until installation to prevent damage to the flange faces.

PART 2 - PRODUCTS

2.1 APPROVED MANUFACTURERS

- 2.1.1 Refer to approved manufacturers list in Appendix.

2.2 Domestic cold water below and above grade:

- 2.2.1 All domestic water pipes and fittings must be certified by DWRA & WVDG.
- 2.2.2 Use PPR, polypropylene (Random polymer), and as specified by DIN 8078 & 1354991 having density 0.909 / cm² / PPR pipes are joined using fusion welding.
- 2.2.3 Use PPR PN-16 for all Horizontal Pipe and PPR PN-25 for all Vertical pipes (pipe risers)
- 2.2.4 Domestic water Transfer pipe line and fittings from UG water tank to roof tanks shall be PPR PN-25.
- 2.2.5 Use uPVC pipes and pipe fittings to BS 3505 Class E solvent with solvent welded fitting to BS 4346 for below grade installation

- 2.2.6 Cold water pipes embedded in builder's work shall be PEX tubes within polyethylene sleeve as specified in clause 2.8 below
- 2.2.7 The incoming cold water pipe to water tank shall be in uPVC pressure pipe class E to BS EN standards
- 2.2.8 Cooled water supply pipes on parking and inside plant rooms shall be insulated and aluminum clad
- 2.2.9 Cooled water supply pipes on the roof shall be insulated and aluminum foil clad.

2.3 Domestic hot water

- 2.3.1 Reinforced fiberglass PPR, polypropylene (Random polymer), SDR-6, and as specified by DIN 8077 and 8078 pipes for Hot water pipes.
- 2.3.2 Hot water pipes embedded in builder's work shall be PEX tubes within polyethylene sleeve as specified in clause 2.8 below
- 2.3.3 Use PPR PN-16 for all Horizontal Pipe and PPR PN-25 for all Vertical pipe

2.4 SANITARY DRAINAGE, RAIN WATER AND VENT PIPING INSTALLED BELOW GRADE

- 2.4.1 Sanitary drainage, rainwater and vent piping installed Above grade:
 - A. Pipes 32mm to 50mm diameter : uPVC pipe with Rubber ring push fit joints as per BS EN 1329 (BS 5255)
 - B. Pipes 82mm to 160mm diameter: uPVC pipe with Rubber ring push fit joints as per BS EN 1329 (BS 4514).
- 2.4.2 All above grade drainage pipes running above occupied spaces (flats , corridors , retails , amenities ,...)shall be sound proof pipes and fittings as specified in clause 2.5 below.
- 2.4.3 For the shafts and car parking areas, above grade drainage pipes shall be UPVC pipe.
- 2.4.4 Sanitary drainage, rain water and vent piping installed below grade:
 - A. Drainage pipe work below grade within the building and upto manhole and between manholes shall be uPVC to BS EN 1401 (BS 4660).
- 2.4.5 Drain pipework installed below buildings and roads shall be provided with 15cm concrete encasement.
- 2.4.6 Drainage forced main pipe work shall be uPVC to BS 3505 Class E.

2.5 SOUND PROOF DRAINAGE PIPES

- 2.5.1 All above ground soil, waste and rainwater pipes in the building which is running inside the occupied areas shall be soundproof drainage pipes and fittings.
- 2.5.2 The pipes should consist of three layers (polypropylene-mineral additives-polypropylene) with a sound insulation layer, density should not be less than 1.9gr/cm³ for soil and waste, or one thick layer pipe made of mineral-reinforced polypropylene.
- 2.5.3 The sound isolation shall be 20 DB for 4.0 l/s flow rate in accordance with DIN EN 14366 and DIN 4109.

- 2.5.4 The sound proof pipes & fittings shall be fire resistant class B2 to DIN 4102.
- 2.5.5 The sound proof pipes & fittings shall be fungus and bacteria resistant, and shall have smooth surface, corrosion resistant.
- 2.5.6 Soundproof drainage pipe system shall comply with following standards:
- DIN 4109m sound proof, absorption standards.
 - DIN 4102, B2. Self-extinguishing flameless.
 - DIN 19560 / DIN EN 1451. Hot water resistance, 90°C (Long term), 90°C (short term).
 - Physical characteristics
- | | |
|---------------------|--|
| Density | - 1.9 g/cm ³ DIN 53479 |
| Elongation at break | - 50% |
| Tensile strength | - 20N / mm ² |
| E-modulus | - 3800 N / mm ² for 1 Layer and 1500 N/ mm ² for 3 Layer |
| Linear expansion | - 0.09 mm / mk |
- 2.5.7 The sound proof pipes and fittings shall be self extinguishing flameless for the fire behavior and complies with DIN 4102, B2.
- 2.5.8 The floor trap shall also be of sound proof material to DIN 4109.
- 2.6 CONDENSATE DRAIN PIPING
- 2.6.1 uPVC Class E to BS EN 1452-2 (BS 3505) with solvent welded fittings to BS4346 for installation not exposed to view.
- 2.7 FLANGES, UNIONS AND COUPLINGS
- 2.7.1 For pipe sizes 50 mm (2 inch) nom. dia. and under:
- 2.7.1.2 UPVC , PP and HDPE Piping: Unions of respective materials, solvent or fusion ends.
- 2.7.2 For pipe sizes 65 mm (2 1/2 inch) nom. dia. and over:
- 2.7.2.2 Plastic piping: Flanges of respective materials, solvent or fusion ends.
- 2.7.3 Grooved and shouldered pipe end couplings: Malleable iron housing clamps to engage and lock, designed to permit some angular deflection, contraction, and expansion; "C" shape composition sealing gasket; steel bolts, nuts, and washers; galvanized couplings for galvanized pipe.

2.7.4 All flanges, unions and couplings shall be pressure rated as the pipe joints.

2.8 PEX PIPES & FITTINGS

2.8.1.1 General

Cold and hot water distribution pipework embedded within builder's work inside toilets, bathrooms, kitchen shall be composed of :

- Cross linked polyethylene pipe according to DIN 16892 and 16893 pulled in a corrugated conduit of ordinary polyethylene.
- Fittings and valves of DZR brass construction.
- Pipes and fittings must be certified by WRAS and WVG D laboratories.

2.8.2 Pipes

2.8.2.1 The inner pipe shall be a cross link polyethylene pipe withstanding upto 95 deg. C at a max. pressure of 10 bar without deformation or damage. Short time temperature loading upto 110 degree C.

2.11.2.2 The pipe shall be resistant to all natural constituents of drinking water to DIN 2000, disinfectants and cleaning agents to DVWG – W291 and DIN 2000, anti-corrosion agents to DIN 1988 T4 (E) and common building materials, such as mortar, cement and plaster.

2.8.2.3 The pipe and fittings manufacturer shall offer minimum of 50 years guarantee for the pipes and 10 years guarantee for the complete system from TOC (taking over certificate).

2.8.2.4 All pipes and fittings shall be stamped on equal intervals showing clearly the name of the manufacturers along with the pressure and temperature rating of these pipes.

2.8.2.5 All pipes shall be laid so that the 16mm per water pipe can be replaced, if necessary.

2.8.3 Fittings & Accessories

2.8.3.1 Pipes shall be connected to the different fittings via a wall box of reinforced plastic fitted with DZR brass elbow suitable for 15mm or 20mm threaded connection. The box should be suitably designed in order to enable the replacement of existing pipes in the even of their damage.

2.8.3.2 All valves, box elbows, tees, bends shall be of DZR brass. Contractor shall use proper tools for assembly as recommended by manufacturer, i.e. assembly pliers, Ratchet torque wrench, cutters, etc.

PART 3 - EXECUTION

3.1 PREPARATION

3.1.1 Ream pipe and tube ends. Remove burrs.

3.1.2 Remove scale and dirt, on inside and outside, before assembly.

3.1.3 Prepare piping connections to equipment with flanges or unions.

3.2 INSTALLATION

- 3.2.1 Route piping in orderly manner and maintain gradient.
- 3.2.2 Install piping in a way that conserves building space and does not interfere with the use of space.
- 3.2.3 Group piping whenever practical at common elevations.
- 3.2.4 Install piping to allow for expansion and contraction without stressing pipe joints or connected equipment.
- 3.2.5 Install exposed piping parallel or at right angles to the building walls, except where otherwise shown on the Contract Drawings.
- 3.2.6 Where changes in pipe sizes occur, install tapered reducer fittings. Use of bushings is not permitted. Install eccentric reducer fittings with level crown.
- 3.2.7 Where changes in pipe direction occur install factory manufactured fittings. Bending or forming of piping is not permitted without Engineer's written permission.
- 3.2.8 Provide adequate clearance for installation of insulation and access to valves and fittings.
- 3.2.9 Provide minimum clearances between piping covered by this Section and other services.
- 3.2.10 Provide access doors where valves and fittings are not exposed. Coordinate size and location of access doors.
- 3.2.11 Slope water piping and arrange to drain at system low points.
- 3.2.12 Establish elevations of buried piping outside the building to ensure not less than 90 cm (3 ft) of cover.
- 3.2.13 Where pipe support members are welded to structural building framing, scrape, brush clean and apply one coat of zinc rich primer to welding.
- 3.2.14 Use main sized saddle branch connections or directly connecting profiled branches (weldolets) to mains in steel piping if main is two pipe sizes larger than branch. Do not project branch pipes inside the main pipe.
- 3.2.15 Establish invert elevations; slope drainage pipes to one percent minimum unless otherwise stated. Maintain gradients as shown on the contract drawings.
- 3.2.16 Install test-tees in soil, waste, vent and rainwater pipe risers at minimum 450 mm (18 inch) above the ground floor level, at every alternate level and at other locations as shown on the contract drawings.
- 3.2.17 During progress of work keep the ends of all piping closed with factory manufactured plugs or blind flanges to prevent entry of foreign matter. Ensure the removal of such plugs or flanges on completion of the works. All blind flanges shall have integral indicating flags.
- 3.2.18 Install bell and spigot pipe with bell end upstream.

- 3.2.19 Install valves with stems upright or horizontal, not inverted, except with written permission of the Engineer for each location.
- 3.2.20 All valves shall be same size as pipeline except if otherwise noted on the contract drawings.
- 3.2.21 Provide flanges / unions on above grade horizontal pressure piping at spacing not exceeding 18.0 Ms (60 feet) on straight runs so as to enable taking down of the piping easily, when required.
- 3.2.22 Use valves with flanged ends in valve pits and when directly buried, irrespective of size.

3.3 APPLICATION

- 3.3.1 Use grooved mechanical couplings and fasteners only in accessible locations.
- 3.3.2 Install unions or flanges downstream of valves and at equipment or apparatus connections.
- 3.3.3 Install dielectric unions or flanged fittings wherever joining dissimilar metals.
- 3.3.4 Install drain valves at system low points and at each riser base. Use ball valves with flanged connections for drainage of below grade mains in valve pits.
- 3.3.5 Install gate valves for shutoff and to isolate equipment, part of systems, or vertical risers.
- 3.3.6 Install butterfly valves for isolation of life support system underground mains.
- 3.3.7 Install globe valves for throttling, bypass or manual flow control services.
- 3.3.8 Install spring loaded check valves on discharge of water pumps.
- 3.3.9 Install lever and weight type check valves on discharge of sump pumps and sewage ejectors.

3.4 SETTING OUT AND EXCAVATION OF TRENCHES FOR BELOW GRADE MAINS

- 3.4.1 Trenches shall be excavated only after completion of site development work like backfilling, if any. Care shall be taken to ensure that no heavy equipment shall be worked or moved over piping which is not designed to take up such loads.
- 3.4.2 Clear the existing ground levels along the line of all mains, particularly at all manholes, building connections etc.
- 3.4.3 Where long lengths of piping mains are to be laid in trenches at slight slope, sight rails should be fixed across the trench at intervals of at least 5 meters at a height equal to the length of the boning rod to be used above the required invert level of the drain or sewer at the point where the sight rail is fixed. There shall be not less than two sight rails in position on each length of the main under construction.
- 3.4.4 The trench shall be excavated to the actual required depth such that the pipe shall bear uniformly on undisturbed ground at every point between joint holes.
- 3.4.5 The Contractor shall examine any unfitting or weak ground material, which may be found below the pipe laying level and report the same in writing to the Engineer, before laying any pipe. In such cases, the unsuitable material shall be removed and made up by backfilling, at additional cost to the Employer.

3.5 INSTALLATION – PARTICULAR FOR BELOW GRADE MAINS

- 3.5.1 Handle pipes with utmost care so as not to subject the pipes to shocks, which may crack or break the internal lining, if any. Cement lined pipes which are cut to length on site shall be tapered at their spigot ends by a grinder and a fast drying coal tar coat applied at the uncoated areas.
- 3.5.2 The pipes shall be laid directly on the undisturbed bottom of the trench with special recesses made for the joints or placed on a concrete bed. The pipes must be aligned carefully both in line and level. Wooden supports must not be placed under the pipes. Precautions shall be taken to prevent dirt from entering the pipe.
- 3.5.3 Ductile iron pipes shall be wrapped up with polyethylene sleeving before laying. Adhesive tapes shall be used to secure the sleeving at socket and spigot ends. Plastic coated wire shall be used to secure the encasement at every one third length of the pipe. After jointing is done and testing completed, encasement shall be pulled to close the whole joint and be secured with plastic coated wire strapping. The pipes shall be laid with the folded part of the polyethylene encasement on the crown of the piping.
- 3.5.4 Piping shall be protected against the ingress of foreign material before the pipe is placed in the trench. If the case is such that the pipe cannot be laid, in the trench and in place, without getting earth into it; each end shall be covered with a heavy, tightly woven canvas bag of suitable size before lowering the pipe into the trench. The bag shall be left there until the connection is to be made to the adjacent pipe.
- 3.5.5 When jointing pipes, a mark shall be made to check the position of the end of the barrel. Leave a gap of 5 to 10mm between the ends of the pipes, for piping with flexible joints.
- 3.5.6 The joints shall be left exposed until the line is pressure tested and approved by the Engineer.
- 3.5.7 Deviations from given levels may not be greater than + or - 1 cm at any point and gradients not greater than 1/20 of the given value.

3.6 BACKFILLING OF PIPE TRENCHES

- 3.6.1 All surplus, suitable and approved excavation material shall be used for backfilling the trenches.
- 3.6.2 All backfill material shall be free of cinders, ashes, refuse, vegetable or organic matter, boulders, stones or other material which are unsuitable for the purpose, in the opinion of the Engineer.
- 3.6.3 Excavations shall be backfilled without unnecessary delay, but not before completion of testing of the piping. The first 30 cm fill shall only be compacted by hand tamping; subsequent layers being compacted by power tampers as approved by the Engineer. No heavy mechanical equipment shall be used for backfilling or compaction.
- 3.6.4 The pipe trenches shall be backfilled and compacted to a minimum of 95 % of AASHTO T - 180 density, in layers of not exceeding 15 cms and fills shall be brought up simultaneously on the full width of the trench.

3.7 INSTALLATION OF MANHOLE ACCESSORIES

- 3.7.1 Frame of manhole cover shall not be cast in together with roof slab of manhole.
- 3.7.2 Covers and frames shall be fixed in such a way as to match adjacent surface levels with tolerances not exceeding + or - 3 mm. Before the maintenance period expires, check all manhole covers and readjust them if they are in excess of above mentioned tolerances.
- 3.7.3 Fix all cast iron accessories to concrete manholes using cement and sand mortar (1:3).
- 3.8 PROTECTION
- 3.8.1 Give the following protective coating / wrapping to piping. All piping above louvered ceilings shall be painted black.
- 3.8.7 Buried plastic piping in areas subjected to traffic loading:
- 3.8.7.1 Encasement with minimum 10cm thick, class K 140, plain concrete.
- 3.8.7.2 Colored metal coated plastic foil warning grids.
- 3.8.8 Concealed (in shaft, walls, above false ceiling) HDPE Piping:
- 3.8.8.1 Plastic foil as sound insulation.
- 3.8.9 Plastic piping buried within/below raft foundation:
- 3.8.9.1 Monolithic concrete channel enclosing the pipe with water proofing membrane extended under the same when pipe work is below the raft level.
- 3.8.10 Plastic piping exposed to direct sunlight:
- 3.8.10.1 One layer of self adhesive, aluminium foil backed kraft tape with 10% overlap.
- 3.8.10.2 One layer of "Densopol 60" with 50% overlap.
- 3.8.10.3 Heavy canvas jacketing with Benjamin Foster finish.
- 3.8.11 Insulated plastic piping in technical areas and below grade pipe trenches.
- 3.8.11.1 Heavy canvas jacketing with Benjamin Foster finish.
- 3.8.12 HDPE storm/drainage piping in areas where the piping is diverting and running horizontally should be acoustically insulated.
- 3.9 TESTING OF PRESSURISED PIPING
- 3.9.1 The pipes shall be jointed, plugged and shall have been in position for at least 24 hours, before the tests are carried out.
- 3.9.2 The piping shall be tested for line, gradient and water tightness. The Contractor shall furnish all labour, and necessary testing instruments such as gauges, pumps etc. as directed by the Engineer. A minimum of two (2) nos identical pressure gauges shall be installed at extremities of the piping circuit to be tested.

- 3.9.3 Fill the piping with clean fresh water, leaving all high points open to allow for purging of air. Allow a soaking period of at least 24 hours for cement lined or concrete pipes.
- 3.9.4 Pressurize the system using manual pumps in increments of 25 percent of the test pressure. Allow a standing period of 10 minutes after each pressure increment. Pressure testing using motor driven pump is not permitted.
- 3.9.5 Do not over pressurize the system under any circumstance. If the test pressure is close to the maximum permissible pressure of any of the system components, install calibrated pressure relief valves in the tested circuit.
- 3.9.6 Prior to testing keep all valves and control devices in open position. After completion of pressure test, close each valve one at a time starting from the pressure release end, so as to ensure tightness of the valve.
- 3.9.7 The piping shall be hydrostatically tested at a pressure of 690 Kpa (100 psig) or 1.5 times the system working pressure, whichever is greater, but limited to 1379 Kpa (200 psig) for copper and steel piping and 1034 Kpa (150 psig) for UPVC piping. The pressure shall be maintained for at least 4 hours. In metallic piping, allow for pressure fluctuations due to ambient temperature variations. Record the circuit temperature along with the pressure readings.
- 3.9.8 Check for leaks by swabbing with a dry tissue. Drop in pressure to the order of 10 Kpa (1.5 psig) per hour shall be considered as acceptable.
- 3.9.9 Do not subject sanitary fixture chrome fittings to the test pressure.
- 3.9.10 Piping shall be tested in segments during the progress of the work. Maintain an official log book for recording the tests carried out on sections of piping, including test pressure, date of test and approval signature of Engineer's representative witnessing the test.
- 3.10 TESTING OF DRAIN, WASTE AND VENT PIPING
- 3.10.1 The pipes shall be joined, plugged and shall have been in position for at least 24 hours before the tests are carried out.
- 3.10.2 The piping shall be tested for line, gradient and water tightness. Furnish all labour and necessary testing instruments as directed by the Engineer.
- 3.10.3 Install compression type rubber plugs with bleeding caps to plug branch risers in horizontal piping. Install inflatable pneumatic rubber plugs for vertical risers and other inaccessible areas.
- 3.10.4 Fill the piping with clean fresh water, leaving all the high points open to allow purging of air.
- 3.10.5 All soil, waste, vent and storm piping shall be tested by filling entire system with water to a minimum head of 3 Ms (10 feet) minimum at any part of the system. The system shall hold the water level to within 100 mm (4 inch) drop in 8 hours. Vertical piping shall be tested in sections of two or three floors at a time.
- 3.10.6 Check for leaks by swabbing with a dry tissue.
- 3.10.7 Test completed piping by simultaneous flushing of all water closets. Test drainage branches by simultaneous discharge of filled bath tubs, washbasins etc.

3.11 DISINFECTION AND FLUSHING OF FRESH WATER PIPING SYSTEM

- 3.11.1 Prior to starting work, verify that system is complete, flushed and clean.
- 3.11.2 Ensure pH of water to be treated is between 7.4 and 7.6 by adding alkali (caustic soda or soda ash) or acid (hydrochloric acid).
- 3.11.3 Add disinfectant, free chlorine in liquid, powder or tablet form to obtain 50 to 80 mg/L residual.
- 3.11.4 Fill the system piping with chlorinated water and bleed water from outlets to ensure distribution and test for disinfectant residual at minimum 15 percent of outlets.
- 3.11.5 Maintain disinfectant in system for 24 hours.
- 3.11.6 During the disinfections period install temporary warning signs in both Arabic and English at conspicuous places throughout the Project against using the chlorinated water.
- 3.11.7 If final disinfectant residual tests less than 25 mg/L, at any outlet, repeat treatment.
- 3.11.8 Flush disinfectant from system until residual equal to that of incoming water or 1.0 mg/L.
- 3.11.9 Take samples no sooner than 24 hours after flushing, from remote outlet of each riser and from water entry, and analyze in accordance with AWWA C601. Submit result of analysis to Engineer for approval.

END OF SECTION

MECHANICAL SECTION: 22 11 23

DOMESTIC WATER PUMPS

PART 1 – GENERAL

1.1 WORK INCLUDED

- 1.1.1 Work of this Section includes all labor, materials, equipment and services necessary to complete the pumps as shown on the drawings and specified herein, including but not limited to, the following:
- 1.1.2 Constant Speed Water Booster pump set
- 1.1.3 Variable Speed Water Booster pump set
- 1.1.4 In-Line water pumps (Cold and Hot water)
- 1.1.5 Manufacturer's factory representative's supervision of installation and start-up.

1.2 RELATED SECTIONS

- 1.2.1 Sound Control
- 1.2.2 Vibration Isolation
- 1.2.3 Insulation
- 1.2.6 Motor Control Centres
- 1.2.7 Variable Frequency Controllers

1.3 QUALITY ASSURANCE

- 1.3.1 Provide pumps with manufacturer name, model and rating/capacity identified.
- 1.3.2 Ensure pumps operate at specified system fluid temperatures without vapour binding and cavitations, are non-overloading in parallel or individual operation. Operate within 25 percent of midpoint of published maximum efficiency curve.
- 1.3.3 Provide pumps having local agents certified by the manufacturers to check, align, test, adjust and commission the pump sets.

1.4 SUBMITTALS

- 1.4.1.1 Shop Drawings: Submit dimensioned shop drawings and performance data for each pump. Submit certified performance curves for each pump.
- 1.4.1.2 Product Data: Manufacturer's latest published data for materials, equipment and installation.
- 1.4.1.3 Submit characteristics such as full load starting and running amps and power factor.
- 1.4.1.4 Submit wiring diagrams for interlock and control wiring, clearly differentiating between portions of wiring that are factory installed and portions to be field installed.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Subject to compliance with the requirements of the specifications of this Section, Client's approved manufacturers and suppliers are acceptable.

2.2 PRODUCT SELECTION

- A. Materials and products supplied by contractor shall match the Client's approved samples provided by the Architect/ Engineer, in all aspects, including but not limited to form, texture, size, shape, aesthetics, performance and technical specification. If a specific sample is not provided by the Architect/ Engineer for any specified product or material, then the Contractor shall submit samples to the Engineer, of such materials and products, for selection and approval.
- B. It is the responsibility of the Contractor to read all the Contract documents together. In case of any conflict or ambiguity, actual or perceived, the more stringent of the specification or conditions will apply, and the Engineers interpretation shall be treated as final and binding. All conflicts or ambiguities, actual or perceived, shall be brought to the attention of the Engineer.
- C. All pump sets shall be factory assembled and tested and witnessed by client/engineer; field assembly or testing of components is not acceptable.
- D. All exposed pumps and accessories in N1 tower roof which are exposed to ambient temperatures shall be selected at operating temperature of 50Deg.C. Contractor shall submit confirmation letter from manufacturer for the capability of operation at 50Deg.c without fail.

2.3 MATERIALS (PUMPSETS)

A. Transfer pump sets

- 1. Vertical multistage centrifugal pump sets - Transfer Pump sets for Domestic Water: Packaged Pump sets. Packaged pump sets shall be self-contained fully automatic package units. The units shall be complete with duty/standby pumps, quantity (as shown on the equipment schedule) and an automatic control panel consisting of the following:
 - a. Starter with overload for each other motor
 - b. TP&N isolating switch, fully enclosed, interlocked with panel externa door and operating handle.
 - c. Hour run meter for each pump
 - d. System pressure gauge
 - e. Residual current circuit breaker 100mA (for each pump)
 - f. Set timing relays
 - g. Hand / Off / Auto switch for each pump
 - h. Duty pump selector switch or system mode selection switch

- i. Run & trip lamps for each motor
 - j. Indicating lamps, "Power On" and "Low Water"
 - k. Main neutral link
 - l. Cyclic controller (2 numbers where standby jockey pump is specified).
 - m. Pressure switch with adjustable setting for each pump
 - n. Dry run prevention relay and level switch in feed tank
2. Where more than one duty pump is specified for a pump set, the duty pumps shall start with a 15 seconds time delay (adjustable) between each pump.
 3. The "hand" button shall be spring loaded and shall override all control circuits (except overload) so that the selected pump can be checked for operation.
 4. Due to limited height of tanks the water levels at which pumps are switched have to be determined accurately. Therefore, level switches with stainless steel drop rods (which act as guides for floats) shall be used. Float switches with plastic floats and directly connected cables shall not be permitted.
 5. The water transfer pump sets shall operate based on level switches located in the tanks. A set of level switches shall be provided for each compartment of the tank. One level switch each shall be provided for starting each of the duty pumps, one level switch for switching off all pumps and one level switch for water high level alarm indication. A three position switch shall be provided to enable isolation of any one of the tanks (or compartment of the tank) from the ON-OFF control circuit for maintenance. The third position shall be for normal operation, i.e. all tanks (or compartments) are included in the ON-OFF control circuit.
 6. The transfer pump set shall include a pressure switch. Pressure switch shall function as a safety cut off for pumps in case the pump cut-off level switch malfunctions and ball float valves in the tank close. In case the pumps are switched off by the pressure switch, the pumps shall not restart for 5 minutes (adjustable), alarm indication shall be provided (indication light on panel fascia and volt free contacts for indication to BMS) and the alarm condition shall persist until it is manually reset.
 7. The transfer pump control panel design shall ensure:
 - a. Automatic operation of selected duty pumps on the basis of level switches (cut in and cut off) or level switch (for cut in) and pressure switch (for cut off). The duty pumps shall be selected at the duty selector switch and H.O.A switch for the standby pump shall be in OFF position.
 - b. Automatic operation of all pumps on the basis of level switches (cut in cut off) or level switch (for cut in) and pressure switch (for cut off) and the cyclic relay. The duty selector switch shall be in 'Auto' position and the H.O.A switches for pumps in 'Auto' position.
 - c. Automatic switchover to standby pump in case of failure of any duty pumps.

- d. Hand position of H.O.A switches is only for testing of pump and shall be spring loaded. Alternatively, non-latching push button shall be used for starting pump in HAND mode.
- e. Indication for water high level.
- f. Dry run protection based on level switch in each compartment of the feed tank including indication. A three position switch will be provided to enable isolation of any one of the tanks (from dry run protection circuit) for maintenance. The third position shall be for normal operation, i.e. all tanks (or compartments) are included in the dry run protection circuit.
- g. Alarm indication when pumps cut off due to pressure switch as detailed above

B. Booster pump sets : (with variable frequency motors)

1. General: The specifications of the pump set shall be same as Packaged Transfer pump set except that:
 - a. The pump motors shall be suitable for continuous operation with Variable Frequency Drives (VFD) and also constant speed operation
 - b. Pump set shall be provided with VFDs.
 - c. Pump set control shall be performed by a microprocessor based controller.
 - d. A hydraulic accumulator with replaceable diaphragm.
 - e. The Control Panel shall ensure:
 - 1) The packaged booster pump set shall maintain a constant system pressure by continuously varying the speed of one pump via a VFD. The other pumps shall cut in/out on mains operation, as required by the system flow demand. At low demand, the first pump in duty sequence shall operate through the VFD to maintain system pressure. As the demand increases to more than the capacity of one pump, the second pump in sequence (and so on for other duty pumps) shall be switched on at constant, full speed and the first pump which is being operated through the VFD shall continue to maintain the set system pressure. On drop in demand, the constant speed pumps will be cut-off first and VFD operated pump will be switched off last.
 - 2) The VFD controlled pump shall always start first and stopped last.
 - 3) All pumps shall be controlled by the VFD in rotation.
 - 4) Pump duty changeover shall be automatic and depends on load, time and fault to ensure equal operating time for all the pumps of the booster pump set. The changeover shall also ensure equal operating time for all pumps with the VFDs.
 - 5) All the pumps shall be of equal size and suitable for parallel operation and connected for automatic cascade control.

- 6) Automatic switchover to standby pump in case of failure of any duty pump.
- 7) The controller shall maintain a constant set pressure by the closed loop control principle i.e. continuously comparing the set value (pressure) with the feedback signal from the pressure transmitter and maintaining zero deviation by adjusting the speed of one pump via the VFD. Industry standard signals shall be used e.g. 0-10V DC from controller to VFD corresponding to 0 – 100% speed and the pressure transmitter output shall be 0-20 mA etc.
- 8) The controller shall automatically match the pump set performance to the system demand by cutting in/out the required number of pumps. The controller shall operate the system with as few pumps as possible. The frequency of starts and stops shall be limited by the setting of minimum and medium switching sequences on the controller.
- 9) The controller shall be set to site adjustable the maximum and minimum permissible operating pressure. If the actual pressure exceeds the set value for 0.5 secs, the fault conditions shall be annunciated.
- 10) Dry run protection based on level switch in each feed tank (for each compartment of the tank) with indication. A three position switch will be provided to enable isolation of any one of the tanks or compartments (from dry run protection circuit) for maintenance. The third position shall be of normal operation, i.e. all tanks (or compartments) are included in dry run protection circuit. In case this feature is not available with control panels supplied by the manufacturer, a separate panel with necessary switches shall be provided and inter locked with the pump set control panel.
- 11) A minimum two numbers of “Variable Frequency Drives” (i.e. VFDs) shall be provided – 1 duty and 1 standby. The VFDs shall be set to operate in rotation on preset (adjustable) time basis, to ensure equal operating time for the VFDs.
- 12) Star/Delta starters shall be provided for all other duty pumps of 5.5 kW or above (DOL starters for less than 5.5kW). Alternatively, VFDs may be used for all pumps.
- 13) The controller shall be capable of communicating with BMS, where specified.
- 14) The complete pump set including the controller shall be factory assembled and tested at the manufacturer’s works. Sets assembled locally by the manufacturer’s agents shall not be accepted.
- 15) Booster pump if installed exposed to external environment, pump supplier to provide a fabricated custom built enclosure along with pump sets to protect the pumps and controls from exposure to excessive solar radiations and harsh environment. Details of the enclosure to be reviewed and approved by the consultant.
- 16) VFD shall be capable to operate at 50 DegC operation at ambient for the exposed panel (placed inside Non-AC) .

17) De-rated VFD panel sizing shall be considered for the external exposed VFD.

18) The deration temperature shall be considered at 50 degC operation for external exposed VFD panels.

C. Water circulating pumps :

1. Water circulating pumps shall be installed where shown on drawings and schedule of equipment.
2. The pumps shall be of equal size located on a common centre line, 1800 apart for mounting in the pipeline. The motor and shaft assembly shall be easily removable from the casing without disturbing pipe work.
3. The duty and standby pumps shall be controlled from an automatic control panel (linked to the BMS where specified). A two position mode selection switch shall be provided on this Control Panel. In 'Auto' position of the mode selection switch, the operation of pumps shall be controlled by the BMS. In 'Manual' position of the duty selection switch, the H.O.A switches for pumps shall be in 'Hand' mode and the pump shall be switched ON/OFF using push buttons.
4. The control panel shall operate the pumps continuously, equalize run hours in the Auto mode, record the run time, carry out auto-switchover to standby pump, monitor switches and annunciate remote alarm at BMS in case of any fault in the system, including switches not being in normal operating positions. 'Run' and 'Trip' indications will also be provided on the pump control panel.

D. Pumps and accessories details:

1. Materials
 - a. Casing: Stainless steel casing with drain connection fitted with plug.
 - b. Impellar: Stainless steel, multistage
 - c. Shafts: Stainless steel
 - d. Seal: Mechanical
 - e. Seal housing : As per manufacturer's standard
2. Temperature Range: Up to 50 Deg C for cold water, up to 120 Deg C for hot water
3. Ambient Conditions: Pumps must be suitable to operate at an ambient temperature range of 55 Deg. C.
4. Pressure: PN 16 minimum or 150% of maximum or as per the system requirement.
5. Motors: Single speed, with oil-lubricated bearings, unless otherwise indicated. Select motor that will not overload through full range of pump performance curve. Comply with Division 22 Section "Motors." Vertical multistage high efficiency centrifugal pumps directly coupled to a TEFC motor. The pump motor shall meet or exceed ASRAE 90.1 standard for motor efficiency.
6. Control Valves: Adjustable, automatic, pilot-operated pressure regulator on each

pump discharge.

7. Relief Valves: Adjustable, pressure relief type on pump discharge.
8. Piping: For piping materials and installation inside pump room. ASTM A 53/A 53M, Schedule 40, galvanized-steel pipe with threaded, cast-iron fittings and threaded joints with plastic powder coating inside and outside.
9. Header End Connections:
 - a. DN 50 and Smaller: Threaded.
 - b. DN 65 and Larger: Flanged.
10. Piping, including valves and other components, may have grooved ends for grooved joints.
11. Control Panel: Factory installed and connected as an integral part of unit complying with IEC and UL 508; automatic for multiple-pump, with load control and protection functions. Protected to IP 55 and wired to IEE wiring regulations.
12. Enclosure: IP55.
13. Hydro pneumatic Tank: Pre charged, bladder tank with replaceable natural rubber, diaphragm or bladder.
14. Flexible Connectors
 - a. Description: Corrugated, bronze inner tubing covered with bronze wire braid. Include copper-tube ends or bronze flanged ends, braze-welded to tubing. Include 1200-kPa minimum working-pressure rating and ends according to the following:
 - b. DN 50 and Smaller: Threaded. Provide flanged ends if pump has flanged connections.
 - c. DN 65 and Larger: Flanged.
15. Pump Enclosure : Pump supplier to provide a fabricated custom built enclosure along with the pump sets to protect the pumps and controls from exposure to excessive solar radiations and harsh environment. Details of the enclosure to be reviewed and approved.

2.4 ACCESSORIES

- A. General: Use only materials that comply with referenced standards and other requirements specified.
- B. Building-Automation-System Interface (As Applicable)
 1. Provide auxiliary contacts in pump controllers for interface to building automation system.
Include the following:
 - a. On-off status of each pump.
 - b. Alarm status.

2.5 INLINE PUMPS

- 2.5.1 Furnish and install as shown in the drawings and as described in these specifications, vertical in-line bronze fitted, single stage, single or double suction centrifugal pumps. Each pump shall be capable of delivering litres per second against a total head as scheduled.
- 2.5.2 Pumps shall each be close-coupled to a ball-bearing, drip proof totally enclosed BS 5000 Parts 10 and 17 vertical electric motor. Motor shall be of such size that it will operate continuously without exceeding its kW rating, exclusive of service factor, at the design capacity and head.
- 2.5.3 Pump casing shall be constructed in cast iron with PN16 flanges for working pressure up to 12 bars at 65 deg C and ductile iron with PN25 flanges for working pressure of 25 bars at 65 deg C. Models 80mm and larger shall have balanced double volute design to minimise radial shaft deflection.
- 2.5.4 Suction and discharge connections shall be the same size flanged and 180° opposite on centerline for pipeline mounting. Casing shall have bronze replaceable wear ring.
- 2.5.5 Impeller shall be bronze, fully enclosed, statically, dynamically, and hydraulically balanced. Motor shaft shall be one-piece stainless steel. Pump shall have stainless steel mechanical shaft seal of the type multi-spring outside balanced type with secondary seal, carbon rotating face and silicon carbide stationary seat which are to be properly vented to the suction connection.

PART 3 - EXECUTION

3.1 GENERAL

- 3.1.1 The pump manufacturer is to be responsible for aligning in the field prior to the start-up of all flexibly coupled pumps. Alignment to be with dial indicator with an accuracy of plus or minus 0.05mm. The pump manufacturer must submit a written report certifying that the alignment work has been performed by his personnel and that the pumps are ready for operation.
- 3.1.2 All base mounted pumps to be leveled up on tapered steel wedges in such manner to permit a minimum of 18mm of grout between the pump base and the top of the concrete base, unless otherwise noted.
- 3.1.3 All pumps are to be installed in such a manner as to provide access for periodic maintenance as well as removal of motors, impellers couplings etc.
- 3.1.4 All in-line pumps are to be independently supported of the connecting pipe work. Base mounted pumps are to be carefully aligned on the concrete/inertia bases prior to bolting down and grouting.

END OF SECTION

MECHANICAL SECTION: 22 12 00

WATER STORAGE TANKS

PART 1 – GENERAL

1.1 REFERENCE

Conform to General Requirements for Mechanical Services of Division.

1.2 DESCRIPTION OF WORK

Furnish and install Water Storage Tanks as indicated on drawings.

1.3 QUALITY ASSURANCE

a) The manufacture and design of hot press moulded GRP sectional water storage tank shall be to the quality standard requirements of BS EN ISO 9001 and shall comply with the superseded version of BS 7491-3 1994 glass fibre reinforced plastic cistern for cold water storage

b) Manufacturer's Qualifications: Firms regularly engaged in manufacture of specified items with characteristics, sizes and capacities required, whose products have been in satisfactory use in similar service for not less than 5 years. Certification, test certificates under specified operating conditions shall be provided by manufacturer.

c) Certificate of WRC approval shall be provided

1.4 SUBMITTALS

Product Data: Submit manufacturer's specifications and installation instructions.

Shop Drawings: Submit manufacturer's assembly-type shop drawings indicating dimensions, weight loadings, required clearances and methods of assembly of components. Maintenance Data: Submit maintenance data and parts lists for each type of equipment, control and accessory, including "trouble - shooting" maintenance guide. Include this data, product data and shop drawings in maintenance manual.

1.5 PRODUCT DELIVERY, STORAGE AND HANDLING

Handle equipment and components carefully to prevent damage, breaking, denting and scoring. Do not install damaged equipment or components, replace with new. Store equipment and components in clean dry place. Protect from weather, dirt, fumes, water, construction debris, and physical damage.

PART 2 - PRODUCTS

2.1 APPROVED MANUFACTURERS

2.1.1 Refer to approved manufacturers list in the Appendix.

2.2 WATER TANKS

2.2.1 Tanks shall be supplied with internal division plates, central where sizes permit, to create two independent equal compartments for maintenance purposes. The internal division plates are to be adequately braced to enable drain down of one compartment whilst the other compartment remains operational.

2.2.2 The separate compartments of the tank shall be provided with an interconnecting valved pipe manifold from which the various distribution services shall be taken.

- 2.2.3 Each compartment shall be provided with a connection for draining down the contents of the tank, which shall be provided with a lock shield pattern valve and shall be piped to a convenient discharge point on a drainage system to discharge in open air at least 150 mm above the discharge point.
- 2.2.4 Each compartment shall be provided with an overflow of the appropriate size to carry away all water discharged to the tank in the event of the inlet control device, ie float operated valve becoming defective, without the inlet control device becoming submerged. In addition a warning pipe of not less than 28 mm diameter shall be provided at least 50 mm below the lowest point of the overflow where tanks exceed 1,000 liters.
- 2.2.5 For tanks exceeding 10,000 liters a device may be fitted to give audible or visual alarm in substitution for a warning pipe as detailed in the Project Specification.
- 2.2.6 Overflow and warning pipes shall be piped independently to a discharge point on a drainage system to discharge in open air at least 150 mm above the discharge point. Pipes discharging to a common point shall be labeled such as to identify the source of the discharge.
- 2.2.7 All overflows and warning pipes shall be labeled throughout their length and fitted with screens to prevent the ingress of insects and vermin. The device shall be fitted with a corrosion resistant mesh of maximum perforations of 0.65 mm diameter and still maintain the cross sectional area of full overflow or warning pipe.
- 2.2.8 Each compartment shall be provided with an air inlet ventilator fitted with a corrosion resistant mesh screen having maximum perforations of 0.65 mm diameter to prevent the ingress of insects and vermin.
- 2.2.10 Each compartment shall be provided with access covers hinged and lockable ABS lid, suitably sized for the maintenance of the float operated valve.
- 2.2.11 Each compartment shall be provided with access covers hinged and lockable ABS lid in a secure manner, suitably sized 600 X 600 mm for man access to enable interior inspection and cleansing of the tank. All tanks shall be provided with internal ladders constructed from GRP where man entry is required for tanks 1.2 meters or more in depth.
- 2.2.12 Tank covers shall be heavy duty flat covers bolted down constructed from materials which will not contaminate any water which may condense on its underside.
- 2.2.13 Galvanized mild steel or aluminum ladders constructed in accordance with BS 4211: 1987 shall be provided to the tank for easy access to float operated valve positions.
- 2.2.14 The entire tank shall be insulated against heat and frost in accordance with this Specification.
- 2.2.15 Insulation shall be at least 40mm thick of expanded polystyrene, density grade 30 or polyurethane foam 25mm thick 40kg per cubic meter insulation material. Insulation shall not be exposed to external weather condition
- 2.3 GLASS REINFORCED PLASTIC SECTIONAL
- 2.3.1 Tanks shall be constructed from hot press mouldings of glass reinforced plastic from standard 1.0 m x 1.0 m, half plates 1.0 m x 0.5 m and corner angles.

- 2.3.2 The panels shall be manufactured by hot tool moulding at a temperature of 140 C at 1.7 bar using chopped BS 3691 'E' glass rovings and BS 3496 continuous filament 'E' glass matt, UV stabilized BS 3532 unsaturated isophthalic modified polyester resin, calcium carbonate filler and a Ditertiary butyl peroxy trimethyl cyclohexane catalyst with pigment coloring.
- 2.3.3 The panels shall be assembled and bolted together with 12 mm stainless steel nuts and bolts conforming to type 316/A4 in accordance with BS 6105 and shall be provided with flat washers to bolt head and nut.
- 2.3.4 All water tanks shall be externally flanged.
- 2.3.5 The manufacturers sealant strip shall be laid between each joint and shall be non-toxic and prohibitive to the support of bacteriological or fungous growth.
- 2.3.6 G.R.P. Tanks (Particular Specifications)

The Contractor shall supply and assemble at Site sectional water tank of hot pressed moulded GRP insulated panels (Glass Reinforced Plastic) of ivory or grey colour provide with lockable manhole cover air vent with overflow and all piping connections. The panels shall comply with BS 7491 : Part 3 : 1994 and shall have convex shape to assist a complete draining of water of the following characteristics :-

- (a) Specific gravity of 1.8
- (b) Tensile strength of not less than 850 kg/cm²
- (c) Flexural strength of 1800 kg/cm² to ISO 178.
- (d) Impact strength of 80 kg/cm² to ISO 179
- (e) Thermal expansion of 2.1×10^{-5} deg. C to ATM D 696.
- (f) Thermal conductivity of 0.6 W/m² C to BS 7491: Part 3.
- (g) Water absorption less than 0.2% to ISO 62.
- (h) Glass content more than 30%
- (i) No light transmission
- (j) Gasket of synthetic rubber
- (k) Bolts and nuts of stainless steel 316 / A4.
- (l) Frame angles and frame plates of hot dipped galvanized steel.
- (m) External ladder of hot dipped galvanized steel.
- (n) Internal ladder of GRP or S.S. according to Manufacturer's standards.
- (o) Tie rods of stainless steel
- (p) G.R.P. panel thick not less than 8mm

- (q) Resin cover of 2mm
- (r) Glass tube level indicator for all compartments.
- (s) Insulation 25mm.

PART 3 - EXECUTION

3.1 WORKSHOP DRAWINGS

- 3.1.1 The Contractor shall be responsible for ensuring accurate detailed construction drawings are produced. The drawings shall illustrate the size, location and type of all connections required, details of float operated valve fixings, location and provision of access covers, vents, etc together with the detailed requirements and Specification for the base support system.
- 3.1.2 The workshop drawings shall be issued to the CA/Engineer for review to ensure the Employers requirements have been met in full.
- 3.1.3 The issue of the drawings for review shall be well in advance of any works to be carried out on site for the preparation of the Builders work requirements of the tank and shall be issued sufficient to allow the CA/Engineer fourteen (14) days from receipt to comment.

3.2 CONNECTIONS TO COLD WATER STORAGE TANKS

- 3.2.1 Connections to GRP and concrete water tanks shall be carried out strictly in accordance with the manufacturers recommendations using materials and components supplied by the manufacturer and shall in all cases be two piece PVC flange connections with mastic seal and secured with stainless steel nuts and bolts conforming to type 316/A4 in accordance with BS 6105.
- 3.2.2 All such types of tank connections shall be provided with pipe support immediately after the assembly to ensure no load is transmitted directly to the connection assembly.
- 3.2.3 Connections to steel manufactured tanks shall be factory fixed prior to dispatch and shall be tapped flanges for screwed pipe work up to and including 50 mm pipe work and pads with studs for flanged pipe work 75 mm and above.
- 3.2.4 Builders work
- 3.2.5 All materials and works associated with the provision of the tank base or support on which the tank will be erected shall be supplied and executed by the Contractor in accordance with workshop drawings produced by the tank manufacturer.
- 3.2.6 The Contractor shall neither procure materials nor execute the works until such time as the CA/Engineer has completed the review of the workshop drawings.
- 3.2.7 The Contractor shall adhere strictly to the specific requirements as required by the particular tank manufacturer shall conform to the following minimum requirements.

3.3 SOLID CONCRETE PLINTHS

- 3.3.1 Solid concrete plinths shall be structurally designed to support the weight of the tank, including the contents, and shall take into account the ground bearing conditions or load limitations of the floor constructions.
- 3.3.2 Plinths shall be a minimum of 150 mm high and in length and breadth 300 mm longer than the tank overall dimensions including flanges so as to create a plinth 150 mm larger around the extremities of the tank. The top surface of the plinth shall be smooth and level to within 2 mm laterally or diagonally across the extremities of the tank, without any abrupt changes in level.
- 3.4 DWARF WALLS
- 3.4.1 Dwarf walls shall be constructed from engineering bricks or reinforced concrete to a minimum width of 200 mm. The walls shall be constructed at 500 mm centres arranged as required to length or breadth of the tank and shall exceed the overall dimensions of the tank by 150 mm at each end.
- 3.4.2 The walls shall be structurally designed to support the weight of the tank including the contents and shall take into account the ground bearing conditions or load limitations of the floor construction.
- 3.4.3 The tolerances applicable to the level of the walls shall be 2 mm across the length of the wall and between walls farthest from each other as well as diametrically across the walls.
- 3.5 STEELWORK
- 3.5.1 Where steelwork mill rolled sections are to be provided the sections are to be located at 500 mm centers arranged as required to length or breadth of the tank and shall exceed the overall dimensions of the tank by 150 mm at each end.
- 3.5.2 The steel sections shall be selected to be structurally suitable to support the weight of the tank including the nominal contents taking into account the ground bearing conditions or load limitations of the floor construction.
- 3.5.3 The maximum permissible deflection of the steel section shall not exceed 1/ 500th of the unsupported span.
- 3.5.4 The tolerance applicable to the level of the steel sections shall be 2mm across the length of the wall and between sections farthest from each other as well as diametrically across the sections.
- 3.6 ERECTION AND COMMISSIONING
- 3.6.1 The erection of the tanks upon plinths, dwarf walls or steel constructed by the Contractor shall be carried out by the tank manufacturer.
- 3.6.2 The tank manufacturer shall ensure that the plinths, dwarf walls or steel supports for tank are strictly in accordance with the agreed installation/ workshop drawings and that all construction tolerances meet with his requirements and the requirements of this Specification, before commencement of the erection of the tank.
- 3.6.3 The Contractor shall be in attendance at all times to ensure sufficient access, lighting and power is available for the Works and that activities by the tank erectors does not impede

other site activities and also that the highest standard of workmanship is maintained at all times.

- 3.6.4 Upon completion, the erected tank is to be offered for inspection by the CA/ Engineer prior to any further works being carried out.
- 3.6.5 After acceptable inspection by the CA/Engineer the tank may be filled with clean, fresh mains water to determine water tightness of the installation. The water tightness testing of the tank shall be continuous for 24 hours during which time the Contractor shall be in constant attendance to action any failure of joints which may occur during this period.
- 3.6.6 A certificate shall be issued by the Contractor to the CA/Engineer to confirm satisfactory completion of the test. The test certificates shall be included in the Record Documents.
- 3.6.7 Where tanks are required to be insulated, execution of the insulation works shall not be carried out until satisfactory water tightness of the tank has been achieved.
- 3.6.8 Sterilization procedures to be adhered to for the commissioning of the storage tank shall be in accordance with this Specification.

END OF SECTION

MECHANICAL SECTION: 22 13 19

DRAINAGE SPECIALITIES

PART 1 – GENERAL

1.1 WORK INCLUDED

1.1.1 Compliance with relevant General Requirements of Mechanical Services.

1.2 QUALITY ASSURANCE

1.2.1 Manufacturer qualification

Firms regularly engaged in manufacture of floor drains, cleanouts and roof drains of types and sizes required, whose products have been in satisfactory use in the similar service for not less than 5 years.

1.3 SUBMITTALS

1.3.1 Product data

Submit manufacturer's technical product data, installation instructions and dimensioned drawings for each type of items.

1.4 MAINTENANCE DATA

Submit maintenance data as per manufacturer recommendations. Include this data, product data, and certification in maintenance manual.

PART 2 – PRODUCT

2.1 MATERIALS

2.1.1 Floor Drains

- Floor drain body shall be in uPVC with rodding eye (110 dia top inlet, 82 dia outlet) with minimum water seal of 7 cm. with heavy duty gratings and covers made of SS 316 and it shall be subject to client engineer approval.

- Floor drain buried underground at plant rooms: Laquered Cast iron body with heavy duty 150 x 150mm nickel bronze gratings and covers.
- The open area of strainer shall be at least two third of the cross sectional area of the drain line to which it connects.

Note: wherever floor drain is indicated it shall be connected to floor trap unless otherwise indicated.

- Flower bed drain where specified shall be heavy duty, dome type drain coated cast iron body with bottom outlet complete with combination membrane flashing clamp/gravel guard, low silhouette dome and stainless steel mesh screen over dome, no floor trap is required.

- Internal Channel grating shall be stainless steel 316L and the external channel grating shall be heavy duty ductile iron to dimensions as indicated on drawings.

- External channel grating and catch basins shall be in heavy duty ductile iron material.

2.1.2 Roof Drains

- a) Rain water outlets shall be epoxy coated cast iron body and dome, wide serrated flashing flanged, flashing clamp device with integral gravel stop. Rain water outlets shall be selected to suit the type of roof where installed and shall be provided with a no hub outlet for mechanical jointing to rain water pipe. Rain water outlets shall be provided with perforated extension assemblies, where required to match the roof construction.
- b) Scupper type rain water outlets shall be dura coated cast iron body with reversible back or bottom outlet, oblique grate with combination frame and membrane flashing clamp.

2.1.3 Clean Outs

- Floor type: At finished areas, square type 150mmX150mm made of uPVC body and S.S 316 cover.
- At heavy traffic area, coated cast iron body with gasket connection and bronze plug with seal, secured round cast iron cover
- Wall type: uPVC body with square nickel bronze smooth wall access cover and frame, 150 x 150mm.
- Cleanout at high level shall be UPVC screw cap with washer and shall be rigidly vertically fitted with proper support.
- All floor clean out shall be perfectly sealed

2.1.4 Grease Interceptors

Automatic grease separator shall be provided with automated cleaning and disposal system, based on EN 1825-1, manufactured from polyethylene for grease separation from commercial kitchen drain piping.

The grease separator shall be buried underground type with integrated sludge trap, polyethylene quick release odour tight covers, with inspection window, with direct disposal hook up pipe on left or right side for sludge and grease disposal, with disposal flange DN 65, PN 10 (DIN 2501) Storz-B-coupling 2.5 inch for disposal truck connection, automated switch over from cleaning to disposal, twin 1 inch solenoid valves, rinse / refill inlet according to DIN 1988, high pressure cleaning / rinsing system with spray jets and flood jets, exterior (left or right) mounted submersible (IP 68) cutting pump with control unit, with refill equipment, inlet / outlet fits SML pipe (DIN 19522) and plastic pipe (DIN 19560). Sludge and grease capacity shall be as shown on drawings.

2.1.5 Oil Interceptors

Large capacity acid resistant coated interior and exterior fabricated steel lint interceptor non skids secured cover, permanent primary straining baffle. Regularly furnished with high inlet and low outlet.

- a) Material: Epoxy coated fabricated steel.

- b Rough-in: On floor.
- c Accessories: Integral deep seal trap, removable integral flow control, adjustable draw-off assembly, sediment bucket.
- d Cover: Steel, epoxy coated, non-skid with gasket, securing handle, and enzyme injection port, recessed for floor finish.

2.1.6 Manholes/gully traps covers

All covers shall be ductile iron with black bitumen coating.

Manhole covers shall have clear opening of 600 x 600 mm.

All manholes and gully traps shall be vented as per drainage department requirements.

Gully trap covers shall have clear opening of 300 x 300 mm.

All manhole covers and frames shall comply with BS 497:1976 part 1.

Internal Manholes:

All internal manhole covers shall be bolt down air tight double seal recessed type with brass trim, they shall be insitu concrete filled with surface finish to match surrounding floor. Covers shall be medium duty, locking and sealed.

External Manholes:

Covers in paved areas shall be medium duty, double seal, recessed type. They shall be insitu concrete/pavement filled with surface finish to match surrounding floor.

Covers subject to vehicular traffic shall be kite marked heavy duty, single seal, non-rock with square push fit seal plate complete with lifting key-holes.

2.1.7 GULLY GRATES

Where indicated on drawings provide kitemarked kerb gully unit of ductile iron hinged grate, road retaining bar with clear opening size 532x405mm. Unit to be black bitumen coated and shall be designed to deter ingress and help prevent blockage. A GRP grid shall be provided as standard.

Storm water / car park gratings shall be kitemarked, heavy duty ductile iron, non rock clear openings size as indicated on drawings, grey - epoxy coated and shall comply with BS 497.

2.2 APPLICABLE STANDARDS

Generally, the construction of all sewer and storm water manholes with covers, gully traps, road kerb gullies shall comply with the standard details drawings of Abu Dhabi Municipality.

PART 3 EXECUTION

3.1 GENERAL

- 3.1.1 All roof and floor drains shall be installed with water proof membrane extended up to and within the drain, and flashed to prevent any water penetration through the drain. Water proofing shall be carried out as recommended by the manufacturer and as approved by the Consultant.
- 3.1.2 Install roof drains at low points of roof areas according to roof membrane manufacturer's written installation instructions. Maintain integrity of water proof membrane where penetrated.
- 3.1.3 Position roof drains and floor drains for easy access and maintenance.
- 3.1.4 Unless otherwise mentioned provide traps at each drain to stop seepage of sewer gases into occupied spaces.
- 3.1.5 Provide transition piece with mechanical joint connectors to couple drains to pipes specified.
- 3.1.6 Install floor drains at low points of surfaces to be drained, set grates of drains flush with finish floor or as indicated.
- 3.1.7 Clean outs shall be installed at change of direction of drainage pipes inside building and where indicated on the drawings. Clean outs shall be not more than 15m apart in horizontal lines for pipe diameters of 100 mm and smaller, and 30 m apart for larger pipes.
- 3.1.8 A clean out shall be provided at the foot of each vertical waste or soil stack and at each floor, the clean out shall be at 0.6 m height from finished floor level.
- 3.1.9 Clean outs on concealed piping shall be extended through and terminate flush with finished wall or floor. Pits or chases may be left in walls or floors provided they are of sufficient size to permit removal of clean outs plug and proper cleaning of the system.
- 3.1.10 Where it is necessary to conceal a cleanout plug, a heavy duty stainless steel covering plate 200 x 200 mm shall be provided which permits ready access to the plug.
- 3.1.11 Cleanouts shall be of the same nominal size as the pipes up to 100 mm pipe diameters and not less than 100 mm for larger piping.
- 3.1.12 Cleanouts shall be so installed that there is a clearance of not less than 450 mm for the purpose of rodding and cleaning.
- 3.1.13 Protect cleanouts during construction period to avoid clogging with dirt and debris and avoid damages from traffic and construction works.

END OF SECTION

MECHANICAL SECTION: 22 14 29

SUBMERSIBLE SUMP PUMPS

PART 1 GENERAL

1.1 RELATED DOCUMENTS

- A Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A Section Includes:

- 1 Submersible sump pumps.
- 2 Wet-pit-volute sump pumps.
- 3 Sump-pump basins and basin covers.
- 4 Packaged drainage-pump units.

- B Related Section:

- 1 Division 22 Section "Sanitary Sewerage Pumps" for effluent and sewage pumps.

1.3 SUBMITTALS

- A Product Data: For each type of product indicated. Include rated capacities, operating characteristics, electrical characteristics, and furnished specialties and accessories.
- B Wiring Diagrams: For power, signal, and control wiring.
- C Operation and Maintenance Data: For pumps and controls, to include in operation and maintenance manuals.

1.5 DELIVERY, STORAGE, AND HANDLING

- A Retain shipping flange protective covers and protective coatings during storage.
- B Protect bearings and couplings against damage.
- C Comply with pump manufacturer's written rigging instructions for handling.

PART 2 PRODUCTS

2.1 MANUFACTURERS

Subject to compliance with the requirements of the specifications of this Section, Client's approved.

2.2 PRODUCT SELECTION

- A. Materials and products supplied by contractor shall match the Client's approved samples provided by the Architect/ Engineer, in all aspects, including but not limited to form, texture, size, shape, aesthetics, performance and technical specification. If a specific sample is not provided by the Architect/ Engineer for any specified product or material, then the Contractor shall submit samples to the Engineer, of such materials and products, for selection and approval.

B. It is the responsibility of the Contractor to read all the Contract documents together. In case of any conflict or ambiguity, actual or perceived, the more stringent of the specification or conditions will apply, and the Engineers interpretation shall be treated as final and binding. All conflicts or ambiguities, actual or perceived, shall be brought to the attention of the Engineer.

C. Submersible Sump Pumps

1. Description: Factory-assembled and -tested, single / duplex, single-stage, centrifugal, end-suction, submersible, direct-connected sump pumps complying with UL 778 or equal. Sump pumps shall be suitable for handling foul water containing dirt, sand, fibers, solids etc. or waste water containing dirt, sand, fibers etc as appropriate for the application. The discharge connection size of foul water pumps shall not be less than 100 mm diameter and throughlet shall not less than 76 mm diameter. The metallic construction impellers shall be designed to minimize chances of blockages and maximize the size of solids which can be handled. The pumps shall be suitable for "stationery wet" installation i.e the discharge connection shall be fixed to the sump floor (with stainless steel fixings) and the pump shall automatically engage (or disengage) when it is lowered along the guide bar (or raised). All guide bars, guide bar supports, fixings, lifting chains drop rods for fixing float switches, pipe supports etc. within the sump shall be stainless steel. The pump shall be externally pitch epoxy coated. Pump shall be non-clogging type with cutter arrangement.
2. Casing: Stainless steel; with stainless-steel inlet strainer, legs that elevate pump to permit flow into impeller, and vertical discharge with companion flange suitable for piping connection.
3. Impeller: Stainless steel or other corrosion-resistant material.
4. Pump and Motor Shaft: Stainless steel, with factory-sealed, grease-lubricated ball bearings and double-mechanical seals.
5. Motor: Hermetically sealed, capacitor-start type, with built-in overload protection; three-conductor waterproof power cable of length required, and with grounding plug and cable-sealing assembly for connection at pump. Comply with requirements in Division 22 Section "Common Motor Requirements for Plumbing Equipment." The sump pump motor shall be suitable for 400 volts, 3- phase, 50 Hz power supply, have Class 'F' insulation and IP 68 protection. The motor efficiency shall meet or exceed ASHRAE 90.1 standards.
6. Moisture-Sensing Probe: Internal moisture sensor with moisture alarm.
7. Pump Discharge Piping: The discharge pipe within the sump shall be cast with high solids – high build polyamide cured coal tar epoxy coating (internally and externally). Dry film thickness of coating to be 300 microns. or PVC-U to BS-3505 Class E. A flanged connection shall be provided between the cast/ductile iron pipes and the Upvc pressure pipework. The gate valves shall be resiliently seated of ductile iron construction with internal and external epoxy paint protection. The non-return valve shall be swing pattern of ductile iron construction with internal and external epoxy

coating. The discharge pipework shall preferably be arranged such that it crosses the slab above sump vertically in the walkway. The pipe shall be sleeved where it crosses the floor slab. A galvanized steel frame shall be provided around the pipes to protect them from mechanical damage.

8. Controls: IP 55 enclosure, wall mounted; with four mechanical or micro pressure switches in water tight enclosures; mounting rod; and electric cables. Include automatic alternator to alternate operation of pump units on successive cycles and to operate multiple units if one pump cannot handle load. The sump pump control panels shall be IP 55 suitable for wall mounting and equipped with anti-condensation heater. The panel shall incorporate the following:
 - a. MCB
 - b. Starter
 - c. Hand / Off/Auto (HOA) switch
 - d. Phase Failure Trip
 - e. ELS with audio and visual indications
 - f. A cyclic relay and 3 position (P1/P2/, P2/P1 and Auto) Duty selector switch shall also be provided.
9. Panel shall be equipped for each pump with:
 - a. MCB
 - b. Starter
 - c. Hand / Off/Auto (HOA) switch
 - d. Phase Failure Trip
 - e. ELS with audio and visual indications
 - f. A cyclic relay and 3 position (P1/P2/, P2/P1 and Auto) Duty selector switch shall also be provided.
10. Selected duty pump shall operate independently controlled by float level control switch when (HOA) switch in 'Auto' position. Duty selection shall be carried out from the duty selector switch. The standby pump shall assist if required.
11. Comprise with starter for manual power on/off the sum pump.

A set of internal level switches shall be provided to control the operation as under:

 - 1). Low level switch to stop the pumps.
 - 2). Level 1 switch to start the duty pump
 - 3). Level-2 switch to start the 'Duty/Assist' pump.
 - 4). Level 3 switch to raise a "high level" alarm.
12. Pumps shall operate as in (2) above except that duty selection shall be carried out by cyclic relay and duty selector switch shall be in Auto position.
13. Pumps shall be capable of operating independently or simultaneously with HOA switch in 'Hand' position overriding the float level switch.
14. In the event of duty pump failure, stand-by pump shall automatically operate.
15. Panel shall have on front door:
 - a. Duty selector switch

- b. Hand off auto switch for each pump
- c. Run and trip lamps for each pump
- d. Panel alive lamp
- e. Control circuit breaker fail lamp
- f. Door interlocked isolator
- g. Labels for above
- h. Earth leakage

16. Operation:

Each foul water and waste water pump is for 100% duty (i.e. one of the pumps is duty pump and the other is standby). However, in the event of excessive flow into the chamber, if the duty pump is unable to cope the standby pump shall be brought on to assist. Therefore, the panel design shall allow for two pumps working simultaneously

D. Building Automation System Interface

1. Provide auxiliary contacts in pump controllers for interface to building automation system. Include the following:
 - a. On-off status of each pump.
 - b. Alarm status.
 - c. Power on
 - d. All H.O.A switches and duty selection switch in auto position
 - e. Common fault on panel
 - f. High water level alarm

E. Level Regulator (Float Switch)

1. Level regulator (Float switch) for the Sewage/drainage shall be ABS float switch type KS with Micro switch in a plastic casing, shall freely suspended at the desired height from its own cable and counter weight. When the liquid reaches the pre-determined level, the micro switch shall close or open the circuit, hereby starting or actuating an alarm device.
2. Regulator casing shall be made of Polypropylene and cable shall be H07RN-3G1.0 or A05RN-4G0.75.
3. Length of the cable shall be suitably selected to connect the control panel without any joint.

2.3 MATERIALS

- A. General: Use only materials that comply with referenced standards and other requirements specified.

2.4 ACCESSORIES

- A. General: Use only materials that comply with referenced standards and other requirements specified.

2.5 FABRICATION

- A. As applicable as required to complete the work.

2.6 FINISHES

- A. General: Color and finish as selected and approved by the Engineer.

PART 3 EXECUTION

3.1 EARTHWORK

- A Excavation and filling are specified in Division 31 Section "Earth Moving."

3.2 EXAMINATION

- A Examine roughing-in for plumbing piping to verify actual locations of storm drainage piping connections before sump pump installation.

3.3 INSTALLATION

- A Pump Installation Standards: Comply with HI 1.4 for installation of sump pumps.

3.4 CONNECTIONS

- A Comply with requirements for piping specified in Division 22 Section "Facility Storm Drainage Piping." Drawings indicate general arrangement of piping, fittings, and specialties.
- B Install piping adjacent to equipment to allow service and maintenance.

3.5 FIELD QUALITY CONTROL

- A Manufacturer's Field Service: Engage a factory-authorized service representative to inspect, test, and adjust components, assemblies, and equipment installations, including connections.
- B Perform tests and inspections.
 - 1 Manufacturer's Field Service: Engage a factory-authorized service representative to inspect components, assemblies, and equipment installations, including connections, and to assist in testing.
- C Tests and Inspections:
 - 1 Perform each visual and mechanical inspection.
 - 2 Leak Test: After installation, charge system and test for leaks. Repair leaks and retest until no leaks exist.
 - 3 Operational Test: After electrical circuitry has been energized, start units to confirm proper motor rotation and unit operation.

- 4 Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.
- D Pumps and controls will be considered defective if they do not pass tests and inspections.
- E Prepare test and inspection reports.
- 3.6 STARTUP SERVICE
 - A Perform start up service.
 - 1 Complete installation and start up checks according to manufacturer's written instructions.
- 3.7 ADJUSTING
 - A Adjust pumps to function smoothly, and lubricate as recommended by manufacturer.
 - B Adjust control set points.
- 3.8 DEMONSTRATION
 - A Train Owner's maintenance personnel to adjust, operate, and maintain controls and pumps.

END OF SECTION

MECHANICAL SECTION: 22 32 00

DOMESTIC WATER FILTRATION & SOFTENING

PART 1 – GENERAL

1.1 RELATED DOCUMENTS

A Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A This Section includes the following types of water filtration equipment for water systems in buildings:

- 1 Bag-type filters.
- 2 Freestanding cartridge filters.
- 3 Off-floor cartridge filters.
- 4 Carbon filters.
- 5 Circulating sand filters.
- 6 Multimedia sand filters.
- 7 UV Sterilizer
- 8 Water Softener

B Related Sections include the following:

- 1 Division 22 Section "Domestic Water Piping Specialties" for small cartridge-type water filters for plumbing fixtures and for plumbing piping strainers.
- 2 Division 23 Section "Hydronic Piping" for hydronic piping strainers.

1.3 DEFINITIONS

- | | |
|---|---|
| A | ABS: Acrylonitrile-butadiene-styrene plastic. |
| B | FRP: Fiberglass-reinforced plastic. |
| C | NBR: Acrylonitrile-butadiene rubber. |
| D | PE: Polyethylene plastic. |
| E | PP: Polypropylene plastic. |
| F | ppm: Parts per million. |
| G | PVC: Polyvinyl chloride plastic. |

1.4 SUBMITTALS

- A. Product Data: For each type of product indicated.
- B. Shop Drawings: Detail water filtration equipment assemblies and indicate dimensions, weights, loads, required clearances, method of field assembly, components, and location and size of each field connection.
 - 1 Wiring Diagrams: Power, signal, and control wiring.
- C. Certificates of Shop Inspections and Data Reports: For products required to have ASME label, signed by product manufacturer.
- D. Welding certificates.
- E. Manufacturer Seismic Qualification Certification: Submit certification that water filtration equipment, accessories, and components will withstand seismic forces defined in Division 22 Section "Vibration and Seismic Controls for Plumbing Piping and Equipment." Include the following:
 - 1 Basis for Certification: Indicate whether withstand certification is based on actual test of assembled components or on calculation.
 - a The term "withstand" means "the unit will remain in place without separation of any parts from the device when subjected to the seismic forces specified."
 - b The term "withstand" means "the unit will remain in place without separation of any parts from the device when subjected to the seismic forces specified and the unit will be fully operational after the seismic event."
 - 2 Dimensioned Outline Drawings of Equipment Unit: Identify center of gravity and locate and describe mounting and anchorage provisions.
 - 3 Detailed description of equipment anchorage devices on which the certification is based and their installation requirements.
- F. Source quality-control test reports.
- G. Field quality-control test reports.
- H. Startup service test reports.
- I. Operation and Maintenance Data: For water filtration equipment to include in emergency, operation, and maintenance manuals.

1.5 QUALITY ASSURANCE

- A Source Limitations: Obtain each type of water filtration equipment through one source from a single manufacturer.
- B Product Options: Drawings indicate size, profiles, and dimensional requirements of water filtration equipment and are based on the specific system indicated. Refer to Division 01 Section "Product Requirements."

C Welding: Qualify processes and operators according to ASME Boiler and Pressure Vessel Code: Section IX.

D Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.

E Comply with NSF 61, "Drinking Water System Components - Health Effects; Sections 1 through 9," for all components that will be in contact with potable water.

1.6 PROJECT CONDITIONS

Interruption of Existing Water Service: Do not interrupt water service to facilities occupied by Owner or others unless permitted under the following conditions and then only after arranging to provide temporary water service according to requirements indicated:

- 1 Notify Construction Manager no fewer than two days in advance of proposed interruption of water service.
- 2 Do not proceed with interruption of water service without Construction Manager's written permission.

1.7 COORDINATION

Coordinate size and location of concrete bases with Architectural and Structural Drawings.

1.8 EXTRA MATERIALS

Furnish extra materials described below that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

- 1 Bag-Type Filter Bags: Bags for bag-type filters equal to 200 percent of amount installed for each size and media indicated.
- 2 Cartridge-Filter Elements: Elements for cartridge filters equal to 200 percent of amount installed for each size and media indicated.

PART 2 PRODUCTS

Packaged Filtration System

- 2.1.1 A packaged multimedia filtration system complete shall be supplied for turbidity odour and smell removal. Capacity shall be as indicated on the drawings.
- 2.1.2 Filter tank shall be suitable for a working pressure of 689 KPa (100psi) and tested at 50% in excess of the working pressure. Shell height to allow a minimum free board space of 100% of the mineral bed depth for adequate expansion during backwashing. Tank to be equipped with one manhole located in the top head of the tank. Filter tank to be supported by legs and be painted with a rust inhibiting primer. Filter tank shall be of glass reinforced polyester construction.

2.1.3 The collector system shall consist of a plate welded to the complete diameter of the tank and centrally supported by a pier. Plate to accommodate collectors spaced to collect and dispense water laterally. Multiple layers of filter media to be furnished to properly distribute the backwash water and support the mineral bed.

2.1.4 Following media shall consist of four layers as follows:

- a) Top layer to consist of light weight chips to retain large flat pieces of debris.
- b) Second layer to collect the bulk of course turbidity.
- c) Third layer to remove finer particles not collected in the upper layers.
- d) Bottom layer of high density material to polish the water removing particulates down to 10 micron size.

2.1.5 All filter media shall be of good quality possessing characteristics of durability, long life, resistance to attrition and shall not impart taste, odour or colour to the water being treated. Filter media shall be selected for removing particulates down to 10 micron size.

2.1.6 The filter unit shall be equipped with a fully automatic, self contained control valve using a cartridge design permitting instant service. Timer to automatically control the filter clean-up cycle on a pre set schedule. Duration of backwash and rinse cycles to be factory set to accommodate site conditions.

2.1.7 The filtration unit shall be suitable for removal of suspended matters, dechlorination, organic removal, and odour removal.

2.2 Cartridge Type Water Filter

2.2.1 Provide cartridge type water filters where indicated on drawings.

2.2.2 Filter shall be in-line to the appliance and installed in an accessible manner to facilitate cleaning easily. Filtering media shall be food grade polypropylene cartridge to achieve water quality upto 5 microns.

2.2.3 Filter casing shall be stainless steel. Cover shall be fitted with air vent of the spring release type.

2.3 PACKAGED WATER FILTRATION UNIT AND SOFTENING UNIT

2.3.1 Pre Chlorination

The raw water supplied by the Municipality has to be chlorinated by means of Sodium hypochlorite solution. The Sodium Hypochlorite Dosing set shall comprise of dosing tank (PVC), Electronic metering pump (PP).

DATA SHEET FOR THE CHLORINATION DOSING SYSTEM

Description	Data	Chlorination Details
Type of Chemical		Sodium Hypochlorite solution
<u>Dosing tank details</u>		
Number		1
Type		Vertical, Circular
Material of tank		Polyethylene
Capacity		300 Ltrs
Number		1
Type		Electronic Metering
Material of pump		PVC
Capacity		0 – 20 Lph

2.3.2 AUTO MULTIMEDIA FILTRATION UNIT

Type of Treatment for Removal of Suspended impurities

1. Vessel:

The filter vessels shall be fabricated from high grade steel and internally lined with epoxy paint. The External surface of the vessels shall be protected with a primer undercoat and Epoxy paint.

The vessel shall be designed for 100 PSI working pressure and tested to 150 PSI. The vessel shall be supplied internally coated with Zinc Epoxy and externally painted.

2.Valves :

Automatic Solenoid driven Pneumatic Diaphragm / Butterfly valve to control the operating steps. The valves shall be made of Carbon Steel. The water treatment equipment supplier as per the requirement shall arrange necessary compressed airline.

3. Frontal Piping

The frontal piping of the filter unit shall be u PVC surface protected with U.V resistant paint.

4. Media

The Multi Media Sand Filter bed shall consist of a top layer of a uniformly graded granular form of anthracite. The intermediate layer shall consist of three grades of Silica sand. The underbid shall consist of a layer of gravel, which is a granular material of extra ordinary density available in graded sizes for fine filtration.

Charge of carefully graded filtering Sand and Anthracite Media complete with supporting gravel underbid.

5. Pressure Gauge

Inlet / Outlet gauge shall be fitted to give indication of pressure drop through filter.

6. Controls

The filters shall be provided with Specialized PLC Controllers with timer mechanism to enable to program the backwash cycle of the filter at any time of the day. The PLC Controller shall complete with Soft Key Board and LCD display and shall give signals to the Solenoid driven Pneumatic Butterfly valves.

2.3.3 DUPLEX SOFTENER

1. The unit shall consist of softener vessels for continuous supply of the softened water. The plant shall be capable of producing zero temporary hardness at the rate as scheduled.
2. The Duplex softener should consist of two tanks in CS Epoxy Internally and Externally construction. Pressure rating for the tanks is 10 bars.
3. Control valve mounted on the vessel shall be constructed from brass for strength, durability and corrosion resistance to ensure longer life. The controller shall be equipped with volume based Electronic controller in which regeneration is initiated after specified volume of water has passed.
4. Setting conditions for volume based regeneration shall be programmed depending on raw water hardness.
5. Provision for Manual initiation of regeneration shall be provided.
6. The interconnecting pipe system shall be specially designed for the units and hydraulically balanced to allow uniform load on each resin tank. These shall be dimensioned to ensure minimum pressure loss in the pipe system.
7. The unit shall come with a brine tank of PVC / Fiberglass material of 300 litres capacity.

2.3.4 ULTRA-VIOLET STERILIZER

1. The UV unit shall utilize a compact integrated design, incorporating both the UV treatment chamber and electrical base, where all electrical components and instrumentation are located. These compact one piece UV models shall be engineered for indoor installations and controlled operating environments. The unit shall be suitable for the design flow mentioned on drawings.
2. The unit shall have the following features:
 - 316L stainless steel treatment chamber.
 - Endplate, mounted 360 UV.
 - Intensity sensor.
 - 4-20mA Output signal.

- Temperature safety control.
- Lamp out alert circuit.
- Sanitary inlet and outlet fittings.
- UV dosage > 30,000 micro watt-sec/cm².

2.3.5 SOURCE QUALITY CONTROL

A Hydrostatically test before shipment to minimum of one and one-half times pressure rating.

B Prepare test reports.

2.4 Domestic Water treatment:

1. Water treatment (automatic dosing & alarm) for domestic water must be implemented to achieve at least WHO standard.
2. Protection against corrosion and scale formation in all technical circuits.

2.4.1 Treatment of water for domestic use

1. Cold water for domestic use

Objective:

- Total TH (total hardness) not important if below 120 ppm approx.
- PH above 7 and below 8.5
- Bacteriological quality in accordance with criteria specified in one above.

Treatment equipment

- Softener to adjust total TH where necessary to correct level.
- Dosing pump for injection of alkaline drinkable product (caustic soda or carbonate of soda) to bring Ph above 7.
- Dosing pump with supply of chlorinated solutions.
- give bacteriological treatment of the water if necessary, or approved
- System of sterilization by ultra violet rays (UVC).

2. Closed circuit for technical equipment (heating circuit, chilled water circuit).

Objectives:

- total TH = 0 ppm
- Ph above or equal to 8.5

Treatment equipment

- Softener to bring down total TH to 0 ppm
- Movement pot for alkaline product balls for correction of Ph below 8.5.

a) **Air cooling circuit**

Objectives:

- TH = 0 ppm
- Ph above 8.5
- Anti-algae

Treatment equipment

- Softener to bring TH to 0 ppm
- Dosing pump for injection of alkaline product in order to bring Ph above 8.5
- Proportional dosing pump for injection of anti-algae product (Bleach or chlorinated product).

b) Hot water for domestic use

- . Objective:
 - TH = 120ppm

Treatment equipment:

- Softener to bring TH to correct value,

Dosing pump for film forming treatment, with reactor tank and pulse meter allowing for dosage in relation to actual water consumption and not in relation to theoretical throughput as for alkaline treatment

PART 3 EXECUTION

3.1 EXAMINATION

A Examine areas and conditions, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of filters.

B Examine roughing-in for piping systems to verify actual locations of piping connections before equipment installation.

C Examine walls and floors for suitable conditions where filters will be installed.

D Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 CONCRETE BASES

A Anchor filters to concrete bases.

1 Install dowel rods to connect concrete base to concrete floor. Unless otherwise indicated, install dowel rods on 18-inch (450-mm) centers around full perimeter of base.

2 For supported equipment, install epoxy-coated anchor bolts that extend through concrete base and anchor into structural concrete floor.

3 Place and secure anchorage devices. Use setting drawings, templates, diagrams, instructions, and directions furnished with items to be embedded.

4 Install anchor bolts to elevations required for proper attachment to supported equipment.

5 Concrete base construction requirements are specified in Division 22 Section "Common Work Results for Plumbing."

6 Cast-in-place concrete materials and placement requirements are specified in Division 03.

3.3 CARBON-FILTER INSTALLATION

A Install carbon filters on concrete bases, level and plumb.

1 Prepare carbon-filter tank distribution system and underbid, if any, for filter media and place specified media into tanks.

B Install seismic restraints for carbon-filter housings and anchor to building structure.

3.4 SAND-FILTER INSTALLATION

A Install sand-filter tanks on concrete bases, level and plumb.

1 Prepare sand-filter tank distribution system and underbid for filter media and place specified sand and other media into tanks.

B Install seismic restraints for sand-filter tanks and accessories and anchor to building structure.

3.5 CONNECTIONS

A Piping installation requirements are specified in other Division 22 Sections. Drawings indicate general arrangement of piping, fittings, and specialties.

B Install piping adjacent to equipment to allow service and maintenance.

C Make piping connections between water filtration equipment and dissimilar-metal water piping with dielectric fittings. Dielectric fittings are specified in Division 22 Section "Common Work Results for Plumbing."

D Install shutoff valves on feed water-inlet and filtrate-outlet piping of each water filtration equipment filter and on inlet and outlet headers.

1 Metal general-duty valves are specified in Division 22 Section "General-Duty Valves for Plumbing Piping."

2 Plastic valves are specified in Division 22 Section "Domestic Water Piping."

3 Exception: Water filtration equipment with factory-installed shutoff valves at locations indicated.

E Install pressure gages on feed water-inlet and filtrate-outlet piping of each water filtration equipment filter. Pressure gages are specified in Division 22 Section "Meters and Gages for Plumbing Piping."

1 Exception: Water filtration equipment with factory-installed pressure gages at locations indicated.

2 Exception: Cartridge water filters.

F Install valve bypass water piping around each water filtration equipment filter.

1 Metal general-duty valves are specified in Division 22 Section "General-Duty Valves for Plumbing Piping."

2 Plastic valves are specified in Division 22 Section "Domestic Water Piping."

3 Water piping is specified in Division 22 Section "Domestic Water Piping."

4 Exception: Bag-type water filtration equipment.

5 Exception: Cartridge water filtration equipment.

G Install drains as indirect wastes to spill into open drains or over floor drains.

H Ground equipment according to Division 26 Section "Grounding and Bonding for Electrical Systems."

I Connect wiring according to Division 26 Section "Low-Voltage Electrical Power Conductors and Cables."

3.6 FIELD QUALITY CONTROL

A Manufacturer's Field Service: Engage a factory-authorized service representative to inspect, test, and adjust field-assembled components and

equipment installation, including connections, and to assist in field testing. Report results in writing.

B Perform the following field tests and inspections and prepare test reports:

1 Leak Test: After installation, charge system and test for leaks. Repair leaks and retest until no leaks exist.

2 Operational Test: After electrical circuitry has been energized, start units to confirm proper unit operation.

3 Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.

C Remove and replace water filtration equipment that does not pass tests and inspections and retest as specified above.

3.7 STARTUP SERVICE

A Engage a factory-authorized service representative to perform startup service for carbon- and sand- filter water filtration equipment.

1 Complete installation and startup checks according to manufacturer's written instructions.

B Sample filter filtrate after startup and at three consecutive seven-day intervals (total of four samples), and prepare certified test reports for required water performance characteristics.

3.8 DEMONSTRATION

A Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain carbon- and sand- filter water filtration equipment. Refer to Division 01 Section "Demonstration and Training."

END OF SECTION

MECHANICAL SECTION: 22 33 30

ELECTRIC WATER HEATERS

PART 1 GENERAL

1.1 WORK INCLUDED

1.1.1 Compliance with relevant General Requirements of Mechanical Services.

1.2 DESCRIPTION OF WORK

This section describes the Electric Domestic Water Heaters as part of domestic water storage and distribution system.

The Contractor shall engineer, supply, install, test and commission all items to comply with the design.

The Contractor shall ensure that the proposed locations of heaters are adequate and enough space is available for removal of elements and vessel itself for servicing and replacement.

All heaters shall be factory assembled and electrically prewired.

For other related plumbing connections like water pipes, pumps, pipe specialties refer relevant sections of mechanical services.

For other related electrical connections like starters, wires and cables refer to relevant sections of electrical services.

1.3 QUALITY ASSURANCE

1.3.1 Manufacturer's Qualifications: Firms regularly engaged in manufacture of water heaters of types and capacities required, whose products have been in satisfactory use in similar service for not less than 5 years.

1.3.2 Codes and Standard:

BS 853, "Clarifiers and storage vessels for central heating and hot water supply".

UL 1453, "Electric Booster and Commercial Storage Tank Water Heaters".

Provide water heater components, which are UL listed and labeled.

Electric hot water storage shall be complied BS 853-1 1990, BS 853-2 1996 for Calorifiers and storage vessels for heating and hot water supply.

1.4 SUBMITTALS

1.4.1 Product Data: Submit manufacturer's technical product data including rated capacities and efficiencies of selected model clearly indicated; operating weights; furnished specialties and accessories; and installation and start up instructions.

1.4.2 Shop Drawings: Submit manufacturers assembly type shop drawings indicating dimensions, required clearances, and methods of assembly of components.

- 1.4.3 Wiring Diagrams: Submit manufacturer's electrical requirements for power supply wiring to water heaters. Submit manufacturer's ladder-type wiring diagrams for interlock and control wiring required for final installation of water heaters and controls. Differentiate between portions of wiring that are factory installed and portions that are to be field installed.
- 1.4.4 Maintenance Data: Submit maintenance data and parts lists for each type and size of water heater, control, and accessory; including troubleshooting" maintenance guide. Include this data, product data, shop drawings, and wiring diagrams in maintenance manual.
- 1.4.5 Certificates: Submit appropriate certificate of shop inspection and data report as required by provisions of ASME Boiler and Pressure Vessel Code.
- 1.5 DELIVERY, STORAGE AND HANDLING
 - 1.5.1 Handle water heaters and components carefully to prevent damage, breaking, denting and scoring. Do not install damaged water heaters or components; remove from site and replace with new.
 - 1.5.2 Store water heaters and components in clean dry place. Protect from weather, dirt, fumes, water, construction debris, and physical damage.
 - 1.5.3 Comply with manufacturer's rigging and installation instructions for unloading water heaters, and moving units to final location for installation.

PART 2 PRODUCTS

2.1 DOMESTIC INDIVIDUAL WATER HEATER

Hot water heater shall be horizontal type to suit the location and space availability. Where specified these shall be suitable for under sink installation. Vessel shall be glass lined steel. Heating element shall be copper sheathed electric immersion type.

Each heater shall be provided with a thermostat with adjustable setting in the range 32 to 82 deg. C and a safety thermal cut out operable at 90 deg. C with manual reset.

The vessel and electrical components shall be encased in a sheet metal enclosure suitable for a working pressure of 600 Kpa with baked enamel finish. Tank shall be insulated with polyurethane insulation. A temperature indicator mounted on the casing shall display the temperature of water. Each heater shall be supplied with drain cock and 20 dia safety valve, discharging to the building waste water drainage system via a floor trap.

PART 3 EXECUTION

3.1 INSTALLATION

- 3.1.1 Install water heater in accordance with manufacturer's installation instructions. Install units plumb and level, firmly anchored in locations indicated, and maintain manufacturer's recommended clearances.
- 3.1.2 Support: Place units on concrete pads orient so controls and devices needing services and maintenance have adequate access.

- 3.1.3 Piping: Connect hot and cold water piping to units with shut off valves and unions, connect recirculating water line to unit with shut off valve, check valve, and union. Extend relief valve discharge to closest floor drain.
- 3.1.4 Gages: Provide thermometers and pressure gauges on inlet and outlet piping of water heaters and recirculating pump, in accordance with Basic Mechanical materials and methods section "Meters and gages"
- 3.1.5 Electrical wiring: Install electrical devices furnished by manufacturer but not specified to be factory mounted. Furnish copy of manufacturer's wiring diagram submittal to electrical installer.

Verify that electrical wiring installation is in accordance with manufacturer's submittal and installation requirements of electrical specifications. Do not proceed with water heater start up until wiring installation is acceptable.

- 3.1.6 Start up : Start up, test and adjust electric water heaters in accordance with manufacturer's start up instructions. Check and calibrate controls.

END OF SECTION

MECHANICAL SECTION: 22 40 00

PLUMBING FIXTURES

PART 1 - GENERAL

General conditions apply to this section

1.1 RELATED DOCUMENTS

- A. This Section includes plumbing fixtures and trim, faucets, other fittings, and related components.
- B. Related Sections: The following Sections contain requirements that relate to this Section:

1.2 DEFINITIONS

- A. Accessible: Plumbing fixture, building, facility, or portion thereof that can be approached, entered, and used by physically handicapped, disabled, and elderly people.
- B. Fitting: Device that controls flow of water into or out of plumbing fixture. Fittings specified in this Section include supplies and stops, faucets and spouts, shower heads and tub spouts, drains and tailpieces, traps and waste pipes. Pipe fittings, tube fittings, and general-duty valves are included where indicated.

1.3 SUBMITTALS

- A. General: Submit each item in this Article according to the Conditions of the Contract and Division 1 Specification Sections.
- B. Product Data for each plumbing fixture category and type specified. Include selected fixture, trim, fittings, accessories, appliances, appurtenances, equipment, and supports. Indicate materials and finishes, dimensions, construction details, and flow-control rates.
- C. Wiring diagrams from manufacturer for electrically operated units.
- D. Maintenance data for plumbing fixtures and components to include in the operation and maintenance manuals specified in Division 1.

1.4 QUALITY ASSURANCE

- A. Source Limitations: Obtain plumbing fixtures, faucets, and other components of each category from one source and by a single manufacturer.

Exception: Where fixtures, faucets, or other components are not available from a single manufacturer, obtain similar products from other manufacturers specified for this category.

- B. Regulatory Requirements: Comply with requirements of CABO A117.1, “Accessible and Usable Buildings and Facilities”; Public Law 90-480, “Architectural Barriers Act”; and Public Law 101-336, “Americans with Disabilities Act”; regarding plumbing fixtures for physically handicapped people.
- C. Regulatory Requirements: Comply with requirements of Architectural and Transportation Barriers Compliance Board’s (ATBCB) “Uniform Federal Accessibility Standards (UFAS), 1985-494-187” regarding plumbing fixtures for physically handicapped people.
- D. Energy Policy Act Requirements: Comply with requirements of Public Law 102-486, “Energy Policy Act,” regarding water flow rate and water consumption of plumbing fixtures.
- E. Listing and Labeling: Provide electrically operated fixtures and components specified in this Section that are listed and labeled.
 - The Terms “Listed” and “Labeled”: As defined in the National Electrical Code, Article 100.
 - Listing and Labeling Agency Qualifications: A “Nationally Recognized Testing Laboratory” (NRTL) as defined in OSHA Regulation 1910.7.
- F. Select combinations of fixtures and trim, faucets, fittings, and other components that are compatible.
- G. Product Options: Drawings indicate size, profiles, dimensional requirements, and characteristics of plumbing fixtures and are based on specific types and models indicated. Other manufacturers’ fixtures with equal performance characteristics may be considered. Refer to Division 1 Section “Substitutions.”

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Deliver plumbing fixtures in manufacturer’s protective packing, crating, and covering.
- B. Store plumbing fixtures on elevated platforms in dry location.

1.6 PROJECT CONDITIONS

- A. Field Measurements: Coordinate roughing-in and final fixture locations and verify that plumbing fixtures can be installed to comply with original design and referenced standards.

PART 2 - PRODUCTS

2.1 Materials

- A. Refer Architect Specifications, Fixture Selections and Schedules. In the Absence of such table refer the following values.

2.2 FAUCETS

- A. Lavatory Faucets, <Refer Architecture fixture selection and Specifications>:
 - 1. Two-handle mixing pressure type valve.
 - a. Body Material: Commercial, solid brass (with non-metallic coatings or Chrome finish as required by the architect.
 - b. Finish: Polished chrome plate.
 - c. Maximum Flow Rate: Maximum [6.0 L/min.),
 - d. Mounting: As per Architecture Finishing schedule.
 - e. Spout Outlet: Aerator/ Spray
- B. Sink Faucets, <Refer Architecture fixture selection and Specifications>:
 - 1. [Kitchen faucet with spray, three-hole fixture].
 - a. Body Material: Solid brass (with non-metallic coatings or Chrome finish as required by the architect).
 - b. Finish: Polished chrome plate
 - c. Maximum Flow Rate: 7.0 L/min.), unless otherwise indicated.
 - d. Mixing Valve: Two-lever handle.
 - e. Backflow Protection Device for Hose Outlet: Not required.
 - f. Mounting: Deck, Back/wall as recommended by the architect.
 - g. Spout Outlet: Aerator/ Spray
 - h. Drain: Stopper with chain.
- C. Showers, <Refer Architecture fixture selection and Specifications>:
 - a. Finish: Stainless Steel
 - b. Maximum Flow Rate: 8.0 L/min.), unless otherwise indicated.

2.3 TOILET SEATS

- A. Toilet Seats:
 - 1. Toilet seat for water-closet-type fixture.
 - a. Material: Molded, solid plastic.
 - b. Configuration: front with cover.
 - c. Size: Regular.
 - d. Class: Standard commercial.

2.4 FIXTURE SUPPORTS

- A. Water-Closet Supports:
 - 1. Combination carrier designed for standard mounting height of wall-mounting, water-closet-type fixture.

2.5 WATER CLOSETS

A. Water Closets, <Refer Architecture fixture selection and Specifications> :

1. Floor-mounting, floor-outlet, vitreous-china
 - a. Style: [Close coupled] [One piece].
 - 1) Bowl Type: Round front design. Include bolt caps matching fixture.
 - 2) Height: Standard.
 - 3) Design Consumption: Dual Flush Design with Major (6.0 L/flush) Minor (3 L/flush)].
 - 4) Tank: Gravity type. Include cover.
 - b. Supply: NPS 1/2 (DN 15) chrome-plated brass or copper with wheel-handle stop.
 - c. Toilet Seat: Included

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine roughing-in for potable, hot- and cold-water supply piping systems; soil, waste, and vent piping systems; and supports. Verify that locations and sizes of piping and locations and types of supports match those indicated, before installing and connecting fixtures. Use manufacturer's roughing-in data when roughing-in data are not indicated.
- B. Examine walls, floors, and cabinets for suitable conditions where fixtures are to be installed.
- C. Do not proceed until unsatisfactory conditions have been corrected.

3.2 APPLICATIONS

- A. Include supports for plumbing fixtures according to the following:
 1. Carriers: For wall-hanging water closets and fixtures supported from wall construction.
 2. Chair Carriers: For wall-hanging urinals, lavatories, sinks, drinking fountains, and electric water coolers.
 3. Heavy-Duty Chair Carriers: For accessible urinals, lavatories, and other fixtures where indicated.
 4. Reinforcement: For floor-mounted lavatories and sinks that require securing to wall and recessed, box-mounted, electric water coolers.
 5. Fabricate reinforcement from 2-by-6-inch fire-retardant-treated-wood blocking between studs steel plates attached to studs, in wall

construction, to secure fixtures to wall. Include length that will extend beyond ends of fixture mounting bracket and attach to at least 2 studs.

- B. Include fitting insulation kits for accessible fixtures according to the following:
 - 1. Lavatories: Cover hot- and cold-water supplies, stops and handles, drain, trap, and waste to wall.
 - 2. Lavatories: Cover hot-water supply, stop and handle, drain, trap, and waste to wall.
 - 3. Sinks: Cover hot- and cold-water supplies, stops and handles, drain, trap, and waste to wall.
 - 4. Sinks: Cover hot-water supply, stop and handle, drain, trap, and waste to wall.
 - 5. Fixtures with Offset Drain: Cover hot- and cold-water supplies, offset drain, trap, and waste to wall.
 - 6. Fixtures with Offset Drain: Cover hot-water supply, offset drain, trap, and waste to wall.
 - 7. Other Fixtures: Cover exposed fittings below fixture.

3.3 PLUMBING FIXTURE INSTALLATION

- A. Assemble plumbing fixtures and trim, fittings, faucets, and other components according to manufacturers' written instructions.
- B. Install fixtures level and plumb according to manufacturers' written instructions, roughing-in drawings, and referenced standards.
- C. Install floor-mounted, floor-outlet water closets with closet flanges and gasket seals.
- D. Install floor-mounted, back-outlet water closets with fittings and gasket seals.
- E. Install wall-hanging, back-outlet water closets with support manufacturer's tiling frame or setting gage.
- F. Install toilet seats on water closets.
- G. Install wall-hanging, back-outlet urinals with gasket seals.
- H. Install flushometer valves for accessible water closets and urinals with handle mounted on wide side of compartment. Install other actuators in locations that are easy for handicapped people to reach.
- I. Install tanks for accessible, tank-type water closets with lever handle mounted on wide side of compartment.
- J. Fasten wall-hanging plumbing fixtures securely to supports attached to building substrate when supports are specified, and to building wall construction where no support is indicated.

- K. Fasten floor-mounted fixtures to substrate. Fasten fixtures having holes for securing fixture to wall construction, to reinforcement built into walls.
- L. Fasten recessed, wall-mounted fittings to reinforcement built into walls.
- M. Fasten wall-mounted fittings to reinforcement built into walls.
- N. Fasten counter-mounting plumbing fixtures to casework.
- O. Secure supplies to supports or substrate within pipe space behind fixture.
- P. Set shower receptors and mop basins in leveling bed of cement grout.
- Q. Install individual stop valve in each water supply to fixture. Use gate or globe valve where specific stop valve is not specified.
 - 1. Exception: Omit stop valves on supplies to emergency equipment, except when permitted by authorities having jurisdiction. When permitted, install valve chained and locked in OPEN position.
- R. Install water-supply stop valves in accessible locations.
- S. Install faucet, laminar-flow fittings with specified flow rates and patterns in faucet spouts when faucets are not available with required rates and patterns. Include adapters when required.
- T. Install supply, flow-control fittings with specified flow rates in fixture supplies at stop valves.
- U. Install faucet, flow-control fittings with specified flow rates and patterns in faucet spouts when faucets are not available with required rates and patterns. Include adapters when required.
- V. Install shower, flow-control fittings with specified maximum flow rates in shower arms.
- W. Install traps on fixture outlets. Omit traps on fixtures having integral traps. Omit traps on indirect wastes, except where otherwise indicated.
- X. Install disposers in sink outlets. Install switch where indicated, or in wall adjacent to sink if location is not indicated.
- Y. Install hot-water dispensers in back top surface of sink or in counter with spout over sink.
- Z. Install escutcheons at wall, floor, and ceiling penetrations in exposed, finished locations and within cabinets and millwork. Use deep-pattern escutcheons where required to conceal protruding pipe fittings.
- AA. Seal joints between fixtures and walls, floors, and counters using sanitary-type, 1-part, mildew-resistant, silicone sealant according to sealing requirements

specified in Division 7 Section "Joint Sealants." Match sealant color to fixture color.

3.4 CONNECTIONS

- A. Piping installation requirements are specified in other Division 15 Sections. Drawings indicate general arrangement of piping, fittings, and specialties. The following are specific connection requirements:
 - 1. Install piping connections between plumbing fixtures and piping systems and plumbing equipment specified in other Division 15 Sections.
- B. Supply and Waste Connections to Plumbing Fixtures: Refer to plumbing fixture schedules at the end of this Section for fitting sizes and connection requirements for each plumbing fixture.
- C. Supply and Waste Connections to Equipment Specified in Other Sections: Connect equipment with supply inlets, supply stops, supply risers, and traps specified in this Section. Use fitting sizes required to match connected equipment. Connect fittings to plumbing piping.
- D. Ground equipment.
 - 1. Tighten electrical connectors and terminals according to manufacturer's published torque-tightening values. Where manufacturer's torque values are not indicated, use those specified in UL 486A and UL 486B.
- E. Arrange for electric-power connections to fixtures and devices that require power. Electric power is specified in Division 16 Sections.

3.5 FIELD QUALITY CONTROL

- A. Verify that installed fixtures are categories and types specified for locations where installed.
- B. Check that fixtures are complete with trim, faucets, fittings, and other specified components.
- C. Inspect installed fixtures for damage. Replace damaged fixtures and components.
- D. Test installed fixtures after water systems are pressurized and demonstrate proper operation. Replace malfunctioning fixtures and components, then retest. Repeat procedure until units operate properly.

3.6 ADJUSTING AND CLEANING

- A. Operate and adjust faucets and controls. Replace damaged and malfunctioning fixtures, fittings, and controls.
- B. Operate and adjust disposers, hot-water dispensers, and controls. Replace damaged and malfunctioning units and controls.
- C. Adjust water pressure at drinking fountains, electric water coolers, faucets, shower valves, and flushometer valves having controls, to produce proper flow and stream.
- D. Replace washers and seals of leaking and dripping faucets and stops.
- E. Clean fixtures, faucets, and other fittings with manufacturers' recommended cleaning methods and materials. Include the following:
 - 1. Remove faucet spouts and strainers, remove sediment and debris, and reinstall strainers and spouts.
 - 2. Remove sediment and debris from drains.

3.7 PROTECTION

- A. Provide protective covering for installed fixtures and fittings.
- B. Do not allow use of fixtures for temporary facilities, except when approved in writing by Owner.

END OF SECTION